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ANTHROPOLOGY

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ANTHROPOLOGY

STUDY MATERIAL

PAPER - I

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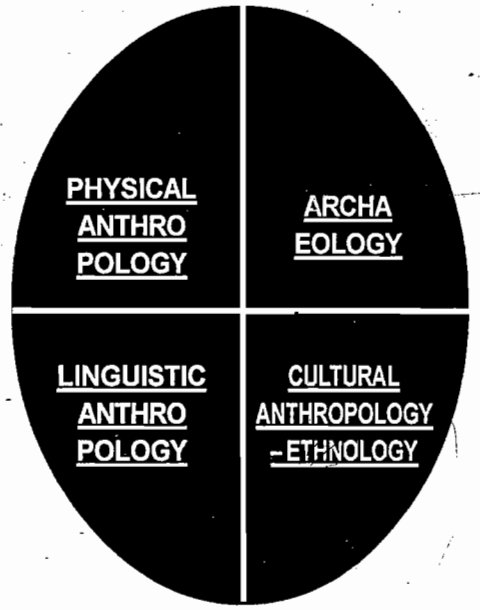
CHAPTER - 1

ANTHROPOLOGY AND ITS SUBFIELDS

The word *'anthropology'* stems from the Greek words *'anthropo, meaning human beings or humankind, and 'logia'*, translated as knowledge of or the study of. Thus, we can define **anthropology** as the systematic study of humankind.

Historians, Psychologists, economists, sociologists and scholars in many other fields systematically study humankind in one way or another. Anthropology stands apart because it combines four subdisciplines, or subfields, that bride the natural sciences, the social sciences, and the humanities. These four subdisciplines – physical anthropology, archaeology, linguistic anthropology, archaeology, linguistic anthropology, and ethnology or cultural anthropology – give anthropologists a board approach to the study of humanity the world over, both past and present.

- Biological Anthropology
- Forensic Anthropology
- Paleoanthropology
- Human Anatomy
- Human Taxonomy
- Paleopathology
- Primatology
- Ethology
- Population Genetics
- Human Ecology



- Prehistoric Archaeology
- Historical Archaeology
- Classical Archaeology
- Demographic Archaeology
- Biblical Archaeology
- Maritime Archaeology
- Underwater Archaeology
- Urban Archaeology
- Ethnoarchaeology
- Industrial Archaeology
- Cognitive Archaeology
- Cultural Resource Management

- Structural Linguistics
- Historical Linguistics
- Phonology
- Morphology
- Comparative Syntax
- Ethhnosemantics
- Cognitive Linguistics
- Pragmatics
- Sociolinguistics

- Ecological Anthropology
- Demographic Anthropology
- Economic Anthropology
- Social Anthropology
- Political Anthropology
- Legal Anthropology
- Anthropology of Religion
- Psychological Anthroplogy
- Medical Anthropology
- Urban Anthropology
- Applied Anthropology
- Ethnomusicology
- Anthropology of Art
- Ethnopoetics

1. PHYSICAL ANTHROPOLOGY

Physical anthropology is the branch of anthropology concerned with humans as a biological species. Physical anthropologists conduct research in two major areas: human evolution and modern human variation. Research has now traced the African origins of humanity back over six million years. Fieldwork in other world areas has traced the expansion of early human ancestors throughout the world. Much of the evidence for human origins consists of **fossils,** the fragmentary remains of bones and living materials preserved from earlier periods. The study of human evolution through analysis of fossils is called **paleoanthropology.**

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The second branch is **primatology,** the study of primates. **Primates** are mammals that belong to the same overall biological classification as humans and therefore share similar physical characteristics and a close evolutionary relationship with us.

Another group of physical anthropologists focuses their research on the range of physical variation within and among different 'modern' human populations. These anthropologists study human variation by measuring physical characteristics, such as body size, variation in blood types, differences in skin colour, or various genetic traits.

An increasingly important area of research for some physical anthropologists is genetics, the study of the biological 'blueprints' that dictate the inheritance of physical characteristics. Genetics has become an increasingly important complement to paleoanthropological research. Through the study of the genetic makeup of modem humans, physical anthropologists have been working on calculating the genetic distance among modern humans, thus providing a means of infering rates of evolution and the evolutionary relationships within the species.

2. ARCHAEOLOGY

Through **archaeology,** the branch of anthropology that seeks out and examines the artifacts of past societies, we learn about the culture of those societies – the shared way of life of a group of people that includes their values, beliefs, and norms.

Archaeologists have examined sites the world over, from campsites of the earliest humans to modern landfills. Some archaeologists investigate past societies whose history is primarily told byy the archaeological record. Known as *'prehistoric archaeologists'*, they study the artifacts of groups such as the ancient inhabitants of Europe or the first humans to arrive in the Americas specialisation in Archaeology.

a. *Historical archaeologists,* work with historians in investigating the artifacts of societies of the more recent past.

b. *Industrial archaeologists,* biblical archaeologists, medieval and postmedieval archaeologists, and Islamic archaeologists.

c. *Underwater archaeologists* work on a variety of places and time periods the world over; they are distinguished from other archaeologists by the distinctive equipment, methods and procedures needed to excavate under water.

Still other archaeologists have turned their attention to the very recent past. For example, in 1972 William L. Rathje began a study of modern garbage. A careful study of garbage reveals insights about modern society that cannot be ferreted out in any other way.

3. LINGUISTIC ANTHROPOLOGY

Linguistic Anthropology focuses on the relationship between language and culture, how language is used within society, and how the human brain acquires and uses language. Linguistic anthropologists seek to discover the ways in which languages are different from one another as well as how they are similar to one another. Two wide ranging areas of research in linguistic and historical linguistics.

a. **Structural linguistics** explores how language works. Structural linguistics compare grammatical patterns or other linguistic elements to learn how contemporary languages mirror and idffer from one antoher. Structural linguistics between language and thought patterns among different groups of people.

b. **Linguistic anthropologists** also examine the connections between language and social behavior in different cultures. This specialty is called **socio linguistics**. Sociolinguistics are interested both in how language is used to define social groups and how belonging to particular groups leads to specialized kinds of language use.

c. **Historical linguistics** concentrates on the comparison and classification of different languages to discern the historical links among languages.

4. CULTURAL ANTHROPOLOGY OR ETHNOLOGY

Cultural anthropology or ethnology, is the subfield of anthropology that examines various contemporary societies and cultures throughout the world.

Cultural anthropologists or ethnologists use a unique research strategy in conducting their fieldwork in different settings. This research strategy is referred to as **participant observation,** which involes learning the language and culture of the group being studied by participating in the group's daily activities.

The results of the fieldwork of the cultural anthropologists are written up as an **ethnography,** a description of a society. A typical ethnography reports on the environmental setting, economic patterns, social organization, political system, and religious rituals and beliefs of the society under study. This description of a society is based on what anthropologists call ethnographic data. The gathering of *'ethnographic data'* in a systematic manner is the specific research goal of the cultural anthropologist.

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RELATIONSHIP OF ANTHROPOLOGY WITH OTHER DISCIPLINES

The scope of anthropology is much broader than that of any other disciplines. Eg ; Economics deals only with economic behaviour, political science only with political behaviour, and so on. Anthropology, on the other hand,is concerned with all of those areas, from a comparative perspective, and specially with the interrelationships between these areas of behaviour in any particular society. It overlaps with every other disciplines in at least some areas of interest, yet it stills retains its individuality.

The commonality of all disciplines along with anthropology is that, all share a common interest in understanding human behaviour and differences exist in the field of its approach. Anthropology utilizes the holistic approach to investigate all aspects of human behaviour, in all places and at all times due to the recent advancement in science and technology and due to the wider use of anthropology in developmental processes and curing the ills of traditional societies, it also has the relationship with life science, medical sciences and development science.

1. ANTHROPOLOGY AND HISTORY

Anthropology	History
■ This study associated with past, present and future	■ Study of past
■ It studies the biological evolution of man	■ It studies only the cultural evolution not biological

No anthropologist can work without an awareness on the past, of what the particular sequence of events was that led to the situation under study i.e., History In describing antoher culture, or in comparing aspects of two cultures, the anthorpologist does not ignore the historical background.

Yet history and anthropology are distinct disciplines. Historians focus on past events, and their investigation of values, motivations, and behaviour is directed toward the explanation of why things occurred the way they did in one particular sequence of events. Anthropology seeks to generalize from such historical explanations.

It is difficult do distinguish history from anthropology in many respects, except to point out the historian's retrospective approach to past events, in contrast to the cultural anthropologists emphasis upon understanding contemporary events as they are happening.

2. ANTHROPOLOGY AND SOCIOLOGY

Anthropology	Sociology
■ Basically Related with the study of primitive societies	■ Basically Related with the study of modern societies,
■ It also studies the bio-logical aspect of man,	■ It has nothing to do with the biological aspect of man

Both the subjects share a common interest in human society, in the behaviour of humans in groups, and in networks of social relations. However, there are differences between the practise of sociology and the practise of Anthropology which result from the different kind of societies which each discipline has traditionally selected for study. Thus sociologists are best known for their studies of only one kind of society, the industrial or modern society. Anthropologists, on the other hand, have tended to concentrate on primitive and peasant societies. Following from the different kind of society which the two disciplines have studied are differences in the techniques of date collection and analysis employed. Sociologists have extensively used questionnaires and other means of gathering quantifiable, or countable data. Accordingly, expertise in statistical techniques and formal methods of analysis has been more widely valued in sociology than in anthorpology. Sociologists have concentrated on societies in which their informants or respondents - the poeple who answer their question and their objects of study are often literate. Anthorpologists, on the other hand, have worked most of non-literate societies or who, as peasants within a nation-state are illiterate. In addition, the anthropologists traditionally have worked in small, simple societies, while the sociologists have been concerned with analyzing nation-states, which are usually much more complex.

Today there is increasing covergence between anthropology and sociology. Both in subject matter and in techniques, sociologists are now carrying out research in areas of the world that were once almost exclusively within the anthropological orbit, and anthropologists are undertaking studies in areas of industrial society, for example in urban ghettos in the United States. As anthropologists extend the scope of their techniques of data colletion used be sociologists. Similarly, sociologists have found that there is something of value-in

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anthropologic techniques and are increasing their use of informal interviews.

3. ANTHROPOLOGY AND ECONOMICS

Anthropology	Economics
■ Basically related with primitive societies,	■ Basically related with modern societies,
■ Approach of economic anthropologists is holistic	■ Approach of economists is inductive

The social science of economics concentrates on particular areas of human behaviour. Economists have tended to limit their studies to modern societies. In the primitive and peasant societies which the anthropologists have traditionally studied economics do not stand out as distinct areas of human activity amenable to separate analysis, as they do in modern societies. In primitive societies the economic orders are usually submerged in the social order.

In comparison to Anthropology, Economics is one of the oldest and theoretically most sophisticated disciplines. In its long tradition, economists have developed a number of technical terms. In recent decades, few cultural anthoropologists have tried to borrow some general ideas from economics; other strongly felt that it would be irrelevant to explain the Economic behaviour of pre-industrial (non-literate) societies in term of formal economics which were developed for the industrial societies. Anthropologists study of physical aspect of man which has no use of Economics.

4. ANTHROPOLOGY AND PSYCHOLOGY

Anthropology	Psychology
Basically concerned with the study of human behaviour in cultural perspective of primitive societies,	Basically concerned with the study of human behaviour in all kind of societies.

Psychologists have a tendency to gather their data and make their observations in the society of which they are themselves members. Anthropologists have placed on a comparative basis psychological statements founded on the observation of behaviour in a single society or a single type of society. The most important in this regard is the subdivision of socio-cultural anthropology known as the study of personality in culture. Anthropologists, psychologists, and psychiatrists have collaborated in studies of child rearing and the formation of personality type in non-western societies. Those anthropologists studying personality in culture have shown that different cultures interested inculcating different values to train their children differently and that the personalities of adults in a culture reflect its child rearing practices.

A major contrast between anthropology and psychology involves attitudes towards the individual in the two disciplines. Psychologists are interested in the individual as a individual. But anthropologists often are not. Rather, many anthropologists study individual behaviour in order to determine behaviour typical of certain positions and relationships in that society.

5. ANTHROPOLOGY AND LAW

Anthropology	Law
Deals with all the concerned aspects in general	Primarily deals with legal aspect, which is an ingredient of Political Anthropology

Anthropology in the wider perspective deals with all the social, political, ethical, economic and legal apects in the case of primitive societies. The basic function of law in the modern societies is to maintain the ethical order by the set of social as well as legal obligations. Legal institutions are different in primitive societies, where as the essence of law is more or less similar to that of modern societies i.e. to maintain the legal order. Anthropologists study these legal aspects in the science of legal anthropology.

Political anthropologists, in the role of legal anthropologists have discussed the topic of law at length, for they believe, it is one of the essential functions provided by the political system of all societies. If it is agreed that all societies have political system it would mean all societies do have 'law' and have some sorts of legal mechanism to maintain ethical order. Law is usually thought of as consisting of written regulations enforced by specialized agencies of political (government) machinery. From this point of view pre literate/ simple societies lack the law. Anthropologists generally prefer a broader aspects of law (legal norms) which is considered as an aspect of social control.

Societies at all level of cultural evolution have some sorts of legal insituttions. A simple but effective form is to be found in the institution of 'self-help' or the principle of reciprocity. Other forms are the informal meeting or 'moot' and formal meeting or specialities of 'Court System' .Both are highly developed in parts of Africa and Asia. Modern police and court systems are complex institutionalized form of legal institutions.

The law in primitive societies has certain different dimensions as that it is believed, that is the law of God, who so ever will not obey the instructions of elder will be punished in form of natural calamities. Moreover the law is inforced

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among them in the form of totems, taboos, religious regulations as well as ritualistic obligations. Now it can be concluded that the legal aspects is undifferentiated from over all activities of primitive man.

6. ANTHROPOLOGY AND POLITICAL SCIENCE

Anthropology	Political Science
■ Basically related with primitive societies	■ Basically related with modern societies
■ Here the political power is 'fused' is within the socio-economic and religious authority	■ Here the political power 'segregated and 'specified'
■ Approach of political anthorpologists is holistic.	■ Approach of political scientists is inductive

Anthropology is primarily concerned with evolution and systematic growth of mankind, whereas political science is primarily concerned with the 'political behaviour' of mankind assuming man to be a political entity.

On the basis of availability of examples it is quite easy to establish that in primitive societies political order has been generally apart of social and religious order. i.e to say that political power is fused with the religious authority but in study of political science 'power and politics' (in one form or another) has been the prime concern of study.

In primitive societies 'power' is generally inherited from ones father or grandfather with religion and divine sanctions. But in modern societies 'powers is acquired through self struggle, as it has been highly democratically institutionalized. Division of power in primitive societies is generally merely or horizontal nature'. However in modern societies power is distributed with 'vertical fashion'.

7. ANTHROPOLOGY AND LIFE SCIENCE

Man is an animal biologically, therefore all the biological characteristics like growth, adaptation, reproduction, digestion, respiration and all other metabolic activities are the subject matter of life sciences. Moreover anthropology is the science in which we study many biological characteristics of a human being which have a greater role in determining the various race and racial types, capabilities/ capacity of pre-historic man with repect to the modern man. It is possible for the anthropologists that the nutritional ability is one of the important factor for well being of human being and are studied with life sciences. Primatology is the branch of Zoology which is also the part of life sciences is the study of apes and monkeys. Primatologists sutdy the evolution, behaviour and social life with their natural environments.

By and large anthropology is the study of all the aspects of human being including physical, social, cultural and linguistic except, while biology or life sciences is the study of human origin, human evolution and human variability.

8. ANTHROPOLOGY AND MEDICINE

Although anthorpology is an independent field of study but interdisciplinary approaches are quite common while we study the applied aspects of it. As it has the relationship with many fields of knowledge, it also has relationship with medicine. An anthropologist does not study the prelevant medicine and medical system. They emphasizes much on the folk medicine. There are many kind of medicines existing in the non-European societies like conventional medicine, environmental medicine, psycho-somatic medicine and so forth and so on. In case of anthropology and its sutdy with the medicine it is very important to notice that medicine does not exactly mean, what the medicine in practise in the civilized societies, but the medicine prevalent in the primitive societies. Anthropologists have find out that in primitive societies too, there is a system which looks after the ills which are of the many kind. Shaman and medicine man are the functionaires which are very common in curing the ill.

In today's perspective, the serological research has brought a belief that has encountered the racial conflict in the world. The blood group and other blood proteins are similar in all the people of the world irrespective of the geographical, nutritional and ecological factor. Anthropologists look after at the formal health care system.

In primitive societies medical care is not an independent aspect. It is inherently related with the faith of the people concerned. Faith plays a very important role at the time of the treatment, while in the case of the civilized societies medical system is not influenced by the faith but depend upon the nature of the diseases and the medicine prescribed for it. Anthropologists say that at a place only one person looks after the health care system and their is no specialization in his field of activity. When we study the life and the activities of the people concerned the medicine obviously is the way of their life. Therefore, it can be concluded that medicine and anthrpology are interlinked and intertwined to each other.

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TERMS AND CONCEPTS OF EVOLUTIONARY BIOLOGY

DOLO'S PRINCIPLE OR IRREVERSIBILITY

Evolution is irrevocable. Once an animal lineage has passed through a number of stages, a reversion, stage by stage, to the original ancestral condition does not occur.

"A structure that changes its form in evolution will not revert to its earlier form." This is called Doll's principle.

This principle is very important in studies of primate evolution. Eg : 1) Dentition is a crucial evidence of ancestral or descendant status. Once a tooth of a particular series (incisor, premolar, molar) is lost, it does not recur again in the same series in the same form. 2) After the extinction of flying reptiles, the combination of wings and reptile living did not come together.

The doctrine of irreversibility is only a generalization, not law. Major evolutionary steps are compounded of many smaller steps, each preserved by natural selection. The exact reversal of such sequences is a rare case to occur. But many times in history, advanced organisms have returned to ancestral habits and modes of life. This is possible due to commonality of environment and resultant functional adaptation and a gross similarity between ancestral and descendant structures is achieved without any genuine reversal at all. Eg : 1) Many reptiles and mammals have reverted to an aquatic mode of life. They assumed a fish like, streamlined form and the limbs have become shortened, webbed and finlike. Yet, the skeleton remains same as the class to which the animal belongs. 2) Many angiosperms returned to water and assumed alga-like appearances but their morphology is still that of flowering vascular plants.

These evidences indicate that major evolutionary changes once appeared are never reversed.

COPE'S RULE

The paleontologist Edward Drinker Cope proposed two laws in evolutionary biology.

1) Population lineages tend to increase body size over geological time.

2) Animals which seem to be less specialized are found for a much longer time in the fossil record.

The former is discussed the most. Based on his studies among fauna in North America, Cope proposed that the earliest fossil primates are very tiny and the later ones are large. This feature is common among mammals. Large size enhances the ability to avoid predators and capture prey, enhances reproductive success and improves thermal efficiency. Moreover, in large carnivores, interspecies competition for food tends to relatively intense and bigger species tend to dominate and kill smaller competitors. Progenitors of hyper carnivorous lineages may have started as relatively small-bodied scavengers of large carcasses, similar to foxes and coyotes, with selection favouring both larger size and enhanced carniodental adaptation for meat eating.

He extended his studies to the Eocene ancestors of modern horses that were about the size of a dog. Gradually the horses evolved to as much as ten times heavier.

Evidences from paleontology suggest that largest representatives of any group are not its earliest ones, though not necessarily its latest ones either.

The tendency towards size increase has been by no means universal. For eg : herbs and shrubs are of recent origin. They originated from trees and other large plants. According to A.J.Arnold who studied planktonic Foraminifera founded that the newer species are of small size because smaller organisms are less specialized and they have more structural flexibility and were able to branch towards new lineages. There is a short-lived disadvantage to being large during periods of high stress (mass extinctions) that is the reason for smaller populations to be found in the subsequent generations.

Hooijer has pointed out that progressive size decrease has been characteristic of many vertebrates during the Quaternary period, which is now in progress.

GAUSE RULE (COMPETITIVE EXCLUSION PRINCIPLE / GAUSIAN EXCLUSION PRINCIPLE)

This principle explains the phenomenon between populations within an ecosystem. If two species occur at the same trophic level, then they are likely to compete with each other for food. The competition may result either of the following.

1) Adaptive radiation of one or both species may lead to restricting them to separate niche within the trophic level. This minimizes the competition.

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2) If the competitors occupy the same niche, or strongly overlapping niches, and equilibrium situation may be reached in which neither succeeds (i.e absence of competitor) nor one of the competitors declines in numbers to the point of extinction. The later phenomenon is called "*competitive exclusion*".

This phenomenon was studied by Gause(1934) on competition between several species of Paramecium.

When the two species of Paramecium -> Paramecium Aurelia and Paramecium caudatum were cultured together. The former has a competitive advantage over the later for gaining food and after 5 days the numbers of P.caudatum start decreasing to the level of extinction.

This principle was later confirmed in further animal experiments. Ex: Increased cultures of duckweed (Lamna) species; L.gibba was capable of excluding L.polyrrhiza.

The study of natural populations is made more complex by 1) the larger no. of interacting populations 2) Uncontrollable environmental conditions like temperature, moisture and food supply etc.,

ADAPTIVE RADIATION

Adaptive radiation was defined by Buattner Janusch as the rapid increase in numbers and kinds of any evolving group of animals (a species, a genes/a super family). These animals many take advantage of environmental changes and exploit a number of new places – econiches – to which they are adapted.

Adaptive radiation may occur as a planet-wide event or as a regional phenomenon. The large scale change in the life forms during the Paleocene is an example for the fomer.

During the Paleocene, as the temperatures dropped, reptiles (particularly the large dinosaurs) became extinct as they are poikilothermic. They had a selective disadvantage of metabolism and all physiological processes slowing down when external temperatures drop.

While the mammals being homiothermic go adopted to the changing climates. They evolved into distinct types in adapting to new environments. Of them, primates took to trees, bats to air and whales to water. Each mammalian order again developed into different forms, each form acquiring a new mode of life to adjust itself to the environment. This was a planet-wide adaptive radiation.

The local/regional adaptation could be witnessed in the spread of arboreal primates, the Old World Monkeys into trees of the tropical forests. In such a case adaptive radiation is said to have occurred when a group of organisms fits into a part of the planetary living space into which it could not have moved earlier, as a result of changes in the group's relation to the environment.

But it is difficult to reconstruct a very detailed account of local adaptive radiation. However inferences from the morphology of fossils and comparative studies of the living forms help to some extent.

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CHARACTERISTICS OF PRIMATES

The order 'primates' is a group of highly variable organisms making it difficult to list the features. But, Le Gros Clark opines that it is possible to rank this order based on the ideal features. For eg., the fore-limb structure, the habitat features, vision, facial snout, dentition, etc.

1) The fore-limb structure of the primates exhibits presence of a clavicle, two separate bones in the lower arm and pentadactylism. They are more adapted to arboreal living. For such locomotion digits are mobile and thumb is opposite. In most primates the claw has evolved into the nail, the tips of the fingers in primates have tactile pads which act as friction pads in grasping and also convey information about the environment by means of a refined sense of touch.

2) Unlike in terrestrial mammals, in the arboreal primates the olfactory senses become useless. Hence the olfactory regions of the brain have been reduced and the muscle/snout becomes smaller.

3) The vision is highly developed. Most primates see in color and in three dimensions. Color-vision helps in distinguishing details and identifying the stationary objects. This was also to help arboreal living.

The eyes are enlarged with a changed structure. They have retina with cells, rods and cones. Eyes are enclosed in bony eye-sockets, a feature not found in other mammals. The eyes have rotated to the front of the face, permitting 3D stereoscopic vision.

4) With reduction of snout and flattening face, facial expressions (that are very necessary for social living) were made possible by differentiation of the muscles of the face. The upper palate is unattached to the upper gum. This permits a wide range of gestures.

5) The generalized mammalian dentition includes 44 teeth (3-1-4-3 dental formula) while in primates the number is reduced, reaching 32 in hominids. The teeth are small and simple in structure and are differentiated into incisors, canines, premolars and molars to serve different functions like cutting, holding, grinding etc.

6) Primates have evolved a more efficient form of reproduction. In the other placental mammals there is epitheliochorial placenta while in primates there is a hemochorial placenta. i.e. the uterus in primates undergoes changes to become a spongy, blood-filled mass, the foetal blood vessels are surrounded by maternal blood and materials pass through a single vessel wall in moving from one blood system to the other.

7) Primates generally have two mammary glands, in pectoral position. The position is determined by the nourishing habits of the mother. The female bears one or at most a few offsprings at a time, instead of a litter.

8) Adults are sexually active althrough the year. The primates possess a pendulous penis shown by some of the mammals too. The testis are descended into a scrotum.

9) Primates have greater intelligence. Their brain is large in relation to the body size. It always contains the calcarine fissure and a posterior lobe.

10) Most primates live in large social units and have a large childhood period, the factors which facilitate learning and call for long periods of dependence on the parents.

PRIMATE BEHAVIOUR

Primate behaviour is one of the ways of primate adaptation. A large part of primate behavioural patterns is learned rather than innate. Behavioral patterns differ not only from species to species but also from group to group within the same species, though they live in similar environmental conditions. Despite this diversity, primatologists identified the following common primate behaviors that were put across in the studies of

a) Baboon by Irven DeVore and Sherwood Washburn
b) Chimpanzee by Jane Goodall
c) Langur by Phyllis Dolhinow
d) Mountain Gorilla by George Schaller
e) Gibbon and Howler Monkey by C.R.Carpenter

Various aspects of primate behavior are :

1) Group Living : Most primates congregate in social groups known as 'troops'. Such a living offers benefits such as defense against predators, enhanced food gathering, more intensive social learning, more assistance in rearing offspring and increased reproductive opportunities.

In some primates like Baboons, the animals live within the troop for their entire life. While in Chimpanzees the social unit is an ever-changing association of individuals. Gibbons and Siamang live as mated pairs or small family groups. In Baboon, the basic sub-unit within

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the troop is a small harem consisting of an adult male, two or more adult females and their young ones.

2) **Social Grouping:** It is unique to primates. While living in groups, grooming through interpersonal behavior such as hugging, kissing, sex, cleaning debris, fleas and ticks from hair provides social cement. It promotes hygiene and reduces conflict and friction between males and females and between young and old. In Chimpanzee female-female relations are expressed in 'grooming activity'.

3) **Communication:** Primates communicate with one another through body movements, vocalizations, olfactory signals and facial gestures. Expressions like fear, affection, threats, greetings, courtship signals, warnings of impending danger are communicated through grunts, hoots, ground-slapping, head-bobbing, screaming, scent-marking and facial gestures.

4) **Dominance - Hierarchy and Dominance-Submission:** Dominance-hierarchy refers to ranking of primates in social status determined by good physical strength, age, confidence, aggression and ability to attract the support of other males. Generally males dominate females of the same age, older and stronger individuals acquire a higher rank than the younger and weaker ones. Females with higher ability to get food dominate and they have enhanced reproductive success.

In some primates like Baboons, once established, the hierarchy is rigidly followed with changes occurring due to age and death. In Chimps, the relation between adult-males is described as dominance-submission; i.e., if two males go after the same fruit the subordinate male holds back. Sometimes they exhibit submission through gestures. The dominance is more rigid in the plains than on the trees.

5) **Physical Aggression:** Aggression in primates ranges from simple gestures to violence and warfare. In Baboons, most aggression is expressed through gestures alone. Threat gestures include staring, raising the eye-brow to expose the distinctively colored eye-lid, slapping the ground and jerking the head back and forth.

According to Goodall's conclusions on Chimpanzees, when dominance-hierarchy is disrupted or well established, violence and warfare may result. For eg., adult males patrol the perimeters of their home ranges and trespassers are violently attacked. Sometimes violence is the result of fight for supremacy.

6) **Sexual Behavior:** The male-female sexual behavior among primates does not follow a single pattern. It varies according to environmental circumstances and the particular species of primate. In Baboons, male and female form 'a consort pair', Gibbons have monogamous sexual bonds between one male and one female. In Chimpanzees, the consort pairs are not formed and mating is quite promiscuous. Sometimes, female gorillas side with a young dominant male against an older one. Such social nuances distinguish primate behavior from other species.

7) **Mother-Infant Relation:** This relation is the corner stone of primate life. Unlike other animals, primates tend to have longer periods of maturation than other animals. During infancy, primates are taken care by their mothers forming a lasting bond. This bonding is strong in the anthropoid apes. Among chimps, though the troop has temporary members, the mother-infant bond is never broken irrespective of their ages.

8) **Training and Tool Making :** Training and hunting and 'other activities' by parents is witnessed in pre-human primates as well. The method of feeding among Chimpanzees is one such example. Mother trains the offspring in making the termite stick and using it to identify and catch termites. They are trained in building new nests in which they sleep. Such a practice is also seen among the Gorillas.

Signs of labour based on sex and sharing clusters are also seen. In Chimpanzees, hunting is a male activity and a group activity. The meat is shared among the group. Such a division of labor is found among all the anthropoid apes.

These studies in primate behavior show that most of the behavior patterns are learned and hence great differences are found in the behavior of different primates and individual groups within primates.

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DATING TECHNIQUES

The Dating Techniques deal with measurement of age of tangible (ancient) objects. The temporal studies are necessary to assess evolutionary change, cultural developments and technological innovations. The dating methods can be divided into 2 types.

A. Relative dating B. Absolute dating

A. RELATIVE DATING

It refers to dating methods that determine whether one particular fossil, artifact, fossil locality or site dates before or after another.

According to Wheeler (1956) - "The relative chronology is the arrangement of the products of non- historic societies into a time relationship, which may not have any dates, but which has a sequence....."

Various methods of Relative Dating include -

1. Stratigraphy
2. Paleontology
3. Polynology or Pollen Analysis
4. Patination
5. Rate of accumulation of cultural or natural deposits.
6. Chemical Analysis of Collagen; Fluorine, Uranium & Nitrogen contents of bones. (FUN TRIO)

1. Stratigraphy : It is the basic relative dating technique; pioneered by Niels Stensen in the 17th C. The method is based on the 'law of supraposition' proposed by Stensen. It states that, "in any succession of rock layers, the lowest rocks have been there the longest & the upper rocks have been in place for progressively shorter periods". The Paleo anthropologist/ archeologist must ensure that there has been no disturbance of layers. The succession gets reversed if the depositional agency has the power of constantly getting lower in level through time. (eg: in river banks)

2. Palaeontology/Dating by using Animal Remains : This method is based on the assumption that, changed climate will bring about the occurrance of different animals and plant species. i.e with change in climate, some species become extinct. This assumption helps in establishing relative dates.

Eg : Evidence of Elephas antiquus (a forest elephant) indicates temperate climate & that of E. Primigenius (a steppe elephant) indicates a steppe/ tundra environment. In the later stages of Pleistocene, the presence of forest and steppe varieties of reindeer suggest the alternation of warm & cold periods in Wurm glacial period in France.

Small species like rodents & birds, some molluscs and snails are very sensitive to climatic changes. Their presence/ absence in archaeological sites indicates climatic change.

This method provides approximate age of objects, if not accurate. Because, the complete species may not disappear at the same time. Some might live in isolated areas.

The limitations of Palaeontological dating :

a. It is risky to interpret past climate on the basis of bones because they are not a random sample around the site, but brought by the hunters mostly. Man must have hunted animals from around his habitation. So, only few species are represented in the sample.

b. Several animal species can live in a wide range of climates.

c. Inferences about climatic tolerance of species, that they have not changed over millennia is not a safe assumption.

d. Though, some animals prefer a particular environment, they may be able to live in different environmental conditions. In the absence of proper information related to preferences & tolerances, it is difficult to conclude on the environment of the time.

3. Polynology or Pollen Analysis : Polynology is the study of "Pollen grains", the minute male reproductive parts of plants. They have excellent preservative ability and are different in structure for different genera or types of plants. Their study helps in tracing the ancient climate as well as the dates of the deposits. A level-wise microscopic study of the ancient pollen samples obtained from a vertical section of a prehistoric site helps to trace the past vegetation history. This if combined with C - 14 dating gives accurate date of the times also.

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Lenhar Von Post (of Sweden), was the 1st to develop this method in 1916. It was initially confined to the study of pollen from forest trees, later extended to all flowering plants that scatter pollen.

The types of soils in which pollen can be preserved:

- peat bog deposits are ideal (eg : N. Europe)
- dry sites, sands & clay
- acidic soils with p^H less than 5.5 (G.W. Dimblely's study)

Limitations of Polynology :

- Majority of scientists held that dry environments do not suit for preservation of pollen.
- Even when pollen samples are found from the banks & streams as they are inorganic, cannot be dated unlike bogs (which are organic)
- Pollen frequencies may be disturbed by remains of domesticated plants or by pollen brought by animals on their fur.

Inspite of these limitations, pollen analysis is useful as relative & absolute dating. If a chronology of pollen profile is prepared, it is easy to find exact number of years of the formation.

4. Patination : It indicates the chemical alterations of rock surface exposed to atmospheric conditions. The chemical alteration of the stone is due to action of iron oxide. A.J.H. Goodwin studied different factors leading to patination (1960).

The amount of patina on the stone is an index of its age. Different types of tools from river gravels, terraces of rivers or lakes can be differentiated based on the relative amount of patina.

5. Rate of accumulation of Cultural or Natural deposits: It involves rough estimation of time on the basis of thickness of the habitation deposits. It was one of the earliest methods of dating.

This method was used in USA and Near East. R. Pumpelly used it in Egypt. N.C. Nelson, W.E. Schenck & E.N. Gifford used in Sacramento Valley. The approximate ages calculated by them, closely matched with C - 14 dates. Rate of accumulation ranged between 1.43 - 1.9 feet per century.

L.L. Loud & M.R. Harrington worked out approximate age dates of culture deposits in Lovelock & Gypsum Caves, Nevada by estimating the rate of filling of cave deposits.

Limitations of the Method :

The rate of growth of cultural deposits of any site is not constant and is found to be subject to factors such as increase/ decrease in population, the use of several debris dumps, the lateral expansion of site etc.

Eg: M.L. Fowler's study of Modoc Rock Shelter in Illinois of 27 feet revealed that

Period (B.C)	deposition
8000 - 5000	1 feet/500 yrs
5000 - 3600	1 feet/300 yrs
3600 - 3300	1.7 feet/100 yrs
3300 - 2700	1 feet/ 400 yrs

- Fowler failed to identify the factors like climatic conditions, variations in living patterns, number of occupants etc.
- Wheeler applied this method in dating Harappan citadel excavations and stated that - though this method is not absolutely useless, it is only of academic interest.

6. Chemical Analysis of Collagen :

(i) Collagen Analysis of Bones : Collagen is a substance that contains fats and proteins present in the bones. When the bones are burried in the soil and as they undergo fossilisation. They start losing collagen at a particular rate. The rate of disintegration of collagen is directly proportional to the rate of fossilisation of the bone. The older the bone is, the lesser the collagen content, and Vice Versa. This helps in identifying the relative age of the given bone.

Laboratory methods to extract collagen from the ancient bones have been revised by Sinex & Faris in 1959.

Radio Carbon dating is also possible on collagen. K.P. Oakley applied it on 'Piltdown bones' which gave a date of 500±100 yrs for the mandible, whereas for the skull a date of 620±100 yrs was obtained. This led to the exposure of the hoax.

(ii) FUN TRIO : This measures the relative age of bones by measuring the elements of Fluorine, Uranium and Nitrogen in the fossil specimens.

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F,U occur naturally in groundwater and gradually collect in bones after they are burried. The bones contain hydroxy apatite. Fluorine removes hydroxyl ions from hydroxy apatite and change into fluorapatite which does not dissolve in water and so remains in bones. Bones that are burried at the same time have same amount of fluorine. The longer the age of the bone, the higher will be the fluorine content. The amount of F content can be determined by chemical analysis or through the X-ray crystallographic method.

Similarly, U ions remove Calcium ions from hydroxy apatite present in the bones & occupy that space. The U content in measured by the beta radiations per million per minute that are given off due to its radioactive decay.

Unlike U & F, N content in bones decreases with prolonged burial, due to the disappearance of the collagen in the bones. The bones of living animals contain 4% nitrogen, and when bones start to decay, the concentration of N steadily deceases. By calculating the % of N remaining in a fossilized bone, the approximate age of it can be calculated. This technique suits well to identify relative age of fossils from the same area.

Limitations :

FUN TRIO Technique is influenced by local environmental factors. The amounts of F,U in ground water differ from region to region. Variables like temp & chemicals present in the surrounding soil affect the rate at which N dissipates. This makes it difficult to relate fossils from different areas.

7. Seriation :

It is a relative dating method, used to date archaeological objects. It is based on the assumption that any particular artifact, attribute or style will appear gradually, increase in popularity until it reaches a peak, and then progressively decreases. Using this principle, archaeologists are able to place categories of artifacts in a relative chronological order.

The technique of seriation is used by James Deetz in studying & dating the gravestones in New England. The study indicated gradual emergence & replacement of several motives on grave stones which indicates the changes in local beliefs & trade patterns.

B. ABSOLUTE DATING METHODS

Unstable radio active isotopes are used in absolute dating. The decay of these isotopes takes place at a set rate, regardless of environmental conditions. (eg : temperature fluctuations, amount of ground water.) Hence, gives accurate results. The amount of decay, is measured by "mass spectrometer". Based on the decay the age of the object can be calculated.

Different types of Absolute/ Numerical dating techniques are:

1. C - 14 dating.
2. Potassium - Argon dating
3. Fission - Track dating
4. Dendro chronology/ Tree ring dating
5. Thermoluminescent dating
6. Archaeomagnetic dating
7. Varve Analysis

1. C - 14 Dating :

It was 1st used by Willard Libby (1960-Nobel Prize in Chemistry). The method is based on the decay of radio carbon (C-14), that eventually decays into Nitrogen. Concentration of C - 14 in a living organism is comparable to that of the surrounding atmosphere and absorbed by the organism as CO_2. When the organism dies, the intake of CO_2 ceases & decay of Carbon. begins. The decay of C - 14 is measured by using its half life (5730 yrs). The technique can be used to date materials less than 50,000 yrs old. It can be used to date organic matter, including fragments of ancient wooden tools, charcoal from ancient fires and skeletal material.

Limitations :

It cannot be used to date materials of millions of years ago. But for dealing with materials of more recent age C-14 dating is of great importance.

2. K - Ar Dating :

During volcanic eruptions, any Ar present in the rock is released, leaving K. As the material cools. K_{40} disintegrates (@1.3 bill yrs of Half - life) into Ar_{40}. By measuring the amount of K_{40} & A_{40} the age of the fossil can be calculated.

The K - Ar dating is based on the following assumptions.

a. The sample at the time of its formation did not contain any Ar.

b. No Ar was added or lost during the life time of the sample

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c. The measuring techniques are accurate.

The method suits best for sites with volcanic activity. This method was used for dating the material from Olduvai George in the Tanganyika, where Leaky found the remains of the Australopithecines.

Limitations :

It cannot be used to find age of recent sites (<500,000 yrs). It is best suitable for samples between 1 mill & 4.5 bill yrs old.

3. FISSION - TRACK Dating :

It is based on the principle that U atoms decay by emitting alpha particles which causes fission track damage on the surface of the material. These tracks can be seen under the microscope. If the total amount of U present in the sample and the density of the tracks can be counted, the ratio between the two gives the age of the sample. It is used to determine the age of geological samples between 300,000 and 4.5 bill years old. Though this method is more useful for palaeo anthropologists, it is successfully used to date glass & pottery glazes of less than 2000 yrs old. Therefore, this method can also be used for dating objects of the recent past.

4. Dendro Chronology :

It is an absolute dating method based on the annular growth rings found in some species of trees. As each ring corresponds to an year, the age of the tree can be determined by counting the number of rings. This principle was used 1st by Reverend. Manasseh Cutler in late 18th C. Modern dendro chronology was pioneered by A.E. Douglass in early 20th C.

The pattern of rings and thickness between the rings varies, based on rainfall and other environmental conditions. Trees of the same species of a single locality exhibit similar annual rings. By overlapping these patterns with successively older trees "master dendro chronology sequence" can be prepared. Age of fossil trees can be found by comparing the sample with the master sequence.

5. Thermoluminescent dating :

This method was initiated by George C. Kennedy (1960) in finding the age of terracotta figurines of Mexico. It is based on the principle that, many materials including clay and stones can store energy by trapping electrons from impurities. On heating, this energy is emitted as a glow, that cannot be detected with the naked eye. The glow is termed thermoluminescence implies the presence of small amounts of radio active elements in all earth materials, both natural and man - made. As a consequence of long periods of time, the radioactivity of these elements gives off rays which push electrons from their normal paths. These electrons thus trapped in the substance can be brought back to their normal positions only by the forces of heat. During this process, each electron while moving back to its normal path sends out a certain amount of light called a photon of light. Millions of release electrons give rise to millions of photons of light. Such one photon of light that is produced is resulted by the heating of a substance and this glow is known as thermoluminescene.

It can be used for such materials which did not undergo any alteration since their initial date of firing, whereas in reality most ceramics have undergone some chemical alteration that affects the thermoluminescent glow. Such an alteration cannot be distinguished from the glow produced by the radiation damage since the pot was fired. This method is of a very good use in detecting modern ceramic fakes from the ancient ones although dating is not possible to give.

6. Archaeomagnetic Dating :

Both artifacts as well as components can be dated by this method. It is a recently developed dating technique which also establishes dates for events in human history. It depends on the fact that the earth is a magnet the force of which operates all the time in, on, or around the earth.

The angles of dip and declination vary from place to place. These measurements indicate that the magnetic field of the earth has changed in direction at a particular rate. A number of studies have shown that a record of past angles of declination and dip is trapped in baked clay. The magnetic and haematite grains in the fired clay align with the magnetic field of the earth surrounding them. When they cool they become magnetic again. Such permanent alignment of grains is called the *'thermo - remnant magnetism'*, which simply means the magnetism remaining in the object. The direction and strength of the second magnetism are determined by the earth's magnetic fields in which cooling takes place.

Geomagnetic north pole which guides compass points drifts constantly at the rate of about 600 miles for every 100 years. By finding out the path of the earth's magnetic

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field at the time when the clay sample was fired and by comparing it with the present magnetic direction, the age of the sample in exact number of years can be calculated.

The archaeomagnetic dating technique is very valuable for prehistoric anthropology partly because it gives absolute dates and partly because baked clay is present in most prehistoric sites. It is best used for complimenting dates obtained by C - 14 method.

The reliability of this method depends on -

1. The availability of good records of change of magnetic paths for areas of near prehistoric sites within a radius of about a max of 1000 miles

2. A series of samples of baked clay from a single area which have known dates against which one can easily measure those without dates.

3. To have such a material which has not been shifted from the place where it was first fired.

7. Varve Analysis :

This is the oldest method used for dating prehistoric objects from excavations. It was described by Gerard D.E. Geer in 1878. Varve analysis like dendrochronology demonstrates seasonal variations and also throws light on the climatic conditions of ancient time.

Varves are annual layers of sediments deposited at the bottom of the lakes by the run off from melting glacial ice. This method is based on the measurement of the relative thickness of varves and their comparison to new sections as in tree-ring analysis.

During the glacial periods when freezing takes place water does not flow and hence there is no sedimentation resulting in narrow bands in the core during winter seasons. During the interglacial periods characterised by warm climatic conditions, the ice sheets melt and the melted water flows into the lakes and deposits sediment in their bottom leading to the formation of wide band in the core. The seasonal variations are also reflected in the nature of the sediments being coarse or fine. The varves range in thickness from less than 0.5" to more than 15", though these sizes are rarely found.

Limitations of this method :

1. The varves form only near ice and so in most parts of the world there are no varves.
2. Even in places where ice was present during the Pleistocene there was no evidence of sediments in lake basins and thus speak of recession of ice sheets.
3. It has been found to be a problem to witness varves till the present outsite Scandinavia.
4. The melting of ice does not occur at uniform rates and may be deposited as varves more or less frequently than annually.

Inspite of the above limitations, the method has been successfully applied in Baltic area, North America, south America and Africa though the correlations of these sequences are not convincing.

There are a number of other methods used for dating archaeological materials particularly depending on the context in which they are found. They consist of the Pleistocene features such as beaches, terraces, dunes, moraines loesses etc.; astronomical dating based on the relationship between the variations in the solar radiation and climatic change; sequence dating based on a generalised temporal sequence of artifact types or associations of artifacts; cross dating which implies the tracing of relationships between different areas with the help of culture sequences etc. All these help in drawing relative chronologies.

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EUROPEAN MESOLITHIC CULTURE

(10,000 B.C - 5 to 4,000 B.C)

First discovery of mesolithic was in Masd'Azil France, where microlithic evidences were found in caves of the areas. Further studies on mesolithic cultures in Europe brought out variants of this culture. The Variation was in dimensions and shapes of tools, the life forms they depended on which is directly linked to ecosystem of locality, method of hafting, existence of art forms and level of advancement in tool technology. Mesolithic Europe is classified into the following nine culture areas based on the above characteristics.

1. Azilian 2. Tardenoisean 3. Austurean
4. Sauveterranean 5. Larnian 6. Maglamosean
7. Kitchenmidden 8. Champignean 9. Ertboll

1. Azilian :

This was the culture area first discovered at Masd'Azil in France.It falls in the layers between magdelenian culture and Neolithic.

Geographical extent : Microliths of Azilian are found in France, Spain, Germany, Belgium, England and Switzerland.

Features : This area lies between Magdelanian and neolithic in tool technology and usage of ecosystem. Blade tools particularly truncated blades (backed blades), triangles, points and lunates are found. Unique feature of Azilian culture is the existence of painted and engraved pebbles. The use and reason for their presence is unexplained.

2. Tardenoisean :

Location : Alp mountains to Britain, Pyrenees to Belgium, also found in parts of eastern Europe.

Features : They have many similitaries with Azilian but are characterised by presence of Geometric microliths predominated by Trapezes while micro burins and backed blades are rarely used.

3. Austurean :

Location : Coast of Canterbury, Spain, France, Northern Portugal.

Tool typology : Consisted of big to medium sized pebble tools which is unusual in mesolithic. Rough axes, scrappers, picks, choppers are found. This culture area presented cave as well as open living.

4. Souveterranean :

Location : Belgium, Holland, England, some sites of Germany.

Tools : Blades (retouched), lunates, triangles, trapezes, Borers, Scrapers. Large number of borers and serappers of paleolithic type indicate the evolution of same population of upper paleolithic to mesolithic.

5. Larnian :

Named after the Ist site Larnae in Northern Ireland.

Tool typology : Blades, Scrappers, points, borers. This culture area continues to have a large number of upper paleolithic traditions.

6. Maglamosean :

Location : Largely focussed in Denmark on Coast of Baltic Sea.

This phase in marked by a transformation from hunting fishing and food collection economy of Mesolithic to agriculture and domestication of animals. It heralded a new beginning in economic and social change, giving way to civilization.

Tool Typology :

Flakes, points, cleavers, burnt microliths, small axes, knives, fishing hooks, throwing spheres. This region presents largest variety of tools ranging from medium sized tools to microliths. Flint, Wood, bone and horn are majorly used. Large variety of fishing tools are found indicating their habitation adjacent to water sources. But there are no clues of domestication of animals.

7. Kitchen midden culture :

It is a special born of European Mesolithic studied from fossil evidences of heaps from disposed kitchen material.

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Area : Baltic sea, North sea region.

Features : Garbage of Kitchen was studied which revealed presence of fish, sea shells in large quantities. Tools of flint, bone, including fish bone, horns were found.

Tools : included trapezes, lunates and other mesolithic material.

8. Champignean :

Location : I[st] Discovered in Belgium later in Spain.

Features : Medium ot large scale flaked adzes, retouched axes picks, erude arrow heads, scrappers, blades. This area is known for pitdwelling.

9. Ert boll :

Denmark, North Germany.

It is the most advanced mesolithic culture area in Europe. Tools of this area include, microliths, trapezes, burins, scrappers, points, apart from Neolithic tools like axes and adzes (celts). This area presented nascent neolithic culture.

Thus mesolithic in Europe is a respense to environmental adaption to particular areas. This in later times gave rise to variants in Neolithic and other metal ages.

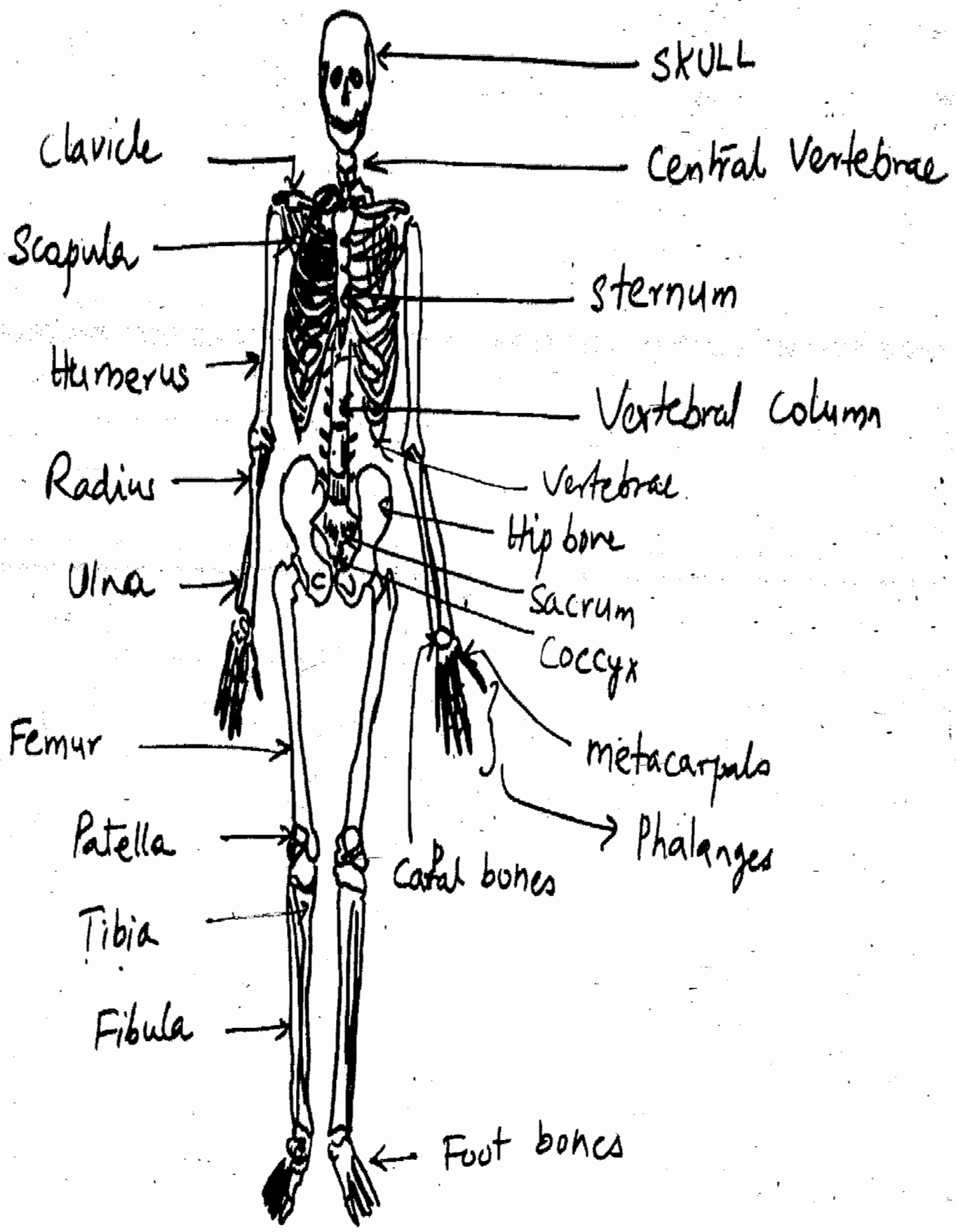
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HUMAN SKELETON



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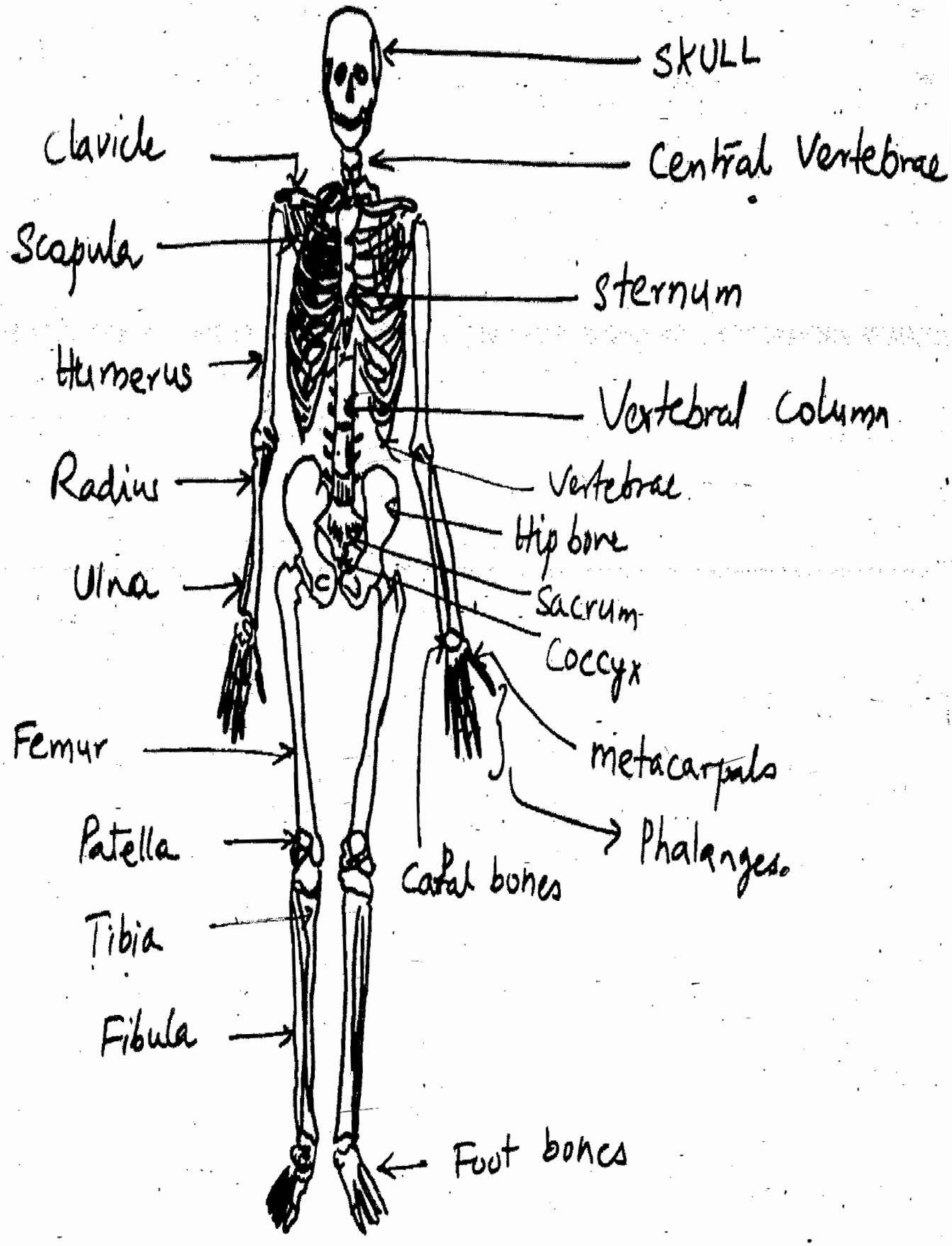
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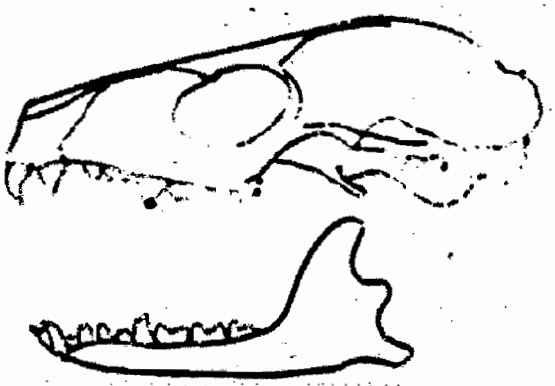
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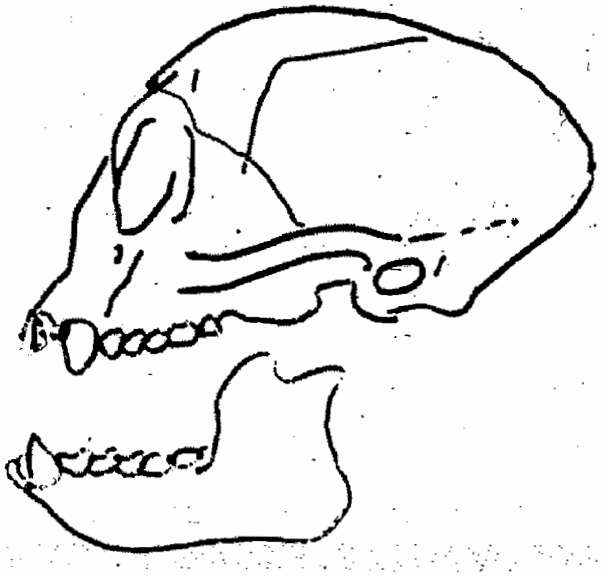


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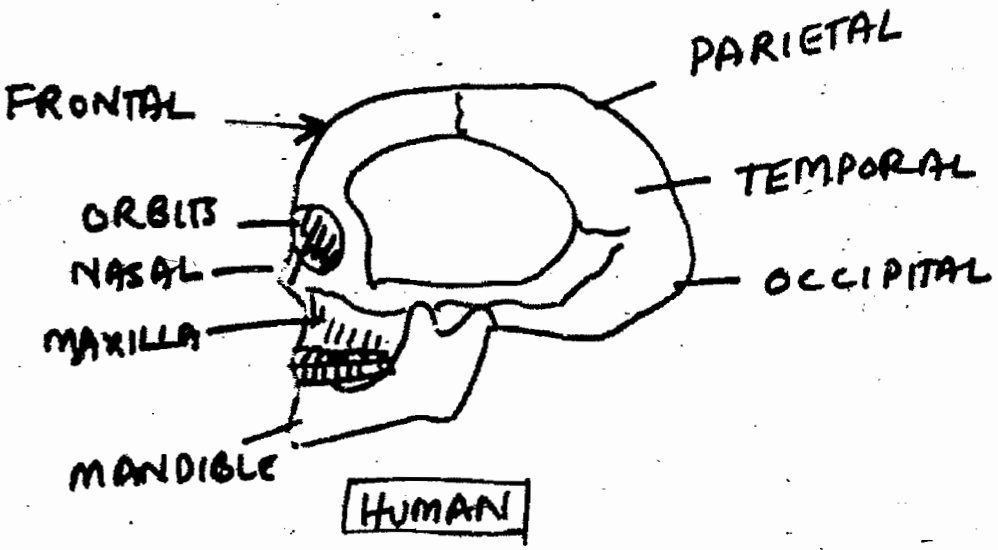


TREE SHREW



MONKEY

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HUMAN

Reducing in evolution

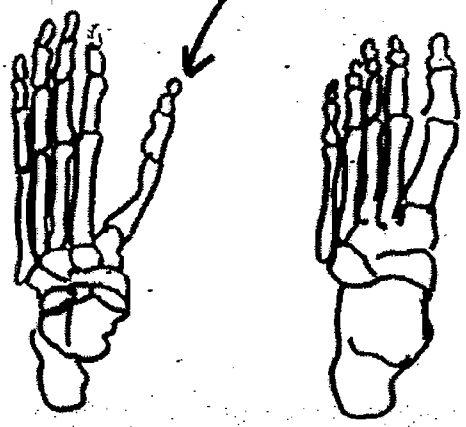
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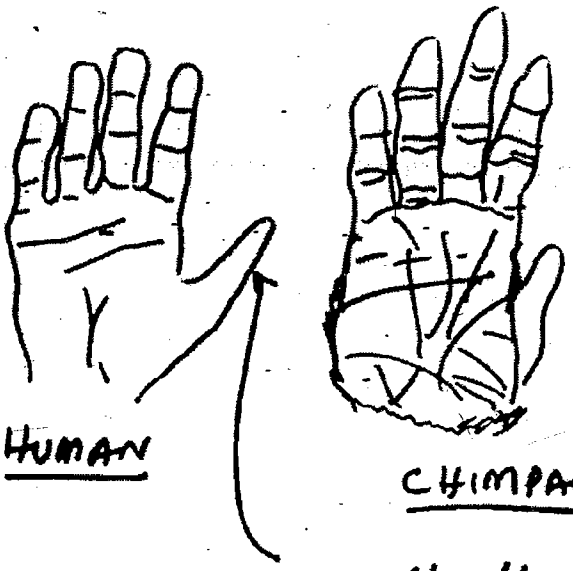
COMPARISON OF APES & HUMANS

opposable hallux



GORILLA HUMAN

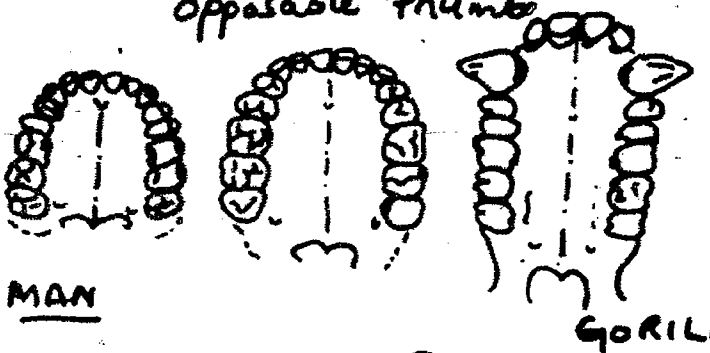
FOOT SKELETONS



HUMAN CHIMPANZEE

HANDS

opposable thumb



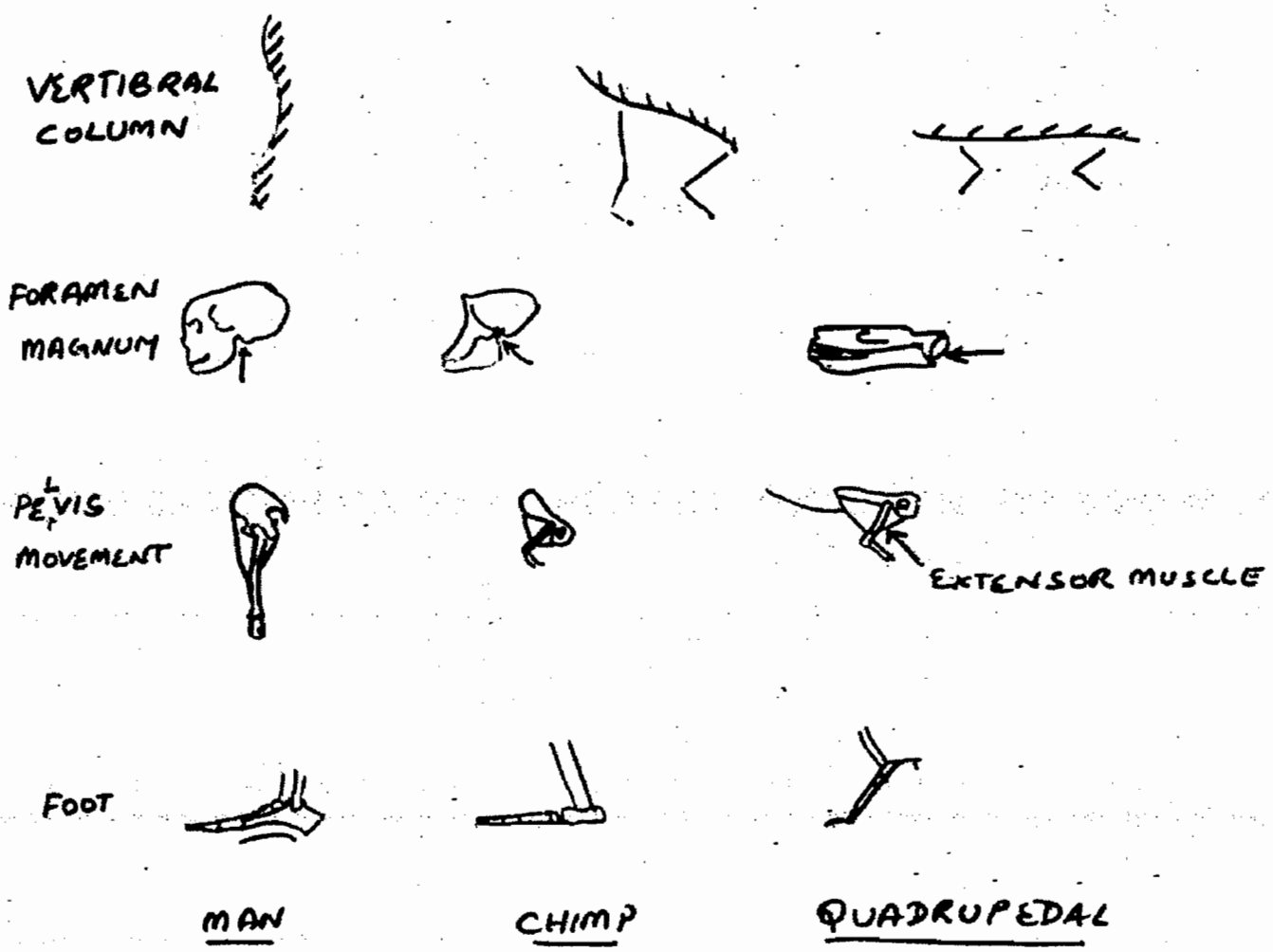
MAN GORILLA

Australopithecus

Changes in dental arcade

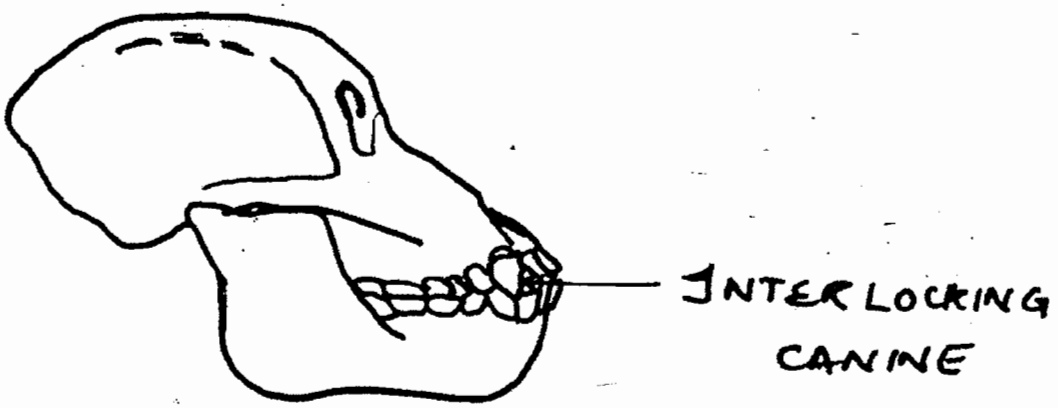
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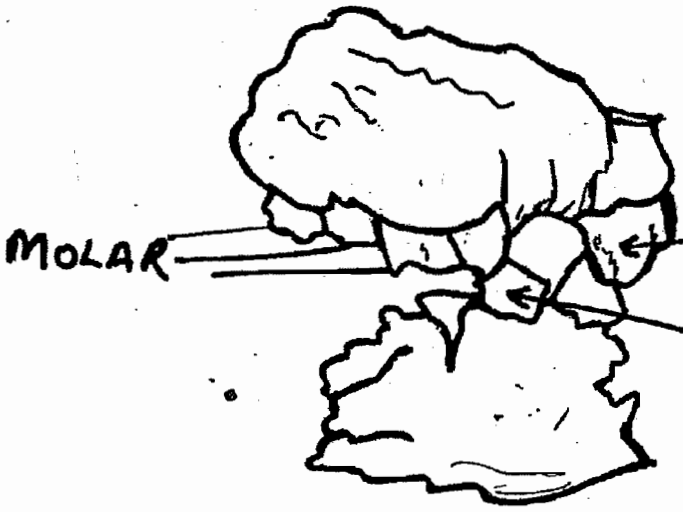
BIPEDALISM – structural changes



DRYOPITHECUS AFRICANUS

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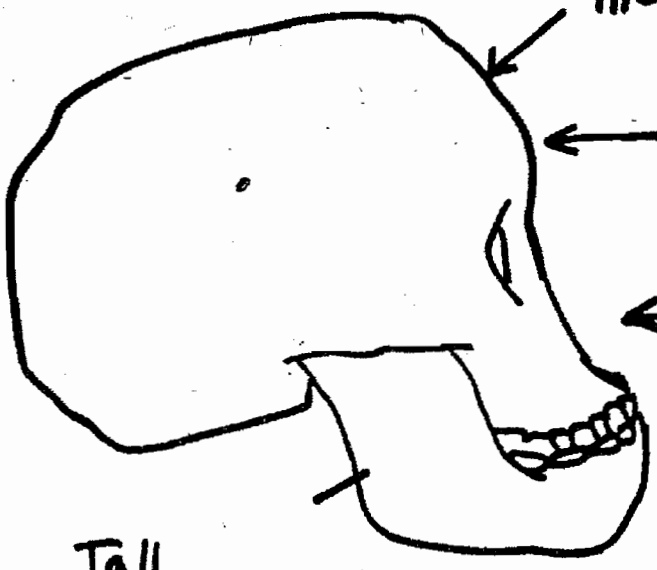
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SIVAPITHECUS

china - Lufang
India - Sivalik
Africa - Kenya
Europe - Hungery

TAUNG BABY (A. africanus)

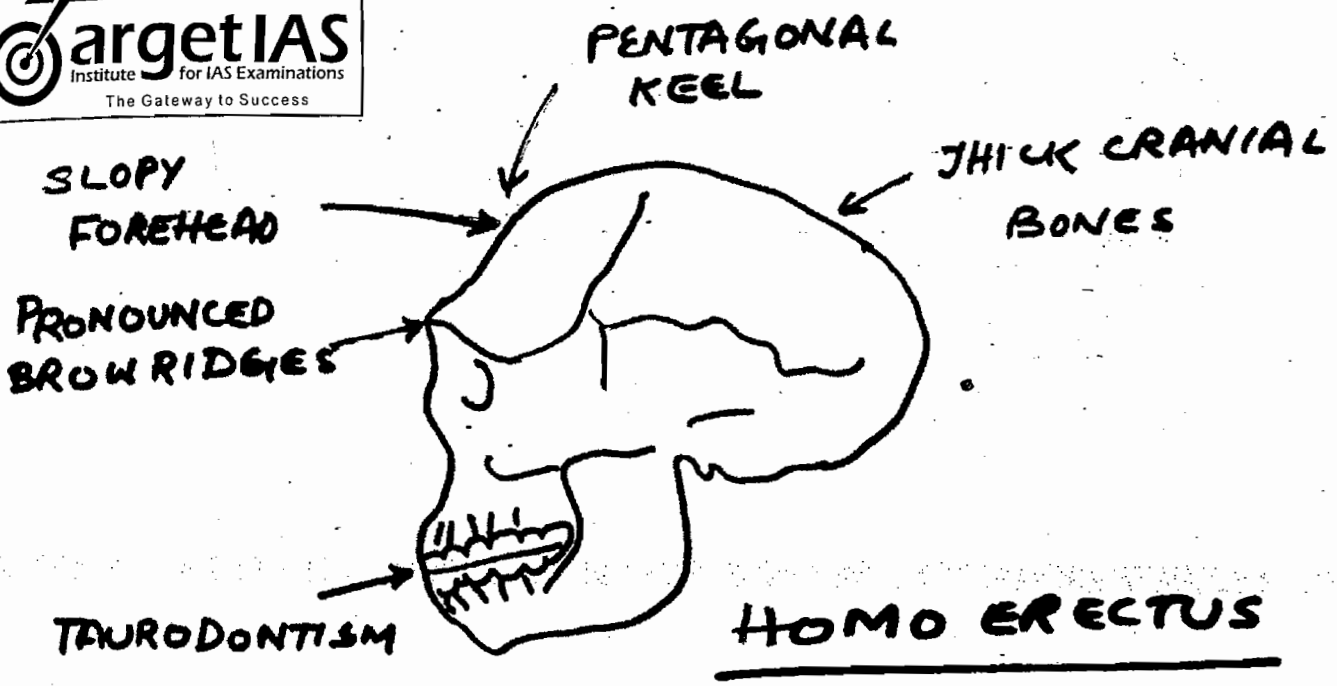


- small canines
- Vertical incisors

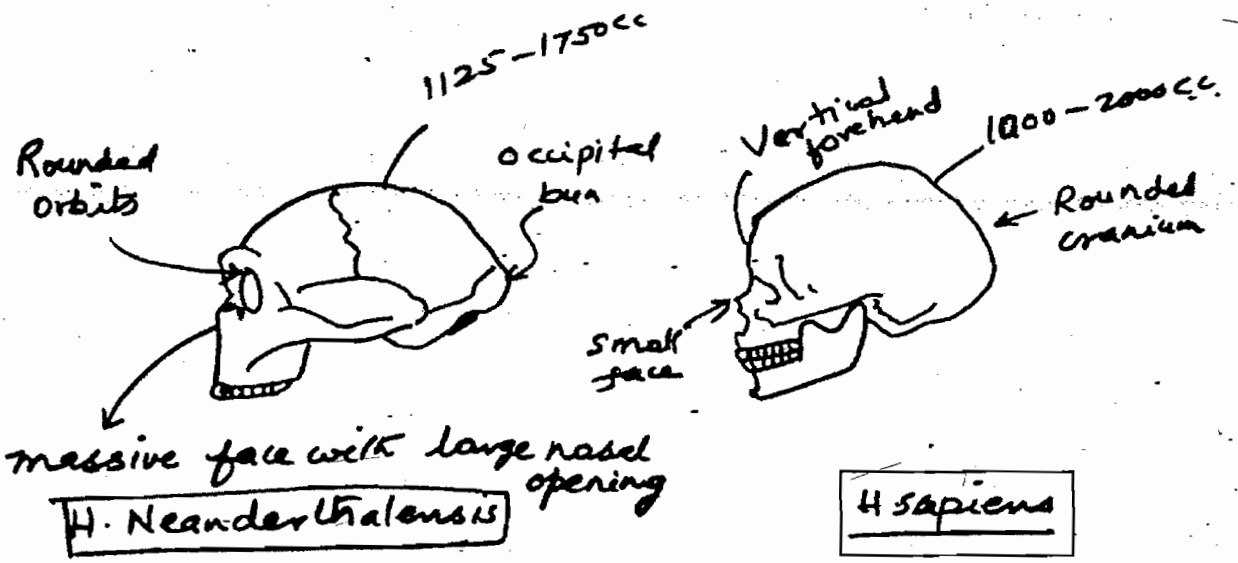
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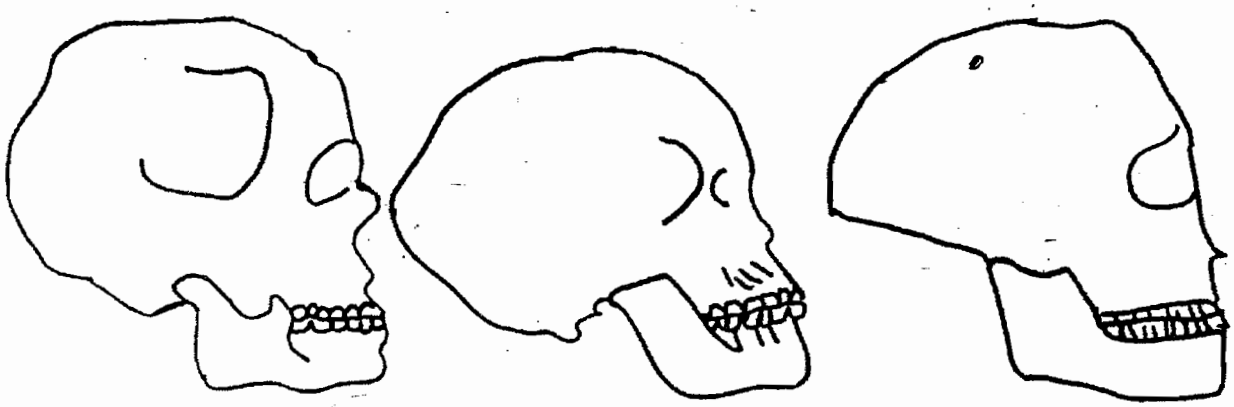


HOMO ERECTUS



H. Neanderthalensis

H sapiens



CRO-MAGNON GRIMALDI CHANCELADE

EARLY HOMO SAPIENS

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HOMO SAPIENS

INTRODUCTION :

Whether man has evolved form Neanderthal or not is still a question. Some Anthropologists believe that none of the Neanderthals disappeared without producing the offspring leading to H. Sapiens. The connecting link (lineages) between N and H. Sapiens are -

1. Grimaldi man
2. Cromagnon man
3. Chcelade man

Of these, Grimaldi man is believed to be oldest (30 to 40,000 years old). Though, all of them are grouped together, they have subtle differences.

Fossil	Time	Technology	Physical Similarities
Cromognon	Upper Paleolithic	Aurignacian	Caucosoid
Grimaldi	Upper Paleolithic	Aurignacian	Negroid
Chancelade	Upper Paleolithic	Megdelenian	Eskimo

1. GRIMALDI MAN (of late pleistocene) :

1. *Discovered* in *Grotte des Enfants care* near Grimaldi Village on the mediterranean coast.
2. *Skeletons* - a women of 30 years and a boy of 15 years, stained in *red ochre*. This indicates *intestional* burial.
3. Grave material - consisted of tools of aurignarian culture - i.e. double edged (retouched) flints, knives, scrapers, gravers, saws etc.

4.Physical features -

Skull & face -

i. cranial capacity varied between 1250 and 1450 c.c. Sexual dimorphism evident in cranial capacity. Women averaged te 1265 cc & men 1455 c.c.

i. Skull had elliptical contour , with small mastoid process (the part of temporal bone behind ear). Skull was long, narrow & high (hyper dolicocephalic)

iii. Face was large and short with face becoming narrow at chin (chin was poorly developed). Skull & face features presented a *disharmonic face*.

iv. Feable supra oribital ridges, with bulbous forehead (Similar to Negroids)

v. large and rectangular orbits.

vi. broad and short nose (platyrrhine), snubbed nose similar to Negroids.

vii. Alveolar prognathism (i.e prognathism of teeth) similar to Negroids & Australoids).

viii. Strong & broad jaws.

ix. Teeth - in general large with upper molars of 4 cusps and lower madars of 5 cusps (similar to Australoids)

Post Cranial features -

Lower limbs much longer than lipper limbs, indicating less - energetic bipedalism. The limb proportions resem Negroids. Prof. Sollas and Prof. Boule associate "Grima man" with Bushmen & Hotten tots.

Though in the post cranial features (particularly, no energetic bipedalism and efficient hand manipulations) similar among Cromagnon & Grimaldi, the features (Grima - elliptical & cromagnor- pentagonol) place them apa Negroid features are well - developed in Grimaldi m whereas they are diluted in Cromagnon.

Differences between Grimaldi & Cromagnon :

Cromagnon	Grimaldi
1. Skull - pentagonal contour	1. Elliptical contour
2. Prognahism	2. more than cromagnin (alveolar prograthism)
3. Dolicocephalic	3. Hyper dolicocephhalic
4. Leptorrhine nose	4. Platyrrhine nose
5. Stature : tall	5. Short
6. Caucosoid features	6. Negroid features

2. CRO - MAGNON MAN** (Old man of Cromagnon)

Placed in the same genus and Species as t contemporary Homo Sapients, the hominid Cromagnon w first discovered by *Louis Larlet* in 1868, in Cro-magnon ca in les Eyzies, France. The *cro magnon rock - shelter* produc 5 skeletons, first studied by Paul Broca and Pruner B *Quatrefagues & rock - Hamy*. named them *'Cro-magnon ra* Based on the physical similarities with Cro-magnon, differ populations & present day are co wordered the descende & cromagnon.

Eg :

According to Collignon	- French people
According Quatrefagues & Hamy	- Dal Race of Swede
According Paul Broca	- Basques, Kabyles

Culture :

i. The tools are of aurignacian culture, consisting of bo & stone tools, cave - paintings, sculptures, stone statu of female figurines etc. Ivory was also used. The gra tools indicated change in ecosystem & lifeforms. C Magnon lived in dryer & warmer climates, dominated reindeer. Hence this period was called *'Reinder Age*

ii. The economy was characterised by *hunting & fishing*

iii. Habitat - caves & rock - shelters.

Physical features -

Skull - Cranial capacity - 1660 CC (by Keith)
- 1590 CC (by Boule)

- long & narrow skull (dolicocepholic) with low vault
- vertical and broad forehead (similar to humans)

} indicative of disharmonic face

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- Pentagonal cranium, bulging occiput, wide and rectangular orbits, strong & prominent check bones;
- nose - long, narrow
- maxillar prognathism
- strongly developed chin

Post Cranial Features -

- Femur with well developed linea aspera, indicating muscular individuals
- Pelvis of females indicated birth of less matured babies and thus longer socialisation.
- Height 168.4cm (According to Boule 182 cm)
- Limb proportions indicate Negroid features (leg is long as compared to femur & Forearm long as compared to arm)

Though in certain aspects Cro - Magnon resembles the modern Negroids, on the whole he *resembles* the contemporary *Caucasoids.*

Progress of Cro Magnon -

The physical features of Cro - magnon continuing in several populations indicates that it was not extrict by the end of Upper Paleolithic; but continued in Neolithic period upto the present day.

3. CHANCELADE MAN

1. **Discovery** - Skeleton of an old man found in a *rock shelter* in *chancelade,* Dordogne, France (1888). It was buried with arms folded on breast and knees touching the laws. The posture of burial and use of *red ochre* indicate *intensional burial.*
2. **Culture** - Magdalenian of Upper Paleolithic, with tools of ivory, bone and antler. Blades and harpoons were dominant. There was evidence of cave paintings.
3. **Physical Features -**
   - Cranial capacity : 1530 cc (according to keith) & 1710 cc (according to Testut).
   - Dolicocephalic with sagittal crest - an *Eskimo feature* Sagittal crest is more developed than Cro - magnon.
   - *Eskimo features* in chancelade are very clearly evident. Some of them are - short stature (5"-the height of the chancelede old man); large and high dolicocepholic head, wide and long face, prominent check bones, leptorrhine, limb structure. Based on these features chancelade is considered ancestor of Eskimoss.

Other Features :

- quadrilateral orbits, body of lower jaw being narrow but broad at the ramil. (i.e joint of jaw with skull); well - developed mastoid mastoid process almost vertical forehead; harmonic bread (face being long & broad)
- strong and massive limb bones indicating muscular body; upper limbs comparatively longer than lower limbs (Eskimo feature); Femur bent and linea aspera well marked; distance between 1st metatarsal and great toe resembling Neanderthal.

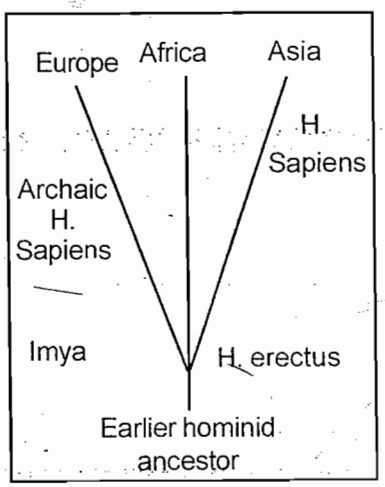
Apart from physical features. Chancelade & Eskimo share *Cultural Similarities* evident in similarity in tools used. According to the supporters of this view, with the end of glacial period, chancelade man moved northward and reached N. America, and their descendants are the modern Eskimoes. This view was apposed by Hooton.

THEORIES OF ORIGIN & DISTRIBUTION OF MODERN HUMANS

Although paleoanthropologists generally agree that H.erectus evolved into H. sapiens, they disagree about how, when & where the transition occurred. The theories in this regard are classified into

1. Multiregional evolutionary model
2. Replacement model (single - source model)

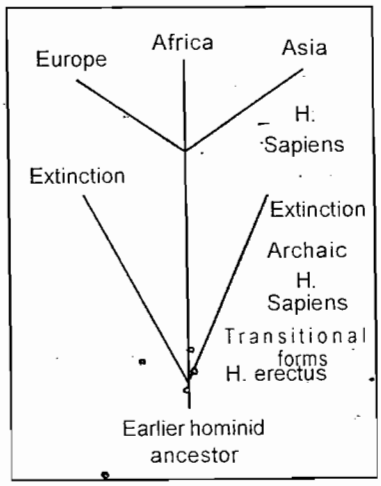
1. Multi regional evolutionary model :



According to this model, the gradual evolution of H. erectus into modern H. sapiens took place in many regions of Asia, Africa, Europe at the same time. Through natural selection & gene pool particular to the region, H. erectus developed into populations of particular traits. i.e. emergence of H. sapiens was a wide-spread phenomenon, though different regional populations continued to exhibit distinct features. The continuity between the H. erectus and modern man is evident from skeletal features.

Eg : Cromagnon had Caucosoid features
Grimaldi man had negroid features
Chancelade man had eskimo features

2. Replacement Model (Single - source model) :



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This model holds that Homo sapiens evolved in one area of the world first and migrated to other regions.

This theory assumes that H. sapiens were contemporaries of the earlier H. erectus but eventually replaced them. The H. Sapiens who evolved in Africa, spread to other region replucing the earlier homimids.

This theory has a significant lacuna. Though many fossils of modern Homosapiens, archaic H. Sapiens and early hominids are found in Africa, we cannot pinpoint the precise stratigraphic position of these specimen. In the absence of any accurate methods to date different fossils, it is difficult to conclude that African fossils of other areas.

There is another set of anthropologists, who postulated that H. sapiens may have originated in South West or East Asia. But these hypothesis too, cannot be confirmed through available fossil evidence.

Partial Replacement Hypothesis :

The two theories (multi regional & single source models) may not be completely correct. The process of human evolution is much more complex, involving physical diversity, behavioural, social and linguistic differences. Migration may not have been a undirectional affair. Many routes of migration recolonisation of previously occupie territories, gene flow may be some factors. This thinkir formed basis of *"Pal Replacement hypothesis"*. In or partial replacement model, it is presumed that H. Sapier first appeared in S. Africa (100,000 yr. ago) and migrate When they came across Archaic H. Sapiens of oth places, as a result of hybridisation, gave rise to new forms of H. Sapines. But this hypothesis does not tal into consideration, the possibility of multiple regional orig

Mitochondrial DNA Research :

The study of mt DNA Berkeley (1987), conducted Africa, Asia, Europe, Australia & New Gunea indicat that mt DNA of African women tended to be more diver (heterogeneous), a sign that they had a long time mutations while women of other regions had mt DNA mc homogeneous indication for less time of mutations. Th interpretation favoured a replacement model, with Afri as the ultimate place of origin of all later anatomica human populations.

Though the method of study was challenged no mc concrete study challenging Berkeley's study is present so far. On the other hand, his study opened evenues f new studies in tracing the origin of man.

* * *

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CHAPTER - 2

MARRIAGE

I. PROBLEMS RELATED TO UNIVERSAL DEFINITION OF 'MARRIAGE'

Q. Discuss the problems related to universal definition of 'marriage'?

Ans :

Marriage is generally considered as 'a union between a man and a woman such that the children born to woman are recognised legitimate offsprings of both parents'

– Notes & Quaries

But this definition given by Notes & Quaries is not universally applicable to all types of 'marriages' the world over various attempts to revise the definition have fallen short of the requisite.

Various definitions of marriage

Traditional Definitions	Modern Definitions
– Westermarck	– Madan & Majumdar
– Murdock	– Stephens
– Seligman	– Kathleen Gough
	– Malinowski
	– Edmurd Leach

Westermark's definition – 'Marriage is a religious accepted unon of a woman & a man living together, with mutually agreed sexual rights".

'Living together seems to' be a crucial aspect of **Watermarck's definition** But among Nayars of Kerala wife & husband do not live together. Moreover, they do not have secual rights. Among **khasas** the sexual behaviour of wife varies – when she is at husband's place he has complete sexual rights over her. But, she can mate with others while at mother's place and even permitted to become pregnant. Among **Dinka** tribe in-case of not having a progeny the husband may depute another male to have a child. The next important aspect of this definition is marriage between 'a male & female'. This does ot explain 'boy-boy' marriage among **Azande** of Africa where the husband even pay bride prize to get boy-wife. Cheyanne Indians too have this practice. Similarly it also ignores "female marriages' among **Nuer** of Sudan.

Murdock – 'Marriage is a universal institution consisting of individuals' establishing a famiy by living together with economic cooperation.

The Major points of importance in this definition are:

a. Universality of institution – which may not be accepted Infact 'family' seems to be a universal institution that 'marriage'. This institution of family can be identitied even among prehuman mammals.

b. 'Living Together' – is another feature which is not applicable to 'Nayar' families : which also has no economic cooperation. The major flaw of this definition is negligence of sexual aspects.

Seligman's definition :

"Marriage is a union of a man and woman, where in the children bor to the woman are accepted to be the legal children of the couple'.

The definition does not take into account polygamous marriage. The sexual rights of individuals and social acceptance are considered. Marriage is seen as a union of only male & female. In this respect it does not explain 'male-male' marriages & 'female–female' marriages in 'Azand and Nuer' societies.

The definitions proposed in 20th century are called modern definitions The major change observed in these definitions is

Monogamy ----------to----------> Polygamy

Some even spoke of divorce (eg : Stephens)

1. *Madam & Majumdar's definition* – 'Marriage consists of social and economic relations arising out of sexual relations between man and woman, which is accepted by religion or civil law'.

2. *Stephens* – (In "The Family in Cross – Cultural Perspective") 'Marriage is a legally accepted sexual union. It is a more or less permanent relationship between the couple with mutually accepted economic rights & responsibilities between the couple and between the couple & their future children' This definition does not specify whether the marriage between male-female/ otherwise. Hence it explains unisexual marriage. It also encorporates possbility of divorce But originating factor of marriage is 'sexual union'

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In **Chukchees** of Siberia, shaman establishes relation with the boy before marriage. Among Okinawa & Trutskee premarital children are permitted. Stephen's definition gives primacy to economic aspects. But in Nayars there is no emphasis given to them.

According to **Malinowski** 'A legal marriage gives a woman a socially accepted husband and to her children a socially accepted father'.

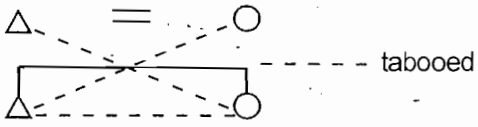
Similarly **R.C. Brown** defines it as a social arrangement by which a child is given a legitimate position in a society". A question arises from these definitions – i e is marriage necessary for children"

Kathaleen Gough studies Nayar society and gave a better accepted definition 'Marriage is a union of a woman with one or more other individuals establishing reciprocal rights & responsibilities between them and their to be children giving all the rights to children as a member of that society'.

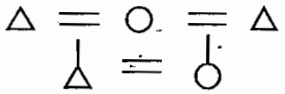
Studying all these definitions **Edmund Leach** concluded that it is difficult to have a universaly accepted definition marriage. Beyond doubt. Marriage depends on principles of legitimacy. The necessity of this institution has come about to give a legitimate position to parents as well as to a child. Hence, among human societies the institution of marriage can be said to be universal. Due to cultural variations in its form a universal definition is far from possibility.

II. INCEST - TABOO

'Incest refers to sexual relations with a close relative'. All cultures have taboos against it although the taboo is a cultural universal, cultures define incest diffrently. Incest taboos force people to marry outside their own group, to form alliance & to maintain trade networks. To begin with, incest tabu meant the ban of tintercourse between close kins beyond a nuclear family. The list included parallel & cross cousin marriages. Parallel cousins belong to the same generation & same descent group and hence considered to be siblings in many societies. Hence, Tabooed, But sexual contacts within the nuclear family are almost every where tabo.



The variation in 'taboo' are seen in. Lakher (South East Asia; Tibet); where half-sibling are tabooed.



The above type of marriage is permitted in Lakher. Hence, it can be stated that different cultures vary in definition of relatives. The kinship terminology dictates the definition of taboo. But there iare certain exceptions: Inca, Hawaiian & Egyptian royal families do not have taboo on incest. This is to kept up purity of blood, property/ kingdom etc. There are different theories which explain the origin causes of incest taboo.

A. Childhood Familiarity theory :

Propounded by Edward Westermarck in 'The History of Human Marriage'(1891). According to this theory children reared in the same household are naturally averse to having sexual relations with one another in adulthood. The logical flow of the theory is 'if there was a natural aversion to incest, the taboo would the unnecessary'.

Westermarck's theory of aversion is based on Israeli 'Kibbutz', where children who have been raised together tend to avoid marriage. There are many societies which can prove this theory wrong. Ir Arapesh of New Guinea Studied by Margaret Mead & the Eskimo, young children are betrood & raisec together, usually by the boy's parents, before the marriage is consumated.

In Northern Taiwanese, bride enters her futur husband's home as a child. The theory of Westermar is supported by Lowie. If this theory is extended t non-human priorities, There is no evidence for a instrinct against incest. Hence taboo is more cultur than biological.

B. Theory of Biological Degeneration ;

Proposed by Morgan – the taboo emerged due early Homo's noticing abnormal offsprings emergir due to incestuous relations, the offspring born lat were successful. But this theory of inbreeding is recent origin. Inbreeding produces offspring wi exhibiting harmful gene which were hitherto recessiv The modern genetics proved that the closer the degr of inbreeding, the more harmful the genetic effec According to Theodoscus Dobzhansky, incestuo marriages yield in still births, Infantile deaths etc. E the increase of hereditary diseases between clo

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relatives & unknown persons is very high. Hence they cannot be declared disgenic. Hence this theory fails to explain taboo.

C. Demographic theory :

Proposed by Wilson Wallis & Mariam Slater. According to this, in early times mating within a family (between parents & children) was unlikely due to short span of time. Mating between siblings was also unlikely because of large gap between them (dur to post partem sex taboo & high infant mortality). But this does not explain mating in a nuclear family when the life span increases.

D. Co operation theory :

Proposed by E.B. Taylor & elaborated by Leslie White & Claude Levi' strauss Poeple are compelled to marry outside their groups in order to form unions with other groups & promote harmony. Incest taboo also breaks down the suspicion and hostility between family groups – hence the theory can be stated as 'marryout or Die out'.

E. Freud's psycho analytical theory

Taboo is a reaction against unconscious unacceptable desires : Children naturally have attracton towards their parents of opposite sex. To repress & renounse this feeling incest is needed. This is quite contradictory to Westermarck's theory.

F. Family Disruption Theory :

Associated with Malinowski. Sexual competition among family members would create rivalry, tension & leads to disruption of family. As family is the cornorstone of society, family must function effectively. But this does not explain the extence of brother, sister incest in ancient Egypt. This theory also does not explain the origin of incest taboo.

Incest Taboo prevents localism & economic provincialism.

None of the explanasions for incet taboo is sufficient to explain the existence of taboo in humans. Whether incest taboo is atypical of human society is not definite. (Eg. chimpanzee – mother tries to keep away her adolescent male offspring).

III. MARRIAGE PAYMENTS

Marriage, in many societies, is a group affair and involves exchange of goods and/or services between bride's & groom's kingoups. All these translations that take place before, after or at the time of marriage have been termed 'marriage payments'.

In many societies marriage involves economic considerations. According to Germans, 'It's not a man that marries maid, but field marries field, vineyard marries vineyard, cattle marries cattle'.

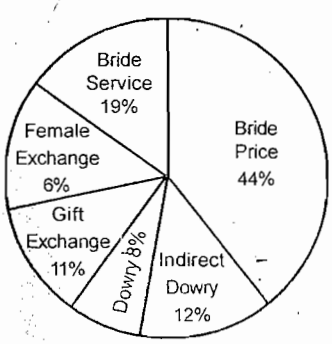
In about 75% of societies one/ more explicit economic transactions take place before or after the marriage.

The various forms of economic marriage payments are

Form of Payment	% of Societies Practicing	Groom's Group △	Bride' Group ○
Movement of bride			
1. Bride wealth/ Bride price	44%	←	→
2. Dowry	8%	←	
3. Sultor Service	19%		→
4. Gift Exchange	11%	←	
5. Exchange of females	6%		→

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Based on the data from Alice Schiegel & Tohn Eloul, Marriage Transaction : Labor, Property & Status', the distribution of Economic Marriage Transactions among societies that have them :



1. Bride Price/ Bride Wealth :

It is a gift of money or goods from the groom or his kin to the bride's kin. Descent-group members have to contribute to bride wealth. Previously, Bride wealth was equated with the purchase of bride & was termed "bride price". But now, it is known as a compensation paid to

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the bride's kin for loss of 2 working hands. Bride Price makes the children born to the woman full members of her husband's descent group. Hence, it is also called 'progeny price".

Why Bride Price?

- It guarantees sexual rights to the groom over the bride.
- It serves the function of cementing the relationship between two groups established through marriage.
- It grants the groom the right to marry the bride and the right to her children.
- Bride Price works as a security to the bride's family. If the marriage fails for no fault of her they need not return. Bride wealth is insurance against divorce. Because when bride price is easy to repay, divorce can take place. The more difficult it is to reassemble the bride wealth, the more stable is marriage & the rarer is divorce.
- Among Ba-thonga of South East Africa if bride's kin fails to repay the bride wealth, groom may claim her brother's wife.

Where is Bride Price found?

- Commonly seen in patrilineal societies.
- In horticulture societies and those societies which lact social stratification.
- Where women contribution more to primary subsistance activities & where they contribute more than men to all kinds of economic activities. Even if she contributes more, her status is not superior to men in the same society.
- Occurs in societies in which men make most of the decisions in the household–indicates lower status for women.
- Bride Price is very common in Africa & Oceania. Payments are made in different currencies – livestock & food are more comon.

  Nandi – Cattle, goats, sheep, Cowrie Shells etc.

  Subanun (Phillipines) – Several times the annual income of groom + 3 to 5 years of bride service.

2. Dowry :

"A substantial transfer of goods or money from the bride's family to the bride'. It is not a transaction between the kin of the bride to the kin of the groom. A family has to have wealth to pay dowry. No wealth comes back to the family that gave dowry.

Functions :

- Dowry is meant for the bride to help her establish her new home after marriage or help the groom establish himself if professionally.
- In societies like India, where women are perceived as burdens, it compensates for the added responsibility of husband.
- It is intended to guarantee future support for a woman and her children.
- Dowry is intended to attract the best groom for a daughter in monogamous societies with high degree of social stratification.

The practice of dowry was common in Medieval &Renaissance Europe. It is still practiced in E. Europe, Italy, France, India, China & West Asia.

In Indian Hindu society dowry can be traced back to types of marriages. According to mythology, Arsha, Asura & Prajapatya marriages involve payment of money by bride's father to the groom. In Hindu philosophy every 'DANA' has to be accompanied by 'DAKSHINA' 'Kanya Dana' being the best kind of 'Dana' it has to be accompanied by 'Vara dakshina'.



3. Suitor Service :

It generally acts as a replacement for bridewealth among the poorer sections of the societies. Here the groom or any of his close relatives works for a number of years (specified) to have rights over the bride and her progeny to be. In suitor service the payment is not in cash or goods, but in the form of service.

Among the North Alaskan Eskimo the boy works for his in-laws after the marriage is arranged. He simply has to catch a seal for them. Among Hopa and Yoruk Indians half of the service can be paid in cash.

4. Gift Exchange :

This involves exchange of gifts of about equal value by the two kin groups about to be linked by marriage. This indicates 'reciprocity'. The gifts help a comfortable living for the couple. Among Andaman Islanders as soon as marriage is fixed, the two parties stop interacting & start sending gifts through a third party. Among them an important item of exchange is 'Food Items'.

In Kwaito – of Soloman Islands – gifts are paid in 3 different stages –

i. a part is paid as a 'down payment'
ii. another part as a 'feast'
iii. a third part given as 'clothes' being a gifted to parents of bride & groom.

Whatever could be the item of exchange, the basic idea is strengthening relationships with other groups.

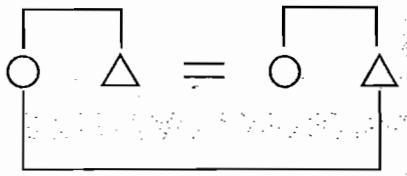
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5. Exchange of females :

A very less % of societies have this type of exchange. In this practice a sister of female relative of the groom is exchanged for the bride. Ex : The Horticultural, egalitarian societies like TIV of West Africa & Yanamamo of Venezuela – Brazil have high status of women. The contribution of women is higher & hence a women is compensated for the other.

This practice is also observed in societies with dowry/ bride price. i.e. when economic aspects dominate, compensation has to be paid Nirankaris have sister exchange.

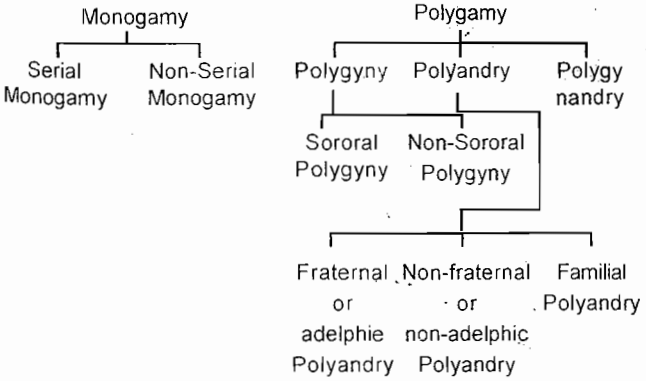


All these are different types of economic exchanges prevalent the world over, which indicate the important of economics in social social & cultural relations.

IV. FORMS OF MARRIAGE

Anthropologists have found that marriage is a universal institution found all over the world and its institutional form has been accepted by all, though it is not the same all over the world. It varies from culture to culture and from society to society. It varies even within a culture and a country, and to some extent within the same people spread over different parts of the country.

However, some of the commonly known forms of marriage all over the world are monogamy, polygyny and polyandry. Among these, however, monogamy is the most popular institution. If one marries a single person at a time, the marriage form is called monogamy. (Greek : Poly-multiple; gamy – marriage). In other words, the form of marriage depends upon the number of spouses involve in the marriage. The following table shows the various forms and sub-forms of marriage.



1) Monogamy

Monogamy is a form of marriage in which an individual has a single wife at any given time. Serial monogamy is a sub form of monogamy in which an individual has several spouses in succession. The Semangs of Malay forests, for example, limit themselves to one wife at a place. In case of divorce or death of wife, a Semang marries again and again and remains to be monogamous. Non-serial monogamy is a sub-form of monogamy in which an individual has the same single spouse life long.

Monogamy occurs in all types of societies in different parts of the world. It occurs among the food gatherers and hunters like the Andamanese, Chenchus and Juang of India, Semang and Sakai of Malya, Ituri of Congo forests and Arunta of Australia. Several horticultural societies like the Basseri of Iran, Kalmyk of Mongolia, Lapps of Finland and Norway and Dinka of Africa practise monogamy along with polygyny. Likewise numerous agricultural societies like the Dafla, Oraon and Purum of India, Dusun of Borneo and the peasant groups dispersed all over the world practise monogamy and polygyny. Advanced industrialized societies in different parts of the world are monogamous.

Reasons

Different societies practise monogamy for demographic, economic, biological, social, legal and psychological reasons.

(i) Monogamy is a response to balanced sex ratio. Even though sex ratio is not uniform, it is approximately 1:1 all over the world. It is commonly said that a man cannot have two wives except by preventing somebody else from having one.

(ii) Monogamy offers a solution to the problem of adapting to the limited availability of various resources in the environment.

(iii) Monogamy favours almost everyone a chance to have at least one spouse.

(iv) Monogamy provides effective sexual gratification for both men and women.

(v) Monogamy keeps intra-sex jealousies and quarrels at minimum.

(vi) Monogamy facilitates relatively easy rules of inheritance, succession and membership in kin groups.

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(vii) Monogamy helps to maintain effective child rearing procedures and augment close emotional ties between parents and children.

2) Polygamy

Polygamy is a form of marriage in which an individual has multiple spouses at any time. Polygamy has many sub-forms; Polygyny and Polygynandry. Polygyny is the sub-form in which an individual has multiple wives at any time. Polygyny exists in two specialized variations : Sororal Polygyny and non-sororal polygyny. Sororal polygyny is a variety of polygyny in which the multiple wives of an individual are sisters. Non-sororal polygyny is a variety of polygyny in which the multiple wives of an individual are not sisters.

Polyandry is the sub-form of polygamy in which a woman has multiple husbands at any given time. It appears in three specialized variations : fraternal or adelphic polyandry, non-fraternal or non-adelphic polyandry and familial polyandry. Fraternal or adelphic polyandry is a variety of polyandry in which the multiple husbands of a woman are own brothers (off-spring of the same mother). Non-fraternal or nonadelphic polyandry is a variety of polygynandry in which the multiple husbands are either clan brothers (those who belong to one clan) or unrelated men. The Todas of Nilgiri Hills in Tamil nadu and the Khasas of Jaunsar Bawar in Uttar Pradesh practise both fraternal and non-fraternal polyandry in which the multiple husbands of a woman were own brothers and clan brothers respectively. However, among the Nayars, the multiple husbands of a woman were several unrelated men, Familial polyandry is a variety of polyandry in which the husbands of a woman are father and son. This rare variety of polyandry occurs among the Tibetans.

Polyandry is a sub-form of polygamy in which a man has multiple wives and a woman has multiple husbands at any given time. It shows the coexistence of polygyny and polyandry. It is a rare sub-form of polygamy existing among the Todas of Nilgiri Hills, the Khasas of Jaunsar Bawar and the Marquesans of Polynesia. Some men in these societies practise monogamy as well as polyandry. As a result each man will have one wife exclusive to himself and another wife shared along with brothers or clan brothers. That means each man has two wives at any time but at the same time he along with his brothers or clan brothers are multiple husbands to one woman. Therefore, polygyny and polyandry occur simultaneously.

* * *

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FAMILY

I. TYPOLOGY OF THE FAMILY

Typology of the families means systematic classification of the families into several types. A typology of the families can be constructed on the basis of the criterion or form of marriage. The forms of marriage may be monogamy, polygyny and polyandry. They are based on the number of male or female spouses involved in each union. Taking the different forms of marriages as the basis, the families can be classified into different types : Monogamous families, polygynous families, polygynous families and polyandrous families.

Another typology of families can be built on the basis of the criterion of descent. The term descent refers to the way in which one acquires membership of a kinship group. If the descent is traced through father in every generation it is called patrilineal descent or agnatic descent; if the descent is traced through mother in each generation, it is called bilateral descent. If one traces descent through mother in one generation but through father in another generation it is called ambilineal descent. According to the types of descent we may classify the families into several types: patrilineal families, matrilineal families, bilateral families and ambilineal families. If it is a natrilineal family, one may trace his ancestry through his father if it is a matrilineal family, one may trace ancestry through one's mother; if it is abilateral family, one may trace your ancestry through both father and mother; and if it is an ambilineal family, one may trace one's ancestry through father in one generation but in the next generation one's son may trace ancestry through his mother, that is, his wife.

A different typology of the families can also be constructed with the help of the criterion of residence. The term residence refers to post-marital residence. If residence is with or near the groom's parents it becomes patrilocal or virilocal (Latin:vir=male, locus=place); if it is with or bride's parents it becomes matrilocal or uxorilocal (Latin; uxoriwife, locus=place); if it is either with or near groom's parents or with or near bride's parent it is called bilocal or ambilocal (Latin; ambi=both, locus= place); of it is with or near groom's mother's brother, it is called avuneulocal, and if the couple sets up a new household removed to some degre from the relatives of either the bride or the groom, it is called neolocal (Greek : neo=new, locus=place) residence. According to these patterns of residence the families can be classified into several types: patrilocal families, matrilocal families, bilocal families, avunculocal family and neolocal families.

A distinct typology of the families can be built by using authority as a critirion. The term authority refers to the right to have power or commanding influence over the members of the family. If the authority is in the father, it is called paternal authority and if it is vested in the mother, it is called maternal authority. According to these two types of authority the families can be classified into two types : paternal families and maternal families. In paternal families authority runs in the male line. It is vested generally in the eldest male living or in the eldest son. In maternal families authority runs in the female lines. It is vested in the eldest female living or in the eldest daughter.

Types of succession also can be used for constructing a typology of the families. The transmission of office or rank is called succession. Succession may be patrilineal or matrilineal. Matrilineal succession means the passing of titles or ranks from females to females. Patrilineal succession means the passing of titles or ranks from males to males. Accordingly the families may be classified into patrilineal families and matrilineal families.

The families can also be classified into several types on the basis of composition of relatives or relatives or relatives or kinship structure. That means the families can be classified by determining how many father-offspring or mother-off-spring units are linked with particular women or men respecitively. A family that has one father-offspring unit linked with a woman or mother-offspring unit linked with a man is called single family, elementary family, simple family, atomistic family, immediate family or nuclear family. A family that has more than one father-off spring unit or mother-offspring unit linked with particular woman or men is called a composite family. The composite family may contain several sub-types. If the composite family has more than one mother-offspring unit linked with several men, it becomes polyandrous family; if it has more than one father-offspring unit linked with several women; it becomes plygynous family; if it has more than one father- offspring unit each linked with a woman or more than one mother-offspring unit each linked; it is called an extended family.

The following table gives the various types of families calssified on the basis of specific criteria such as marriage, descent, residence, authority and composition or kinship structure.

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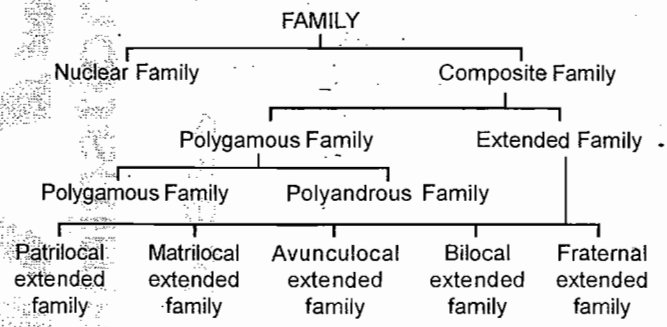
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Specific Criterion	Types of the families
Marriage	
Monogamy	Monogamous family
Polygyny	Polygynous family.
Polyandry	Polyandrous family.
Descent	
Patrilineal	Patrilineal family.
Matrilineal	matrilineal family.
Ambilineal	Ambilineal family.
Bilateral	Bilateral family.
Residence	
Patrilocal	Patrilocal family.
Matrilocal	Matrilocal family.
Bilocal	Bilocal family.
Avunculocal	Avunculocal family.
Neolocal	Neolocal family.
Authority	
Paternal	Paternal family.
Maternal	Maternal family.
Composition	
One father, Offspring or mother Offspring unit linked with a mother or a father	Nuclear family
More than one father one or mother-offspring unit linked with women and men.	Composite family.

Typology Based on Multiple Criteria

The typologies of families, each based on a single criterion, are not complete ones for two reasons. First, each single criterion employed to classify the families into distinct types takes into consideration only one aspect of the family, either marriage or descent of residence or authority or structure at a time, without simultaneously taking into consideration the various aspects of the family. Second, each single criterion cannot be considered as having universal applicability. That is, each criterion cannot be successfully employed to classify the families existing in all types of human societies in all places, at all times. Therefore, multiple criteria are to be employed for evaluating as satisfactory typology of the families.

George Peter Murdock, Radcliffe–Brown and several other anthropologists have constructed separate typologies of families on the basis of multiple criteria, such as forms of marriage, patterns of residence and networks of kin relations (Murdock and Radcliffe-Brown). The typology of families envolved by Murdock is most commonly used by anthropologists in studying household arrangements in different societies across the globe. The following chart shows the types of families constructed by George Peter Murdock.



The nuclear family consists of a married man and woman with their offspring although in-individual cases one or more additional persons may reside with them.

One composite family is an aggregation of two or more nuclear families. It may be divided into two types: The polygamous family and the extended family. The polygamous family consists of two or more nuclear families affiliated by plural marriages, i.e., by having one married parent in common. In this context the plural marriage may be polygyny or polyandry.

An extended family consists of two or more nuclear families affiliated through an extention of the parent-child relationship. Depending upon post-marital residence, an extended family may be patrilocal, matrilocal, bilocal and avunculocal. The patrilocal extended family consists of two or more nuclear families affiliated through an extention of father-son relationship. The Matrilocal extended family consists of two or more nuclear families affiliated through an extension of mother-daughter relationship. The bilocal extended family, which is also known as ambilocal extended family, is a combination of patrilocal extended family and matrilocal extended family. The avunculocal extended family consists of two or more nuclear families affiliated through an extension of maternal-uncle and sisters son relationship. It includes the nuclear family formed by a man and his wife and daughters, and the nuclear families formed by his sister's son and their wives and children.

The patrilocal extended families are most common form which involves each three generations of kin living together, viz, parents, their children and the families of their children. If the parents die, their children may remain together often with several brothers heading the family. This form of extended family, which involves the joining of two or more sibling's families is known as the fraternal joint families. The fraternal joint families are also found in

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China and Pakistan. In Pakistan it is called Khumbah; in Yugoslavia, Romania and Bulgaria, it is known as Zadruga. The joint family among the Tanala of medagascar is similar to a patrilocal extended family.

Apart from these most common types of extended family, there are some uncommon types of extended family, such as matrifocal families, consanguineal families and stem families confined to a few society in different parts of the world. A matrifocal family is an aunts, nieces, cousin sisters or grandmothers. A matrifocal family is characterized by the absence of permanent adult male members. This type of family is found in Jamaica, Naity, Guyana and other Carabean countries.

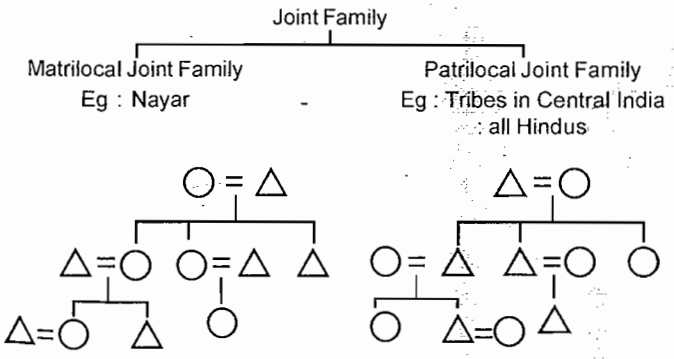
The consanguineal family is an extended family which includes a woman, her son, her daughters, her daughter's children. However, it does not include son's wife and their children. Thus, a consanguineal family consists of a woman, her brothers and her offspring.

The stem family is a kind of patrilineal extended family, which includes an old couple, one of their son, one of their son's children. It is, however, found among some of the farming population of Europe.

II. Q. Define joint family. Explain the changes coming up in Hindu Joint Family system due to modernisation.

Ans :

'Joint Family' is a collection of more than one primary family, one the basis of close blood ties and common residence.



In a patrilocal joint family father, mother married sons, their wives and unmarried children live under a common roof.

In a matrilocal joint family, mother, father, married daughter, their husbands and unmarried children live under a common roof.

In mainstream Hindu society, Bhils, patrilocal joint families are prevalent. Khasis have matrilocal joint family an individual lives along with his wives and children. Eg : Kipsigi, Lakhar, Tam ala, Mariya, Baiga. In these societies eldest wife has special powers.

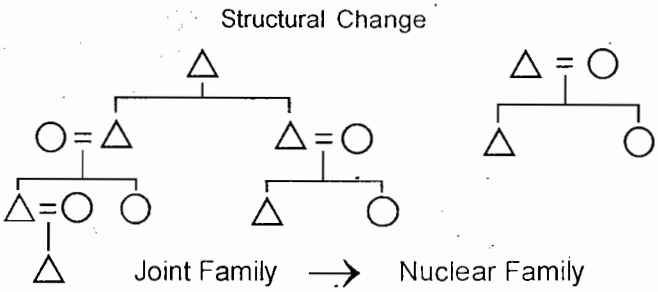
A typical Hindu Joint family consists of a man, his parents, brothers, his and their wives, married sons, unmarried children. The Indian-joint family is patrilocal & patrilineal. The power – authority flows through male line and sex i.e., Father or eldest brother will be the fountain head. Marital ties are less significant than blood ties. Males and brothers operate as a unit with all authority. Though women leave their family of origin, to stay with the husband, she is still considered a stranger; with no role and power. The male collective decision making keeps the women aloof. Women have menial position at home & in society due to her relatively less contribution to economy.

Traditional joint family was performing a large range of services.

1. It was a mechanism of redistribution of wealth & income. The joint family was functioning as a unit of production as well as distribution. The non-earning members and old & child, disabled were taken care off. But today family is left more as a unit of consumption rather than production.
2. Culturally, joint family helps in transmission of traditions. It is a learning place for the child, It even teaches the gender bias-in-behaviour patterns, dress patterns etc.

   Child is better taken care off. This is a powerful incentive given to the children. The Joint family initiates and continues the socialisation process. Love affection i.e., emotional security is provided to the child, the aged and the diseased.
3. Economically – it trains the children in traditional skills. Apart from this, it offers a source of security. Ofcourse, in the process, the efforts of few will be used by others and the later may turn out to be lazy.

The Hindu joint family system has been subjected to Industrialisation, mobilization, changing role of women etc. These factors are bringing structural & functional changes in joint family.



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Industrialization has brought with it, job opportunities in distant places. Hence, there is a pressure due to mobility. Diversification of work leads to differential incomes. In contrast to the earlier situation, wherein all the members of a family produced together and had equal access to the produce, the present system is individual oriented. The rights depended on achievement. There is hardly any wealth to be inherited (due to reduction in value of land – as India particularly depends on Agriculture). Income is related to performance. The unwillingness of members to share fruits with others has induced pressures in the structure of Joint family. Hence, the size is diminishing. Neolocal residence and nuclear family system is on an increase.

The role of women & changing trends in man – woman relationship is another important feature. Universally, man was at a higher position than woman economically and authority wise. But the perilisation & division of work among women & men (child care & earning) provided for a strong bond of marriage between man & woman. But education, particularly female education has increased the earning capacity of female. This resulted in less dependence of female over male. In other words, modern education taught equality. There is more recognition of rights by women and more assertiveness. The growing faminism inareased consciousness of equal rights.

All these led to break up of joint family. Apart from these, alternative institutions are emerging for various functions traditionally performed by joint family. Eg : Hospitals to take care of the diseased, old age homes, crusche, schools etc. The role of family in socialisationary process also declined. Newspapers, peergroups, media are giving a lot of information. Hence, for information the child does not solely depend on family. Coupled with this, children are kept in hostels (making them occasional guests to their own house), parents spending a lot of their time outside house is reducing the importance of family.

Depending on the above facts it cannot be concluded that joint family has disappeared from Indian social scene. It is sattaining newer forms through adaptive changes.

- Even when some brothers are living in urban areas may not be totally aloof from his family. He may be helping in education of other kins and improving agriculture.
- Worldwide industrial organization are running along family lines. Ex : TATAs & Birlas. This calls for pooling up of resources, through the members do not stay together.

Hence Hindu Joint Family on the one hand is losing its importance, on the other it is attaining never functions.

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DOMESTIC SPHERE OF CULTURE

The conventional way to describe the domestic life of simpler societies is in terms "of family or household types, marriage and the relational patterns of the members of the group, i.e. kinship. Very often, these are described under social organisation. In this chapter, our task is to present some relevant findings of the domestic domain of societies as reflected in ethnographies. Societies all over the world have domestic unit of one or other kind. Activities of this unit include preparation and consumption of food; cleaning, grooming and enculturating the young; sleeping and adult sexual intercourse. (There are a few societies, in which sexual tasks are performed away from their dwelling places!) Cultural anthropologists try to understand the very organisation and functions of social groups structured around a domestic unit.

THE NATURE OF DOMESTIC GROUP

Study of social forms is centrally important in socio-cultural anthropology. Anthropologists have been analysing them for more than a century. It is easy to describe the range of social forms in terms of 'family', 'marriage', residence', and 'kinship' but when we come to look at them on a grand comparative scale, confusions do arise. The basic issue posed here is how to look at the social form. Many texts begin with either a definition of the family or a proposition that the family is universal, Murdock, for example, opens his discussion in Social Structure (1949) by stating:

"The family is a social group characterised by common residence, economic cooperation and reproduction. It includes adults of both sexes, at least two of whom maintain a socially approved sexual relationship and one or more children, own or adopted, of the sexually cohabiting adults".

Since Murdock. anthropologists have focused their discussions mainly on three topics–the presence or absence of nuclear of elementary family, typologies of family and the development cycle in family. In his cross-cultural studies, Murdock argued that the nuclear family was found in every human society. Talcott Parsons (1955) also claimed that the small conjugal family provided the most satisfactory framework for socialising children.

In recent decades, however, cultural anthropologists have been critical of this line of thinking. Studies by Gough (1959), Spiro (1954), Fox (1966) and Goodenough (1970) among others, have shown that the nuclear or conjugal family is not necessarily the core of a society. These studies, however, do not preclude the existence of small domestic units. Rather, they show how erroneous are terms like 'elementary' or 'extended' family when the boundaries for these social groupings have not been defined, and when it has not been well established, how these social units operate within their context.

FAMILY, HOUSEHOLD AND DOMESTIC GROUP: CONCEPTUAL ISSUES

It would be profitable if we first clarify the meaning of the terms family, household and domestic group. Anthropological usages of these terms have been ambiguous because each of these concepts may carry more than one meaning at a time, and the users of these terms seldom specify the context in which they are used. Many times, the terms family, household and domestic group are viewed as synonymous even when it appears that the household and domestic groups are not based on the family.

The frequently cited works on Taravad of the Nayar and on Okka of the Coorgs of south India offer excellent examples.

MEANING OF FAMILY AND HOUSEHOLD

Some scholars have delineated systematic distinctions between family and household. The generally accepted meaning of the term 'family' has already been discussed. A 'household' is composed of people who unlike family, though not necessarily related, live in the same house or compound. Bohannan (1963, 1968) uses the term 'household' for this

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kind of arrangement. He defines the family in terms of kinship and marriage; and the household in terms of residence and propinquity. He uses household as a synonym for domestic group. He states: "A family is a kinship group, marked by kinship (genealogical) relationships. A household is a 'local' or spatial group, marked by propinquity. A family need not live together as a household. A household need not be made up solely, or even primarily, of members of the family. Thus, the family and the household must not be confused (even though such is a norm in most of the societies)" (Bohannan).

Household is a group of people who live together and form a functioning domestic unit. Bohannan cautions that 'the members of a household' may and may not constitute a family; and even if they do so, it may or may not be a simple nuclear family.

Household is an unfortunate term because it carries more than one meaning. In common parlance, it means a 'house' or a 'hut' or a dwelling place where people live together. The 'living together' implies residing under the same roof. It does not deal with the question of whether or not the members form a common, consumption unit (hearth), a production (economic) unit and a resource-holding unit. Nor does it refer to who are the members. For a western (industrial) society, the household is treated as the same unit as that of a family. In many non-western societies, the situation is less clear-cut. For this reason, many ethnographers describe a non-western society without making appropriate qualification in their use of these terms. This should be avoided.

In specific sense, the term household refers to a number of activities common to a group of person. At least three, such activities are commonly recognized–residence, food consumption and production. To put it differently, a statement concerning the household organisation of a society implies that some people (not necessarily related to one another) live together (under the same roof, or in adjacent buildings or huts forming a common compound.), sharing and contributing to a common hearth. Here, sharing and contributing to a hearth means that these members hold a common resource/fund/budget and their consumption needs in part are met by this fund. The variations in household criteria are of particular concern to census enumerators who, while taking a household census, must decide what criteria should be followed. When not specified in explicit terms, confusion arises and the analysis becomes dubious.

THE MEANING OF DOMESTIC GROUP

Some scholars, among whom Meyer Fortes (the British social anthropologist) appears to be the forerunner, have encouraged ethnographers to emphasise on 'domestic group' rather than on 'family'. Because of paucity of space, it is not possible here to examine relevant ethnographic works reflecting this trend. It suffice to say that the investigators are now much concerned with defining the criteria of the domestic group and separating it from the family and household. Fortes defines domestic group in the following words:

"The domestic group is essentially a householding and housekeeping unit organised to provide the material and cultural resources needed to maintain and bring up its members".

Fortes further remarked that in the realm of domestic group "kinship, descent and other bonds enter into the constitution of the group, whereas the nucleus is formed purely by the direct bond of marriage, filiations and siblingship" Thus, he maintains that the family and the domestic group are analytically distinct, although actual composition of the two may be identical. The major features of the domestic group are co residence, production consumption of food and participation in day-to-day affairs in the domestic life. It is the task of the ethnographers not to use these terms synonymously if the situations are different. Jack Goody has supported these criteria in the light of ethnographies.

To state briefly, a domestic group refers to a group of people related by marriage, filiations and siblingship, who

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concurrently form a residential, reproductive, consumption, production and a resource-holding unit. This statement needs further clarification. Related by marriage means the relationship among the members by the bond of marriage. Filiation refers to the relationship between the parents (father, mother) and children. If relation extends over two generations, then father (male) or mother (female) is usually regarded ancestor or ancestress and this is called descent. Siblingship refers to the relationship between brothers and sisters of the same parents.

The people forming above-mentioned unit may not actually live under the same roof; there may be a cluster of huts forming a large compound. Further, it may not be the case that the members would cooperate in all these ways. Ethnographers find one or more reproductive/consumption/ production units in a society upon a given moment of time. In a society, domestic group containing more than one reproductive/ consumption/ production unit represents the phases of development which will be described shortly under 'processual analysis'.

DIVERSE FORMS OF DOMESTIC GROUP

Typological Approach

In anthropological textbooks, chapter on the 'institution of family and marriage' are full of statements concerning types of families. Some anthropologists consider the number of persons including the number of mates as the basis of classification, others stress the degree of genealogical relations (primary or secondary, etc.), the relative importance of kinship versus conjugal ties (affinal, congenial, etc.), and still others consider the activities of members for classification (orientation, procreation). In fact, the categorisation of family (household/domestic group) is culture specific. We must not forget that a particular group enters into another type along with changes in the individuals and their role-behaviour. In other words, some other variables are affecting the organisation of a social group. However, it has become a tradition to depict the social life in popular typological terms. The following are some varieties and sub-varieties of the social life:

I. Conjugal/nuclear variety
II. Consanguineal
III. Composite conjugal
   (a) Polygynous
   (b) Polyandrous
IV. Composite unilineal/extended
   (a) Patrilineal
   (b) Matrilineal
   (c) Joint

In many societies, the living arrangement is around a coresident husband and the wife and their dependent unmarried children. This is called nuclear family. It is to be noted that the core of the family, in comparative perspective, is the unit of mother-plus-dependent children. When a male (husband) joins to this unit, the resultant group is nuclear family. But, his roles are as spouse-cum-sexual partner, begetter of children, bread provider, legal guardian is not always subsumed. Consider the following example:

The existence of matrifocal family among many societies of the Caribbean, British Guinea, Puerto Rico and elsewhere clearly emphasises on 'mother-children unit", Matrifocal or matri-centric family is a form of domestic unit in which husband (father) is not regularly present. The cases of Nayar and the Caribbean societies are archaic type matrifocal domestic groups. Probably matrifocal family it is the result of division of labour separating adult males and adult females in a community.

Among the Nayar (a Hindu caste group of Malabar, south India) the domestic group (Taravad) consists of a cluster of brothers and sisters related matrilineally. Soon after the puberty, a ritual is performed which allows the sisters to bear children by the males of another taravad. The brothers act as guardians and economic providers. This kind of arrangement is locally termed as Taravad. A taravad is a corporate social group (Gough, 1958).

In many African societies such as the Ashanti and the Tiv, the living arrangement takes the form of an extended social group. Among the Ashanti, reports Fortes, a man has several wives, each of whom would normally reside with her

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pre-adult children in a separate hut in the same husband's compound. She is also the owner of the granary attached to her hut. Each co-wife and her children constitute separate nuclear units consisting of mother and children and thus, form independent units among the Caribbean societies as discussed above.

The above examples clearly indicated the nature, composition and organisation of social group called domestic group. A simple domestic group, i.e. a nuclear unit develops or expounds into a larger composite unit through time.

PROCESSUAL ANALYSIS OF DOMESTIC GROUPS

From the above account, the crucial points emerging are: why, when and how a domestic group would change into other forms. To put it differently, how a domestic group develops? To answer this, some anthropologists have turned towards 'processual analysis' of clustering social group (family/domestic group). A processual approach is the systematic way a group changes or develops through passaged of time. In a society, for example the conjugal pair forms concurrently a single residential, reproductive, consumption, production and resource-holding unit. In course of time, some members are added and deleted due to birth, marriage, death and migration. Thus, the domestic unit started by the parents changes over a period of time and includes a number of additional reproductive, consumption, production and resource-holding units. In other words, a domestic group undergoes a series of transformation. Ethnographers of family-researches referred to this as 'developmental cycles' in domestic group.

PHASES OF DEVELOPMENTAL CYCLES

Sociologist Charles Loomis used the phrase 'developmental cycle' for the first time. In anthropology, Meyer Fortes popularised this in the analysis of family, marriage and kinship of non-western societies. He delineated their phases/stages/paths in any domestic group. They are: phases of expansion, dispersion and displacement.

The first phase starts from establishment of a conjugal pair, i.e. family of procreation,-i.e. the unit within which people have or adopt children of their own. In others words, the expansion of the original group continues till the birth of the last child to the couple. The phase of dispersion is marked by the first marriages of all the children. In this situation, some or all of the married children are coresidents of the parents' unit. The death of the parents' and establishment of family of orientation by the offspring characterise the third phase of replacement. Family of orientation refers to the unit into which people are born and receive socialisation.

To the processual lists, one phase is a dynamic part of the original unit; to the typologists, one particular 'type' or 'form' is present at ā given moment of time. Processualists are more interested in carving out all possible phases through which a group passes. They suggest that every society has characteristic demographic properties: and that these properties manifest in conjunction with economic variable in the functioning of domestic group. In short, pattering and development (phases/ paths/cyclings) are reflected through the distribution of surviving sons, their relative ages, their ages relative to their father/mother, marriage age, age at the birth of the first and the last sons, spacing of birth, distribution of economic resources such as land and forest, among others

DEVELOPMENTAL CYCLES, RESIDENCE PATTERNS AND INHERITANCE CUSTOMS

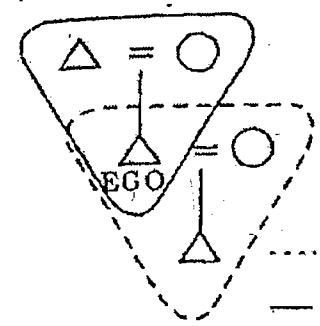
We have discussed above the three popular stages in developmental cycles, viz. expansion, dispersion and replacement. At any point of time, the current phase of domestic group may influence its residential patterns. For example, in societies like the Korwa of North-central India, if one of the spouses is the last child remaining in the group after the others have married and left, he/she stays on as prospective heir.

The pattern of residential decision may depend on factors like residence customs and succession (Netting, 1984). In some societies, the oldest child inherits the property. This is known as primogeniture In other societies, the youngest' hold remains in the domestic group and a inherits the property. This is referred to as ultimogeniture. In the latter case, the child begins a new family cycle after his father.

Thus, the processual analysis may hope understanding the changes in the social structure, which is beyond the scope of typological description of a social group. The domestic groilp is not static; it follows a series of stages- expansion, dispersion and replacement. These patterns are outcome of a number of variables–demographic, economic and cultural norms such as inheritance customs. A consideration of these variables is absolutely essential if we are to go beyond structural description of social structure.

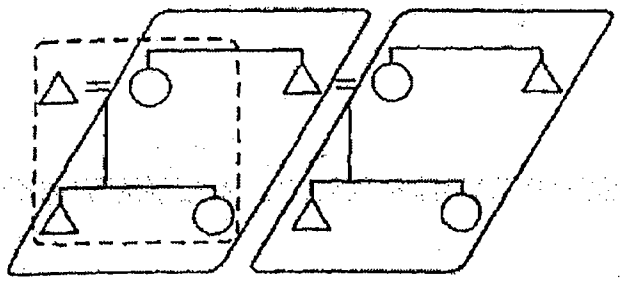
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---- Family of orientation

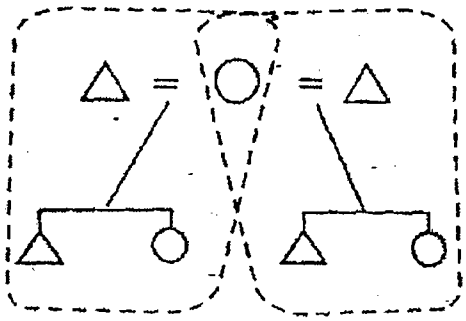
—— Family of procreation



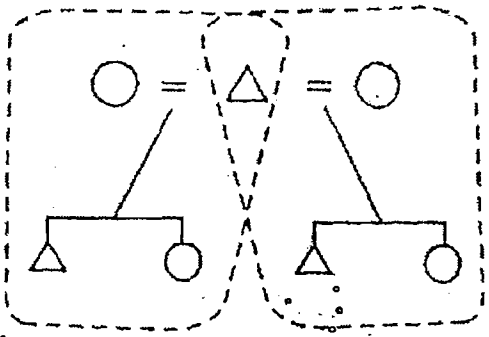
---- Conjugal family

—— Consanguineal family

1. Composite conjugal

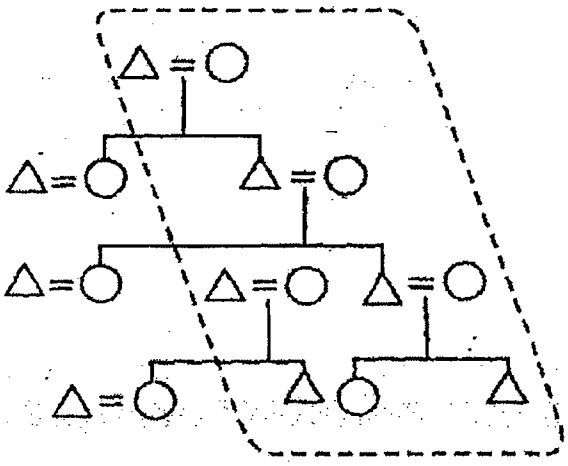


a. A polyandrous composite family

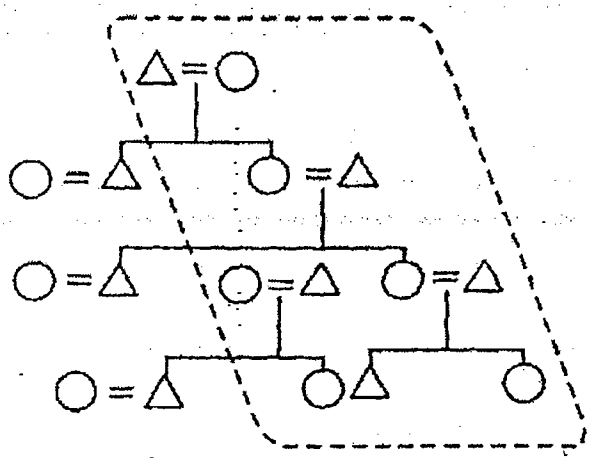


b. A polygamous composite family

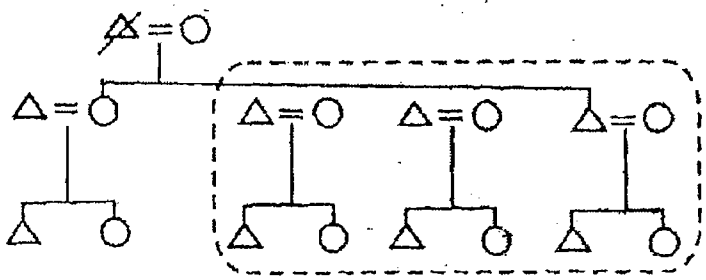
2. Composite unilineal or extended family



a. A patrilineal extended family



b. A matrilineal extended family



c. Joint family

Fig. I Diverse forms of domestic groups. (Dotted lines indicate conjugal family segments)

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CHANGES IN THE INDIAN FAMILY

'From joint family to elementary (or nuclear or individual) family'. This has been a slogan to summarise changes in the family in India during modern times. Recently, some sociologists have pointed out that the changes implied by the slogan are not taking place. What is emerging, they say, is not the elementary family but a new form of the joint family. Even these sociologists, however, do not question the correctness of describing the traditional family system of India as the joint family system. This belief about the traditional Indian family subsumes a set of some other beliefs: (a) Traditional India was village India, and the joint family was therefore a characteristic of village India; (b) contrari-wise, urban areas are new and characterised by the elementary family; and (c) urbanisation, therefore, leads to disintegration of the joint family. Some sociologists have pointed out that urbanisation does not lead to disintegration but only to transformation of the joint family. They do not, however, question the belief that the joint family was always a characteristic of village India.

To distinguish between 'household' and 'family' has now become common in sociology and social anthropology. In common English parlance the word 'family' has several different meanings, including 'household'. The common Indian word for the family, viz., *kutumb,* has likewise several different meanings. However, for the sake of technical analysis, 'household' should be distinguished from the other referents of 'family'. For example, two brothers and their wives and children may live in two separate households, but they may be bound by a number of relationships of many kinds. For the sake of clarity, it is necessary to consider such inter-household relationships as distinct from relationships within the household. Although the aim of the study of the family should be to study it in all its aspects, a beginning has to be made with the study of the household. An examination of ideas about the past of the Indian household is, therefore, crucial in a comprehensive study of changes in the Indian family.

If we take a census of households in any section of Indian society– a village, town, or caste – and examine their numerical and kinship composition, we find a number of types of composition, ranging from the most simple single-member household to a very complex household of many members. A 'simple' household is composed of a complete elementary family or a part of an elementary family. A 'complex' or 'joint' household is composed of two or more elementary families, or of parts of two or more elementary families, or of one elementary family and parts of one or more other elementary families.

One of the first tasks in an analysis of households is to formulate the types of composition. The structure of the household becomes more complex as more categories of relatives are included. In a one-member household there is no relationship; in a two-member household there is one relationship; but beyond this the addition of one relative means an addition of more than one relationship. For example, the addition of a son's wife to a household of father, mother and son, means the addition of relationships not only between the son and his wife but also between father-in-law and daughter-in-law and between mother-in-law and daughter-in-law. Addition of relationships tends to create conflict between roles. For example, the conflict between a man's loyalty to his wife and loyalty to his parents is proverbial. Each person in a household is involved in a complex pattern of behaviour with every other member. Everyone in a household has his own likes and dislikes, habits, tastes and idiosyncracies. Life in a household is marked by sentiments and emotions, and cooperation as well as conflict. Therefore, if our aim is to understand household life in its entirety, the formulation of types of household composition should take into account all the various members of a household. After the types have been formulated, it is necessary to examine the frequencies of households of various types.

Classification of households according to types of composition is not, however, an end in itself. The types are not discrete and haphazard but are interrelated in a developmental process. This process may be in progression or in regression. Progressive development of a household takes place due to increase in membership, mainly by marriage, death and partition. It may be mentioned in passing that there is always some pattern in the developmental process, but it is not cyclical in nature as considered by Fortes and his associates.

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One of the determinants of the developmental process is a set of explicitly stated rules or norms governing the formation of households. In most sections of Indian society, the bride after marriage leaves her parental home and goes to live in her conjugal home. A son and his wife are required not only to start their married life in his parental home but also to continue to live there afterwards. This norm has a number of implications. For instance, if a man has more than one son, each of the junior sons and his wife will have to live not only with his parents but also with his senior brothers and their wives and children. In fact, frequently people state that a man and his wife should live with his brothers and their wives. Furthermore, they say that brothers and their wives should live together not only during the lifetime of the brothers' parents but also after their death, and the brothers' sons and their wives should also live in the same household. Sometimes the norm is extended still further.

Taking all these norms together, the central idea is that while female patrilineal descendants of a male ancestor go away to live with their husbands, the male patrilineal descendants and their wives should live together. The wives should be so completely incorporated into their husbands' kin-group that they should not be divorced and that even after their husbands' death they should stay on in the same household. Unmarried children should be with their parents; in the event of divorce or death of their mother, they should stay with their father or his male patrikin. The central idea behind these norms can be briefly called 'the principle of residential unity of patrikin and their wives'. It is necessary to clarify that this principle is normative in nature, and that there are always deviations from it as in the case of all norms. The measurements of conformity to and deviation from the principle is, therefore, an important problem of inquiry.

While the principle is common to almost the entire Hindu society, there are differences between different sections of the society in the extent to which it is observed. First of all, there are differences in the maximum extent to which the developmental process goes in progression along the path set by the principle. For example, in a Gujarat village I have studied, there is no case of two or more married brothers living in a single household after the death of their parents. In a nearby village, on the other hand, there is a considerable number of households of this type. Such differences in the maximum extent may also exist between villages and towns, between one caste and another caste, and between one region and another region.

Secondly, while the maximum extent of the developmental process may be the same in two sections of society, there may be differences in the frequencies of the cases in which the norm is observed within this extent. For example, in the Gujarat village just cited, only about 5 per cent of the total number of households are composed of one or both parents and two or more married sons and their wives and children, while 19 per cent of the households are composed of one or both parents and one married son. This is mainly due to the fact that married sons tend to live separately from parents even before the death of parents. Out of 41 cases of parents having two or more married sons, only in 12 cases (29.26 per cent) all the sons live with the parents in a single household, while in 29 cases (70.73 per cent) all or some of the sons live in separate households (in the village itself). And out of 50 cases of parents having one married son, in 38 cases (76 per cent) the parents and the son form a joint household, while in 12 cases (24 per cent) the parents and the son live in separate households.

One of the factors affecting the degree of extension of the principle of residential unity of patrikin and their wives appears to be the degree of Sanskritisation of a caste. This is due to the great emphasis which the Hindu scriptures lay on the high degree of observance of this principle. The ritually higher castes, who are under the greater impact of the scriptures, tend to have a higher degree of extension of the principle. Another factor affecting the degree of extension of the principle appears to be the custom of cross-cousin and uncle-niece marriages.

Whatever be the maximum extent to which the principle goes in progression in a particular section of the society, it is important to note that the processes of progressive and regressive developments go on simultaneously in the society taken as a whole. While one household may be undergoing progression, another may be undergoing regression. As a result, there are always households in the society which are small and simple in composition, along with households which are large and complex in composition.

When a complex household, say, of two or more married brothers, is partitioned, two or more separate

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households come into existence, but at the same time a number of other relationships continue to operate. They would cooperate in economic pursuits, hold and manage property jointly, help each other on many occasions, celebrate festivals, rituals and ceremonies jointly, and so on. This is also a normal process, which highlights the importance of technical distinction between 'household' and 'family' mentioned at the outset. Thus, two or more households may be separate but they may constitute one family.

For a long time, students of the Indian family have used ancient Indian literature for information about its past. This information is of two main kinds: (a) pertaining to the property aspect of the family, which is generally included in the study of Hindu Law, and (b) pertaining to certain family rituals, such as the *shraddha*.

The Hindu legal text *Mitaksara* first defines a coparcenary: it comprises only those males who are entitled by birth to an interest in the joint or coparcenary property, i.e., a person himself and his sons, son's sons, and son's grandsons. As each son acquires by birth an interest in coparcenary property, even a father and his unmarried son are sufficient to constitute a coparcenary. Under the *Dayahhaga,* there is no coparcenary between a man and his son(s), married or unmarried, even though they may be living in a single household. The legal definition of the joint family is based on that of the coparcenary: it consists of all males included in the coparcenary, plus their wives and unmarried daughters. The latter are not coparceners but have only a right to maintenance.

The main points here are two: (i) The legal definition of the joint family is a highly specialised one and has nothing to do with the sociological distinction between elementary family and joint family. A joint family of the legal conception can exist even within an elementary family of the sociological conception. For example, a father and an unmarried son, or a widow and her unmarried son, are sufficient to constitute a joint family according to the law. (ii) The law does not lay down the rule that the joint family of the legal conception should always be a joint household. A son may live separately from his father, and one brother from another, but they continue to be members of their respective joint property group. In brief, the law is concerned primarily with the rights of constituting a property-holding group and of maintenance therefrom, but not with the constitution of the household group.

In the sacred texts *(shastras),* the question as to who should hold and inherit property is discussed usually in relation to the question as to who should perform the *sbraddba* ritual for whom. In other words, the legal definition of the constitution of the joint family tends to coincide with and is sanctioned by the definition of the circle of persons required to perform the *shraddha.* This circle of persons need not live in a single household, just as the persons constituting the joint property group need not live in a single household. It seems to me that it was because of the coincidence of the legal and the ritual definitions of the joint family that the definition given in the *shastras* came to be accepted as the general definition of the Hindu joint family. I shall call it the Indological definition. As most of the early studies were carried out by Indologists (including historians, Sanskritists, and Orientalists) on the basis of sacred literature, and as both Indologists and lawyers were dominant in the academic field in India, the Indological idea of the joint family carried a lot of weight and gained popularity.

The indological-cum-legal material on the Indian family was used by Henry Maine in his general theory of the evolution of the family. He compared and contrasted the joint family of India with the individual family of the West, and considered the latter as later in evolution than the former. He thus laid the foundation of the sociological study of the Indian family, and through him the Indological view of the Indian family came to be accepted in sociology.

I have pointed out the fact that the Indological definition is not concerned primarily with the composition of the household. Insofar as it was concerned with the household, it laid down only the definition of the ideal household, or in my terms, only the maximum extent of progression of the developmental process of households. The Indological literature does not provide any information on the various types of households, nor on the frequency of households of each type. Obviously there was no census of households in any section of Hindu society, at any period of time. Furthermore, the literature provides information more about the Brahmins and a few other higher castes whose property relations and rituals were governed by the sacred texts. With regard to the household also, the ideal that the texts emphasised was high– a household of four generations– and it seems only the higher castes tried to emulate the

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ideal to a higher degree. The texts do not provide any idea of differences in the family life of different sections of the society.

We begin to get more precise data on the household only with the beginning of the British administration in India. The Office of the Census Commissioner of India planned to publish a series of volumes on population estimates for the period from the 18th century to 1871 on the basis of censuses conducted in different parts of India during the period. The first volume in this series, dealing with the decade 1820–30, was published in 1965 (Bhattacharya and Bhattacharya 1965). It seems to me that it will be extremely useful if attempts are made to find and then analyse the original schedules of all these censuses. My experience in Gujarat indicates that such attempts are likely to be fruitful. During my search for sources for the study of social history of villages in central Gujarat, I found in the taluka headquarters the original schedules of a census carried out by Captain Cruikshank and his associates, the same schedules that he used in compiling general reports on different parts of Gujarat, which have in turn been used by the Bhattacharyas in their volume on population estimates for 1820–30 mentioned above. In this census of Gujarat, a Census Register was prepared for every village, in which were listed the names of heads of households in the village according to their religion and caste, and against each name were given the following details: (i) houses and huts, (ii) men, (iii) women, (iv) servants and slaves, and (v) total number of persons.

I have made a detailed analysis of the 1820–30 census data on the household composition in a Gujarat village. It shows (a) that the average size of the household was 4.5, and (b) that progressive development of households rarely, if ever, went beyond the phase of co-residence of two or more married sons during the lifetime of their parents. It is significant that the census data on a village in Maharashtra of the same period, which Professor Ghurye has presented in his book *After a Century and a Quarter*, shows the same average size of the household as in the Gujarat village. The early nineteenth century data, thus, indicate that we cannot start the study of changes in the family in India with the assumption that villagers in traditional India always lived in large and complex households of three or four generations. We need a more realistic base line, and the early nineteenth century census data are a very useful source for reconstructing such a base line. As regards differences in the family between rural and urban areas also, it is important to keep in view the position at the beginning of the nineteenth century. The early nineteenth century census data indicate that there was a higher proportion of the population of higher and more Sanskritised castes in towns than in villages. The bulk of the population in villages consisted of lower and less Sanskritised castes. If this point is related to my earlier point that the higher castes were under the greater impact of scriptures and there was, therefore, a higher degree of observance of the principle of residential unity of patrikin and their wives among higher castes, it follows that there was a higher degree of observance of this principle in towns than in villages. The reality was, thus, quite the reverse of the widely prevalent belief that there was a greater emphasis on joint households (i.e., on the principle of residential unity of patrikin and their wives) in villages than in towns in traditional India.

Although the censuses upto 1871 had certain obvious drawbacks–they covered only small areas in the country, and even within a small area the census was not taken simultaneously in all the villages and towns–it is remarkable that they collected and recorded a great deal of information of demographic and sociological value. The later censuses covered wider areas and became uniform in techniques, terminology and classification, but there was retrogression in the case of information concerning the household. In all the censuses from 1867 upto 1941, the distinction between 'household', 'house' and 'building' was not followed uniformly in all the Provinces and States and sometimes even in all the parts of a single Province and State, and hardly any attempt was made to collect information either on the kinship or on the numerical composition of households. Nevertheless, even the meagre information they provided is highly suggestive.

All of them show that the average size of the household was rather low, between 4.5 and 5. E.A. Gait, the Census Commissioner for 1911, stated the position very succinctly: 'the average population per house is 4.9 or much the same as in European countries. In the British Isles it ranges from 4.8 in Scotland to 5.2 in England and Wales'. This suggests that the people in India lived mostly in small and simple households. It is noteworthy, however, that almost all the census officials interpreted their figures unrealistically. They assumed that Indians in pre-British

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days always lived in large households, and therefore, they concluded from their figures that the traditional household system was disintegrating due to modernisation. The assumption about the past was a great obstacle in a realistic interpretation of the figures before them.

In a recent paper (1961), Henry Orenstein has used the census data to examine the usual general idea about changes in the Indian household. He has posed the problem as follows: If the widespread belief that the modern processes of industrialisation, Westernisation, etc., of traditional societies such as India lead to smaller households, then the increasing industrialisation, Westernisation, etc., should be reflected in a decreasing average size of the household. He presents figures to show that the average size of the household has not only not decreased from the 1867 to the 1951 census, but there has been on the contrary a slight tendency–very slight indeed–towards increase. This suggests that the modern processes of industrialisation, Westernisation, etc., have not brought about the so-called disintegration of large and joint households. Orenstein conjectures, I think quite rightly, that the belief about the wide prevalence of large and joint households in pre-British India is false.

During the last thirty years or so, professional sociologists and social anthropologists have studied the problem of changes in the family in India. Some of these studies are concerned only with the household and some with the household as well as other aspects of the family. I would refrain from commenting here upon those studies which are not concerned with the household, as well as upon those studies of the household which are not concerned with the problem of change. As far as the studies of changes in the household are concerned, I submit that most of them have suffered from the assumptions I have examined in this essay. It seems to me that we need a new and more sophisticated framework for our study.

First of all, we have to distinguish the normal developmental process from change. This is a difficult task, but we have to face it. Studies of the early nineteenth century data on the household seem to me to be very necessary for this purpose.

Secondly, there is no point in postulating a single line of change for the entire Indian society. I suggest that it would be profitable to bring into the study of the household Srinivas' ideas on Sanskritisation and Westernisation. In this context it is worthwhile to recall the slight increase in the average size of the household indicated by Orenstein. It is possible that this increase may be due to some demographic factor. Orenstein himself has suggested that it may be due to a rise in the average number of children or in the average number of adults per household. (If the latter is true, it strengthens the argument in favour of an increasing overall emphasis on large and joint households.) Notwithstanding the influence of demographic factors, it seems worth inquiring whether the Sanskritisation of lower castes and Adivasi tribes that has been going on on a massive scale in the country has contributed anything to an overall greater emphasis on the principle of residential unity of patrikin and their wives. It cannot be denied that Westernisation of the higher castes has contributed to a lesser emphasis on the principle, but the countervailing influence of Sanskritisation might have led to an overall tendency in favour of greater emphasis on it. What I plead is that let us not assume that there has been an inevitable trend from large and complex (or joint) to small and simple (or nuclear) households.

Finally, we need to have a fresh look at the problem of the impact of industrialisation and urbanisation on the household. It has already been pointed out that the situation in the past was possibly quite the reverse of what we have been assuming it to be. That is to say, there was greater emphasis on large and joint households in towns than in villages. Migration of rural people to towns, therefore, does not necessarily mean migration from a social environment of large and joint households to that of small and simple households. It seems to me that the study of the long-established population of older towns, and cities is extremely important. I have in mind for example the walled cities of Delhi, Agra and Ahmedabad, certain sections of the population of even Bombay, Calcutta and Madras, and the large number of small towns. We should also examine the extent to which migrants to a town tend to be its permanent residents, and whether they practise the old norms when they are permanently settled. The whole question of relation between migration and the household in India seems to be complicated. In any case, we begin to understand it better if we get rid of the old established assumptions.

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ECONOMIC DEVELOPMENT AND THE CHANGING FAMILY SYSTEM

The relationship between economic development and changes in the family system have been studied by anthropologists, sociologists and historians. In India, several studies have reported close association between changes in economy, emerging entrepreneurial structures and processes of change in the family system. Similarly, studies have also delineated the relationship between family systems and their impact on the nature and direction of economic development. Much speculation persists whether traditional societies at pre-industrial level of economic development with everwhelming base of joint family systems could successfully accomplish industrial development and modernization. Some Marxist historians and economists have postulated a weakening of relationship between family system and family mode of production with the rise of capitalism. A dissociation between family system and modern industrial mode of production is assumed, and contrariwise it is believed that nuclearization of family system may be a universal process in the industrial pattern of economy. A general assumption has prevailed that industrial economy must essentially imply the decline of all traditional communitarian institutions such as joint family, caste and community and religious beliefs and styles of life.

The studies on the processes of industrial development in India have revealed that most these assumptions are not universally valid, and do not apply to the Indian situation. Most often, such assumptions are based on historically specific instances which do hot apply to other societies. The nuclearisation of family structure as an essential and universal consequence of changes following industrialisation is held as valid by a number of sociologists and social anthropologists. Yet, the evidence from the Western societies suggested that rather than industrialisation being a cause of widening nuclear family pattern the latter may be a result of preexisting nulrcar family system in society. Comparative analyses *of* family systems and their role in economic modernization do not support the nuclearization of family as a universal process.

Industrialisation, Enterprenuership and Family System

An important aspect in the treatment of the Indian and some other developing societies by the Western social scientists is the assumption that traditionalism in the social structure of these societies, such as family jointness, extended kinship bonds and caste and community cleavages would not be conducive for industrial and cultural modernization. It is also believed that as processes of economic modernization gain momentum, the traditional institutions, the joint family system, etc. would give ground to modern forms, e.g. the nuclear family system. Studies on the relationship between industrial entrepreneurial development and the family systems in India question the validity of these propositions. There is enough evidence adduced by economists, sociologists and social anthropologists to suggest that processes of industrial and entrepreneurial growth in India have received support from the joint family system. The studies which postulate that industrial and economic modernization would lead to nuclearization of family systems are often based not on systematic structural study of family system but on attitudinal data. A time-depth analysis of family system over successive generations by I.P. Desai revealed a cyclical rather than a linear movement from jointness to nuclearity.

No doubt, industrial and urban developments in India are contributing to changes in many rules of jointness in the family system, such as rules of residence, forms of marriage and authority system. But these changes are adaptive in character. Rule of residence is of prime significance for changes in family structure resulting from industrial and urban growth. Murdock rightly opines: "It is in respect to residence that changes in economy, technology, property, government, or religion first alter the structural relationships of related individuals to one another, giving an impetus to subsequent modifications in forms

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of the family, in consanguineal and compromise kin groups, and in kinship terminology". Migration to cities and towns for work has increased during the past decades, and depending upon the nature of work it affects the joint family system. Also migration from villages to cities and from one city to another for technical and professional occupations results into setting up of nuclear families comprising husband, wife and children, occasionally aged mother or father is invited to live or nieces and nephews may be resident for education. In most such cases, the form that family takes is that of supplemented nuclear family. Even in nuclear families, nuclearity is only residential and functionally the joint family obligations in performance of rituals, kinship and marriage obligations and in ownership of assets and properties and sharing of economic responsibilities are maintained. Those who migrate to cities from rural areas seeking jobs in informal sectors of economy, the pattern is collective and rarely individual. Several members of the joint family including wife, children and brothers, etc. migrate. Earlier, mainly the male members of the joint family migrated to cities for work. This continues but the new tendency is of family migration to cities for work. This results into urban slums and hutments where families of workers establish settlements and continue with traditional family obligations, rituals, values and beliefs.

There are many other ways in which industrialization, urbanisation and diversification of occupations have affected the family systems in India. Differentiation in occupational structure, increase in education of women and urbanization have changed the pattern of mobility, the character of jointness in the family in terms of residence, kinship relationship and social and economic ties. The studies on regional variations in family patterns in India by Pauline Kolenda can indirectly be treated as indicators of the emerging relationship between economic development, caste and joint family structures. Kolenda, based on twenty-six studies on family systems in rural and urban India writes: "while these studies suggest that most Indians do *not* live in joint families, as defined here, they do suggest that most (rural people at least) live in either a joint or a supplemented nuclear family. While the majority of people may live in such joint and supplemented nuclear families, the majority of families defined here as commensal units, are probably nuclear in structure. Emerging during the research were other hypotheses. First, there appear to be regional differences in proportion of joint families. It may be that there are higher proportions of joint families on the Gangetic Plain than in Central India (represented by studies in Maharashtra, Madhya Pradesh, and Western Rajasthan), or in West Bengal. There may be higher proportions of joint families among most castes in north western Mysore state. There also appear to be places such as Gokak Taluk, Belgaum District, Mysore, in which all castes appear to have higher proportions of joint families, and places such as Badlapur, Thana District Maharashtra, where almost all local caste groups have low proportions of joint families". Using a statistical analysis of census data Kolenda and Lorraine Haddon offer further confirmation of regional variations in family patterns.

There is no direct evidence in these regional variations which could correlate with levels or patterns of economic development, such as industrialization and urbanization. But the data do suggest joint family being more frequent in agricultural, upper and middle caste groups and supplemented nuclear family among the urban professional and business classes. If one combines the frequency of joint family house holds with supplemented nuclear families, which only manifest a variation on the joint family systems, the dominant family structure in India tends to be that of the joint and not of the nuclear family system. Its economic correlates are substantial.

The process of economic development has a variable impact on the family systems. Urbanization and industrialization have no doubt led to some breakdown in the traditional joint family and have contributed to the growth of supplemented and other varieties of nuclear families. Apart from the general growth in education, the education of women in India has increased substantially during the past decades. It has increased the chances of their professional employment. Women's enrolment of university education (all faculties) doubled during 1950-51 and

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1970-71. Their employment in professions, especially teaching has also more than doubled during the same period. By categories of employment ironically in non-skilled and semi-skilled categories the share of women has gone down. Their share of employment in factories and organised public or private sector industries has also declined. It fell from 11.43 per cent in factory employment in 1951 to 9.1 per cent in 1971. la public and private sector organized sector the women's employment level was 65 per cent in 1962; it went down to 52.8 per cent in 1973. This uneven pattern of growth in women's employment indicates tensions of modernization in economic technological and educational fields which India has been experiencing. Whereas the rise in higher education of women has increased their share in professional and service sector jobs, the technological advancements in mining, factory production and other industrial sectors including agriculture has had a negative effect on their employment. The impact that it has on the family system in India can only be hypothesised in the absence of systematic studies. It has meant urban-ward migration of couples in nuclear or supplemented nuclear family settings. It may have also increased numerically the single member family units in cities, especially of working men and women, migrating from outside. The educational, technological and industrial advancement and growth in professional, organised-sector skilled jobs and other ancillary services have had many consequences for the family system which are as yet transitional. These do not seem to have caused a breakdown in the joint family bonds, and yet these promote a variety of forms of nuclearity.

Culturally, given the strong caste, community and kinship ties in India, the culture of jojnt family system continues to prevail.

This is evident from a study that Michael M.Ames conducted on the factory workers in the Steel City of Jamshedpur to measure their ideal family preferences. He reports: "The large majority of respondents in the subsample quite clearly consider the traditional single residence joint family as the most respectable, honourable, or "ideal" type of family to maintain, even if they do not themselves have such family or consider one now possible". For example, a proposition that–"if it is possible then people should live together in a joint family"–is accepted strongly or moderately by almost 91 per cent of the respondents, even though in reference to their *own* particular families only half the respondents prefer common residence". This confirms the variable impact that industrialization has upon the family systems. It is mediated by "a number of factors, such as education, income, caste, ethnicity, which themselves may vary independently of industrialization". Similarly, "the several concrete dimensions of family structure may "respond" differently to the pressure of an industrial environment...The residential dimension, where the worker lives, is of course, closely related to, or interdependent with, the man's occupation; it is therefore hardly surprising that residential practices should reflect occupational pressures. Property ownership, on the other hand, is susceptible to pressures that range far more widely in both space and time". The patterns of change is such that the joint family norm continues to ideologically dominate.

Apart from the impact that industrialization has upon the family system, the latter also influences the nature and direction of industrial and economic growth in societies. This has been brought out by several Indian studies dealing with social correlates of industrialization and entrepreneurship. The early studies demonstrated the role of joint families and traditional caste occupations in the rise of business classes, capital formation and the nature of industrial management. The joint business families contributed to the establishment of textile mills, steel factories and other industrial entreprises apart from trade and banking as noted by economists and historians. In early phases of industrial entrepreneurship these families played a role in mobilization of capital, procurement of managerial and technical skills and in establishing marketing networks. Recently, the relationship between family structure and the emergence of business enterprises has been the subject of many studies. The caste, community and religious backgrounds of the successful entrepreneurs in addition to their family structure have been analysed.

The focus of these studies varies, but all of them deal with the interface between industrial enterprise and family systems. First type of studies deals generally

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with the role of caste and community in the rise of modern capitalist entrepreneurship and show its relationship with traditional family occupations to explain the success in this domain. These studies analyse the nature of institutions such as joint family, occupational traditions and kinship and community ties either in a single business caste or community or study interaction of these factors with the successes achieved in industrial enterprises. Some studies also attempt to explore how traditional occupation and joint family system of these groups contributed to their evolution from small traders to pariah capitalists and finally into big industrial entrepreneurs. An important common ground covered in most these studies is to establish a continuity between modern business aptitudes and traditional roles and values related to family, kinship, community and caste, etc.

The second type of studies, though fewer, examine the relationship between family system and organization of industiral firms in the context of modernization of management, distribution of authority and inter-generational relationships. Consolidation of family firms and expansion in their business through marital ties constitutes an important aspect of these studies. Finally, there are studies which analyse transitional processes of movement from agricultural to commercial-industrial enterprises as a result of economic development. In these studies aspects of social mobility and the emerging technological and entrepreneurial orientation among the peasantry have been analysed. These **bear** relationship with the family system not directly but by inference since most wealthy peasants having extended or joint families generate agricultural surpluses working through a family mode of production. This mode of production generates surpluses that are made available for investment in business and industrial enterprises as also for education which adds to their ability for social mobility.

The relationship between economic development and family system dealing with changes in the authority structure in the family such as status withdrawal of heads of household and its consequent impact on personality system of children, a precursor to their growth as innovative personality types associated with innovative entrepreneurial culture and role-sets in society, which is supposed to have been a prime factor in the modernization of society and economy in the west, has not been deeply studied in India. Very few studies of the industrial entrepreneurs have dealt with their cultural and social modernization through changes in their personality system brought about by crisis in the authority structure of their families. One study which comments on this issue while studying industrial entrepreneurs in a Punjab town is critical of the proposition of "status withdrawal". It suggests that what happens more often in case of the entrepreneur families of Punjab is not status withdrawal but "status accrual" through industrial advancement. Similarly, McClelland's theory of achievement orientation and successful entrepreneurial behaviour has also been found to be of only marginal relevance in entrepreneurial performance. It is the other factors, especially entrepreneurial environment which govern the success or failure of entrepreneurs in India as revealed by a study of entrepreneurs in Howrah, Calcutta. On the other hand, Hanna Papaneck considers that "in the Indian context the requirement for a successful entrepreneurial personality include autonomy and personal responsibility. These are supposed to be fostered by a "mother less indulgent than the average and a father who is physically absent or relatively uninvolved in the upbringing of his son". The significance of these studies lies in the implicit recognition of the role of specific cultural and historical factors in India (as against the universalised constructs of Hagen and McClelland) in the growth of entrepreneurial personality, its roles and stuctures. We witness once again the elements of condnuity and historicity in the impact that family system has upon the growth of economic and cultural modernizadon.

The relationship between family system and industrial development is more directly postulated in the studies of business castes, communities and occupational groups. Marwaris in the north and eastern India, Chetdars in the south and Parsis in the western India are examples of caste or community who contributed to the growth of industrial entrepreneurship in India. Their transition from

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traditional merchants to industrial capitalists has been studied in the contexts of their family community and ideology. The biographies of the leading entrepreneurs from these communities have also been evaluated. Timberg who studied the rise of the Marwaris to industrial capitalism refers to the role that joint family system played in enabling these traditional traders to become industrial entrepreneurs. He writes: "In connection with their specialization in trade, these communities have developed various institutions which enable them to trade more easily. If wives and children are left at home so that males can travel on business, the joint family home provides shelter for family dependents. Wandering traders find, wherever they go, support and housing provided them by their communal fellows".... Then with specific reference to the role of joint family in the success of the Marwaris as industrial entrepreneurs Timberg writes:"

We are thus led to re-examine the functions of traditional commercial and social values and institutions as they are used by trading communities in the transition to "modern" forms of economic activity. Perhaps it is the presence rather than the absence of these values, which made the Marwaris and some other commercial communities successful. It may be that many institutions, such as the joint family and strong particularistic caste loyalities are the secret of success in Indian business and industry. For example, the retention of the joint family allegedly restrains entrepreneurs. The individual is constrained by the group and cannot move forward. On the other hand, there is strong indication that a joint family may be a useful institution for mobilizing otherwise scarce factors, such as capital".

Similar conclusions have been drawn by researches on other business communities. Milton Singer (1968) found that traditional joint family among entrepreneurs in Madras played important role in their success through adaptive changes including adjustments in the system of authority, decision-making in the enterprises and in the processes of management. This also covers issues related to training and education of the younger members of the family to take over supervision and management of firms. Compartmentalization of responsibilities and authority tends to be an important strategy of these industrial families. Intensive analysis of adjustments in the structure of industrial families in the process of growth in their enterprises has been conducted by Allan Cohen based on data from eight firms in India. He has observed role-conflict in family business in situations where role pressures of membership of more than one organizations generates incompatabilities and conflicts. The conflict in family firms is greater when there is more emphasis on modernization of the management style. To maintain harmonious relationships within the family firm, according to Cohen adaptive strategies in management styles are adopted. The findings of Cohen under specific circumstances have been confirmed by other studies of family firms in India. A combination of factors maintain the link of modern family firms with tradition. The *Mtiakshra* law of inheritance which sanctions joint ownership of property contributes to the unity of family. Also the cultural factors such as ideological preference for joint family as a measure of the ethic of accommodation and reconciliation sanctified by tradition sustains its unity despite tensions and role conflicts in the process of economic modernization. This has been confirmed by a comparative study of three business families one each belonging to the Parsi, the Gujarati and the Marwari community. Several other studies also support this viewpoint.

Modernization of Agriculture and Family System

Important changes have been taking place in the rural family system since the onset of agricultural modernization. Considering the substantial place that rural society occupies even today in the Indian social system, and the increasing linkages that it is developing with urban and industrial sectors of economy, these changes have a far reaching significance to modernization of society. They are also significant because of the present direction of changes and its challenge as demonstrated by social processes in the structure of the rural family systems. The nuclear families are predominant among the working classes, marginal farmers and lower castes. Pauline Kolenda's study of regional patterning in the family structure

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shows that "it is least characteristic of local *jatis* of Untouchables. It may be somewhat more characterisuc of higher "twice-born" castes"; moreover, she also states "that most (rural people at least) live in either a joint or supplemented nuclear family". In caste terms, it would appear that most upper castes have a predominance of joint or supplemented nuclear families, most middle castes with medium to large landholding have joint families and the landless labourers or lower castes mostly live in nuclear families.

This patterning of family structure in rural society has assumed significance in economic development of villages, particularly in modernization of agriculture known as the "green revolution". Small nuclear families of agricultural workers make it possible for them to migrate from less developed agricultural regions such as Bihar, Uttar Pradesh, Orissa, etc. with the entire family to agriculturally developed states such as Punjab, Haryana and Gujarat, etc. The joint family system of the peasant castes has played a very basic role in modernization of their agriculture and their economic prosperity. The investments made in agriculture during the 1950"s brought about "green revolution" in agriculture during late 1960's. This triggered new social, political, economic and cultural processes in the countryside. Studies on the social structure of this agricultural modernization reveals that in most regions of India, such as Punjab, Haryana, Western Uttar Pradesh, Gujarat, Maharashtra, etc. the traditional peasant castes contributed predominantly to the green revolution. The Jats in Punjab, Haryana and Western U.P., the Kurmis and Ahirs in parts of Eastern U.P., the Patidars and Kunbis in Gujarat, the Marathas in Maharashtra, the Kammas, Reddys and Vokkaligas in the peninsular India were directly involved in the process of this agricultural modernization. These were also the groups which had substantial landholdings and had taken maximum advantages of land reforms introduced after Independence. They also had a long tradition of cultivation in India and were known for hard work, pragmatic outlook and traditional form of rationality typical of the Indian society.

These agricultural groups had strong joint family system and family mode of work on the farms which involved all members of the household. This probably has been a feature of all peasant societies which had led to much discussion on the "family mode of production" among social scientists. Particularly Chayanov's thesis of positive correlation between family size and volume of average production has been a subject for debate. In most peasant castes in India, however, joint family system is a common feature. Studies show that a large size of landholding and a well-knit joint family are the mark of social prestige and bring political power to such families. These factors also reinforce joint family ties. The viability of economic status that joint family farms promote adds to its persistence: The extensive use of family labour (with some additions from outside as the need may be) contributes to the maximization of economic surpluses. This, together with peasant culture of austerity in consumption styles, especially for the women and the junior members in the family, adds to savings and strengthens the economic status of such families.

The green revolution has not only contributed to strengthening the bonds of jointness in the peasant family, it has also introduced economic and technological modernization. There has been a reciprocal reinforcement to a large extent. The technological demands of new agriculture have brought into wake new social conditions in the functioning of the family system. An important change seems to be in the structure of authority and the process of decision making. Traditionally, the elder head of-household took decisions in all matters of the family–social, cultural, economic and technological. With modernization of agriculture and its new linkages with administrative and developmental agencies spread far and wide, a change has taken place. The younger educated or semi-educated members in the households look after the most technical and managerial activities of farming its credit and marketing side, etc., where the actual head of the house-hold plays only a minor role, but is social matters such as marriage rituals, distribution of goods and resources etc., the traditional head of the household continues to play his role. Thus, an adaptive change, which varies from region to region does seem to be taking place in the authority structure of the peasant family system in India.

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Changes are taking place in other aspects of the joint family system among the peasantry, not only among the middle but among the upper caste peasants also. The economic development in the villages has brought about some changes in consumption patterns even though there is no association between family size and increase in consumption. Although as yet only a marginal change in consumption style of the peasant families has taken place, more changes in course of the agricultural development may, however, be anticipated. These will have an impact on the joint family system, especially on its' cohesion and norms of conformity. Studies on the variation of family structure in course of the capitalist development in agriculture in the regions of green revolution establish that processes both of nucleation in some agrarian strata and cohesion of joint family system in others are taking place. The viability of the joint family system, despite capitalist development remains intact because the Indian experience of agrarian capitalism has principally land as asset for investment. Ramkrishana Mukherjee's study of tile *"West Bengal Family Structure, 1946-66* also reiterates the viability of joint family system. Studying the relationship between agrarian structure and the family formation based on the rate of partitioning of households belonging to different agrarian strata in Kerala, N.Krishnaji finds that despite in some cases nucleation of families in the process of agrarian transition taking place, the joint family system prevails as a dominant form.

Migration may bring about break with the rural parent families in case of the landless households, but this is not the case for the families with, land holdings. "This can be expected to be true, especially in the case of families owning large land-holdings. In such cases while the family is subdivided in a physical sense, its different branches may continue to invest jointly in activities allied to agriculture and centred round the family holding". "In sum, we may hypothesise that nucleation of families, while being associated with capitalist development in general, need not affect all agrarian classes uniformly. The processes of dispossession and pauperization of the peasantry can promote nucleation at one end of the peasant spectrum while consolidating the joint family system (or rather retarding its decay) at the other, especially when ownership of land remains the principal means of acquiring wealth". When we read these findings along with the role that the joint family system has played in strengthening the family firms in the industrial sector, we see the continuing significance of the joint family system in the process of Indian economic development. We have already noticed how industrial investment is aided by the joint family system which helps in raising capital, labour and other resources (right connections for getting licences, raw materials and permits, etc.) and contributes to the managerial supervision. The same parallel though in a different form also exists in case of the peasant joint family systems.

The developments in the agrarian structure are not uniform and vary from region to region. Therefore, the effect of agrarian development on the family system must show a variety of responses. In Punjab and Haryana social developments resulting from green revolution have generated stresses in the joint family system from accelerating subdivision of family farms. There has been a downward shift in the acreage structure of the operated holdings over the period of a decade. The distribution of operated land and assets has moved in favour of the big farmers. Intensive pressure on land due to absence of rural industries has contributed to de-industrialization and increase in the proportion of workers in agriculture. All this implies that although joint family might constitute the dominant structure in case of the peasants with big landholdings, the pressure of population in the family leads to structural nucleation despsite the family remaining functionally joint as we found in the case of Kerala. These social stresses of agrarian development have set into motion processes of restructuration in family system and its relationship with economy.

However, green revolution has in most cases reinforced the joint family system which, as in the realm of industry, plays a significant role in economic modernization of the rural economy. There is, however, also a process of nucleation going on, which economists associate with streses resulting from growth of capitalist agriculture. This proposition might sometimes imply the fallacious inference that joint family system is a passing phase in the process of

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industrialisation. However, the attempt to understand the processes of change in the family system merely through statistical measurements or through structural differentiation in the family system is inadequate for grasping its true dynamics. Milton Singer rightly observes: "it is no longer possible to support conclusions about the breakdown of the joint family by citing statistics on the frequency of nuclear households. The burden of proof has shifted: one must distinguish the cultural ideal of joint family from its actual occurrence, and attitudes and sentiments from behaviour. Different types of joint families (lineal, collateral and others) must be distinguished, as must different types of nuclear families". The scrutiny of *kula-vritanta* or family history among the Chitpavan Brahamans of Maharashtra reveals increasingly "that family organization is felt to be good in itself and to be an end in itself. Within the context of a hostile world, some of the authors seem to be saying, that it is comforting to know who one's family members, *kula-bandhu* are, and to watch how they are adjusting to changing circumstances". The strength of the joint family ideology as it confronts the forces of economic modernization shows itself through its adaptive resilience.

Modernization, Economic Development and Family System

The Indian experience of economic modernization and its impact on social structure and cultural values have a historicity of their own. We witness it at many levels of adaptive responses that traditional institutions have made in India in the process of economic, political and cultural modernization. Institutions like caste, religion, joint family system and strong bonds of kinship and community, which many western social scientists held to be obstacles in the path of India's modernization, were found to respond positively to accommodate the forces of modernization. Not that the process of adaptive transformation has been painless or without deviant manifestations. In the system of joint family itself the contradictions between goals and means of rapid economic advancement have led to distorted aspirations and contributed in recent days to rising incidence of demand for dowry in marriages. Many deaths of brides in the event of dowry not being satisfatory or not being paid have been reported. Victimization of women in many other spheres of family, their exploitation by in-laws, husbands and other members of the groom's family is widespread. The rate of divorce though still low in comparison to other countries is rising as more and more women are asserting their legitimate rights. Many women's organisations are engaged in the work of protection of 'Women and restoration of their due rights. These processes demonstrate how the forces of moderhizadon and economic development are tending to generate tensions of restructuration in society. But the over-arching social development remains adaptive and resilient in respect of the transition from tradition to modernization.

This we have found in how the joint family system has contributed to the process of indigenous development of industrial economy, a transition from traditional mercantile trade to industrial family firms. But we also notice that in the process of evolution of this kind conflicts and tensions have also arisen between the forces of nucleation and jointness in the family structure. This takes place on issues of styles and scope of management, rationalization of firms and on its inheritance and control. But more often, business and industrial families succeed in striking reconciliation or balance between principles of primordiality and rationality in control and management. In the field of agrarian modernization we find a similar process taking place. The joint family structure of the peasant households has served as a reservoir of manpower and resources to carry forward agricultural modernization. Both in cases of the family firms and peasant farms changes in the authority structure of the joint family system have taken place wihout a major break from the traditional values and norms. Even the process of nucleation, which is no doubt going on alongwith reinforcement of the joint family structure at different levels, does not portend a linear process of change from jointness to nuclearity. On the contrary, the dominating cultural framework within which these changes in the family structure are taking place is grounded in the idealization of joint family norms in society. Thus, there are changes but many continuities as well. The process reflects not a break from tradition towards modernity but modernization of tradition itself.

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KINSHIP

KINSHIP TERMINOLOGY AND KINSHIP GROUPS

Any system of organisation people who are relatives into different kinds of groups, be they descent-based or ego-oriented, is bound to have an important effect upon the ways in which relatives are labelled in any given society. The fact is, the kinship terminologies of other people are far form being the arbitrary and even capicious ways of labeling relatives that Westerners all too often take them to be, Rather, they reflect the position individuals occupy within their society. In particular, kinship terminology is affected by, and adjust to, the kinds of kinship groups that exist in a society. There are, however, other factors at work as well in each system of kinship terminology, which help differentiate one kin from another. These factors may be sex, generational differentes or genealogical differences. In the various systems of kinship terminology, any one of these factors may be emphasized at the expenses of others, but regardless of the factors emphasized all kinship terminologies accomplish two important tasks. First they classify particular kind of persons into single specific categories second they separate different kinds of persons into single specific categories second they separate different kinds of persons into single specific categories second they separate different kinds of persons into distinct categories. Generally, two or more kin are merged under the same term when similarity of status exists between the individuals. These similarities are then emphasized by the application of one term to both individuals.

Six different systems of kinship terminology result from the application of the above principles : The Eskimo Hawaiian, Iroquois, Crow, Omaha and descriptive systems, each indentified according to the way cousins are classified.

Eskimo System

Eskimo kinship terminology comparatively rare among all the system of the world, is the one used by Anglo-Americans as well as by a number of food foraging peoples. The Eskimo or lineal system emphasizes the nuclear family by specifically identifying mother, father brother and sister, while merging together all other relatives such as maternal and paternal aunts, uncles, and cousins without differentiating among them. For example, one's father is distinguished from one's father's brother (uncle); but one's father brother is not distinguished from one's mother's brother (both are called uncle). In addition one uses the term cousin to refer to all the sons and daughters of aunts and uncles, without distinguishing the sex of the family to the which they belong.

Unlike other terminologies, the Eskimo system provide, separate and distinct terms for each members of the nuclear family. This is probably because the Eskimo system is generally found in societies where the dominant kin ship is the bilateral kindered in which only the closest members of the family are important in day to day affairs. This is especially true of our own society in which the family is independent, living part from and not directly involved with, other kin except on ceremonial occassions. Thus, we distinguish between our closest kin (our parent and siblings) but lump together (as aunts uncles a cousins) other kin on both sides of the family.

Hawaiian System

The Hawaiian system of kinship terminology common in Hawai and other Malayo-Polynesian-speaking areas, but found elsewhere as well, is the least compelx system in that it uses the fewest terms. The Hawaiian system is also called the generational system, since all relatives of the same generation and sex are referred to by the same term. For example, in one's parents generation, the terms used to refer to one's father is used as well for father's brother and mother's brother. Similarily one's mother, her sister, and one's father's sister are all lumped together under a single term. In ego' generation males and female cousins are distinguished by sex and equated with brothers & sisters.

The Hawaiian system reflects the absence of strong unilineal descent and is usually associated with ambilineal descent. Because ambilinea rules allow one to trace descent through either side of the family, and memberss on both the father and the mother's side are looked upon as being more or less equal a certain degree of similarity is created among the father's and the mother's siblings. Thus they are all simultaneously recognized as being similar relations and are merged together under a single term. In like manner, the children of the mother's and father's siblings are related to oneself the same way as one's brother and sister are. Thus, they are ruled out as potential marriage partners.

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Iróquois System

In the Iroquois system of kinship terminology one's father and father's brother are referred to by a single term, as are one's mother and mother's brother are given separate terms. In one's own generation brothers, sisters, and parallel cousins (offspring of parental the father's brother) of the same sex are referred to by the same terms, people who are classified in the same category as ego's actual mother and father. Cross-cousins (offspring of parental siblings of opposite sex, that is , the children of the mother's brother or the father's sister) are distinguished by separate terms. In fact cross-cousins are often preferred as spouses, for marriage to them reaffirms alliances between related lineages.

Iroquois terminology is very widespread and is usually found with unilineal descent groups. It was, for example, the terminology in use until recently in rural Chinese society.

Crow System

In the preceding systems of terminology, some relatives were grouped under common terms, while others of the same generation were separated and given different lables or terms. In the Crow systems, another variable enters the picture the system ignores the distinction that occurs between generation among certain kin.

The Crow systems, found in many parts of the world, is the one used by the, Hopi Indians. Associated with strong matrilineal descent organization it groups differently the relation on the father's side and, mother's side. Cross-Cousins on the father's side are equated with relatives in the parental generation, while those on the mothers side are equated with the generation of ego's children. Otherwise the system is much like Iroquois terminology.

From out point of view, the Crow systems seems terribly complex and illogical. Why does it exist. In societies like that of the Hopi, where individual identity is so dependent on descent is group affiliation and where descent is matrilineal, it makes sense to merge father's sister, her daughters, and even her mother together under a single term, regardless of generation. These are women through whom descent is traced in the lineage that sired ego, just as ego's own children, along with those of his mother's brother, were sired by men of ego's own lineage. Thus, it makes sense for ego to equate his maternal cross-cousins with the generation of his own children.

Omaha System

The Omaha system is the partrineal equivalent of the matrilineal Crow system. Thus, one's mother's sister are designated by a single term one's father and father's brother are merged together under another, While one's parallel cousins are merged with brothers and sisters. Cross-Cousins on the maternal side are raised a generation while these on the paternal side are equated with the generation of ego's children. Thus, children borns of women from one patrilineage, for the men of another, are lowered by one generation.

Descriptive System

Although it has replaced Iroquois terminology among the rural Chinese, the descriptive system is found among few of the world societies. In this system, one's mother's brother is dintinguished from one's tfathers brother, who is distinguished from one's father, one's mother's sister is distinguished from one's mother, as well as one's father's sister. Each cousin is distinguished from all others, as well as from siblings. It is therefore more precise than any of the other systems, including our own, which may be one reason it is so rare. Infew societies are all one's aunts, uncle cousins, and siblings treated differently from one another.

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KINSHIP SYSTEM

RULES OF DESCENT

Kinship can be basedon bilateral descent as well as *unilateral descent*. Generally, there are three rules of descent from both the parents and the later through either of the parent - thus *patrilineal* and *matrilineal;* hence these rules of bilaeral descent assign a person to a close group of kinsmen to whom he is related through both sides, males and females. *Patrilineal descent* affiliates him with a group in which descent is traced only through males. *Matrilineal descent* connects him with a group related only through females in those societies, which follow double descent, a person may be assigned simultaneous to two groups, one patrilineal and the other matrilineal.

It is not yet determined that which of these basic three systems developed first in the cultural history of man. It has only been possible to identify some of the cultural circumstances with which each system may have been recurrently associated, thereby suggesting a casual co-relation that appears to be significant.

While discussing the family, we have seen that the rule of post-nuptial residence that determines the domestic group is primarily determined by the patterns of a people's technoeconomic adaptation. The hunters and some of the developed frarmers may have found that in the male work solidarity is more important for them. This is, clearly seen in their control of allocation of goods and their role in offence and defence, that are most critical to the maintenance and protection of their ways of life. This seems to be directly reflected in the high incidence of the virilocal residence rule which, ultimately produces the patrilocal family among all such peoples.

Among the horticulturists, where women have to do most of the gardening and other associated domestic work, matrilocal families are found. The threat of warfare and inter-tribal of inter-communal feuds is less in such societies. Therefore, the need for male solidarity is comparatively less emphasized. The rarer rule of uxorilocal post-nuptial residence occurs most commonly in such societies.

The residence rule has a tendency to be bilocal where the techno-environmental adaptation is unstable enough or where the techno-economic positions of men and women are so nearly equivalent that a more flexible rule of post-nuptial residence is found to be most adaptative.

These observations indirectly prove the thesis that rules of descent appear to be solely determined by rules of residence. Residence rules and descent rules both relfect the pattern of a people's techno-environmental and techno-economic adaptation. They are formed by these conditions and are reinforced by cultural conditions and tradition; incourse, they are particularly governed by the political system and by the ideology behind it. They may act back upon these other aspects of culture, sometimes inhibiting and sometimes directing their development. In some case the rules of residence and descent themselves give way to the pressures for technological innovation and economic change, and they begin to change.

1. Bilateral Descent

Bilateral kinship is significant particularly because of the fact that it recognizes both kinds of kindred, paternal as well as maternal. Moreover, it may extend to more than four generations. This is more so if the descent is symmetrical and the individual is related to four grandparents. The principle can even be potentially, extended further laterally and multiplied geometrically in each ascending generation, thus relating him to all third and fourth cousins and all eight great-grandparents.

In practice, this potentially enoromous aggregation is very often 'too inclusive' to be socially useful and pared down, and usually too small a group of close kinsmen to whom the person can accurately trace a genealogical tie.

The modern family, both of the West and the East, generally recognizes kinship through both the parents. It regmains, however, a fact that this recognition may not be taken into cognisance or perpetuated in the names of the off-spring.

In primitive and preliterate societies, however, other kinds of descent are also recognized. Thus, there may be sibs in a society which completely ignore one of the two lines of descent and are, consequently, called unilateral group.

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2. Double Descent

As distinct, and almost opposite, to such groups are double descent and bilineal kin groups. The latter groups consists of only those persons who are related to ego through both patrilineal and matrilineal ties. In a double descent group, it is necessary that some kin representing each line is included. And individual may belong to a pair of such groups. He is affiliated through the agnatic line with his consanguineous kinsman who shared descent from a common male ancestor. At the same time, he may also be related through the uterine line, with a group of kinsmen who trace their descent from a common ancestress. As a result of this twofold recognition, the individual belongs to two, rather than one, consanguineous kin groups, one patrilineal and the other matrilineal.

In tracing the nature of relationship some rules are followed in the society. For example, if the relationship is traced through a common ancestor, the person bound together through this relationship is aman, they are called agnates or agnatic kin, sometimes also as patrilineal kin. The descendents of a woman ancestress are called uterin kin or matrilineal kin. The kin, who are related to each other directly through some descent are called lineal kin and those who branch out from the main group like uncle and cousins, are called collateral kin. Bilateral descent is not somuch widespread. The unilateral descent is, however, more prevalent.

3. Unilateral Descent

The unilateral descent reckons descent through only one parent, the father or the mother. Thus special social significance is attributed to the biological tie through one parent. The individual in such society is assigned membership of a particular consanguineous kin group right from his birth. If the descent is traced unilineally through the father, the individual is affiliated with a consanguineous kin group through the male or agnatic line and the rule of descent becomes patrilineal. If the descent is traced unilineally through the mother, the individual is considered related with a consanguineous kin through the femal eor uterine line, the rule of descent become matrilineal.

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KINSHIP BEHAVIOUR

Kinship usage or behaviour refers to definite & comparatively permanent patterns & behaviour between different members of a society. Such types of behaviour can be verbal/ non-verbal. They guide the type of relationship between the kins.

1. Avoidance :

In all societies some kind of avoidance or another is found between different kins. Avoidance is the behaviour pattern which prohibits touch, or talking or seeing particular relatives.

Reasons for avoidance:

1. One of the earliest theories for avoidance was given by *Tylor*. He related Mother-in-law, Son-in-law avoidance to matrilocal residence. The restricted contact was to protect the authority of mother-in-law. But later Anthropologists opined that, it may only be a coincidence.
2. *Freud & Frazer* based their theories on the repugnance for tabooed sexual relations. Frazer reported brother-sister avoidance in a Ceylonese tribe the 'Vedda' & in Trobriand Islanders. Among the Trobrianders, if the brother happens to see his sister making love with a man, all three will have to commit suicide.

   Hence, according to Frazer the purpose of avoidance is to prevent sexual intimacy & incest. The avoidance between members of the same sex is a later development. Freud's psycho- analytical theory also supports Frazer's theory.
3. In many societies using spouse's name is tabooed for fear of witchcraft & sorcery.
4. *Turney-High* - "avoidance is a mechanism to preserve peace in the family. Parents-in-law, daughter-in-law avoidance is to avoid conflict between loyalty of a woman as wife & as a daughter-in-law.
5. *R.C. Brown* - "avoidance is a social fact & must have a *social explanation"*. Family is a social institution. Wherever people come in contact, there exios a possibility of a co-operation or conflict. The best way to *prevent conflict is* 'avoidance'. *Chapple & Coon* agree with this view. *"To preserve social structure 'avoidance' is essential"*.
6. Robert Lowie - *"Avoidance is to preserve one's own culture*. Daughter-in-law represents an alien culture with different social & moral values. To prevent her from making cutural inroads avoidance with in -laws is needed.

2. Joking Relationship :

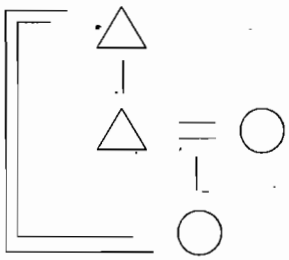
It is a behaviour just opposite to 'avoidance', where the relatives are free to exchange even the abuses. Joking may amount to *abuse and banter, obscene & vulgar references* to sex. It is termed as "*mutual disrespect"* by R.C. Brown as noticed in North America, E. Africa & Asia. It also *indicates a potential sexual relationship*. Eg: Brother-in-law, Sister-in-law; (A man & his wife's younger sister; a woman & her husband's younger brother). In South India, joking relationship exists between a woman & her maternal uncle indicative of uncle-niece marriage.

Such behaviour also exists *with one's maternal uncle's wife,* indicative of inheritance of property of one's maternal uncle. *Matrilineal societies exhibit this*. Eg: *Hopi & Trobriand Islanders.*

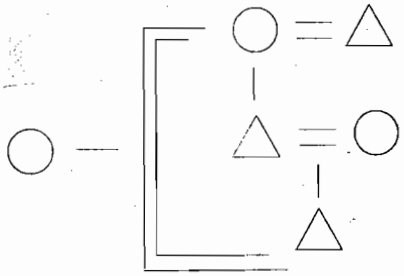
JokingRelationship can be of two types.

a. Symmetrical joking relation - The parties involved have equal rights to joke/redicule etc. Eg: Jija-sali; Devar-Bhabi
b. Asymmetrical joking relation - only one side exerts the right. Generally it is seen between elders & children, wherein elders use it to correct the wrong behaviour of younger ones.
c. Joking relationship is found in the behaviour of grandparents & grand children. Eg : Oraon, Baiga, Chamar.

S.C. Roy's study of Oraon reported joking relationship which gave way to marriage between grandfather & grand daughter.



Verrier Elwin reported marriage between grand mother & grand son among the Baigas. Among Chamars, a depressed class of leather - tanners in North India, marriage between grand father & grand daughter is reported which resulted in birth of a child.



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According R.C. Brown, like avoidance relationship, this also indicates a social significance. It fortifies the kinship ties with redicule, abuse etc. According to Chappel & Coon, this brings people together which may not be possible otherwise.

Hence joking relationship is a verbal/non-verbal relationship which gives right to counter attack in a mutually reciprocal manner.

3. Teknonymy :

When two relatives do not address each other directly, but through a third person or a symbol, the usage is known as teknonymy. A common practice in India, where women generally do not utter the names of the husbands or elderly relations.

Ex : "Ramu-ki-maa"; "Ramu-ke-bapu".

This indicates respect. Among tribal societies uttering husbands's name is tabooed for fear of sorcery & witchcraft. Eg : Among Santhals wife is not supposed to call her husband by his name. According to *Tylor, technonymy can be traced back to matrilineal, matriarchal societies,* where woman never accepted the son-in-law as one of them in their residence & recognised through the daughter or any other secondary relationship.

Technonymy is also a symbol of avoidance, indicating most of the times respect through cultural variation do occur.

4. Couvade :

"Imitation by a husband of the behaviour of his wife during pregnency & delivery". The practice consists in making a husband lead the life of an invalid along with his wife whenever she gives birth to a child. He refrains from active life, goes on sick diet & observes certain taboos. Eg : African tribes, Indian tribes like Maler, Khasi, Toda. Among Khasi-husband, like his wife, cannot cross a stream or wash clothes until all the birth related cerimonies are over.

Why Couvade? :

1. It is the survival of the transitional stage of maternal - paternal complex. During maternal stage, the paternity is known and is considered unimportant. But in the transition stage- (when residence may be patrilocal & inheritance - patrilineal) ascertaining paternity became important. Among Todas, the conventional method of ascertaining paternity includes keeping the father confined in a room or the customary bow & arrow ceremony.
2. Todas, couvade is found associated with patriarchal clans & there is no evidance to show that these *claims* at any time followed a system of inheritance other than patrilineal.
3. Malinowski - "convade cements the bond between husband & wife". It is also a social mechanism designated to secure paternal affection.
4. Psycho - analytical explanation - This usage indicates husband's desire to lighten the wife's discomforts by a process of participation.
5. Modern explanation - Couvade is not based on any supertition, but on a fact. A woman who has been delivered of a child undergoes certain chemical processes within her body which affects the husband, who also stays along with her. But critics of this view argue that illness would result only in few cases, and is only anticipated in all the remaining cases.

Raghan regards this as an irrational belief which might have originated prior to "marriage". It might have been a contributory cause for the emergence of marriage as an institution.

5. Avunculate :

The usage is peculiar to matrilineal societies, where mother's brother has a permanent place in the family.

He has a prior right over their loyalties.

He transmits his property to his nephew.

Nephew works for the maternal uncle rather than for his own father.

[Note : The maternal uncle's authority is known as "avuncupotestality". If nephews & nieces are broughtup in their maternal uncle's family it is called "avunculocal residence".**]**

"Though this behaviour is identified with matrilineal societies; may also be found in patrilineal societies. It may be due to *diffusion* or a *survival of the past.*

In South Africa every brother is paired with a particular sister and here 'avunculate' becomes more specific & precise.

Among *Tsonga & Nyoro* - all non-sense activities of sister's son are ignored.

The Patrilineal version of avunculate is called "amitate". Here father's sister plays a prominent role.

All the kinship usages help in defining kinship and the associated roles.

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CHAPTER - 3

ECONOMIC ANTHROPOLOGY

DEFINITIONS OF PRIMITIVE ECONOMY

On the basis of these meanings and scope of primitive economy specially in the field of social anthropology, scholars have defined precisely the primitive economy, and some of these definitions are given below for the benefit of the students :

Ralph Piddington (1952) says "Economic system is designed to satisfy material wants of the people, to organise production, to control distribution and to determine the rights and claims of ownership within the community".

Raymond Firth (1952) is of the opinion that 'Economic organization is a type of social action. It involves the combination of various kinds of human services with one antoher and with goods in such a way that they serve the given ends."

Majumdar and Madan (1956) believe that 'It consists of the ordering an organisation of human relation and human efforts in order to procure as many of the necessities of day to day life as possible, with the expenditure of minimum efforts. It is attempted to secure the maximum satisfaction possible through adapting limited means to unlimited ends in an organized manner."

George Dalton (1971) argues that 'All societies have structured arrangements to provide the material means of individual and community life. It is these structured rules that we call an economic system."

In brief, it can be said that the concept and meaning of economic system with special reference to the tribals may be defined that economic system may have two important relations, and the process of distribution existing and operating in a given socio-political set up. The mode of production implies technique and organisation of economic activities relating to production. The structure of production means social-relations in the performance of production activities and in the process of distribution between different social groups of the tribal societies. Finally, it may be concluded that the mode of production in tribal economy is traditional, indigenous and culturally predominant. The tribal people are culturally a social unit and at the same time system of distribution in linked to the barter system or to the mutual exchange or the least monetary system. The tribal people work hard to get ehrir livelihood to meet the basic needs of life, like food, shelter, etc., as well as the social needs like the materials for *rites-de-passage*, through their economic performances. Herskovits has rightly said that an individual operating as a member of his society in terms of the culture of his group is the economic unit.

CHARACTERISTICS OF THE PRIMITIVE ECONOMY

There are many characteristics of the primitive economic system which have been enumerated by some of the scholars which are given below for the benefit of the students.

According to Dalton (1971), there are three important characteristics of the primitive economy, such as :

(1) *Small Economy :* It is this smallness of scale which is the fundamental characteristic of primitive life (Wilson, 1941; 10) that most (but not all) resources, goods and service transactions take place within a community of persons numbered in hundreds or thousands. There are two other factors which make tribal economics small in scale, (i) Frequently one or two staple items comprise usualy large proportion of total produce. It is common for these important staples to be produced within the small framework of a tribe (ii) A relatively small number of goods and services is produced and acquired.

(b) *Simple technology compared to the industrialized economics :* The tool are either made by the user himself or are acquired free from a craftsman or from a manufacturing group.

(c) *Geographical or Cultural Isolation (comparatively):* Majumdar and Madan (1956) have found nine important traits of a primitive economy as noticed in the tribal India and elsewhere.

There is an absence of technological aids in a tribal economy which results in inefficient, inadequate or even wasteful exploitation of nature. Consequently, the bare minimum necessary for sustenance is raised with great difficulty. An economic surplus is rare in their community.

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The economic relations among the tribals themselves are mostly based on barter and exchange, money as a store and measurement of value and medium of exchange is not used widely. Institutions like banking and credit are used only in dealing with non-tribal groups which depends upon the nature and frequency of contacts with them.

The profit motive in economic dealings is generally absent. The role of an incentive is fulfilled by a sense of mutual obligation, sharing and solidarity.

- Co-operative and collective endeavour is a strongly developed feature of their economy.

The rate of innovation, internal or induced, is very low and consequently they are stabler and make hardly any progress.

The regular market as an institution along with its conditions of market like perfect competition and monopoly is absent what comes nearest to it is the weekly market or festival and seasonal meets.

The manufacturer of consumer rather than capital goods is common and the same are consumed, nothing being saved or exchanged in trade.

Specialization based on specially acquired specific technical abilities is absent. However, division of labour based on factors other than specialization, like sex, is widely present.

The notion of property is closely related to display and expenditure of wealth rather than to its accumulation. Material goods, movable and immovable may be referred to as property and this entails the existence of some rules of inheritance. Both types of ownership, collective and individual, are known.

The characteristics of the tribal economy may broadly be viewed in three ways, viz., (i) the structure of the tribal economy, (ii) the tribal economy as a socio-economic and cultural system, and (iii) economic characteristics of the tribal economy.

The structure of the tribal economy is generally based on forest and sea and forest coastal and island tribals. The simple technology and absence of technological aids is the other structural feature of the tribal economy. At the socio economic and cultural level the family is a unit of both production and consumption. The community itself acts like a co-operative unit, and the tribal communities living in a village or locality are economically independent. The distribution is generally based on gift and ceremonial exchange.

Lastly, in analysing the pure economic characteristics as general economics prescribes, the features are two, viz., absence of profit in economic dealings and presence of periodical markets.

Finally, keeping in view the close association of indigenous tribals production and social organization, a nine point framework has been given to illustrate the fundamental characteristics of tribal economy, (i) Forest-based economy (ii) unit of production, consumption and pattern of labour being the family, (iii) simple technology, (iv) absence of profit in economic dealings, (v) the community, a co-opeative unit, (vi) gift and cremonial exchange (vii) periodical markets, (viii) interdependence, and (ix) the minor institutions related to tribal economy.

Vidyarthi (1977) suggested that the first and foremost characteristics of the tribal economy is the close relationship between their economic life and the natural environment or habitat which is usually the forest. Besides the forest, the existing natural environment moulds their economy to a great extent. For example, at Lakshadweep Islands, the existing ecology of sea and the coconuts forest exerrt significant influence on the economy of the islanders (Jha, 1992). The Birhors of bihar, the Chenchus of Andhra Pradesh, the Juangs of Orissa, the Kadars of Kerala, the Paliyans and Panians of Tamil Nadu, etc., depend on the forest and in these areas, the flora and fauna predominates as the primary source of food. The Bhils of western India depend on forest of Mahua and Biri leaves to a great extent. The pastoralist Gaddis of Himachal Pradesh depend on forest for the pasture for Dhar for their goats and sheep. In this way, the tribal economy is usually forest-based. However, some changes have occured among the agriculturist tribes like the Mundas, the Oraon, the Santhals, the Raj Gonds, etc., specially after independence of India and their economy is now no more forest-based.

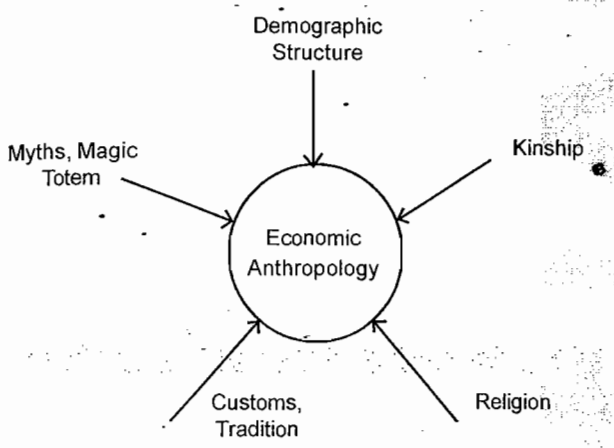
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SCOPE OF ECONOMIC ANTHROPOLOGY

Economic anthropology studies economic activities of man in his holistic framework, which is socio-cultural frame-work. Various aspect of tribal life affects his economic activity.



Majumdar and Madan says in his book (An introduction to Social Anthorpology) it "Consists of the ordering and organization of human relations and Human efforts in order to procure as many of the necessities of day to day life as possible with the expenditure of minimum efforts. It is attempted to secure the maximum satisfaction possible; through adopting limited means to limited ends (needs) in an organized manner."

In the view of Ralph Piddington "Economic system is designed to satisfy material wants of the people, to organize production, to central distribution and to determine the rights and claims of ownership within the Community."

George Dalton's (Economic Anthorpology and development) opines that "all societies have structured arrangement to provide the material means of individual and community life. It is these structured rules that we call an economic system."

Economic anthropology is the comparative; cross–cultural study of economic systems, both economics and Economic anthropology laid emphasis on the problems involved in the production, distribution, and consumption of the material goods. Economic anthropology also concerns with gift-giving and sharing on which the economists attah less importance. Economic anthropologists are basically concerned with two questions:

a. How are production, distribution, and consumption organize in different societies? This question focuses on systems of human; behaviour and their organization and

b. What motivates people in different cultures to produce, distribute or exchange, and consume? Here the focus is not on systems of behaviour but on the individuals who participate in those systems.

DISTRIBUTION AND EXCHANGE

Karl Polanyi, an economist, classified the cultural systems of distributing material goods into three modes: reciprocity, redistribution, and market exchange.

1. RECIPROCITY

Reciprocity refers to a transaction between two parties wherby goods and services of roughly equivalent value are exchanged. This may involve gift giving, but in non-Western societies, pure altuism in gift giving is as rare as it is in our own society.

Social customs dictate the nature and occasion of exchange. When an animal is killed by a group of hunter in Australia, the meat is divided among the families of the hunters and other relatives. Each person in the camp gets a share, the size depending on the nature of the person's kinship tie to the hunters. The least desirable parts may be kept by the hunters themselves. When a kangaroo was killed, the left hind leg went to the brother of the hunter the tail ot his father's brother's son, the loins and the fat to his father-in-low, the ribs to his mother-in-low the forelegs to his father's younger sister, the head to his wife, and theentrails and the blood to the hunter. If there were arguments over the apportionment, it was because the principle of distribution wer not being followed properly.

Types of Reciprocity

a. Generalised reciprocity : refers to exchange without involvement of money.

In agriculturist tribes we can see the general reciprocity, which happens of the time of transplantation. In Munda, Oraon, Gond (all agriculturist tribes) close and distant kins come together to help each other at the time of transplanting paddy. Dhumkuria of Oraon helps the villagers for token gift at the occasion of festivals, wedding, ceremonial expedition at the paddy field. By the Jajmani system of different festivals of Hos and Santhals, they pay visits to each other. In Paharias of Palamau (shifting cultivators) construction of a house is a matter of co-operation from family, extra family to lineage level. The family in need, serves Satto and Sarbat to the members of service such co-operation is known as Badliva.

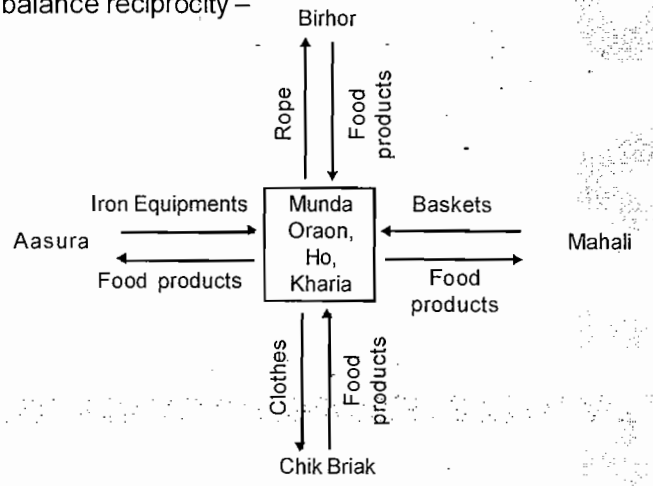
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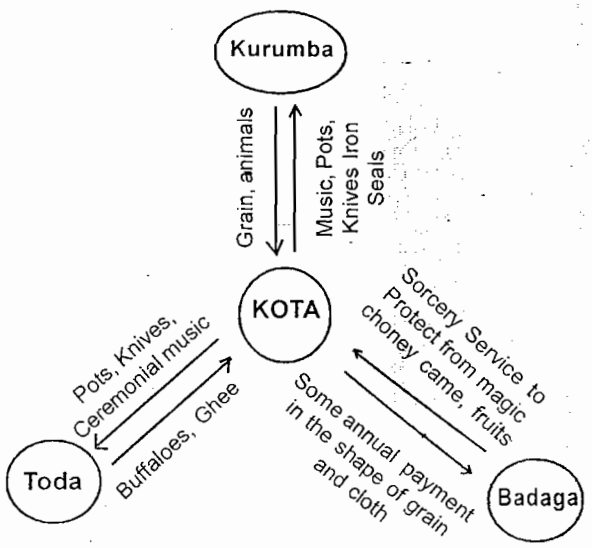
Balanced Reciprocity

This type of reciprocity could be seen in the many type of agriculturist tribes of India like Munda, Oraon, Ho and Kharis. Given diagram is helpful in understanding the balance reciprocity –



Balanced reciprocity

In the given diagram al the tribes situated in block are agriculturists by profession and are aboriginals of Chhotanagpur. Others are also in habitants of that area. Asura provides them iron equipments for harvesting prupose and gets food products for their own use. Birhor gives them rops and gets food products. Mahal provides baskets for their use and gets food products. Chik, Briak gives them clothes for their use and get food products for their own use.



Balanced reciprocity

Negative reciprocity is the mode of peripheral exchange particularity between the tribals and non-tribals, at the weekly markets, where the baniyas or local traders bargains their goods at high cost. But such a state of affairs is absent in the inter tribal dealings.

Barter and Trade

Exchange that takes place within a group of people generally takes the form of generalized or balanced reciprocity. When it takes place between two groups, there is apt to be at least a potential for histility and competition. Therefore, such exchange may well be in the form of negative reciprocity by which scare items from one group are exchanged for desirable goods from another group. Relative value is calculated, and despite an outward show of indifference, sharp trading is more the rule when compared to the reciprocal nature of the exchanges within a group.

An arrangement that took partly of balanced reciprocity and partly of barter existed between the Kota, in India, and three neighbouring tribes who traded their surplus goods and certai services with the Kota. The Kota were the musicians and artisans for the area. They exchanged iron tools with the other three groups and provided the music essential for ceremonial occasions. The Toda furnished to the Kota ghee (a kind of butter) for certain ceremonies and buffalo for Funerals relations between the Kota and the Toda were friendly. The Badaga were agricultural and traded their grains for music and tools. Between the Kota and Badaga there was a feeling of great competition, which sometimes led to some one-sided trading practices; it was usually the Kota who procured the advantage. The forest-dwelling Kurumba, who were dreaded sorcerers, had honey, canès, and occasionally fruits to offer, but their main contribution was protection against the spernatural. The Kota feared the Kurumba, and the Kurumba took advantage of this fact in their trade dealings, so that they always got more than they gave. Thus there was great latent hostility between these two tribes.

Silent trade is a specialize form of barter in which no verbal communication takes place. The earliest and best-known description was given by Herodotus, who told of silent trade between the Carthaginians and the people who lived on the western coast of Africa, beyond the Pillars of Herculean. In California, an instance of silent trade occurred between the Tubatulabel and the other tribes with whom they had peaceful relations. They would trade pinion nuts and tabacco for lenghts of white clam shell disks which passed for currency among all the tribes of that region.

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Silent trade need not involve actual face to face contact. Such is the case with many forest-dwelling people in the world. Cerleton Coon has described how this system works.

The Kula Ring

Kula of the Trobriand Islanders, first described by Malinowski. The Kula is trading system that involves the exchange of scarce goods, comption for prestige and the all important ceremonial exchange of highly valued necklaces and arm shells. Some of the first-class necklaces and arms shell have names and histories analogous to Stradivarius violins, and, as Malinowski said, 'always create a sensation when they appear in a district. "The trade takes place throughout the islands, with the necklaces travelling in a clockwise fashion from island to island and the arms shells travelling counter clockwise. No one man holds these valuables very long and each employs strategies to improve his position. It is the Trobriand Island version of how to win without actually cheating. But a man cannot be too crafty or no one will trade with him.

A few men on each islands participate in the Kula and have trading partners on the other islands. An important man may have as many as 100 trading partenrs. These partnerships are lifetime relationship. Much ceremony and magic are attached to the kulla.

The Kula involves ocean voyages of 50 miles or more, and, as might be expected, more than necklaces arm shells are exchanged. A man may simply exchange necklaces and arms shell with his Kula partner, but he is free to haggle and barter over other goods he has brought along on the trip for trading. In this way these islands people have access to a whole range of material goods not found on their own islands.

The Kula is a most elaborate complex of ceremony, social relationships, economic exchange travel, magic, and social integration. To see it only in its economic aspects is to misunderstand it completely. The Kula demonstrates once more the close interrelationship of cultural factors that is especially characteristic of non-Western societies.

2. REDISTRIBUTION

In nonliterate groups, where there is a sufficient surplus to support a government, income will flow into the public coffers in the form of gifts, taxes, and the spoils of war; then it will be distributed again. The chief or king has three motives in disposing of this income; the first is to maintain his position of superiority by a display of welath; the second is to assure those who support him an adequate standard of living; and the third is to establish alliances outside of his territory.

The administration of the Inca empire in Peru was highly efficient both in the collection of taxes and method of control. A census was kept of the population and resources. Tributes in goods and, more important, in services were levied. Each craftsman had to produce a specific quota of goods from materials supplied by overseers. Forced labour might be used for agricultural work or work in the mines. Forced labor was also employed in a program of public works, which included a remarkable system of roads and brides throughout the mountainous terrain, aqueducts that guaranteed a supply of water, and storehouses that held surplus food for use in times of famine. Careful accounts were kept in income and expenditures. A governmental bureaucracy had the responsibility for seeing that production was maintained and that commodities were distributed according to the regulations set forth by the ruling powers.

Through the activities of the government, redistribution took place. The ruling class lived in great luxury, but goods were redistributed to the common people when necessary. Redistribution is a pattern of distribution by which the exchange is not between individuals or between groups, but rather, by which a proportion of the products of labor is funnelled into one source and is parceled out again as directed by a central administration. The progressive income tax is one method of redistribution in the United States; wealthy people theoretically pay relatively higher taxes and this money is redistributed through programs such as Social Security and aid to dependent children. For a process of redistribution to be possible, a society must have a fairly complex system of political organization and substantial economic surplus.

3. MARKET EXCHANGE

The flow of goods passing from owner to owner in the marketplace, as form of exchange, is familiar to North Americans and is readily observable in foreign place.s In fact, in the United states, there has been a revival and proliferation of 'flea markets' where anyone, for a small

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flee, may display and sell handicrafts, secondhand items, farm produce, and paintings. There is ecitement in the search for bargains and an opportunity for haggling. A carnival atmosphere prevails with eating, laughing, and conversation, and items may even be bartered without any cash passing hands. These flea markets, or farmers markets, are similar to the markets of non-Western societies.

The first and most basic characteristics of the market in non-Western societies is that it always means a marketplace, a specific site where the actual goods are exchanged. In non Western societies, the marketplace is totally different from what is known in moder economy as the principle of markets exchange. The market principle involves the systems of establishing price by the powers of supply and demand, no matter where the transactions made. Although some of our market transactions do take place in a specific identifiable location – much of the trade in cotton, for example, takes place in New Orleans' Cotton Exchange – It is also quite possible for a North American to buy and sell goods without ever being on the same side of the continent.

The chief goods exchanged in non-Western markets are materials items produced by the people. The people bring to the market the produce and animals they have grown and raised and the handicrafts they have made. These they sell or exchange for items they want and cannot produce themselves. Land, labour, and occupations are not brought and sold as they are through the Western market economy.

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CHAPTER - 4

POLITICAL ANTHROPOLOGY

KINDS OF POLITICAL SYSTEMS

Political organization is the means through which a society maintains social order and reduces social disorder. It assumes a variety of forms among the peoples of the world, but scholars have simplified this comples subject by identifying four basic kinds of political systems: bands, tribes, chief - doms, and states. The first two forms are uncentralized systems; the latter two are centralized.

I. DECENTRALIZED POLITICAL SYSTEMS

Until recently, many non - Western peoples have had neither cheifs with established rights and duties nor any fixed form of government, as we understand the term. Instead, kinship and descent from the principal means of social organization among such peoples. The economies of these societies are of a subsistence type, and populations are typically very small. Leaders do not have real authority to enforce the society's customs or laws, but if individual members do not confirm, they may be made the target of scorn and gossip, or even ostracized. Important decisions are usually made in a democratic manner by a consensus of adults, often including women as well as men; dissenting members may decide to act with the majority, or they may choose to adopt some other course of action, if they are willing to risk the social consequences. This form of political organization provides great flexibility, which in many situations confers an adaptive advantage.

Band Organization

The **band** is a small autonomous group, and is the least complicated form of political organization. Bands are usually found among food foragers and other nomadic societies in which people are organized into politically autonomous groups, numbering anywhere from about 50 to a few hundred. Bands are kin groups, composed of men and / or women who are related (or assumed to be), with their spouses and unmarried children; the closeness of the group is indicated by the usual presence of rules that prohobit marriage between band members. Bands may be characterized as associations of related families who occupy a common territory and who live there together, so long as environmental and subsistence circumstances are favorable. The band is probably the oldest form of political organization, since all humans were once food foragers and remained so until the development of farming and pastoralism over the last 10,000 years.

In egalitarian groups, where everyone is related, there is reduced potential for conflicts to develop in the first place. Many of those that do arise are settled informally through gossip, ridicule, direct negotiation, or mediation. Decisions affecting a band are made with the participation of all its adult members, with an emphasis on achieving consensus, rather than a simple majority. Leaders become such by virtue of their abilities and serve in that capacity only as long as they retain the confidence of the community. Thus, they have neither a guaranteed hold on their position for a specified length of time nor the power to force people to abide by their decisions. Eg: Kurg Bushman

Tribal Organization

The decentralized or multicentric authority system, in which separate bands or other social units are integrated by pantribal factors, such as clans that unite people in separate bands or communities, or age grades or associations that cross-cut kinship or territorial boundaries.

Greater population density in tribes than in bands brings a new set of problems to be solved, at the same time permitting new kinds of solutions.

As in the band, political organization in the tribe is informal and of temporary nature. Whenever a situation requiring political integration of all or several trbal groups arises, they join to deal with the situation in a cooperative manner. When the problem is satisfactorily solved, each group then returns to its autonomous state.

Among the Navajo Indians, for example, the individual did not think of government as something fixed and all-powerful, and leadership was not vested in a central authority. A local leader was a man respected for his age, integrity, and wisdom.

Among the social mechanisms that induced members to abide by group decisions were withdrawal of cooperation, gossip, criticism, and the belief that disease was caused by antisocial actions.

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Kinship Organization

In many tribal societies the organizing unit and seat of political authority is the clan, an association of people who consider themselves to be descended from a common ancestor. Within the clan, elders or headmen are responsible for regulating the affairs of members and represent their clan in relations with other clans. As a group, the elders of all the clans may form a council that acts within the tribe or for the tribe in dealings with outsiders. In some societies the strategic and tractical planning for warfare rests in the hands of the clan.

Age - Grade Organization

Age-grade systems provide a tribal society with the means of political organization beyond the kin group. Under this system, youths initiated into an age grade, following which they pass as sets from one age grade to another at appropriate ages. Age grades and sets cut across territorial and kin groupings and so may be important means of political organization.

Ex : Tiriki of East Africa

Among them, the warrior age grade guarded the country, while judicial elders resolved disputes. Between these two age grades were elder warrior, who ere in a sense understudies to the judicial elders. The oldest age grade, the ritual elders, advised on all matters involving the well-being of all the Tiriki people. Thus, political matters of the tribe were in the hands of the age grades and their officers.

Association Organization

Common-interest associations that function as politically integrative systems within tribes are found in many areas of the world, including Africa, Melanesia, and India. Ex : Cheyenne

The Melanesian Big Man

The Big Man combines a small amount of interest in his tribe's welfare with a great deal of self-interested cunning and calculation for his own personal gain. His authority is personal; he does not come to office, nor is he elected. His status is the result of acts that raise him above most other tribe members and attract to him a band of loyal followers.

Typical of this form of political organization are the Kapauku of West New Guinea. Among them, the Big Man is called the *tonowi*, or "rich one". To achieve this status, one must be male, wealthy, generous, and eloquent; physical bravery and skills in dealing with the supernatural are also frequent characteristics of a tonowi, but they are not essential. The Tonowi functions as the headman of the village unit.

II. CENTRALIZED POLITICAL SYSTEMS

As a society's social life becomes more complex, as population rises and technology becomes more complex, and as specialization of labor and trade networks produces surpluses of goods, the need for more formally defined, centralized leadership becomes greater. In such societies, political authority and power are concentrated in a single individual - the chief - or in numerous groups of individuals - the state. The state is a form of organization found in societies in which each individual must interacton a regular basis with large numbers of people with diversified interests, who are neither kin nor close acquaintances.

Chiefdoms

A chiefdom is a ranked society in which every member has a position in the hierarchy. An individual's status in such a community is determined by membership in a descent group; those in the uppermost levels, closest to the chief, are officially superior and receive deferential treatment from those in lower ranks.

The office of the chief may or may not be hereditary. Unlike the headman of bands and lineages, the cheif is generally a true authority figure, and his authority serves to unite his community in all affairs and at all times. For example, a chief can distribute land among his community and recruit members into his military service. In chiefdoms, there is a recognized hierarchy consisting of major and minor authorities who control major and minor subdivisions of the chiefdom. Such an arrangement is, in effect, a chain of command, linking leaders at every level. It serves to bind tribal groups in the heartland to the chief's headquarters, be it a mud-and-dung hut or a marble palace

On the economic level, a chief controls the economic activites of his people. Chiefdoms are typically redistributive systems; the chief has control over surplus goods and perhaps even the labor force of his community. Thus, he may demand a quota of rice from farmers, which he will redistribute to the entire community. Similarly, he may recruit laborers to build irrigation works, a place, or a temple.

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The chief may also amass a great amount of personal wealth and pass it on to his heirs. Land, cattle, and luxury goods produced by specialists can be collected by the chief and become part of his power base. Moreover, high-ranking families of the chiefdom may engage in the same practice and use their possessions as evidence of status.

Ex : Kpelle of Liberia

State Systems

The state, the most formal of political organizations, is one of the hallmarks of civilization. In the state, political power is centralized in a government, which may legitimately use force to regulate the affairs of its citizens, as well as its relations with other states. Although their guiding ideology pretends that they are permanent and stable, the fact is that since their appearance some 5,000 years ago, states have been anything but permanent. Whatever stability they have achieved has been short term at best; over the long term, they show a clear tendency to instability and transience. Nowhere have states even begun to show the staying power exhibited by more decentralized political systems.

The state is found only in societies with numerous diverse groups, social classes, and associations ; it brings together under a common rule many kinds of people. In all known states, past and present, society is divided into social classes, and economic functions and wealth are distributed unequally. A market economy is an integral part of most states, as are vast surpluses of goods and services and the intense specialization of labor.

Ex : Swazi of Swaziland

They are primarily farmers, but cattle raising is more highly valued than farming : the ritual, wealth, and power of their authority system are all intricately linked with cattle. In addition to farming and cattle raising, there is some specialization of labor : certain people become specialists in ritual, smithing, woodcarving, and pottery. Their goods and services are traded, although te Swazi do not have elaborate markets.

The Swazi authority system is characterized by a highly developed dual monarchy, a hereditary aristocracy, and elaborate rituals of kinship, as well as by statewide age sets. The King and his mother are the central figures of all national activity, linking all the people of the Swazi state; they preside over higher courts, summon national gatherings, control age classes, allocate land, disburse national wealth, take precedence in ritual, and help organize important social events.

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CHAPTER - 5

RELIGION

DEFINITION

E.B. Tylor defined Religion as the belief in supernatural beings'.

Durkheim categorised everything in this world into two - sacred and profane. To him sacred is related to supernatural. And he feels all the sacred things are also found inside the society.

Malinowski defined religion taking a functional position, 'Religion is a mode of action as well as a system of belief and a sociological phenomena as well as a personal experience.'

Clifford Geertz in his essay 'Religion as a culutural system' gave an elaborate definition: "Religion is a system of symbols which acts to establish powerful pervasive, and long lasting moods and motivation in men by formulating conceptions of a general order of existence and clothing these conceptions with such an area of factuality that the mood and motivations seem uniqely realistic.

THEORIES & FORMS OF RELIGION

TOTEMISM :

Totemism implies a tribal social organisation of the sib or clan pattern which is associated with a form of supernaturalism consisting of some typical attitude towards animals, plants or some other natural objects. Totem is related both to religion and clan organisation.

Tylor opined totemism emerged in the form of religious worship and ancestor worship. Durkheim saw in totemism the collective emblem of society and worship of society is the source of totemism according to him. Boas, Swanton etc. who regarded totemism as an extension of the personal relation of an individual with an animal and plant. Golden-Wieser regarded totemism as a socio-religious institution. Herbert Risley pointed out, in India the religious aspect of totemism is almost dead to a considerable extent, and the social aspect is more operative.

The primary reasons behind the development of totemism is a social order.

Eg : Traditions regarding the origin of certain clans among the Tamaria. indicate that the totemic animal or plant has helped to protect the human ancestor of the clan. Most of the explanations in relation to the origin of clans among Ho, Munda and the santhal conclude that totems originated out of necessities at a time when human help failed to provide required relief. Thus in most cases totems have been conceived as the substitute of representative of the divine spirit thereby linking them with religion.

FETISHISM :

If a particular natural object is imaginatively attached with some supernatural power and individual being to have faith in it directly, this type of association is known as fetishism. The objects used for this purpose is known as 'fetish'. The idea of fetishism is much related with magic. This is entirely a faith in something unexplainable and uncontrollable in general. It's test is it's efficacy, i.e. for how many times or in how many occasions the object could be used successfully.

ANIMISM :

E.B. Tylor believed that religion originated and was maintained on the beliefs of sour, ghosts, ancestral spirit and some more things which are imagined and illusioned without much real rationality, in them. He named this as

The concept of 'soul' varied from one place to another, from one tribe to another but a universal faith in the existence of 'soul' can be found. Tylor proposed an essential connection between this 'animism' and religion of the early men while the later scholars like R.H. Lowie argued that animism and religion largely overlap and they are not identical at all.

Tylor believed that an attitude of reverence towards something intangible and non-material spiritual beings forms the core of te earliest form of primitive religion 'Ancestor worship was the earliest form of worship.

Tylor was criticised for being so certain about the logical mind of the primitive people or identifying a philosopher in the primitive men. Tylor's theory over emphasised one side of the primitive religion i.e. *'belief in*

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soul and spirit'. However, we can not entirely overlook the role of this aspect, rather it could have been formed one important part in developing religion.

ANIMATISM :

Codrington coined the term 'Mana' as he found the idea while he was working among the Melanesians. This word implies force that is present in all animate and inanimate objects. in a person or in an object some form of supernatural power may be believed to exist. The theory evolved by Marett on this concept of 'mana' is known as animatist theory.

Animism was concerned with soul or ghost or some other form of existence of the dead ancestors etc. 'animism' is designating something more real, which can be visible also. For example Ho people believe in an all pervading life force and behind every object of nature there is the presence of God as well as of evil spirit. so they try to please this spirit for their own benefit and betterment. Their chief deity based on this belief is known as 'sing Bonga'. This Ho concept of Bonga is reflecting the theory of animatism or manaism. Some Nothe American tribe calls this power as Orenda.

NATURISM :

Supernatural power is often identified with natural objects like thunder, sun, earthquakes etc. And any change in the nature is believed to be the expression of the God or supernatural force.

Max Muller, the German scholar and Indologist was the main propounder of the theory of naturism. The primitive man, according to him, was inspired by the natural phenomena to find out the reason behind some natural calamities and catastrophies.

The origin of religion, according to MaxMuller lies in the attempt to explain and describe the reasons behind and implications of natural phenomena. Some religions are very close to this naturalism, specially the vedic literature of Hindu religion has references about drawn, ocean, sun, moon as Gods and Goddesses. Among the Mundas a practice was prevalent, as a Munda goes to the tops of hills and throw down stones of all sizes and shapes, so that the rumbling sound of stones falling would resemble the rumbling of thunder, and they believe that rains would follow. The Ho would burn faggots so that smoke may cover the sky resembling cloud and rain. Thus, the outward events that affect life have been explained and accounted for by primitive people on the basis of knowledge that they gained through experiences and gradually this helped them to develop religion. Nature, thus, acted as a catalyst in creating the idea of religion though it was not hte only one. Among the Mundas, there is 'Burubonga', the deity of the hills and mountains and 'Ikir Bonga' is the God of ponds and lakes or Nage-era is the deity of canals and streams.

FUNCTIONALIST THEORIES

Durkheim, Malinowski and Radcliffe Brown have proposed the functional explanation of primitive religion. Malinowski considers religion as a device to secure mental and psychological stability in an individual's life.

Human mind always suffers from pain and tension and it seeks a relief from the same, then religion is made to bring about a readjustment between man and the supernatural in upset states of existence.

For Radcliff - Brown religion emerged to instill a sense of dependence in it. To him, the individual should learn to depend on society through some rituals and behavioural norms.

He feels religion is not a device to save individual from his individual stress and strain rather it is to assure a social solidarity and homogenity. All the members of a particular society come close to one another and share their failures and expectations by practising religion.

Durkheim also emphasises the functional aspect of religion and he stresses on the beliefs and rites. He offered two important concepts viz. sacred and profane. In every tribe a distinction is made between these two. He is of the opinion that the positive belief's referred to God related to sacred and profane consists of magic or pseudo-science. Some activities are performed as compulsory in every religion and some others are forbidden so the first one is the sacred and the second one is the profane.

RELIGION, MAGIC AND SCIENCE
FRAZER'S TYPES OF MAGIC

Magic is a way to discover the mystery of the world and the explanation is not based on reason but it is in terms of 'hidden force'. Some people are believed to have

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some extra power to control evil forces or to extract benefit from good spirit. Thus, 'magic represents an exaggerated notion of human potency." according to D.N. Majumdar.

James Frazer made a detailed analysis of magic-and religion in his book 'Golden Bough'. He believed in the process of evolution of human knowledge and attempted to explain the evolution of culture through the evolution of magic. religion and science. From his point of view, the stages of development of human civilization correspond to magic, religion and science. The most primitive men explained and controlled natural happenings around human beings with the help of magic and gradually came religion and science. Frazer says, "the magician does not doubt that these same causes will always produce the same effects, that the performance of the proper ceremony, accompanied by the appropriate spell, will inevitably be attended by the desired results...".

Frazer categorised magic into- (a) Homeopathic magic and (b) Contagious magic. The first one is based on the principle of 'Law of similarity' i.e. like produces like. For example, the Khond believe that for rain human sacrifices are essential. As the tear rolls down from the sufferer's eyes and blood comes out of the wounds, the rain will come in similar way. The causal connection between two events is sought but is is established in an irrational and imaginary condition. Effigy burning is also a way to show the same principle as the pain of the effigy is understood as on the pain of the real person concerned.

(b) The second principle is based on the law of contract. This implies if something is directly associated with a person or an animal it becomes a part of him / it and that should not be visited by any other as it may cause harm to the latter. This association is extended to clothing, nail and hair-trimmings, utensils, personal effects and soon. It is for this reason primitive people used to give all the belongings of the person after his death along with the dead body.

Frazer viewed the evolution of society through the stages coinciding magic, religion and science, though this kind of a unilinear evolution may not be conceived always.

MAGIC AND RELIGION

Magic is more primitive method to understand and explain natural and supernatural phenomena. Social adjustment was the primary need which could be materialised either by coercion or by submission. Magic was the means to control the nature and social forces in a coercive manner, though in most occassions coercion did not follow any logic. Religion was supporting the cause of submission to the supernatural forces which ultimately save the man.

Magic always emphasises on secrecy and the individual capability of the magician. Religion on the other hand is public and affair. Generally it involves all the people of the community, it preaches the notion, the objective and the probable answer to the queries of the people.

In religion the priest and the practitioners command respect in the society. It is believed that religion is a positive force, having faith in it empowers man to be uplifted. Whereas magician is feared, he is a distant person for the people.

Religion is always conceived as a benevolent method to achieve something good for the people. It purifies the mind, it expects blessings from the absolute power, it offers welfare of the community as a whole. But, magic is considered harmful for the society because it not only solves a problem, as it causes the problem who is supposed to be the problem.

Some similarities between magic and religion can be identified. Both the religious priest and magicians meditate between this world and the unseen world. The magician's art and religious rituals, both are meant to create an atmosphere of suggestibility, and the technique employed by both is, more or less, similar.

Magic and religion serve the same purpose, viz, that of restoring confidence in time of danger and crises. When magic fails repeatedly, religion helps and both may often jointly contribute towards tiding over social and economic crises.

Magic is often associated with fetishism i.e. some specific objects are added with a force of life. Similarly, some inanimate objects are believed to carry some supernatural power and those are worshipped.

MAGIC AND SCIENCE

Magic assumes an uniformity of nature and that brings magic close to science. The magician believes in the laws

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of nature. Both magician and the scientist assume the succession of events in an orderly manner. Both of them believe that certain number of uniform laws guide the nature towards a specific direction, no one can overlook those laws, rather laws are to be utilised by them to adjust them with the human need.

Science is guided by reasons and examined through observation while magic covers itself behind mysticism and abstraction.

Tylor argued that the development of magic in a systematic way takes the form of science. Frazer does not accept it. However, he too admits that magic is a semi-science.

Science is open to all for common good of the society, magic helps man to utilise the negative power in favour of them. Evans-Pritchard points out, the supernatural power has two faces - one is benevolent and the second is malevolent. Science explores the former and magic, the latter.

Science and magic both depend upon mechanistic procedures. But the scientist observes and experiments, with the natural objects and the magician is concerned with the methods to deal with supernatural. Both science and magic base themselves on observations and experiments.

Failures are possible both in science and magic. Even after series of experiments a scientific conclusion can be proved among afterwards. Similarly, magician may fail to get result even after performing all the required rituals. However, scientific failures can be rectified by further research and magical failures leave no scope of research by the same magician, the alternative is to be searched out.

Magic, religion and science may not always coincide with the evolutionary stages of savagery, Barbarism and Civilization as conceived by Tylor. Even at the present stage of civilization magic, religion and science can be found in a mixed or isolated manner.

Thus it is not the stages of the development of civilization but the variation in the level of human intelligence and differences in belief structure influence the existence of magic, religion and science. Also there is interconnection between all three as people like to discover the unknown - sometimes highlighting chance factors (magic), sometimes emphasising on faith in supernatural force (Religion) and some other time considering experiments and objective factors (science).

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CHAPTER - 6

ANTHROPOLOGICAL THEORIES

1. NINETEENTH – CENTURY EVOLUTION

Modern anthropology emerged from the intellectual atmosphere of the Elightenment, an eighteenth century philosophical movement that stressed social progress based on human reason, and Darwin's theory of evolution. The first professional anthropologist – that is, an individual appointed to an academic position of anthropology – was a ninetteth - century British scholar, Edward B. Tylor. In 1871 Tylor published a major work titled Primitive Culture. At that time, Great Britain was involved in imperialistic expansion all over the world. Tylor thus had access to descriptions of non-Western societies through his contacts with travelers, explorers, missionaries, traders, and government officials. He combined these descriptions with nineteenth - century philosophy and Charles Darwin's ideas to develop a theory of societal evolution.

a. **Unilineal Evolution : Tylor –** The major question addressed by early anthropologists was : Why are societies at similar or different levels of evolution and development? Many people ask the same kinds of questions today. Tylor tried to answer that question through an explanation known as unilineal evolution. **Unilineal evolution** is the view that societies evolve in a single direction toward complexity, progress, and civilization. Tylor's basic argument was that because all humans are bestowed with innate rational faculties, they are continuously improving their societies. Through this process of evolution, societies move toward 'progress' and 'civilization'.

In arriving at this conclusion, Tylor used accounts from Western observers to compare certain cultural elements from different societies, including technology, family, economy, political organization, art, religion, and philosophy. He then organized this evidence into categories or stages, ranging from what he termed 'savagery' to 'barbarism' to 'civilization'. Theoriests like Tylor assumed that hunter-gatherers and other non-Western societies were living at a lower level of existence than were the 'civilized' societies of Europe. This was an ethnocentric view of societal development based on the belief that Western society is the center of the civilized world, and that non-Western societies are inherently inferior.

Tylor and other nineteeth-century thinkers also claimed that 'primitives' would eventually evolve through the stages of barbarism to become civilized like British gentlemen and ladies. However, Tylor believed that these societies would need some assistance from the civilized world to reach this stage.

b. **Unilineal Evolution : Morgan –** Morgan was a lawyer and banker who became fascinated with Native American societies. He gathered information on the customs, language, and other aspects of the culture of the Iroquois – speaking peoples of upstate New York. Eventually, under the auspices of the Smithsonian Institution, he distributed questionnaires to missionaries and travelers to collect information about many other non-Western societies.

Morgan and Kinship Theories Morgan was particularly interested in kinship terms used in different parts of the world. He observed that the Iroquois kinship terms were very different from those of English, Latin, Greek, and other European societies. He also noticed that these kinship terms were similar to those of the Ojibwa Indians, a group living in the midwestern United States. This led him to explore the relationship between the Iroquois and other peoples.

From these data, he speculated that humans originally lived in 'primitive hordes,' in which sexual behaviour was not regulated and individuals didn't know who their fathers were. He postulated that brother-sister marriage then develolped, followed by group marriage, and eventually a matriarchal family structure in which women held economic and political power. Morgan believed that the final stage of the evolution of the family began when males took control of the economy and politics, instituting a patriarchal system.

In addition to explore the evolution of the family Morgan, like Tylor surveyed technological, economic, political, and religious conditions throughout the world. he compiled this evidence in his book Ancient Society

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which presented his overall scheme of the evolution of society. Paralleling Tylor's views, Morgan believed in a hierarchy of evolutionary development from 'savagery' to ' civilization'.

According to Morgan, one crucial distinction between civilized society and earlier societies is private property. He described the 'savage' societies as 'communistic', in contrast to 'civilized' societies, whose institutions are based on private property.

Unilineal Evolution : A Critique

Although these nineteenth–century thinkers shared the view that humanity was progressing through various stages of development, their views were ethnocentric, contradictory, and speculative, and their evidence second hand, based on the accounts of biased Europeans. The unilineal scheme of evolution was much too simplistic to account for the development of different societies.

In general, the unilineal evolutionists relied on nineteenth-century racist views of human development and misunderstandings of biological evolution to explain societal differences. For example, both Morgan and Tylor believed that people in various societies have different levels of intelligence. They believed that the people in so-called 'savage societies' have less intelligence than those in 'civilized societies'. Nevertheless, despite their inadequate theories and speculations regarding the evolution of society, these early anthropologists provided the first systematic methods for thinking about and explaining the similarities and diversity of human societies.

2. DIFFUSIONISM

Diffusionism, which developed in the early part of the twentieth century, maintains that societal change occurs when societies borrow cultural traits from one another. Cultural knowledge regarding technology, economic ideas, religious views, or art forms spreads, or diffuses, from one society to another. Major schools of difussionism:

a. **British Diffusionism :** The British school of diffusionism derived its theory from research on ancient Egypt. Smith and Perry were specialists in Egyptian culture and had carried out research in Egyptology for a number of years. From this experience they concluded that all aspects of civilizations, from technology to religion, originated in Egypt and diffused to other cultural areas. To explain the fact that some cultures no longer had cultural traits from Egypt, they resorted to an ethnocentric view, maintaining that some cultures had simply become 'degenerate'. That is, in contrast to the civilized world, the lesser developed peoples had simple forgotten the original ideas borrowed from Egypt.

b. **German Diffusionism :** The German school of diffusionism differed some what from that of the British. Schmidt and his followers argued that several early centers of civilization had existed, and that from these early centers cultural traits diffused outward in circles to other regions and peolples. In German this view is referred to as the Kulturkreise (culture circles) school of thought. In explaining why some primitive societies did not have the characteristics of civilization, the German school, like the British difussionists, argued that these peoples had simply degenerated. Thus, diffusionist views, like the unilineal evolutionary views, represent ethnocentric perspectives of human societies outside the mainstream of Western civilization.

The Limitations and Strengths of Diffusionism

1. Early diffusionist views were based on erroneous assumptions regarding humankind's innovative capacities.
2. Like the unilineal theorists, they maintained racist assumptions about the inherent inferiority of different non-Western peoples.
3. The diffusionists assumed that some people were not sufficiently innovative to develop their own cultural traits.
4. Its assumption that cultural traits in the same geographical vicinity will inevitably spread from one society to another.
5. Anthropologists find that diffusion is not an inevitable process. Societies can adjoin one another without exchanging cultural traits.

However diffusionism as a means of understanding societal development does have some validity. For example, diffusionism helps explain the emergence of the classical civilizations of Egypt, Greece, Phoenicia, and Rome. These peoples maintained continuous

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contact through trade and travel, borrowing many cultural traits from one another, such as writing systems. Thus, these anthropologists, again like the blind men and the elephant, had some partial explanations to offer on human behaviour and society.

3. HISTORICAL PARTICULARISM OF FRANZ BOAS

Boas had a tremendous impact on the development of anthropology in the United States and internationally. Boas was a vigorous opponent of the unilineal evolutionists. He criticized their attempts to propose stages of evolution through which all societies evolve. He criticized their use of the comparative method and the haphazard manner in which they organized the data to fit their theories of evolutionary stages. He maintained that nineteenth-century schemes of evolution were based on insufficient empirical evidence. Boas called for an end to "armchair anthropology." He proposed that all anthropologists do rigorous, scientifically based fieldwork to collect basic ethnographic data.

Boas's fieldwork experience and his intellectual training in Germany led him to conclude that each society has its own unique historical development. this theory, known as **historical particularism,** maintains that each society must be understood as a product of its own history. This view led Boas to adopt the notion of **cultural relativism,** the belief that each society should be understood in terms of its own cultural practices and values. One aspect of this view is that no societyy evolved higher or lower than another. Thus, we cannot rank any particular society above another in terms of degree of savagery, barbarity, or civility. Boas called for an end to the use of these derogatory, ethnocentric terms.

The Boasian view became the dominant theoretical trend in anthropology during the first half of the twentieth century. Anthropologists began to do rigorous ethnographic fieldwork in different societies to gather sound empirical evidence. Boas encouraged his students to develop their linguistic skills so that they could learn the languages of the peoples they studied.

Boas worked in all four subfields of anthropology : physical anthropology, archaeology, ethnology, and linguistics. Some of his most important works involved taking precise assessments of the physical characteristics, including brain size and cranial capacity, of people in different societies. Boas was one of the firs;t scientists in the United States to demonstrate that brain size and cranial capacities of modern humans are not linked directly to intelligence. His research indicated that brain size and cranial capacity differ widely within all races. Boas's findings challenged the racist assumptions put forward by the unilineal evolutionists. They also repudiated the type of racism that characterized balck-white relations in the United States, as well as Nazi theories of racial superiority.

A direct out growh of the Boasian approach was the emergence of culture-and-personality theory in American anthropology. Boas trained two particularly noteworthy students, Ruth Benedict and Margaret Mead, who to the development of more careful research regarding personality and culture. Additionally, Boas pioneered the study of art, religion, folklore, music, dance, and oral literature, providing the humanistic aspect of the anthropological enterprise.

4. FUNCTIONALISM : BRITISH ANTHROPOLOGY

Functionalism is the view that society consists of institutions that serve vital purposes for people. Instead of focusing on the origins and evolution of society, as the unilineal theorists did, the British functionalists explored the relationships among different institutions and how these institutions function to serve society or the individual. The two prominent figures of functionalism A.R. Radcliff Brown and Bronislaw Malinowski.

Structural Functionalism : Radcliffe Brown

The type of functionalism associated with Radcliffe-Brown is sometimes referred to as structural functionalism. Radcliffe-Brown had done research in Africa and on the Andaman Islands in southeastern Asia. He focused on the structure of society as reflected in the differing institutions that function to perpetuate the survival of society. He argued that a society's economic, social, political and religious institutions serve to integrate the society as a whole. He studied the social institutions that function to enhnace group solidarity in small-scale societies. Radcliffe-Brown suggested that all norms for specific behaviors and obligations among different people in kinship relationships promote order and stability. Thus

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to Radcliffe-Brown, these social institutions serve society's needs.

Psychological Functionalism : Malinowski

Malinowski's functionalism differed from that of Radcliffe-Brown in that it focused on how society functions to serve the individual's interests or needs. This view is known as psychological functionalism. Malinowski did his major ethnographic study in the Trobriand Islands off the coast of Papua New Guinea. He tried to demonstrate how individuals use cultural norms to satisfy certain needs.

Malinowski's analysis of magic among the Trobriand islanders illustrates his psychological functionalism. He observed that when the islanders went fishing in enclosed lagoons where fishing was reliable and safe, they depended on their technical knowledge and skill alone. When they went fishing on the open sea, however, which was more dangerous and highly unpredictable, they employed extensive magical beliefs and techniques. Thus, Malinowski argued that the use of magic arises in situations in which humans have no control over circumstances, such as weather conditions, Magical techniques are used to reduce internal anxieties and tensions for these individuals. In addition to magic, the Trobrianders have an elaborate system of belief concerning death, the afterlife, sickness, and health. These beliefs aid in serving the needs of individuals as they adapt to the circumstances and exigencies of life. In other words, the individual has needs, both physiological and psychological, and cultural institutions, customs, and traditions exist to satisfy them.

The Limitations of Functionalism

Functionalists ignored historical processes. They were not concerned with the historical development of differing institutions but rather focused exclusively on how these institutions serve society and the individual. They could not explain, why British society experienced rapid technological change whereas other societies did not, when all of these societies had similar needs.

Functionalists were unable to explain social and cultural change very well, because they tended to view societies as static and unchanging. They could not explain why, if all institutions perform a particular function, these institutions would need to change.

5. TWENTIETH – CENTURY EVOLUTIONISM

Also called Neo-evolutionism, was championed by Leslie white, Gordon Childe and Julian Steward.

White treated societies, or sociocultural systems, as entities that evolved in relation to the amount of energy captured and used by each member of society. This energy is directed toward the production of resources for their survival. In White's words, 'Culture evolves as the amount of energy harnessed per capita per year is increased, or as the efficiency of the instrumental means of putting the energy to work is increased". In other words, the degree of societal development is measured by the amount of energy harnessed by these sociocultural systems. The greater the energy, the more highly evolved the sociocultural system.

White's hypothesis of cultural evolution explained the differences in levels of societal development by examining differences in technology and energy production. For example, he hypothesized that small-scale hunting-and-gathering societies had not developed complex sociocultural systems because they depended primarily on human energy for production. Because of a limited energy source for producing resources, their societies were simple, meager, and undeveloped. But following the agricultural revolution and the capture of energy through the domestication of plants and animals, sociocultural systems changed dramatically. The agricultural revolution represented an efficient use of human energy in harnessing new energy reserves, such as using draft animals to pull plows. In turn, these technological changes led to the emergence of cities, complex states, powerful political and religious elites, and new ideologies.

According to White, tracing the modern industrial age, as fossil-fuel technology developed, new forms of energy such as coal, oil, and natural gas were used, and sociocultural changes accelerated. Up until the Industrial Revolution, the changes in agricultural societies had been gradual, taking several thousand years. In contrast, the Industrial Revolution has taken less than five hundred years to produce widespread global transformations. Because White focused on sociocultural change on the global level rather than on particular societies, his approach has been called general evolution.

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Steward and Cultural Ecology

Steward was instrumental in establishing the field of cultural ecology. Also called ecological anthropology. **cultural ecology** stresses the inter relationship among the natural conditions in the environment-rainfall, temperature, soils – and technology, social organization, and attitudes within a particular sociocultural system. Steward focused on how specific sociocultural systems adapt to environmental conditions.

Steward's cultural-ecology framework divides sociocultural systems into two different spheres : the culture core and secondary features. The culture core consists of those elements most closely related to subsistence : the environments. The other characteristics, such as social organization, politics, and religion, constitute secondary features of society. Because Steward investigated the detailed characteristics of different environments, his approach is referred to as specific evolution, as opposed to White's general evolution. One of his most illustrative case studies involved the Shoshone Indians of the Great Basin of the western United States.

The twentieth-century evolutionists differed from the earlier nineteenth-century evolutionists in several ways. First they did not assume a unilineal direction of society through formalized stages such as savagery, barbarism, and civilization. Second they were not ethnocentrically biased or racist when it came to understanding why different societies are at various levels of development. Third, they did not assume that sociocultural evolution toward complexity is always equated with 'progress'. as did the nineteenth-century theorists. The neo-evolutionists held that some aspects of small-scale societies are, in fact, better than those of complex societies.

6. CULTURAL MATERIALISM

Marvin Harris refined the neoevolutionary approach of White and Steward neoevolutionary approach of White and Steward as a perspective he called cultural materialism. **Cultural materialism** is a research strategy that focuses on technology, environment, and economic factors as key determinants in sociocultural evolution. Cultural materialists divide all sociocultural systems into infrastructure, structure, and superstructure. The infrastructure, structure, and superstructure. The infrastructure includes the technology and practices used for expanding or limiting the produciton of basic resources such as food, clothing and shelter. The structure consists of the domestic economy (family structure, domestic division of labor, age and gender roles) and the political economy (political organization, class, castes, police, military). The superstructure includes philosophy, art, music, religion, ideas, literature, advertising, sports, games, science and values.

According to cultural-materialist theory, the infrastructure largely determines the structure and superstructure of sociocultural systems. As the infrastructure changes, the structure and superstructure may change accordingly. Technology, energy, and environmental factors are crucial to the development of all aspects of society. All societies must devise ways to obtain food and shelter, and they must possess and adequate technology and energy to provide for the survival and continuity of the population. Although cultural materialists do not deny that superstructural and structural components of society may influence cultural evolution, they see infrastructural factors as being far more important. This theoretical perspective represents and extension of the foundations laid down by White and Steward.

Criticisms of Cultural Materialism

1. Harris focuses too exclusively on environmental factors and ignores social, political and religious values and beliefs.
2. Cultural materialism tends to emphasize the infrastructural mechanisms that strictly 'determine' the structure and superstructure of the society. This results in a form of technological determinism that is muc too mechanistic in analyzing the social and cultural conditions within a society.
3. As critics of technological determinists note, the level of technological develoment doesn't tell us anything about the complexity of religions, kinship, family and marriage systems, art, folklore, etc.
4. Harris's paradigm underplays the importance of symbolism and language and the influence these factors have on people's beliefs and motivations.

Harris admitted that he does emphasize infrastructure as the major determinant of the direction in which a society will evolve. Harris suggests that his critics neglect his attempt to demonstrate how structural and super structural

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aspects of society play a role in influencing conditions for social and cultural evolution. He indicated that when he discusses infrastructural determinism, he means this in a 'probabilistic' sense. In other words, he eschews a strict or rigid form of determinism and suggests that a particular infrastructure will result in the probabilities or tendencies to produce certain forms of structure or superstructure within a aociety. Harris noted that ideas, beliefs, and values can also provide feedback mechanisms and generate change within society.

7. POSTMODERNISM AND ANTHROPOLOGY

This group includes such figures as James Clifford, George Marcus, and Michael Fischer. The postmodernists suggest that traditional ethnographic research is based on a number of unsound assumptions. One such assumption is that the ethnographer is a detached, scientifically objective observer who is not influenced by the premises and biases of her or his own society. Cliefford characterizes one of the models maintained by ethnographers such as Malinowski as that of the 'scientific participant-observer' Malinowski produced his descriptions as if he did not have any involvement with the Trobriand people. It was as if he were some sort of computing machine without feelings, biases, or emotions, who was simply recording everything about native life.

According to postmodernist critics, ethnographies such as those compiled by Malinowski were intended as a type of scientific study in which the subjects behaved and the ethnographer simply recorded this behavior.

Another type of inquiry criticized by the postmodernists is the interpretative model used by symbolic anthropologists such as Clifford Geertz. In contrast with the 'objective' model, the interpretative model focuses on the understanding of symbols from the point of view of the members of the society. It assumes that the people in a particular society have a better understanding than others of their own concepts, and the interpretative anthropologist must try to reconstruct these understanding so that others can also understand the meanings of these symbols.

Postmodernists critique this type of interpretive ethnology because it is also written as if the ethnologist disappears from the setting through empathy with the subjects. In other words, the people themselves are not described as offering their own interpretations, but instead the ethnologist is looking over their shoulders and reading the cultural meanings and texts of their society and interpreting them.

Postmodernists have recommended that ethnographers today adopt a different way of researching and writing. Instead of trying to distance themselves from their subjects, ethnographers should present themselves in dialogue with their subjects. Clifford argues that an ethnography should consist of many voices from the native poulation, rather than just a dialogue between the ethnographer and one of his or her informants.

Reaction to this postmodernist critique has varied within the discipline. On the one hand, some ethnographers have charged the postmodernists with characterizing traditional ethnographic methodologies unjustly.

On the other hand, the postmodenist debate has stirred many ahthropologists to rethink their roles. Currently, more ethnographers are writing about their actual interactions and relationships with the people they study.

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CHAPTER - 7

CULTURE, LANGUAGE & COMMUNICATION

THE EVOLUTION OF LANGUAGE

The evolution of hominids was accompanied by increases in brain size. The expansion of the hominid brain reflects the increasing size of the cerebral cortex, the part related to all of the higher functions of the brain, such as memory and, most likely, symbolic and cultural capacities. Although many capacities of the human brain-including memory, learning and other functions – are not completely understood by modern science.

The Human Brain and Speech

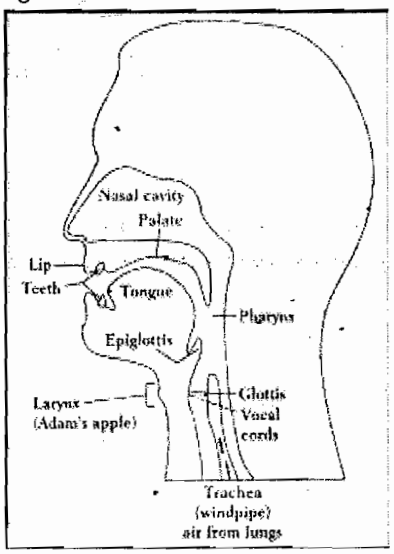
The human brain is divided into two hemispheres. Although neither hemisphere is dominant, and both play important roles in all functions, most human linguistic skills are more closely identified with one hemisphere.

One area of the brain that is located in the left hemisphere and that especially influences human language abilities is known as *Broca's area;* it is associated with the production of sound or pronunciation and with grammatical abilities.

Another area related to linguistic abilities is *Wernicke's area,* also located in the left hemisphere of the brain. Wernicke's area is associated with the ability to understand the meanings of words and sentences, or the semantics of language.

Human Anatomy and Speech

Human vocal organs form an irregular tube connecting the lungs, windpipe, and larynx, the voice box containing the vocal cords. Pharynx, the part of the vocal tract between the back of the tongue and the larynx extending into the nasal cavity is also a vocal organ. The larynx serves to hold air in the lungs and to control its release. These vocal organs work in conjunction with our tongue, lips, and nose to produce speech. the nasal cavity, lips and tongue can constrict or stop the flow of air at any point, enabling us to make vowel or consonant sounds.



The physiology of speech making which is unique to humans.

STRUCTURE OF LANGUAGE

The structure of language is studied as under:

1. Non-Verbal Communication
2. Verbal Communication

Human Communication is the blend of these two. Though Verbal articulation is powerful, sometimes non-verbal acts may also be used as substitutes.

NON-VERBAL COMMUNICATION

It includes a variety of extra-linguistics messages conveyed and received by all the sensory apparatus of human body. Some of the models of non-verbal communication includes.

1.Para linguistics

It is concerned with the association between certain vocal patterns and emotional expressions. It includes the study of all extra speech sounds such as hissing, shushing, whistling, initiation sounds and a variety pitch, hesitations and rate of talking.

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2. Kinesics

Articulation with the help of body movements is termed as Kinesics. It includes all actions, physical or physiological, automatic reflexes, posture, facial-expressions, gestures and other kinds of body movements. Bridewhistell contributed to the study of Kinesics in a contextual approach.

3. Proxemics

Non-Verbal behavior which use space as their channel of communication is called Proxemics. It involves out-of-awareness distance sitting. Edward Hall in his book "The Hidden Dimension" identified four social distances – intimate, personal, social and public.

4. Sign Language

It is used by the deaf to communicate through hand and body movements. Different countries have adopted different signs languages. American- Sign Language (ASL), British Sign Language, French Sign Language are the most common.

ASL is based on four key aspects of visual information – shape, orientation, location and movements of hands and palms. Shape could be flat hand, fist hand, cupped hand etc, orientation could be palm-up, palm-down etc, location of hand may be at chin, waist and movement may be upward or downward. ASL also involves head movement, eye movement and facial expression and so on.

VERBAL COMMUNICATION

The Study of language involves understanding the production and reception of sounds. It requires the appraisal of the following levels :-

Language

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Phonology (Sounds)

↕

Morphology (Meaningful communication of sounds)

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Syntax (Meaningful Contribution of words)

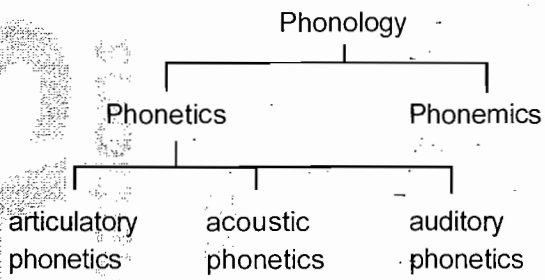
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Symantics (meaning)

Structure of Verbal Communication

1. Phonology :

It is the study of sounds. It is divided into two



Phonetics is the study of production, transmission and reception of speech sounds. It has three sub branches. The study of mechanism of production/ "articulation" of sounds is 'articulatory phonetics'; and the study of the perception of speech sounds (via ears) is 'auditory phonetics'.

The phonetics studies different speech organs because different sounds are produced by different organs. Eg: some are voiced sounds and some are voiceless; some oral and some nasal. Based on these the following characteristics are studied.

Stops are the sounds produced when the airflow is completely blocked off in the oral cavity. (eg: pin,tin, din)

Fricatives – the sounds produced when the air stream is not completely stopped as in the case of 'stops' , but is obstructed from flowing freely from the mouth. Eg: theta,zebra.

Affricates – Produced initially by the obstruction of the airflow as in the case of stops, followed immediately by slow release of air as in the case of fricatives. Eg: Church, Judge.

Liquids – Sounds which are produced with some obstacle to the airflow to the airflow in the mouth, but which is not sufficient or strong enough to cause friction eg: life,right

Apart from these the sounds like glides, short or lax vowels, reduced vowels, phones and allophones are also studied.

Phonemics deals with the study of sound and sound patterns of a specific language.

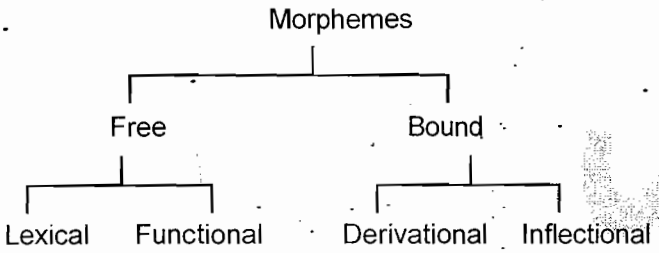
2. Morphology

Focusses on understanding the form of language. The "minimal units of meaning or grammatical function" "Morphemes" are studies in morphology.

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Free Morphemes – can stand by themselves as single words without any attachments. Eg. Boy, man, dog, ox, because etc., These are divided into Lexical Morphemes (which are ordinary nouns, adjectives & Verbs like boy, man etc.,) & functional morphemes (the conjunctions, prepositions, articles, pronouns etc.,)

Bound Morphemes – cannot normally stand and are typically attached to another form. Eg. Many prefixes and suffixes that we use. These are divided into – deviational morphemes which are used to make either a new word or a word of different grammatical category from the stem.

Eg: Kind + "-ness" => Kindness (adj-> noun)

Inflectional Morphemes – do not produce new words but indicate change in the grammatical function of the stem. (eg. Jump + '-ing' => Jumping)

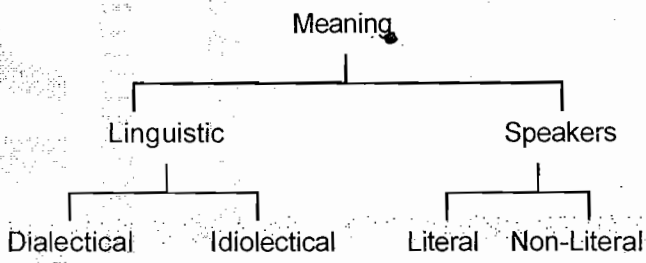
3. Syntax

"The analytical arrangement of words within the sentence" These are certain very definite and stringent underlying rules needed to arrange words to present a correct sentence. Most of the languages have devised their own methods to show the relationship of one word to another in a sentence. Grammar (appropriate use of function words and the proper arrangement of words in a sentence) forms the most important aspect of Syntax.

Inspite of prescriptive rules given in any languages – grammar, Chomsky feels that every speaker works out for himself or herself a set of rules, unconsciously, which constitutes her/his linguistic ability.

4. Semantics

The study of meaning of words, phrases and sentences. Meaning may be of two types.



Linguistic meaning refers to conceptual meaning while speaker's meaning is associated with the context.

Ex: The statement "What a beautiful nose you have" can be taken linguistically or teasingly/ satirically.

The linguistic meaning may also differ according to dialect or according to individual (idiolectical)

The structural study of language also includes different theories of meaning of words and laxical relationship between them.

Thus the structure of language helps us understand production of language, rules associated with them and individual variations on the one hand, non-verbal aspects and their significance on the other.

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CHAPTER - 8

RESEARCH METHODS IN ANTHROPOLOGY

GENESIS & GROWTH OF FIELD WORK TRADITIONS IN ANTHROPOLOGY (L.Q)

Field Work refers to *"first-hand study"* of a people through "personal observation, interview and interaction that involves *prolonged residence* among the people being studied". It involves direct interaction between the field worker and his subject.

1. Beginning of field work :

Anthropological Field Work began is 18th C. In 18th & 19th C. Anthropologists depended on secondary sources. They were *"arm chair Anthropologists"*, who depended on travelogues, history books etc.

But even in 19th C. Anthropologists like *Morgan* used questionnaires for comprehensive work. He visited Iroquois, used *Genealogical techniques to study Kinship.*

2. Royal Anthropological Institute :

Prepared lists of Questionnaires in *"Notes & Queries on Anthropology"*, on various aspects of culture.

3. Franz Boas :

With him began systematic Field Work. He emphasised detailed collection of facts. In 1883 he visited Baffin Island, *lived & travelled with Eskimos.* In 1888, he visited tribes of N.W. Canada.

1898-99 - He led 1st *interdisciplinary expedition* to *Torris Strait.* His students & Anthropologists who were associated in Torris Strait carried forward the tradition of Field Work.

Eg : Rivers – visited, stayed among Todas used genealogical method of study.

4. R.C. Brown

Visited Andaman Islanders, collected facts regarding myths, customs, ceremonies, to *reconstruct* Culture - History

5. Malinowski

Gave new life to Field Work traditions. He spent long time among Trobriand Islanders, studying socio- cultural aspects in totality. (Argonauts of Western Pacific" (1922) & "Coral Gardens and their magic" (1935)). He put forth the requirements of Field Work as -

1. Residence for a long time.
2. Learning & using local language for eliciting information.
3. Using Participant Observation
4. Understanding Cultures from the world view of the natives.

With him Field Work became a professional activity. It came to be recognised that, the object of study is not the place but what happens within it. (Ex : the daily events, expressions of values, attributes). He emphasised the role of informers in providing information.

6. Successors of Malinowski

Malinowski's students conducted field work in various societies & used advanced techniques.

Eg : Leach - Political systems of Highland Burma.
Firth - Tallancy
- Primitive economies of New Zealand 'Maori'
Evanse Pritchard - Azande, Nuer

7. Culture - Personality School -

Championed by American Anthropologists. It was to know the impact of culture on personality & vice verse.

a. Margaret Mead –

- Studied various societies by field work.
- Used comparative method in studying impact of culture on personality in - Mundugumor, Arapesh, Tschamboli.
- Used Field Work techniques of - interviews; case studies, visual anthropology (ie use of photo graphs & video tapes). Visual Anthropology revolutionized understanding & presentation of facts (used in Bali)
- Used small group of informers.
- Used *"Restudy Method"* to study distant & extinct cultures. (in National Character studies).

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b. Ruth Benedict :

- *Content analysis method* by examining historical documents, literature, art forms, news paper articles, dairies of Japanese Prisoners, their life histories, case studies (In National Character Studies of Japan during World War - II).

c. Kora - Du - Bois :

- During her field work in Alor (Indonesia), She recorded traditional ethnographic facts, from biographies, dreams of children, children's games, etc.
- She administered projective tests, thus incorporating psychological techniques in Anthropology.

8. Robert Redfield :

Pioneered study of Peasant Communities, by studying village societies of Mexico

In the field of Physical Anthropology, more direct experiments are conducted by the Anthropologists. Use of Anthropometry, Somato type studies by photographs, study of curing methods by participant observation (Eg : Quasalid in Kwakiutl) to name a few.

Thus, to bring out vital facts modern Anthropologists use Scientific Techniques. It is not simply collection of data, but concluding on the facts is also vital. It is here modern Anthropology offers Etic & Emic ideologies to better understand cultural aspects.

IMPORTANCE OF FIELD-WORK IN ANDHTORPOLOGY

Anthropology is a field-science and field-work is an integral part of this discipline. The early anthropologists, specxially the classical evolutionists did not collect the scientific data by conducting field-work. They depended on the travellers, traders and missionaries who supplied them informations on the curious customs of the savages. and tha's why their data were based on secondary sources and their findings were mostly conjectural.

However, when anthropology as a separate discipline of teaching and research was started in some of the European and American Universtieis in the late half of the 19th century, the importance of carrying out empirical field-work became compulsory. In the beginning, descriptive ethnographic works were taken up and the simple monographs were written by the early ethnographers.

Ethnographic field-work is a research carried out by the anthropologists among the living peoples in different parts of the world. It is generally undertaken to collect basic ethnographic data on a little known society or sub-society. In some of the ethnographic researches some theoretical aims are also examined, besides the descriptive nature of the research work. In many ways the beginning of field-work is a most difficult part of the research. Many serious problems can occur during the first few weeks in the field. First, there is a problem of initial impression one ceates during the first meeting before the members of the society or sub-society one is studying. The problem is greatest among the inaccessible tribal societies where the group is small and the stranger is seen as a potential threat to the group.

Entry into the community necessitates the establishment of a good support. The role one assumes at the beginning of the field-work can affect research work during the later stages of the field-work. The researchers sometimes experience 'culture shock' and they become disoriented when they realise that the people they are studying operate by a different set of cultural rules. Culture shock usually has only a temporary effect. However, after a few days of depression the researchers is now able to take up the field-work seriously. The researcher who is not take up the field-work seriously. The researcher who is not fluent in the language of the native, faces some problm in the beginning.

As field-work is a compulsory paper in the curriculum of anthropology and sociology of the most of the universities of India, field-work is done by the students under the supervision of a teacher who imparts knowledge of the theory, model, designs of research, application of research methods, selection of appropriate topics of research, creation of specific aims, objectives and hypothesises, etc. All these preparations are done at the preliminary stage when some relevant books concerning the research topics are also consulted. However, after doing all these preparations at the headquarters, one now leaves for the field for conducting researches.

APPROACHES IN ANTHROPOLOGICAL FIELD-WORK

In anthropological field-work various approaches are applied to collect data and make analysis of the scientific materials.

i. *Synchronic Approach:*

Synchronic approach is based on empirical field-work where from the first-hand data are collected. The ethnographic details as well as the analytical details

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are analysed in terms of the empirical materias. The nineteenth century scholars of the early anthropologists never applied the synchronic approach to study the primitive people. As referred to earlier, they depended too much on the secondary sources.

ii. *Dychronic Approach*

This approach is not based on field-work, rather it is based on the written documents or other indirect materials, of course, which are relevant for the concerned research work. Some materials are collected through the library work.

However, usually in carrying out a research work a researchers usually combines both the approaches to write a balanced report usually from an analytical point of view.

iii. *Inductive and Deductive Approach*

In anthropology usually the unit of study is always small, say a village, a particular tribe or caste, etc. By studying a village the researchers throws light on the wider perspective of the subject. The most befitting example of this approach is Dube's book – *Indian Village (1955),* which is based on the study of a village named located Shamir Pet, near Hyderabad.

iv. *Textual and Contextual Approach*

The textual approach is based on the study of textual references concerning work. For example, in the study of sacred complexes of Hindu Tirthas (places of pilgrimage), the textual approach becomes an essential part of the research work. When Vidyarthi studied the Sacred Complex of Hindu, Gaya (1961), he first, read several Purans in which the myths and mahatms had been described and thereafter, he collected the field materials. Similarly, in the study of the *Sacred Complex of Kasi* (1979), the authors (Vidyarthi, Saraswati and Jha), consulted many Purans and other historical documents, besides the empirical field-work. Thus, textual approach becomes necessary not only in the study of Indian civilization but also in the study of culture change and other analytical and problem-oriented research work where bench-line data are to be consulted in order to explain the field-data.

The contextual approach is like a synchronic approach which is based on empirical field-work. The contextual approach is applied in the field to collect first-hand data by applying the different field methods.

v. *Etic and Enic Apporoach :*

Etic is a level for a variety of theretical approaches in anthorpology concerned with the outsiders view of culture. Etic involves the careful specifications of the categories, the logical relations between categories and assumptions underlying the uses of these categories by social scientists. In brief, however, it may be said, when an outsider studies the culture with special reference to the natives language, it is called an etic approach.

Emic approach refers to a variety of theoretical field approaches in anthropology concerned with the inside or native view of a culture. This concept was developed by Kenneth Pike (1954). Thus, in the emic approach the researcher usually presents theinsiders' view.

DEFINITION OF HYPOTHESIS AND METHODOLOGY

The anthropological researches, have been popular, through not unhealthy prejudiced against theories and contrasted with experience. However, an established theory is only a generalisation from experience which has been again confirmed by the anthorpological field study.

Research methodology, as a matter of fact, includes everything concerning the research. However I may demonstrate it in a tabular form.

Research Methodology

Research Methods or Field Techniques	Form of Thought. It includes research designs formulation of hypothesis, if any, aims and objectives, selection of suitable approaches etc.

A hypothesis is usually defined merely as an unconfirmed opinion, that judging by what is already known it is reasonable to assume that further facts will be found by the reasearcher to support it or to reject it.

'Methodology' may be used to refer to theorctical discussions concerning the entire research proposal including the forms throught of the research, aims and objectives of the research, application of the the research methods, etc. Thus a distinction must be made between methods and methodology. By the word method, we mean usually the several research methods like interview method or observation method or G.T. method, etc., but when we

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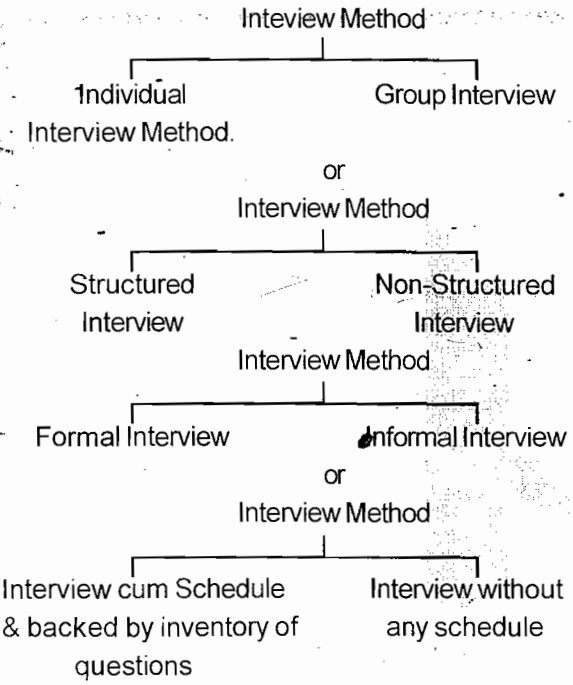
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refer to the word' Methodology', we mean the entire syllabus of the research activities including the application of the research techniques. In other words, the methodology carries a broad meaning and may be used to designate specially the analytical discussions of various research methods, which fall within the ambit of methodology.

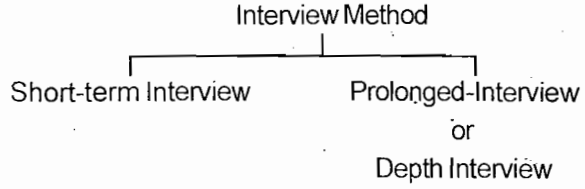
'Methodology' is also often used in a narrow sense to refer to the methods, techniques or tools, employed for the collection and processing of data. But in a broader sense methods or techniques are one of the components of methodology. Finally methodology may be called to designate all the concepts and procedures employed in the analysis of data, however, collected to arrive at conclusion.

FIELD-METHODS IN ANTHROPOLOGY

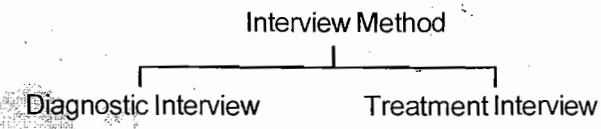
a. **Interview Method** : This is a very simple anthorpological method which is applied by the researcher to collect data. However, selection of the information is very important. An informant should be a reasonable man, who is aware of all the happening taking place around him. Thus, he should be a knowledgeable man.



According to the length of Time :

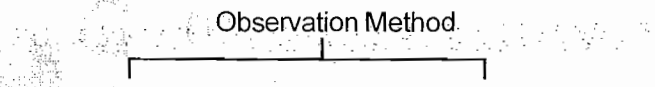


From the pathological point of view :



Thus, there are various types of the Interview Method and the relevance of its application in the field depends on the nature of research one is conducting. An interview is thus, an oral type of questionnaire in which the interviewee gives the needed information in a face to face relationship.

b. **Observation Method** : Observation method is also equally suitable for different types of research works. Broadly speaking it is of two types :



In participation observation method, the researcher has to collect the field data by participating in the activities of the natives. For example, if one is studying the economic activities of the native, he or she will participate in the economic activities of the informants and thereby he/she will observe and collect the data. In casess of marriage studies, or ceremonies, festivals, etc., participant observation method is very useful.

However, when one does not participate in the ceremony or ritual activities, but collects the data by observing from a distance, this is called non-participant method.

c. **Sampling Method** : Sampling method is applied when one is styding a very large population where census and survey of house to house are not possible to apply. Thus, sampling is done keeping in view the size of the population so that the proper represenation of the people under study is made.

Sampling is of two types – Random sampling and stratified sampling. In random sampling the researcher is free to apply sampling selecting the informants from different groups, but under stratified sampling, the population is stratified under different heads, viz., caste/ tribe, age sex, education, profession, economic status, etc., and thereafter the representational samples are selected for study.

In both the types of sampling methods statistical knowledge is required. Through the application of this method, quantitative data are usually collected and

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various diagrams. and graphs are prepared to explain the sampling data. However, the impressionistic and qualitative facts are not explained through this method.

d. Life History and Case Study Method : These two methods are very old in anthropology. When the informants are selected for depth interview, the researchers make a biographical analysis of the life-history of the informants. Usually the researcher gives emphasis as to how to get many insights and knowledge of the informants nder interview. By studying the biographies of many informants, conclusions are drawn accordingly and this type of research is primarily based on life-history method. When cases of violation and noted specially in the cases of extra-martial relations in the society, or violation of incest-taboos, etc. cases are collected by the ethnographers to assess the quantum of stress and strain as well as the nature of disorganisation in the society.

e. Genealogical Method : "Genealogy refers to pedigree or a systematic narration, or description of family tree showing all the ramifications of kinship linkages, kinship terms, various kinship usages & succession etc".

- Nadeem Hasnain

It is a chart connecting an individual (the ego), with his/her ancestors, alive or dead and with relatives who are alive; marital ties and social relations like adopted children.

Origin of the method :

1841 - First used by Sir George Grey in the study of tribal people of W. Australia.

1860 - Morgan investigated kinship systems (terminology) of Iroquois on genealogical basis.

1900 - Rivers developed scientific technique of recording the genealogies. He proved the usefulness of this method in genetic, cultural analysis & social demography. He referred to *qualitative element* - by which genealogy gives meanings of kin terms & appropriate behavior; quantitative element - which gives demographic details (size, sex ratio etc)

Setting Genealogical Chart :

The following symbols are prescribed by International Federation of Eugenic Organisations (1946) & Royal Anthropological Institute (1951).

Δ:	Male	=/ ⌴	= marriage
O:	female	// ⌐	= Consanguineal tie
Δ:	dead male	|	= descent
Ø:	dead female	⌐	= sibling
□/◇:	unknown sex/ person whose sex in not remembered	≠	deceased marriage/ divorce
		I,II,----	generations
	multiple births		

Difficulties/ Draw Backs of the Method :

1. Taboos - against revealing ancestor's names, possess difficulty in obtaining ancestral names.
2. Lack of objectivity - As the information is taken from the subjects themselves, there is a strong possibility of subjectivity & misguiding the anthropologist.
3. Rechecking / Cross checking of information is not possible
4. Overlapping of lines of descent possess difficulty.

Application of the Method :

1. It provides basic data about subsystems of kinship, descent, inheritance, succession, residence, regulation of marriage, insert taboo etc.
2. Used for studying demographic details like sex composition, patterns of fertility, mortality, morbidity and migration, divorce, residence etc.
3. It provides time series data comprising of details regarding present & past generations.

f. Questionnaires & Schedules :

Questionnaires & Schedules are much used tools to gather a variety of data pertaining to personal preferences, social attitudes, belief, opinions, behavior patterns etc.

"A questionnaire is a list of questions sent to a number of persons for their answers and which obtains standardised result that can be tabulated and treated statistically". It secures answers to questions by using a form which the respondent fills by himself.

Origin :

Questionnaire was 1st used by Morgan extensively to study kinship terminology in different societies. Ex : Asia (Tamils), Far East, Australia etc. His Questionnaire consisted of 200 questions, organised in a complicated

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way. Hence, out of 1000s of Questionnaires posted, only few came back.

In the later times Questionnaire method was improved. By the time Culture-Personality School originated, the method attained scientific outlook.

Types of Questionnaires :

(1) Structured Questionnaire (2) Pictorial Questionnaire

(3) Unstructured Questionnaire

Structured Questionnaires –

1. They pose definite, concrete & preordained questions.
2. They are prepared in advance & not constructed on the spot during the questioning period.
3. Additional Questions are used only when need arises to classify vague or inadequate replies by informants.
4. The responses may be

a. **Open - ended** – Free & spontaneous expressions on the part of informants. They are used for intensive studies of a limited number of cases or for preliminary exploration of new problems and situations.

b. **Closed - form** – These responses (Eg : Yes/No; or choosing one from the given alternatives) are sought when categorised data are required.

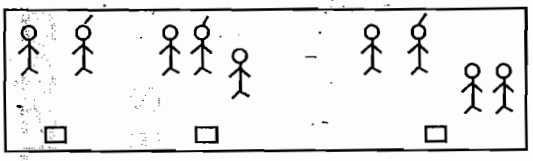
Uses of Structured Questionnaires –

- to initiate a formal inquiry.
- in the study of economic or social problems.
- measurement of opinion on public issues/events.
- studies of administrative policies & changes.
- on the cost of living, child welfare, public health etc.

Pictorial Questionnaires :

Pictures are used in some studies solely for the purpose of promoting interest in answering the questions.

Eg : How large is your Family? (mark in the appropriate box).



These questionnaires are used extensively in studies of social attitudes & prejudices in children.

Unstructured Questionnaires :

Consist of questions in which the stimulus & response both are left free. Flexibility is the greatest advantage of it. It is used for intensive studies, but for a limited number of selected cases.

Limitations of Questionnaires :

- Not applicable to illiterate & less educated respondents.
- Inappropriate where the objective of the survey involves a good deal of explaining.

Precautions to be taken –

- Careful phrasing & structuring of questions.
- Questions not to be ambeguious or confusing.
- Respondents should not be compelled to give socially unacceptable responses.
- Order of questions to be rational.
- Opening Questions should generate interest.
- Before questionnaire is finalised 'pre-testing' to be done.

* * *

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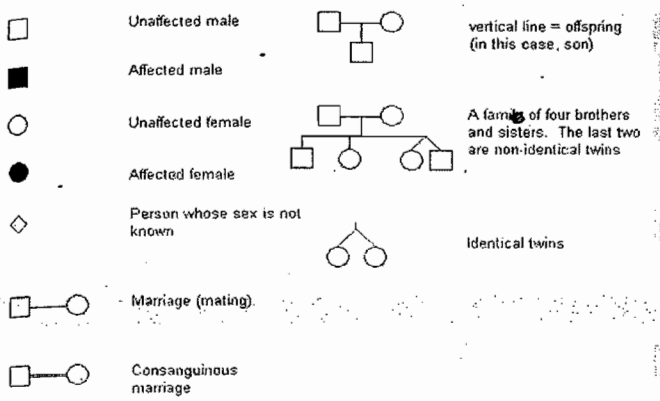
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CHAPTER - 9

METHODS OF GENETIC STUDIES OF MAN

1. PEDIGREE ANALYSIS

Following are the symbols used to draw pedigrees.



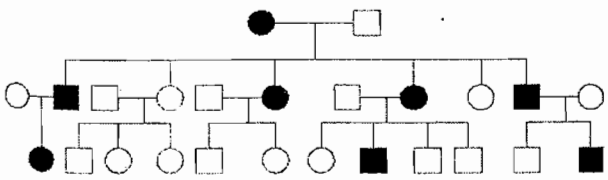
Generations are numberered from the top of the pedigree in uppercase Roman numerals, I, II, III etc. Individuals in each generation are numbered from the left in arab numberals as subscripts, III_1 , III_2, III_3 etc.

Modes of inheritance

Most human genes are inherited in a Mendelian manner. We are usually unaware of their existence unless a variant form is present in the population which causes an abnormal (or at least different) phenotype. We can follow the inheritance of the abnormal phenotype and deduce whether the variant allele is dominant or recessive.

Autosomal dominant

A dominant condition is transmitted in unbroken descent from each generation to the next. Most matings will be of the form M/m x m/m, i.e. heterozygote to homozygous recessive. We would therefore expect every child of such a mating to have a 50% chance of receiving the mutant gene and thus of being affected. A typical pedigree might look like this:

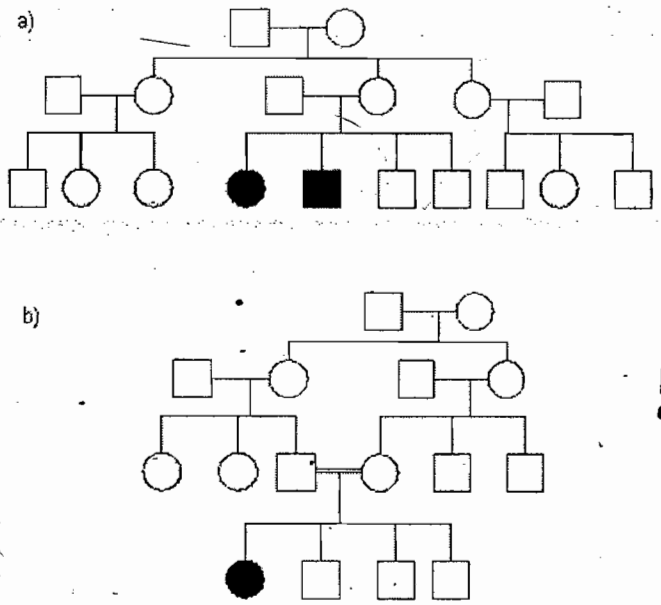


Examples of autosomal dominant conditions include *Tuberous sclerosis*, *neurofibromatosis* and many other cancer causing mutations such as *retinoblastoma*

Autosomal recessive

A recessive trait will only manifest itself when homozygous. If it is a severe condition it will be unlikely that homozygotes will live to reproduce and thus most occurences of the condition will be in matings between two heterozygotes (or **carriers**). An autosomal recessive condition may be transmitted through a long line of carriers before, by ill chance two carriers mate. Then there will be a ¼ chance that any child will be affected. The pedigree will therefore often only have one '**sibship**' with affected members.

a) A 'typical' autosomal recessive pedigree, and b) an autosomal pedigree with **inbreeding**:



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If the parents are related to each other, perhaps by being cousins, there is an increased risk that any gene present in a child may have two alleles **identical by descent**. The degree of risk that both alleles of a pair in a person are descended from the same recent common ancestor is the degree of **inbreeding** of the person. Let us examine b) in the figure above. Considering any child of a first cousin mating, we can trace through the pedigree the chance that the other allele is the same by common descent. Let us consider any child of generation IV, any gene which came from the father, III_3 had a half chance of having come from grandmother II_2, a further half chance of being also present in her sister, grandmother II_4 a further half a chance of having been passed to mother III_4 and

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finally a half chance of being transmitted into the same child we started from. A total risk of

½ x ½ x ½ x ½ = 1/16

2. TWIN STUDIES

The English anthropologist Francis Galton (1822–1911), a cousin of Charles Darwin, conducted some of the first reported twin studies. Advancing his cousin's theories of evolution, he performed twin studies in 1876 to investigate the extent to which the similarity of twins changes over the course of development. He is considered the originator of the field of medical genetics because of his considerable contributions to the nature-versus-nurture debate and the research methods he developed for evaluating heritability. Genetic determination is the combination of genes that creates a trait or characteristic, and heritability is what causes differences in those characteristics.

Twin studies, meaning comparisons of identical (monozygotic) twins to fraternal or nonidentical (dizygotic) twins, are performed in order to estimate the relative contributions of genes and environment—that is, the extent to which environmental versus genetic influences operate on specific traits. Twin studies also help to determine the proportion of the variability in a trait that might be due to genetic factors. The studies aim to identify the causes of familial resemblance by comparing the concordance rates of monozygotic twins and dizygotic twins. Monozygotic twins share the same genetic material—they have 100% of their genes in common—and dizygotic twins share only half their genetic material—they have 50% of their genes in common. Most twin studies report their results in terms of pair-wise concordance rates, which measure the number of pairs, or proband-wise concordance rates, which count the number of individuals.

Monozygotic twins serve as excellent subjects for controlled experiments because they share prenatal environments and those reared together also share common family, social, and cultural environments. Furthermore, studies of twins can both point to hereditary effects and also estimate heritability, a term that describes the magnitude of the genetic effect. The limitations of such studies include the potential to overestimate or underestimate the role of genetics if environmental influences treat twins as more alike or more different than they actually may be. In addition, in some studies it has been difficult to control for other potential causes or sources of variation.

Some of the most conclusive twin study research has analyzed identical and fraternal twins who were raised apart. Researchers have sought to establish whether characteristics such as personality traits, aptitudes, and occupational preferences are the products of nature or nurture. Similar characteristics among identical twins reared apart might indicate that their genes played a major role in developing that trait. Different characteristics might indicate the opposite—that environmental influences assume a much stronger role. By comparing monozygotic and dizygotic twins, investigators can test their hypotheses and confirm the findings of earlier research. For example, if identical twins raised in different homes have many similarities, but fraternal twins raised apart have little in common, researchers may conclude that genes are more important than environment in determining specific characteristics, traits, susceptibilities, and diseases.

A Scandinavian twin study, "Environmental and Heritable Factors in the Causation of Cancer—Analyses of Cohorts of Twins from Sweden, Denmark, and Finland" (*New England Journal of Medicine*, vol. 343, no. 2, July 13, 2000), reported by Paul Lichtenstein et al., looked at nearly 45,000 pairs of twins to determine if the likelihood of developing certain kinds of cancers is more closely linked to environmental exposures than genetics. The results of this large-scale study revealed that environmental factors are linked to twice as many cancers as genetic factors. In fact, the risk of developing only three types of cancer (albeit some of the most common cancers)—breast, colorectal, and prostate cancers—show a significant genetic correlation.

3. KARYOTYPING

Objectives:

1. To demonstrate a microtechnique for reliable chromosomal analysis of leucocytes obtained from peripheral blood.
2. To prepare a karyotype from the chromosomes of human male or female.
3. To use the karyotyping techniques for diagnosing a chromosomal disorder.

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What is Karyotyping:

Chromosomal aberrations are abnormalities in the number or microscopically observable structure of chromosomes. The number of chromosomes in human cells is 46 with 22 autosomal pairs (one of each type contributed by the other and one of each type from the father) and 2 sex chromosomes - 2 X chromosomes for females (one from father and one from mother) or an X and a Y chromosome for males (the X from the mother and the Y from the father). The chromosomes visible only at the metaphase stage of mitosis, 22 homologous pairs of autosomes and two sex chromosomes. Each chromosome has a characteristic size and shape in the "normal" cell. During most of the cell cycle, interphase, the chromosomes are somewhat less condensed and are not visible as individual objects under the light microscope. Mitosis, or nucleus division, is the first part of M-phase and in consists of four stages (prophase, metaphase, anaphase and telophase). However during cell division, mitosis, the chromosomes become highly condensed and are then visible as dark distinct bodies within the nuclei of cells. The chromosomes are most easily seen and identified at the metaphase stage of cell division.

The banding of chromosomes by using dyes was discovered in the late 1960's and before that cytogeneticists depended on chromosome length and position of a constriction to identify the individual chromosomes. The band width and the order of bands is characteristic of a particular chromosome - a trained cytogeneticist can identify each chromosome (1,2,3...22, X and Y) by observing its banding pattern under the microscope. chromosomes are arranged and numbered by size, from largest to smallest. This arrangement helps scientists quickly identify chromosomal alterations that may result in a genetic disorder.

Identifying chromosomes has become easier in recent years by using certain staining techniques. One of the most common staining techniques involves giemsa stain, which gives the chromosomes a banded appearance (hence giemsa banding or G-banding) G-banding is the treatment of chromosomes in the metaphase stage with trypsin (to partially digest the protein) and stain them with Giemsa. Each homologous chromosome pair has a unique pattern of G-bands, enabling recognition of particular chromosomes.

Karyotyping is the process by which doctors and geneticists take pictures of the chromosomes while the cell are undergoing mitosis. The picture is then enlarged. The picture of the chromosomes are then cut up so that each chromosome is removed. The chromosomes are matched up and attached to a paper according to size. The chromosomes pairs are numbered from largest to smallest. There are 22 pairs of chromosomes which match up exactly. Then the sex chromosomes are paired, in the female (XX) the chromosomes match and in the male (XY) the chromosomes do not match. This technique can be used to assess the "normalcy" of an individual's chromosomes and to assay for various genetic diseases such as Down's syndrome and Klinefelter's syndrome. It is estimated that one in 156 live births have some kind of chromosomal abnormality.

Karyotypes are arranged with the short arm of the chromosome on top, and the long arm on the bottom. In addition, the differently stained regions and sub-regions are given numerical designations from proximal to distal on the chromosome arms. For example, Cri du chat syndrome involves a deletion on the short arm of chromosome . It is written as 46,XX,5p-. The critical region for this syndrome is deletion of 15.2, which is written as 46,XX,del(5)(p15.2). Peripheral Blood Karyotyping Medium With Phytohemagglutinin (PHA) is intended for use in short-term cultivation of peripheral blood lymphocytes for chromosome evaluation. It is based on RPMI-1640 basal medium supplemented with L-Glutamine, fetal bovine serum and antibiotics (penicillin and streptomycin). Karyotyping Medium is supplied as frozen medium, which is ready for use after thawing and phytohaemagglutinin supplementation.

The blood cell karyotyping method was developed to provide information about chromosomal abnormalities. Lymphocyte cells do not normally undergo subsequent cell divisions. In the presence of a mitogen (PHA), lymphocytes are stimulated to enter into mitosis by DNA replication. After 48-72 hours, a mitotic inhibitor (colcemid) is added to the culture to stop mitosis in the metaphase stage. After treatment by hypotonic solution, fixation and staining, chromosomes can be microscopically observed and evaluated for abnormalities.

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Types of banding

Q- Banding (Qyinacrine banding) : It uses Fluoresent staining procedure by Quinacrine hydrochloride. It produces bright and dim bands in specific manner. It covers most of the chromosome.

G- Banding (Giemsa banding) : It is used in addition to Q banding. In this technique chromosomes are treated with Trypsin.It produces dark bands.

R- Banding (Reverse banding) : It involves saline heat treatment of chromosomes prior to Giesma staining.

C-Banding (Centrometric banding) : It involves staining of DNA from chromosomes by alkali extraction of DNA from chromosome. It involves harsh pretreatment of chromosomes with dilute acid followed by strong alkali and warm saline treatment prior to Giesma staining.

Cd- Banding (Centrometric dot banding): It stains protiens associated with the centromere.

G-11 Staining: It involves Giesma staining at a high alkaline pH value (11 or above)

AgNOR Staining : It stains the nucleolus regions of ribosomal RNA sites.

Since the banding pattern of each chromosome is specific, it is posible to identify all the chromosomes individually and prepare a Karyotype.

4. BIOCHEMICAL GENETICS

Used to understand

1. Inborn errors in metabolism
2. Abnormal hemoglobins

1. Inborn errors in metabolism

Biochemical pathways play an integral and essential part in the proper functioning of the human body. Defects in the production, breakdown or transport of a specific enzyme or protein can have deleterious consequences. Ultimately, disorders resulting from inborn errors of metabolism (IEM) are caused by the deficiency of a particular enzyme, and/or the accumulation of abnormal metabolites.

IEMs can be categorized as follows:

1. Disorders of protein metabolism: Organic acidopathies, amino acidopathies, urea cycle defects
2. Disorders of carbohydrate metabolism: Galactosemia, glycogen storage disorders
3. Disorders of fatty acid oxidation: Medium-chain acyl CoA dehydrogenase (MCAD) deficiency
4. Mitochondrial disorders
5. Lysosomal disorders: Hurler syndrome, Hunter syndrome, Tay-Sachs
6. Peroxisomal disorders: Zellweger syndrome, X-linked adrenoleukodystrophy (X-ALD)

Many inborn errors of metabolism will manifest symptoms in the newborn period, including the organic acidemias, the urea cycle defects, and certain disorders of amino acid metabolism.

The majority of IEM are inherited in an autosomal recessive manner; although there are a few that have X-linked recessive inheritance. Examples of IEMs with X-linked recessive inheritance:

- Ornithine transcarbamylase (OTC) deficiency - urea cycle defect
- Hunter syndrome - lysosomal storage disorder

2. Hemoglobinopathies

Hemoglobinopathy is a kind of genetic defect that results in abnormal structure of one of the globin chains of the hemoglobin molecule. Hemoglobinopathies are inherited single-gene disorders; in most cases, they are inherited as autosomal co-dominant traits. Common hemoglobinopathies include sickle-cell disease. It is estimated that 7% of worlds population (420 million) are carriers, with 60% of total and 70% pathological being in Africa. Hemoglobinopathies are most common in ethnic populations from Africa, the Mediterranean basin and Southeast Asia.

Hemoglobinopathies imply structural abnormalities in the globin proteins themselves. Thalassemias, in contrast, usually result in underproduction of normal globin proteins, often through mutations in regulatory genes. The two conditions may overlap, however, since some conditions which cause abnormalities in globin proteins (hemoglobinopathy) also affect their production (thalassemia). Thus, some hemoglobinopathies are also thalassemias, but most are not.

Hemoglobin variants

Hb S: **Sickle-cell disease** (**SCD**), or **sickle-cell anaemia** (or **anemia**, **SCA**) or **drepanocytosis**, is an autosomal recessive genetic blood disorder withoverdominance, characterized by red blood cells that assume an abnormal, rigid, sickle shape.

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Sickling decreases the cells' flexibility and results in a risk of various complications. The sickling occurs because of a mutation in the hemoglobin gene. Life expectancy is shortened. In 1994, in the US, the average life expectancy of persons with this condition was estimated to be 42 years in males and 48 years in females, but today, thanks to better management of the disease, patients can live into their 50s or beyond. In the UK, the current life expectancy is estimated to be 53–60 years of age.

Sickle-cell disease, usually presenting in childhood, occurs more commonly in people (or their descendants) from parts of tropical and sub-tropicalregions where malaria is or was common. One-third of all indigenous inhabitants of Sub-Saharan Africa carry the gene, because in areas where malaria is common, there is a fitness benefit in carrying only a single sickle-cell gene (sickle cell trait). Those with only one of the two alleles of the sickle-cell disease, while not totally resistant, are more tolerant to the infection and thus show less severe symptoms when infected. The prevalence of the disease in the United States is approximately 1 in 5,000, mostly affecting Americans of Sub-Saharan African descent, according to the National Institutes of Health. In the United States, about 1 out of 500 African-American children born will have sickle-cell anaemia.

Sickle-cell anaemia is the name of a specific form of sickle-cell disease in which there is homozygosity for the mutation that causes HbS. Sickle-cell anaemia is also referred to as "HbSS", "SS disease", "haemoglobin S" or permutations thereof. In heterozygous people, who have only one sickle gene and one normal adult haemoglobin gene, it is referred to as "HbAS" or "sickle cell trait". Other, rarer forms of sickle-cell disease include sickle-haemoglobin C disease (HbSC), sickle beta-plus-thalassaemia (HbS/â$^{+}$) and sickle beta-zero-thalassaemia (HbS/â^{0}). These other forms of sickle-cell disease are compound heterozygous states in which the person has only one copy of the mutation that causes HbS and one copy of another abnormal haemoglobin allele.

Hb E Disease

Hemoglobin E disease results when the offspring inherits the gen for HbE from both parents. At birth, babies homozygous for the hemoglobin E allele do not present symptoms due to HbF (fetal hemoglobin) they still have. In the first months of life, fetal hemoglobin disappears and the amount of hemoglobin E increases, so the subjects start to have a mild â thalassemia. People with hemoglobin E do not show any symptoms (there is usually no anemia or hemolysis). Subjects homozygous for the hemoglobin E allele have a mild hemolytic anemia and mild splenomegaly.

5.CYTOGENETIC METHOD

Each of our chromosomes has a characteristic structure and displays a specific pattern of dark and light bands when stained with chemicals. These features are conserved in all human beings, making the different chromosomes easy to identify and distinguish from each other under a microscope. They define what is known as a normal karyotype - a normal set of chromosomes. Any deviation from the normal karyotype, in terms of chromosome number or structure, is known as a chromosome abnormality.

While some chromosome abnormalities are harmless variations, most are associated with clinical disorders of one kind or another. Half of all spontaneous abortions are due to chromosome abnormalities but the incidence in live births falls to less than 1 per cent.

Structural abnormalities:

Structural abnormalities can be unbalanced or balanced. The former are similar to numerical abnormalities in that genetic material is either gained or lost. The abnormalities range from the loss or duplication of whole chromosome arms to the deletion or duplication of tiny chromosome fragments barely visible under the microscope. However, even these tiny deviations (microdeletions and microduplications) can encompass several to many genes and have severe effects. Table lists some common deletion and microdeletion syndromes.

Unbalanced structural abnormalities (p = short arm, q = long arm)

Syndrome	Abnormality	Incidence
Wolf-Hirschhorn	Deletion, tip of 4p	1 in 50 000
Cri-du-chat	Deletion, tip of 5p	1 in 50 000
WAGR	Microdeletion, 11p	
Prader-Willi/ Angelman	Microdeletion, 15p	
DiGeorge	Microdeletion, 22q	

Balanced structural abnormalities involve the rearrangement of genetic material but no overall gain or loss. Examples include inversions (where a segment is

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removed from a chromosome, turned on its axis and sealed back in place), translocations (where part of one chromosome becomes attached to another) and ring chromosomes (where the ends of the long and short arms fuse together to form a circle).

Balanced abnormalities can have an immediate effect if a gene is disrupted by the breakpoint, if two genes are fused together, or if the relocation of a gene causes it to be expressed at a higher or lower level than usual. However, the major consequence is to prevent normal chromosome pairing at meiosis, leading to the production of sperm and eggs with incomplete or partially duplicated chromosome sets.

Types of structural abnormalities

- Deletion
- Duplication
- Translocation
- Inversion
- Ring chromosome
- Isochromosome

A. **Deletion** : In genetics, a **deletion** (also called **gene deletion**, **deficiency**, or **deletion mutation**) is a mutation (a genetic aberration) in which a part of a chromosome or a sequence of DNA is missing. Deletion is the loss of genetic material. Any number of nucleotides can be deleted, from a single base to an entire piece of chromosome. Deletions can be caused by errors in chromosomal crossover during meiosis. This causes several seriousgenetic diseases.

Types of deletion include the following:

- **Terminal Deletion** - a deletion that occurs towards the end of a chromosome.
- **Intercalary Deletion / Interstitial Deletion** - a deletion that occurs from the interior of a chromosome.

Small deletions are less likely to be fatal; large deletions are usually fatal - there are always variations based on which genes are lost. Some medium-sized deletions lead to recognizable human disorders.

Deletion of a number of base pairs that is not evenly divisible by three will lead to a frameshift mutation, causing all of the codons occurring after the deletion to be read incorrectly duringtranslation, producing a severely altered and potentially nonfunctional protein. In contrast, a deletion that is evenly divisible by three is called an *in-frame* deletion.

Deletions are responsible for an array of genetic disorders, including some cases of male infertility and two thirds of cases of Duchenne muscular dystrophy. Deletion of part of the short arm of chromosome 5 results in Cri du chat syndrome. Deletions in the SMN-encoding gene cause spinal muscular atrophy, the most common genetic cause of infant death.

B. **Duplication** : **Gene duplication** (or **chromosomal duplication** or **gene amplification**) is any duplication of a region of DNA that contains a gene; it may occur as an error in homologous recombination, a retrotransposition event, or duplication of an entire chromosome.[1] The second copy of the gene is often free fromselective pressure — that is, mutations of it have no deleterious effects to its host organism. Thus it accumulates mutations faster than a functional single-copy gene, over generations of organisms.

A duplication is the opposite of a deletion. Duplications arise from an event termed unequal crossing-over that occurs during meiosis between misaligned homologous chromosomes. The chance of this happening is a function of the degree of sharing of repetitive elements between two chromosomes. The product of this recombination are a duplication at the site of the exchange and a reciprocal deletion.

C. **Translocation** : In genetics, a **chromosome translocation** is a chromosome abnormality caused by rearrangement of parts between nonhomologous chromosomes. A gene fusion may be created when the translocation joins two otherwise separated genes, the occurrence of which is common in cancer. It is detected on cytogenetics or a karyotype of affected cells. There are two main types, **reciprocal** (also known as non-Robertsonian) and **Robertsonian**. Also, translocations can be **balanced** (in an even exchange of material with no genetic information extra or missing, and ideally full functionality) or**unbalanced** (where the exchange of chromosome material is unequal resulting in extra or missing genes).

They are of two types

1. Reciprocal translocations are usually an exchange of material between nonhomologous chromosomes. Estimates of incidence range from about 1 in 500 to 1 in 625 human newborns. Such translocations are usually harmless and may be found through prenatal diagnosis. However, carriers of balanced reciprocal

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translocations have increased risks of creating gametes with unbalanced chromosome translocations leading to miscarriages or children with abnormalities. Genetic counseling and genetic testing are often offered to families that may carry a translocation. Most balanced translocation carriers are healthy and do not have any symptoms. But about 6% of them have a range of symptoms which may include autism, intellectual disability, or congenital anomalies. A gene disrupted or disregulated at the breakpoint of the translocation carrier is likely the cause of these symptoms.

2. Robertsonian translocation This type of rearrangement involves two acrocentric chromosomes that fuse near the centromere region with loss of the short arms. The resulting karyotype in humans leaves only 45 chromosomes since two chromosomes have fused together. This has no direct effect on the phenotype since the only genes on the short arms of acrocentrics are common to all of them and are present in variable copy number (nucleolar organiser genes). Robertsonian translocations have been seen involving all combinations of acrocentric chromosomes. The most common translocation in humans involves chromosomes 13 and 14 and is seen in about 0.97 / 1000 newborns. Carriers of Robertsonian translocations are not associated with any phenotypic abnormalities, but there is a risk of unbalanced gametes which lead to miscarriages or abnormal offspring. For example, carriers of Robertsonian translocations involving chromosome 21 have a higher chance of having a child with Down syndrome. This is known as a 'translocation Downs'. This is due to a mis-segregation (Nondisjunction) during gametogenesis. The mother has a higher (10%) risk of transmission than the father (1%). Robertsonian translocations involving chromosome 14 also carry a slight risk of uniparental disomy 14 due to trisomy rescue.

Some human diseases caused by translocations are:

- Cancer: several forms of cancer are caused by acquired translocations (as opposed to those present from conception); this has been described mainly in leukemia (acute myelogenous leukemia and chronic myelogenous leukemia). Translocations have also been described in solid malignancies such as Ewing's sarcoma.
- Infertility: one of the would-be parents carries a *balanced translocation*, where the parent is asymptomatic but conceived fetuses are not viable.
- Down syndrome is caused in a minority (5% or less) of cases by a Robertsonian translocation of the chromosome 21 long arm onto the long arm of chromosome 14.

D. **Inversion:** An **inversion** is a chromosome rearrangement in which a segment of a chromosome is reversed end to end. An inversion occurs when a single chromosome undergoes breakage and rearrangement within itself. Inversions are of two types: **paracentric** and **pericentric**.

Paracentric inversions do not include the centromere and both breaks occur in one arm of the chromosome. Pericentric inversions include the centromere and there is a break point in each arm.

E. **Inserion:** In genetics, an **insertion** (also called an **insertion mutation**) is the addition of one or more nucleotide base pairs into a DNA sequence. This can often happen in microsatellite regions due to the DNA polymerase slipping. Insertions can be anywhere in size from one base pair incorrectly inserted into a DNA sequence to a section of one chromosome inserted into another.

On a chromosome level, an *insertion* refers to the insertion of a larger sequence into a chromosome. This can happen due to unequalcrossover during meiosis.

F. **A ring chromosome** is a chromosome whose arms have fused together to form a ring. A ring chromosome is denoted by the symbol r. Ring chromosomes may form in cells following genetic damage by mutagens like radiation, but they may also arise spontaneously during development. Disorders arising from the formation of a ring chromosome include ring chromosome 20 syndrome where a ring formed by one copy of chromosome 20 is associated with epilepsy; ring chromosome 14 and ring chromosome 13 syndrome are associated with mental retardationand dysmorphic facial features; ring chromosome 15 is associated with mental retardation, dwarfism and microcephaly. Ring formation of an X-chromosome causes Turner syndrome

G. **An isochromosome** is a chromosome that has lost one of its arms and replaced it with an exact copy of the other arm. This is sometimes seen in some females with Turner syndrome or intumor cells. This may also cause an isochromosome to have two centromeres.

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MENDELIAN GENETICS

LETHAL GENES – SUB LETHAL GENES

[Note : The action of gene at cellular level is unitary (i.e one gene-one action). But some genes produce a broad spectrum of phenotypic changes so that a gene has multiple action. This is called PLEIOTROPY. Lethal genes usually have pleiotropic effect]

Certain genes are known to control the manifestation of some phenotypic trait as well as affect the viability of the organism. Some other genes have no effect on the appearance of the organism but affect the viability alone. These are known as lethals/semilethals depending upon their influence.

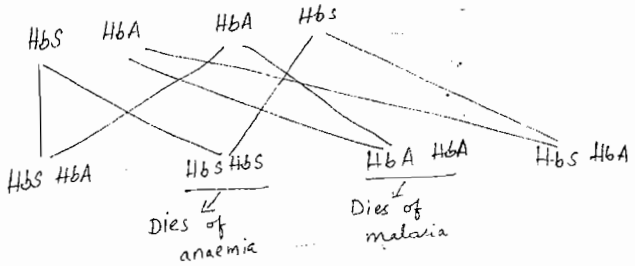
Complete lethal genes in homozygous condition (cc) kill all or nearly all homozygous individuals while in case of semilethal genes (Cc or cC) some homozygous individuals survive. The lethal genes are always recessive for their lethality and express the lethal effect only in homozygous condition.

Lethal genes exert their lethal effect at different times and at different stages in the life cycle. Some cause death of the zygote or early embryo. Others effect later while some lethals have a delayed effect.

Only recessive lethals survive. The dominant lethal die and fail to transmit traits to the next generation. Lethal genes can be – dominant, recessive or conditional.

Dominant Lethal

They are lethal in homozygous condition and produce some defective/abnormal phenotype in heterozygous condition. Their most serious effect is death. Eg: Inheritance of sickle-cell anaemia in man.



Eg 2: Huntington's chorea (muscular failure, mental retardation and finally death)

Eg 3: Epiolia Lethal in man – causes death in early stages of life even in heterozygous condition. (mental retardation, tumors, abnormal skin growth)

Recessive Lethal

Effect only in homozygous condition. Heterozygotes are normal. Hence recessive lethals remain unnoticed and get established in the population. Eg : Tay Sach's disease – (causes death of young children due to failure to produce enzymes needed for normal fat metabolism leading to poor muscular control)

Conditonal Lethal

The genes may be normal to the individual in a particular environment and may prove to be lethal when environment is changed.

Eg : Break off of feathers in poultry.

Multiple factor / multiple alleles

Multiple alleles are defined as " genes that are members of the same gene pair and are located on the same locus. All of them control the same character but each of the allele affects that character somewhat differently than the others"

In any diploid organism, a particular gene pair is represented by 2 alleles of the series. One allele on each of the 2 chromosomes of a homologous pair. This is because an organism has just 2 chromosomes of each type. They form a homologous pair. However in the individuals of a population/species, many alternative forms of the same gene may be present. The grouping of all the possible alleles of a gene pair is defined as a system of multiple alleles and their mode of inheritance is called "multiple allelism" eg: ABO blood groups in man. This system is characterized by the presence of antigens on the surface of RBCs. Persons exhibit 4 phenotypes.

1) Blood group A
2) Blood group B
3) Blood group AB
4) Blood group O

These phenotypes are controlled by a series of 3 alleles represented by 1^0, L^A and L^B. Alleles L^A&$L^A$are re-

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sponsible for the production of their respective A&B antigens and 1^0 is a recessive allele that does not produce any detectable A&B antigens.

Phenotype	Antigen	Genotype
Blood gp O	-	$1^0 1^0$(++)
Blood gp A	A	$L^A L^A$(AA)
		$L^A 1^0$ (A+)
Blood gp B	B	$L^B L^B$ (BB)
		$L^B 1^0$ (B+)
Blood gp AB	AB	$L^A L^B$(AB)

Allele L^0 is recessive and L^A & L^B are dominant to L^0 but themselves are codominant.

CHROMOSOMAL ABERRATIONS

NUMERICAL ABERRATIONS

Numerical aberrations (whole chromosomal aneuploidy) represent a significant proportion of chromosomal changes found in humans. These aberrations can occur as a consequence of chromosome segregation defects during cell division. Segregation errors arising during reductive cell division, or meiosis, are upon fertilisation and subsequent embryo development constitutively present in all cells, resulting in whole organismal aneuploidy. Missegregation in mitosis leads to a mosaic distribution of aneuploidy. Aneuploidy is associated with pathological states in most organisms. Numerical aberrations represent a significant cause of pregnancy loss as well as abnormalities found in live births. Moreover, numerical aberrations are frequently found in ageing tissues or in tumour cells. Although the association of aneuploidy and cancer is known for almost a century, the dispute is still ongoing whether this is a cause or a consequence of cell transformation. Recently, new evidence is emerging that numerical aberrations significantly alter the physiology of eukaryotic cells and might indeed directly contribute to tumorigenesis.

- Numerical chromosomal aberrations result from errors in chromosome segregations.
- Trisomy, monosomy and polyploidy are among the major causes of spontaneous human abortions.
- Trisomies compatible with survival often result in multiple defects.
- Numerical chromosomal aberrations significantly alter physiology of eukaryotic cells.
- Numerical chromosomal aberrations are frequently found in cancer cells and can contribute to tumorigenesis.

SEX CHROMOSOME ABNORMALITIES

The majority of known types of chromosomal abnormalities involve sex chromosomes. In frequency of occurrence, they are only slightly less common than autosomal abnormalities. However, they are usually much less severe in their effects. The high frequency of people with sex chromosome aberrations is partly due to the fact that they are rarely lethal conditions. Like Down syndrome and other autosomal problems, sex chromosome gross abnormalities can be diagnosed before birth by amniocentesis and chorionjc villi sampling.

Sex chromosome abnormalities are gender specific. Normal males inherit an X and a Y chromosome while females have two X's. A single Y-chromosome is sufficient to produce maleness while its absence is necessary for femaleness. Female abnormalities are due to variations in the number of X chromosomes. Male abnormalities are the result of irregular numbers of either the X or the Y chromosome or both.

Female Sex Chromosome Abnormalities

Turner's syndrome:

Turner's syndrome is a chromosomal abnormality involving absense of all or part of one X chromosome. It has been described in many species, including humans, mice and horses. Individuals with this disorder have a female phenotype, but have abnormalities in reproductive function as well as a number of other clinical manifestations of disease. Women with this condition tend to be shorter than average and are usually unable to conceive a child (infertile) because of an absence ofovarian function. Other features of this condition that can vary among women who have Turner syndrome include: extra skin on the neck (webbed neck), puffiness or swelling (lymphedema) of the hands and feet, skeletal abnormalities, heart defects and kidney problems.

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This condition occurs in about 1 in 2,500 female births worldwide, but is much more common among pregnancies that do not survive to term (miscarriages and stillbirths).

Turner syndrome is a chromosomal condition related to the X chromosome. Researchers have not yet determined which genes on the X chromosomeare responsible for most signs and symptoms of Turner syndrome. They have, however, identified one gene called SHOX that is important for bone development and growth. Missing one copy of this gene likely causes short stature and skeletal abnormalities in women with Turner syndrome.

What are the symptoms for Turner syndrome?

Girls who have Turner syndrome are shorter than average. They often have normal height for the first three years of life, but then have a slow growth rate. At puberty they do not have the usual growth spurt.

Non-functioning ovaries are another symptom of Turner syndrome. Normally a girl's ovaries begin to produce sex hormones (estrogen and progesterone) at puberty. This does not happen in most girls who have Turner syndrome. They do not start their periods or develop breasts without hormone treatment at the age of puberty.

Even though many women who have Turner have non-functioning ovaries and are infertile, their vagina and womb are totally normal. In early childhood, girls who have Turner syndrome may have frequent middle ear infections. Recurrent infections can lead to hearing loss in some cases. Girls with Turner Syndrome are usually of normal intelligence with good verbal skills and reading skills. Some girls, however, have problems with math, memory skills and fine-finger movements.

Additional symptoms of Turner syndrome include the following:

- An especially wide neck (webbed neck) and a low or indistinct hairline.
- A broad chest and widely spaced nipples.
- Arms that turn out slightly at the elbow.
- A heart murmur, sometimes associated with narrowing of the aorta(blood vessel exiting the heart).
- A tendency to develop high blood pressure (so this should be checked regularly).
- Minor eye problems that are corrected by glasses.
- Scoliosis (deformity of the spine) occurs in 10 percent of adolescent girls who have Turner syndrome.
- The thyroid gland becomes under-active in about 10 percent of women who have Turner syndrome. Regular blood tests are necessary to detect it early and if necessary treat with thyroid replacement.
- Older or over-weight women with Turner syndrome are slightly more at risk of developing diabetes.
- Osteoporosis can develop because of a lack of estrogen, but this can largely be prevented by taking hormone replacement therapy.

How is Turner syndrome diagnosed?

A diagnosis of Turner syndrome may be suspected when there are a number of typical physical features observed such as webbed neck, a broad chest and widely spaced nipples. Sometimes diagnosis is made at birth because of heart problems, an unusually wide neck or swelling of the hands and feet. The two main clinical features of Turner syndrome are short stature and the lack of the development of the ovaries. Many girls are diagnosed in early childhood when a slow growth rate and other features are identified. Diagnosis sometimes takes place later when puberty does not occur. Turner syndrome may be suspected in pregnancy during an ultrasound test. This can be confirmed by prenatal testing - chorionic villous sampling oramniocentesis - to obtain cells from the unborn baby for chromosomal analysis. If a diagnosis is confirmed prenatally, the baby may be under the care of a specialist pediatrician immediately after birth. Diagnosis is confirmed by a blood test, called a karyotype. This is used to analyze the chromosomal composition of the female. More information about this will be discussed in the section "Is Turner syndrome inherited?"

What is the treatment for Turner syndrome?

During childhood and adolescence, girls may be under the care of apediatric endocrinologist, who is a specialist in childhood conditions of the hormones and metabolism.

Growth hormone injections are beneficial in some individuals with Turner syndrome. Injections often begin in early childhood and may increase final adult height by a few inches.

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Estrogen replacement therapy is usually started at the time of normal puberty, around 12 years to start breast development. Estrogen and progesterone are given a little later to begin a monthly 'period,' which is necessary to keep the womb healthy. Estrogen is also given to prevent osteoporosis.

Babies born with a heart murmur or narrowing of the aorta may need surgery to correct the problem. A heart expert (cardiologist) will assess and follow up any treatment necessary.

Girls who have Turner syndrome are more likely to get middle ear infections. Repeated infections may lead to hearing loss and should be evaluated by the pediatrician. An ear, nose and throat specialist (ENT) may be involved in caring for this health issue.

High blood pressure is quite common in women who have Turner syndrome. In some cases, the elevated blood pressure is due to narrowing of the aorta or a kidney abnormality. However, most of the time, no specific cause for the elevation is identified. Blood pressure should be checked routinely and, if necessary, treated with medication. Women who have Turner syndrome have a slightly higher risk of having an under active thyroid or developing diabetes. This should also be monitored during routine health maintenance visits and treated if necessary.

Regular health checks are very important. Special clinics for the care of girls and women who have Turner syndrome are available in some areas, with access to a variety of specialists. Early preventive care and treatment is very important.

Almost all women are infertile, but pregnancy with donor embryos may be possible.

Having appropriate medical treatment and support allows a woman with Turner syndrome to lead a normal, healthy and happy life.

Super Female/ XXX Syndrome:

A genetic condition where females have an extra X chromosome in each of their cells. Normally female cells have two X chromosomes. This is not usually an inherited condition but a defect that occurs during cell division. Often the condition is asymptomatic.

Incidence-Triple X syndrome occurs in around 1 in 1,000 girls.

Symptoms- Because the vast majority of Triple X females are never diagnosed, it may be very difficult to make generalizations about the effects of this syndrome. The samples that were studied were small and may be non-representative or biased.

Because of the lyonization, inactivation and formation of a Barr body, in all female cells, only one X chromosome is active at any time. Thus, Triple X syndrome most often has only mild effects, or has no unusual effects at all. Symptoms may include tall stature; small head (microcephaly); vertical skinfolds that may cover the inner corners of the eyes (epicanthal folds); delayed development of certain motor skills, speech and language; learning disabilities, such as dyslexia; or weak muscle tone. The symptoms vary from person to person, with some women being more affected than others. There are seldom any observable physical anomalies in Triple X females, other than being taller than average.

Girls with Triple X syndrome are at increased risk of delayed language development, EEG abnormalities, motor-coordination problems and auditory-processing disorders, and scoliosis. They tend to show accelerated growth until puberty. Premature ovarian failure seems to be more prevalent in these women, but most Triple X women seem to have normal fertility. Triple X females are more likely to struggle with personality and psychological problems, and low self-esteem, but these respond well to treatment. Triple X girls are at increased risk of poor academic results at school, and some may need special education. Sometimes, they may suffer from anxiety and be very shy, and this may affect their relations with school peers. They seem to feel much better after leaving school. They benefit very much from a stable home environment. However, much more research is needed to fully understand the effects of this condition.

Triple-X syndrome occurs in women who inherit three X chromosomes—their genotype is XXX or more rarely XXXX or XXXXX. As adults, these "super-females" or "metafemales", as they are sometimes known, generally are an inch or so taller than average with unusually long legs and slender torsos but otherwise appear normal. They usually have normal development of sexual characteristics and are fertile. They may have slight learning difficulties and are usually in the low range of normal intelligence (especially the XXXX and XXXXX individuals). They tend to be emotionally immature for

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their size during childhood. This sometimes results in teachers and other alts labeling them as troublemakers. None of these traits prevent them from being socially accepted as ordinary adult women. Individuals who are genetic mosaics (XX/XXX) have less noticeable symptoms. Triple-X syndromeis less rare than Turner syndrome, but little is known about it. The frequency is approximately 1 in 1,000 female infants and it occurs more commonly when the mother is older.

Male Sex Chromosome Abnormalities

XYY syndrome males inherit an extra Y chromosome their genotype is XYY. As adults, these "super-males" are usually tall (above 6 feet) and generally appear and act normal. However, they produce high levels of testosterone. During adolescence, they often are slender, have severe facial acne, and are poorly coordinated. They are usually fertile and lead ordinary lives as adults. Many, if not most, are unaware that they have a chromosomal abnormality. The frequency of XYY syndrome is not certain due to statistical differences between different studies. It may be as common as 1 in 900 male births to as rare as 1 in 1500 or even 1 in 2,000.

Early studies of XYY syndrome done in European prisons initially led to the erroneous conclusion that these men were genetically predisposed to antisocial, aggressive behavior, below average intelligence, and homosexuality. Contributing to the early view that XYY syndrome men have serious personality disorders was the case of Richard Speck. In 1966, he coldly murdered 8 nurses in a Chicago dormitory. At his trial, his lawyer claimed that he was innocent due to uncontrollable urges caused by his XYY genotype. This novel appeal was akin to claiming insanity or severely diminished mental competence. The jury was not convinced and found him guilty of murder. He was sentenced to life in prison where he eventually died. In fact, Richard Speck did not have an XYY genotype. However, some researchers suggest that the high testosterone levels of XYY men can make them somewhat more prone to violence and that this may cause higher rates of wife beating.

SEX CHROMOSOME ABNORMALITIES

Female Genotype	Syndrome	Male Genotype	Syndrome
XX	Normal	XY	normal
XO	Turner	XXY	Klinefelter

Klinefelter syndrome

Klinefelter syndrome is a disorder that affects only males. Males normally have an X chromosome and a Y chromosome (XY). But males who have Klinefelter syndrome have an extra X chromosome (XXY), giving them a total of 47 instead of the normal 46 chromosomes. People with this disorder develop as males with subtle characteristics that become apparent during puberty. They are often tall and usually don't develop secondary sex characteristics, such as facial hair or underarm and pubic hair. The extra X chromosome primarily affects the testes, which produce sperm and the male hormone testosterone.

Klinefelter syndrome is one of the most common genetic abnormalities. It affects between 1 in 500 and 1 in 1,000 males. The disorder is named for Dr. Harry Klinefelter, who first reported its symptoms in 1942.

How do people get Klinefelter syndrome?

Normally, males inherit one X chromosome from their mother and one Y chromosome from their father. But those who have Klinefelter syndrome inherit an extra X chromosome. Klinefelter syndrome is typically caused by what is called nondisjunction. If a pair of sex chromosomes fails to separate during the formation of an egg (or sperm), this is referred to as nondisjunction. When that egg unites with a normal sperm to form an embryo, that embryo may end up with three copies of the sex chromosomes (XXY) instead of the normal two (XY). The extra chromosome is then copied in every cell of the baby's body.

What are the symptoms of Klinefelter syndrome?

Many people with this disorder have no idea they have it until they hit puberty or try to have children. At puberty, men with this syndrome often develop more breast tissue than normal, have a less muscular body, and grow very little facial or body hair. When men with Klinefelter syndrome try to have children, most discover that they are sterile because they cannot produce sperm. Learning disabilities (not categorized as mental retardation) are also a common problem for them.

Not all males with the condition have the same symptoms or to the same degree. Symptoms depend on how many XXY cells a man has, how much testosterone is in his body, and his age when the condition is diagnosed. The XXY condition can affect three main

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areas of development 1) physical, 2) language, and 3) social.

Physical development

As babies, many XXY males have weak muscles and reduced strength. They may sit up, crawl, and walk later than other infants. After about age four, XXY males tend to be taller and may have less muscle control and coordination than other boys their age. As XXY males enter puberty, they often don't make as much testosterone as other boys. This can lead to a taller, less muscular body, less facial and body hair, and broader hips than other boys. As teens, XXY males may have larger breasts, weaker bones, and a lower energy level than other boys. By adulthood, XXY males look similar to males without the condition, although they are often taller. They are also more likely than other men to have certain health problems, such as autoimmune disorders, breast cancer, vein diseases, osteoporosis, and tooth decay.

XXY males can have normal sex lives, but they usually make little or nosperm. Between 95 percent and 99 percent of XXY males are infertilebecause their bodies don't make a lot of sperm.

Language development

As boys, between 25 percent and 85 percent of XXY males have some kind of language problem, such as learning to talk late, trouble using language to express thoughts and needs, problems reading, and trouble processing what they hear.

As adults, XXY males may have a harder time doing work that involves reading and writing, but most hold jobs and have successful careers.

Social development

As babies, XXY males tend to be quiet and undemanding. As they get older, they are usually quieter, less self-confident, less active, and more helpful and obedient than other boys. As teens, XXY males tend to be quiet and shy. They may struggle in school and sports, meaning they may have more trouble "fitting in" with other kids. However, as adults, XXY males live lives similar to men without the condition; they have friends, families, and normal social relationships.

Clinical diagnosis

Klinefelter syndrome is most often diagnosed in adulthood using a karyotype, an analysis of the patient's chromosomes taken from a blood sample.

Klinefelter syndrome may also be diagnosed during a woman's pregnancy. Doctors can look for the chromosome abnormality in cells taken from the amniotic fluid that surrounds the fetus (amniocentesis), or from the placenta (chorionic villus sampling (CVS)).

How is Klinefelter syndrome treated?

The XXY chromosome pattern can not be changed. But, there are a variety of ways to treat the symptoms of the XXY condition.

- **Educational treatments** - As children, many XXY males qualify for special services to help them in school. Teachers can also help by using certain methods in the classroom, such as breaking bigger tasks into small steps.
- **Therapeutic options** - A variety of therapists, such as physical, speech, occupational, behavioral, mental health, and family therapists, can often help reduce or eliminate some of the symptoms of the XXY condition, such as poor muscle tone, speech or language problems, or low self-confidence.
- **Medical treatments** - Testosterone replacement therapy (TRT) can greatly help XXY males get their testosterone levels into normal range. Having a more normal testosterone level can help develop bigger muscles, deepen the voice, and grow facial and body hair. TRT often starts when a boy reaches puberty. Some XXY males can also benefit from fertility treatment to help them father children.

One of the most important factors for all types of treatment is starting it as early in life as possible.

* * *

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AUTOSOMAL ABERRATIONS

1. DOWN SYNDROME

Article published in 'The journal of Contemporary Anthropology' 2011, titled 'On the Antiquity of Trisomy 21: Moving Towards a Quantitative Diagnosis of Down Syndrome in Historic Material Culture' by John M. Starbuck

Anthropological references from fossils, ancient paintings and other such historical references:

1. **Walker and colleagues** (1991) published an abstract on 7200 year old skeletal remains (SRI-from Santa Rosa Island, CA, which were found in a Native American cemetery. This skeleton is still being studied to prove existance of Down syndrome. If this individual had Down syndrome it would be the oldest recorded and most complete historical skeletal material with this condition to date.
2. Brothwell (1960) published a detailed description of a skull from Leicestershire, England dated to 700-900 A.D, which is probably the most reliable diagnosis of Down syndrome in historical skeletal material due to the level of anatomical detail provided, photographic evidence provided, the caution used by the author during assessment, and the ability of the author to compare this individual to the population from which it is assumed to come from (Berkson 2004). This skull was excavated from the Breedon-on-the-hill burial site – a late Saxon burial ground that may have been associated with a monastery.
3. 'Lady Cockburn and her Children: A cautionary tale'- a painting by Joshua Reynolds (1723-1792)
4. 'The Adoration of the Shepherd' painting by Jacob Jordaens (1593-1678)
5. Andrea Mantegna's (1431-1506) painting titled ' Madonna and Child', which is thought to depict the baby Jesus, had Down syndrome .
6. 'Virgin and Child' painting by Andrea Mantegna (1431-1506)

What is Down syndrome?

Down syndrome (also called Trisomy 21) is a genetic disorder that occurs in approximately **1 of 800 live births**. It is the leading cause of cognitive impairment.

- Down syndrome is associated with mild to moderate learning disabilities, developmental delays, characteristic facial features, and low muscle tone in early infancy. Many individuals with Down syndrome also have heart defects, leukemia, early-onset Alzheimer's disease, gastro-intestinal problems, and other health issues. The symptoms of Down syndrome range from mild to severe.

Life expectancy for individuals with Down syndrome has dramatically increased over the past few decades as medical care and social inclusion have improved. A person with Down syndrome in good health will on average live to age 55 or beyond.

Down syndrome is named after Doctor Langdon Down, who in 1866 first described the syndrome as a disorder. Although Doctor Down made some important observations about Down syndrome, he did not correctly identify what causes the disorder. It wasn't until 1959 that scientists discovered the genetic origin of Down syndrome.

What are the chromosome basics of Down syndrome?

Genes on an extra copy of chromosome 21 are responsible for all characteristics associated with Down syndrome. Normally, each human cell contains 23 pairs of different chromosomes. Each chromosome carries genes, which are needed for proper development and maintenance of our bodies. At conception, an individual inherits 23 chromosomes from the mother (through the egg cell) and 23 chromosomes from the father (through the sperm cell). However, sometimes a person inherits an extra chromosome from one of the parents. In Down syndrome, an individual most often inherits two copies of chromosome 21 from the mother and one chromosome 21 from the father for a total of three chromosomes 21. Because Down syndrome is caused by the inheritance of three chromosomes 21, the disorder is also called trisomy 21. About 95% of individuals with Down syndrome inherit an entire extra chromosome 21.

Approximately 3% to 4% of individuals with Down syndrome do not inherit an entire extra chromosome 21, but just some extra chromosome 21 genes, which are attached to another chromosome (usually

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chromosome 14). This is called a ***translocation***. Most of the time, translocations are random events during conception. In some instances however, a parent is a balanced carrier of a translocation: The parent has exactly two copies of chromosome 21, but some of the genes are distributed to another chromosome. If a baby inherits the chromosome with the extra genes from chromosome 21, then the child will have Down syndrome (two chromosomes 21 plus extra chromosome 21 genes attached to another chromosome).

Characteristic features- Although the severity of Down syndrome ranges from mild to severe, most individuals with Down syndrome have widely recognizable physical characteristics. These include:

- a flattened face and nose, a short neck, a small mouth sometimes with a large, protruding tongue, small ears, upward slanting eyes that may have small skin folds at the inner corner (epicanthal fold);
- white spots (also known as Brushfield spots) may be present on the colored part of the eye (iris);
- the hands are short and broad with short fingers, and with a single crease in the palm;
- poor muscle tone and loose ligaments are also common; and
- development and growth is usually delayed and often average height and developmental milestones are not reached.
- Adolescents with Down syndrome undergo the same hormonal changes during pubertyas typically developing children. Girls with Down syndrome have regular menstrual periods and should receive instructions on hygiene. Although women with Down syndrome are not very fertile, they can become pregnant. Men with Down syndrome have low sperm count, but in some cases have fathered children. Proper education regarding sexual development and contraception is very important

How is Down syndrome treated?

No cure exists for Down syndrome. But physical therapy and/or speech therapy can help people with the disorder develop more normally. Screening for common medical problems associated with the disorder, followed by corrective surgery, can often improve quality of life. Moreover, enriched environments significantly increase their capacity to learn and lead a meaningful life.

2. PATAU SYNDROME

Patau syndrome, Trisomy 13, is the least common of the autosomal trisomies, after Down syndrome (Trisomy 21) and Edwards syndrome (Trisomy 18). The extra copy of chromosome 13 in Patau syndrome causes severe neurological and heart defects which make it difficult for infants to survive. The exact incidence of Patau syndrome is not known, although it appears to affect females more than males, most likely because male fetuses do not survive until birth. Patau syndrome, like Down syndrome, is associated with increased age of the mother. It may affect individuals of all ethnic backgrounds.

Newborns with Patau syndrome share common physical characteristics:

- Extra fingers or toes (polydactyly)
- Deformed feet, known as rocker-bottom feet
- Neurological problems such as small head (microcephaly), failure of the brain to divide into halves during gestation (holoprosencephaly), severe mental deficiency
- Facial defects such as small eyes (microphthalmia), absent or malformed nose, cleft lip and/or cleft palate
- Heart defects (80% of individuals)
- Kidney defects

Diagnosis

The symptoms of Patau syndrome are evident at birth. Patau syndrome may be mistaken for Edwards syndrome, so genetic testing should be done to confirm the diagnosis. Imaging studies such as computed tomography (CT) or magnetic resonance imaging (MRI) should be done to look for brain, heart, and kidney defects. An ultrasound of the heart (echocardiogram) should be done given the high frequency of heart defects associated with Patau syndrome.

Treatment

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Treatment of Patau syndrome focuses on the particular physical problems with which each child is born. Many infants have difficulty surviving the first few days or weeks due to severe neurological problems or complex heart defects. Surgery may be necessary to repair heart defects or cleft lip and cleft palate. Physical, occupational, and speech therapy will help individuals with Patau syndrome reach their full developmental potential.

3. EDWARD'S SYNDROME

Edward's syndrome is a form of rare genetic disorder where a portion of a person's chromosome 18 is duplicated. The majority of people with the syndrome die during the fetal stage; infants who survive experience serious defects and commonly live for short periods of time. Edwards' syndrome is associated with a broad spectrum of abnormalities which consist of greater than one-hundred and thirty discrete defects involving the brain, heart, craniofacial structures, kidneys and stomach.

The syndrome occurs in about one out of every five-thousand births. Edward's syndrome affects more girls than boys - around 80 percent of those affected are female. Women older than the age of thirty have a greater risk of bearing a child with the syndrome, although it may also occur with women younger than thirty. Edward's syndrome was named after Dr. John Edward.

Causes of Edward's Syndrome

Cells in the human body contain twenty-three pairs of chromosomes that are inherited from the person's parents. In human reproductive cells, ovum cells in women and sperm cells in men each have twenty-three individual chromosomes, referred to as, 'XX,' in women and, 'XY,' in men and numbered one through twenty-two. The extra material from chromosome eighteen, obtained after the egg is fertilized, is responsible for causing Edward's syndrome.

Children with Edward's syndrome inherit an incorrect number of chromosomes. Children with the syndrome inherit three (referred to as a, 'trisomy'), instead of two, copies of chromosome eighteen. Ninety-five percent of children with Edward's syndrome have what is referred to as, 'full-trisomies,' while two-percent are due to translocations where only a portion of an extra chromosome is present. Three-percent of children with Edward's syndrome have what is referred to as, 'mosaic tristomies,' where the extra chromosome is there, but not in every one of the child's cells.

Symptoms of Edward's Syndrome

The majority of children born with Edward's syndrome appear both fragile and weak; many are underweight. Their heads are unusually small, while the backs of their heads are prominent. Their ears are low-set and malformed, and their mouths and jaws are small, a condition referred to as, 'micrognathia.' Babies with the syndrome may experience a cleft palate or lip. Their hands are often clenched into fists, with their index finger overlapping their other fingers. Babies with Edward's syndrome can have clubfeet, as well as toes that may be fused or webbed.

Children with the syndrome can experience problems with their lungs and diaphragm, and blood vessels which are malformed. They may present with a number of types of congenital heart disease, to include atrial septic defect, ventricular septal defect, or patent ductus arteriosus. Children with the syndrome might have an inguinal or umbilical hernia, abnormalities of their urogenital system, malformed kidneys, or undescended testicles if they are male.

Diagnosing Edward's Syndrome

A diagnosis of Edward's syndrome may be reached by the physical abnormalities that are characteristic to the syndrome. A physical examination of the infant can show arched-type finger print patterns, for example, while X-rays can show a shortened breast bone. A more definitive diagnosis can be reached through, 'karyotyping,' which involves taking a sample of the infant's blood for examination of their chromosomes. Through use of specific stains and microscopy, identification of specific chromosomes is possible and the presences of an additional chromosome eighteen is shown.

Edward's syndrome is something that may be detected prior to the birth of the child. Potential testing includes maternal serum alpha-fetal protein analysis or screening, amniocentesis, ultrasonography, and

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chorionic villus sampling. A woman who is pregnant with a child who has Edward's syndrome might have an uncommonly large uterus during the pregnancy because of the presence of extra amniotic fluid. An unusually small placenta might be noted during the birth of the child.

Treatment of Edward's Syndrome

Medical science has not found a cure for Edward's syndrome at this time. Babies with the syndrome commonly present with major physical abnormalities and doctors face hard choices in regards to their treatment. Surgery can treat some of the issues related to the syndrome, yet extreme and invasive procedures might not be in the best interests of an infant whose lifespan could be measured in weeks or even days. Treatment today consists of palliative care.

Approximately five to ten-percent of children with Edward's syndrome survive beyond their first year of life, requiring treatments that are appropriate for the various and chronic effects that are associated with the syndrome. Problems related to nervous system abnormalities and muscle tone affect the development of the infant's motor skills, potentially resulting in scoliosis and crossed eyes or, 'esotropia.' Forms of surgical intervention might be limited due to the child's cardiac health.

Infants with Edward's syndrome can experience constipation caused by poor abdominal muscle tone, something that can be a life-long problem. The results can be discomfort, fretfulness, and feeding issues. Special milk formulas, anti-gas medications, laxatives, stool softeners, as well as suppositories are potential treatments a doctor may recommend. Enemas are something that should not be given to a child with Edward's syndrome because they can deplete the child's electrolytes and alter their body fluid composition.

Children with this syndrome exhibit severe developmental delays, although with early intervention through therapy programs and special education they may reach some developmental milestones. They also seem to have an increased risk for the development of a, 'Wilms' tumor,' a form of kidney cancer that affects children for the most part. It is recommended that children with Edward's syndrome have a routine ultrasound of their abdomen.

4. CRY-DU-CHAT SYNDROME

The geneticist Jerome Lejeune identified cri-du-chat syndrome in 1963. He also discovered the genetic abnormality that causes Down syndrome. Cri-du-chat is one of the most common syndromes caused by a chromosomal deletion. It affects between 1 in 20,000 and 1 in 50,000 babies. In 80 percent of the cases, the chromosome carrying the deletion comes from the father's sperm rather than the mother's egg.

Cri-du-chat (cat's cry) syndrome, also known as 5p- (5p minus) syndrome, is a chromosomal condition that results when a piece of chromosome 5 is missing. Infants with this condition often have a high-pitched cry that sounds like that of a cat. The disorder is characterized by intellectual disability and delayed development, small head size (microcephaly), low birth weight, and weak muscle tone (hypotonia) in infancy. Affected individuals also have distinctive facial features, including widely set eyes (hypertelorism), low-set ears, a small jaw, and a rounded face. Some children with cri-du-chat syndrome are born with a heart defect

Incidence: Cri-du-chat syndrome occurs in an estimated 1 in 20,000 to 50,000 newborns. This condition is found in people of all ethnic backgrounds.

Genetic aspects related to it: Cri-du-chat syndrome is caused by a deletion of the end of the short (p) arm of chromosome 5. This chromosomal change is written as 5p-. The size of the deletion varies among affected individuals; studies suggest that larger deletions tend to result in more severe intellectual disability and developmental delay than smaller deletions. The signs and symptoms of cri-du-chat syndrome are probably related to the loss of multiple genes on the short arm of chromosome 5. Researchers believe that the loss of a specific gene, *CTNND2*, is associated with severe intellectual disability in some people with this condition. They are working to determine how the loss of other genes in this region contributes to the characteristic features of cri-du-chat syndrome.

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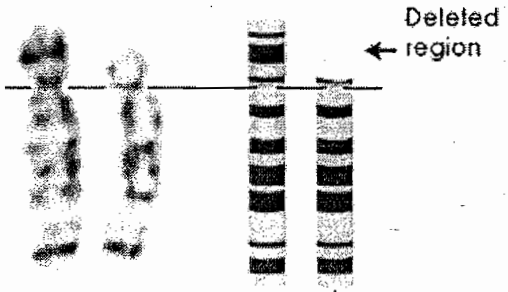
What is Cri-du-Chat syndrome?

The name of this syndrome is French for "cry of the cat," referring to the distinctive cry of children with this disorder. The cry is caused by abnormal larynx development, one of the many symptoms associated with this disorder. It usually becomes less noticeable as the baby gets older, making it difficult for doctors to diagnose cri-du-chat after age two. Cri-du-chat is caused by a deletion (the length of which may vary) on the short arm of chromosome 5. Multiple genes are missing as a result of this deletion, and each may contribute to the symptoms of the disorder. One of the deleted genes known to be involved is TERT (telomerase reverse transcriptase). This gene is important during cell division because it helps to keep the tips of chromosomes (telomeres) in tact.

How do people get Cri-du-Chat syndrome?

A deletion is caused by a break in the DNA molecule that makes up a chromosome. In most cases, the chromosome break occurs while the sperm or egg cell (the male or female gamete) is developing. When this gamete is fertilized, the child will develop cri-du-chat syndrome. The parent, however, does not have the break in any other cells of the body and does not have the syndrome. In fact, the break is usually such a rare event that it is very unlikely to happen again if the parent has another child.

It is possible for a child to inherit a broken chromosome from a parent who also had the disorder.



Cri-du-chat Chromosome 5 pair

Symptoms: Babies with cri-du-chat are usually small at birth, and may have respiratory problems. Often, the larynx doesn't develop correctly, which causes the signature cat-like cry.

People who have cri-du-chat have very distinctive features. They may have a small head (microcephaly), an unusually round face, a small chin, widely set eyes, folds of skin over their eyes, and a small bridge of the nose.

Several problems occur inside the body, as well. A small number of children have heart defects, muscular or skeletal problems, hearing or sight problems, or poor muscle tone. As they grow, people with cri-du-chat usually have difficulty walking and talking correctly. They may have behavior problems (such as hyperactivity or aggression), and severe mental retardation. If no major organ defects or other critical medical conditions exist, life expectancy is normal.

Treatment: Although there is no real treatment for cri-du-chat syndrome, children with the disorder can go through therapy to improve their language skills, motor skills, and to help them develop as normally as possible.

II. GENETIC IMPRINTS IN HUMAN DISEASE

People inherit two copies of their genes—one from their mother and one from their father. Usually both copies of each gene are active, or "turned on," in cells. In some cases, however, only one of the two copies is normally turned on. Which copy is active depends on the parent of origin: some genes are normally active only when they are inherited from a person's father; others are active only when inherited from a person's mother. This phenomenon is known as genomic imprinting.

In genes that undergo genomic imprinting, the parent of origin is often marked, or "stamped," on the gene during the formation of egg and sperm cells. This stamping process, called methylation, is a chemical reaction that attaches small molecules called methyl groups to certain segments of DNA. These molecules identify which copy of a gene was inherited from the mother and which was inherited from the father. The addition and removal of methyl groups can be used to control the activity of genes.

Only a small percentage of all human genes undergo genomic imprinting. Researchers are not yet certain why some genes are imprinted and others are not. They do know that imprinted genes tend to cluster together in the same regions of chromosomes. Two major clusters of imprinted genes have been identified

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in humans, one on the short (p) arm of chromosome 11 (at position 11p15) and another on the long (q) arm of chromosome 15(in the region 15q11 to 15q13).

From the birth of the field of genetics until a decade ago, it was generally assumed that the parental origin of a gene could have no effect on its function. In the vast majority of studies carried out during the last 90 years, this paradigm has appeared to hold true. However, with increasingly sophisticated genetic and embryological investigations in the mouse, important exceptions to this rule have been uncovered over the last decade.

1. The results of nuclear transplantation experiments carried out with single-cell fertilized embryos have demonstrated an absolute requirement for both a maternally-derived and a paternally-derived pronculeus to allow full-term) development
2. In animals that receive both homologs of certain chromosomes or subchromosomal regions from one parent and not the other , dramatic effects on development can be observed including enhanced or retarded growth and outright lethality.
3. Either of two deletions that cover a small region of mouse chromosome 17 can be transmitted normally from a father to his offspring, but these same deletions cause prenatal lethality when they are maternally transmitted .
4. Similar parent-of-origin effects have been observed on the phenotypes expressed by animals that carry a targeted knock-out allele at the *Igf2* locus.
5. Molecular techniques have been used to directly demonstrate the expression of transcripts from one parental allele and not the other at the *Igf2r* locus and the *H19* locus.

The accumulated data indicate that a subset of mouse genes (on the order of 0.2%) will function differently in normal embryos depending on whether they have been inherited through the male or the female gamete, such that one allele will be expressed and the other will be silent. *Genomic imprinting* is the term that has been coined to describe this situation in which the phenotype expressed by a gene varies depending on its parental origin. Further experiments have demonstrated that, in general, the "imprint" is erased and regenerated during gametogenesis so that the function of an imprintable gene is fully determined by the sex of its progenitor alone, and not by earlier ancestors.

With the demonstration of genomic imprinting in the mouse, patterns of disease inheritance in humans have been investigated for the possibility of phenotypes determined by parent of origin in this mammal as well. To date, clear-cut parental effects have been uncovered in the transmission of the juvenile form of Huntington disease, Prader-Willi and Angelman deletion syndromes and certain forms of juvenile familial carcinomas such as multifocal retinoblastoma, Wilm's tumor, embryonal rhabdomyosarcoma and Beckwith-Wiedemann syndrome.

In recent years, there has been an increase in the incidence of various imprinting disorders, especially among children conceived using assisted reproductive technologies, such as *in vitro* fertilization .Because imprinting occurs during gametogenesis, there are growing concerns that various elements of reproductive assistance procedures prevent genes from being imprinted properly or stably transferred during early embryonic development. Understanding the process of imprinting and identifying conditions that interfere with normal imprinting may thus help reduce the incidence of these conditions.

III. GENETIC SCREENING

A **genetic screen** is an experimental technique used to identify and select for individuals who possess a specific genotype.

Genetic Screening usually takes place when an individu al or group shows risk for a disease or trait. Genetic testing can pinpoint a specific allelic interaction or multiple gene interactions, which may lead to a disorder. The common thread of life is DNA and DNA is the only major requirement for genetic sc reening. With knowledge of structure and function of DNA scientists can unlock the mysteries of life.

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Goals of Genetic Screening:

1. medical management or treatment.
2. provision of reproductive information.
3. genetic screening for research.
4. carrier screening, which involves identifying unaffected individuals who carry one copy of a gene for a disease that requires two copies for the disease to be expressed
5. preimplantation genetic diagnosis (see the side bar, Screening Embryos for Disease)
6. prenatal diagnostic testing
7. newborn screening
8. Genealogical DNA test (for genetic genealogy purposes)
9. presymptomatic testing for predicting adult-onset disorders such as Huntington's disease
10. presymptomatic testing for estimating the risk of developing adult-onset cancers and Alzheimer's disease
11. confirmational diagnosis of a symptomatic individual
12. forensic/identity testing

RACIAL CRITERIA

The modern science and technology endeavors to define and classify the various physical types, with reference to their distinctive characters, to determine the elements of which one or the other of the great families of mankind. Any classification of the human species must be based upon observable or measurable bodily characteristics of presumable inheritable type. For ethnological purposes physical characters may be said of two kinds.

1. The superficial or external or outer characteristics, such as skin colour, colour and texture of hair, colour and shape of eyes etc.
2. The internal skeletal or structural features such as texture, Cephalic Index, Nasal Index, facial angle etc.

A. Colour of Skin. Colour is the most striking superficial characteristics of human divisions. Three colors are most universally accepted by anthropologists and ethnologists. Cuvier divided the human groups into three races according to colour of skin.

1. Caucasicus(White colour), or Leucoderms
2. Mongoloid(Yellow colour) or canthoderms
3. Negro (Black colour) or Melanoderms.

But this is a very defective method of judging because of the overlapping of many shades among the races and because of skin colour being evidently adaptive in relation to climate. For examples the white skinned people even within Europe vary from pinkish white among north Europeans to light brown among the south Europeans.

B. Colour and Texture of Hair. Hair may be straight, wooly, red or light brown and wavy. Haddon, in 1924, has made hair the chief basis of classification.

1. Ulotrichi - Frizzly hair, or wooly hair as among most Bushman Negroes and Melanesians.
2. Cynotrichi, or wavy hair as among the Caucasians race.
3. Leiotrichi or straight hair as among most of the yellow skinned peoples of Asia and among the Eskimo.

C. Colour and Shapes of Eyes. The great majority of people have dark-brown eyes due to pigment scattered through the iris. The Pigmentation of the iris varies greatly among peoples of the world. DrBeddoe has made eye colour the chief basis for classification and has combined this with hair colour and skin colour.

Some shade of brown or black is the most common colour, but among the North Europeans blue eyes are prevalent. Another eye distinction is in the fissure or opening of eyes. The fissure is horizontal and normally wide among Europeans but quite narrow among North Asiatic in Southern Europe. The fissure is unique among the yellow skinned Asiatic, the outer angle being higher than the inner, while a fold of a skin, the epicanthic fold covers the inner angle of the eye. This gives the effect known as the 'Mongolian Eye'

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D. **Blood Groups**. Throughout the world, people can be divided into four types according to the kind of substances, their red blood cells contain A, B, AB and 0. Every person can be placed in one of four group. We find that genes responsible for these groups are present through out the world, although the proportions of these different genes differ somewhat from region and from race to race

A new approach to the study of human population is provided by the geographical distribution of the Blood groups. It was discovered by Karl Land Stenier in 1900, that there were definite substances of the serum of some bloods that would agglutinate of clump the cells of certain other bloods. It was further shown that the men can be classified according to particular properties of their blood into four clear cut groups which were represented by the Roman figures I, II, III, IV but these groups are now commonly called 0, A, B, AB, respectively. The blood group to which a person belongs can easily be determined by a competent investigator and it remains the same from birth to death regardless of the condition in which the individual lives. It might seem that mankind could be divided into four races, corresponding to the blood types. But this is not possible at all because, parents and children as well as brothers and sister often have bloods of different types.

E. **Stature** - Topinard's classification of stature, which is generally accepted, comprises four groups.

(i) Tall stature 170 cm. and over.
(ii) Above the average 165 cm. and under 170 cm.
(iii) Below the average 160 cm. and under 165 cm.
(vi) Small stature under 160 cm.

Stature is the largest bodily measurement and represents the sum of the measurements of all the body segments. It is thus no matter for surprise that it varies more widely than almost any other measurement and is readily subject to modification by environmental conditions. Those people who are termed pygmies are not over 148 cm high. Groups classified as short are between 148 cm and 158 cm. Those termed medium stature are between 158 cm and 167.6 cm, tall statured between 170 to 172 cm and those very tall are over 172 cm. Stature, cannot however, be taken as a general criteria of race although in certain cases, it is a good test. For instance, to distinguish the tall Patagonian from the stunted Bushmen.

F. **Nasal Index** - Nasal index is one of the best tests for racial distinction. The nasal index is simply the relation of the breadth of the nose to its height.

This means of classifying races is widely used and claimed by some to be the surest method of distinguishing racial types. The form of the nose is expressed as an index, known as nasal index, which involves two factors:

The **width** (opening in the bony structure between the eyes) and the **length** of the nose (as measured from the root of the anterior) nasal spine. Thus the nasal index expressed as a percentage is again B x 100/L and terms applied as follows:

a) If the person has the nasal index between 55 and 70, he is known as Lepterhinian or long nosed.

b) If between 71-85 as Mesorrhinian or medium nosed and

c) if between 86-100 as Platyrrhinian or flat nosed.

At any rate, this index has proved to be of special use in areas in which the population is of mixed origin. For instance in India it has been claimed that nasal index is an adequate guide to social status.

G. The **Head Index** - The form of the head is ascertained by measuring in a horizontal plane the greatest length from a definite point on the forehead(the glabella) to the back of the head, and the greatest breadth a little above the ears. The proportion of the breadth to the length is a little above the ears. The proportion of the breadth to the length is than expressed as a percentage called the Cephalic Index. The cephalic index is an expression of the breadth of the head or skull in terms of percentage of the length.

Formula for calculation of C.I.

Cephalic index= Maximum breadth/Max.length + 100.

Head shape has been arbitrarily classified, on the basis of cephalic index, into three categories as follows:

	Dry skull	Living head
Delicocephalic or long headed	74.99	76.99
Mesocephalic or mid headed	75-79.99	77-81.99
Brachycephalic or round headed	80 over	82 over

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The difference of the index taken on the living and on the skull is about two units. The head form of skulls is called cranial index. Hence the cephalic index can be converted into the cranial index by the subtraction and vice versa.

H. **Shape of the face and Jaw** - The face usually harmonizes with the form of the head, so that a narrow face accompanies a narrow head. There are notable exceptions, however, for broad faces with long heads occur among sonic Eskimos: Projection of the jaw is found to be an important criterion of the racial classification. The facial angle indicates the angle made by the whole face with brain cap i.e. the angle between the horizontal line and the straight-line joining the centre of the chin and the fore head. If the facial angle is 90 it is called orthognathism. If it is less than 83, prognathism. If it is between 83 90, it is mesognathism. Most of the white races are orthognathists. Black races are prognathists, and yellow races are mesognathists.

Dr Tailor has propounded the theory of classification of human race on the basis of migration. It is known as Migration Theory of Race Evolution. His theory asserts that whole of the human race was born in central Asia and because of climatic changes, physical changes developed and became diversified. According to him Negrito race become first to be evolved and races, which were evolved, afterwards, forced them to go out of the periphery and this process continued. According to this Migration Zone Theory first evolved races are centered towards the poles and the races, which became evolved in the last, are located in the centre. According to Tailor human race may be divided into seven groups.

ANTHROPOLOGY AND RACE 1: THE RISE AND PERSISTENCE OF SCIENTIFIC RACISM

There is little question that we live in a racist society. Economic indicators alone reveal the price that people pay because they have a heavier skin pigmentation than others. The poverty rate among Blacks and Hispanics in the United States is three times that of whites. African Americans account for 10.7 percent of the workforce, but only 6.9 percent of executive or managerial positions. Hispanics account for 9.2 percent of the labor force, but only 4.8 percent of executive positions. You can get addition information on economic inequality in America in the report by Woodrow Ginsburg, Income and Inequality. Yet while racism is obviously pervasive, people often are unaware of the racist attitudes that they hold. One of the questions asked by anthropologists is why do racist attitudes persist? Are people naturally inclined to denigrate people of other groups? Does it have to do with historical attitudes that were used to justify slavery that have persisted? Or is there something about specific cultures that encourage racism?

Most anthropologists recognize that race is a social concept, not a biological one. That is, it stigmatizes some individuals as different and reinforces the privileges of others. There is no evidence that there are large groups of biologically distinct human beings (i.e. subspecies) that correspond to what people refer to when they talk about "race." Furthermore, to base any kind of biological category on a single physical characteristic, such as skin color (which, in itself is incredibly varied and determined by multiple genes), is clearly nonsense. Yet the concept of race persists in our popular culture, and is occasionally given legitimacy by researchers from fields as varied as psychology and political science.

In the nineteenth century scholars such as Samuel George Morton attempted to prove that some "races" were superior to others by measuring the cranial capacity (brain size) of skulls representing different groups (e.g. "Blacks," "American Indians," "Whites"); his concluded, based on his measurements, that Whites were superior to other groups.

Almost a century later, Harvard biologist Stephen Jay Gould described in his book, The Mismeasure of Man, how he remeasured the same skulls used by Morton. He discovered that Morton must have attempted to confirm to himself what he and other Americans already "knew": that whites were superior to blacks and to Indians. He did this by accepting data that confirmed his biases, and rejecting data that did not support them. For example, he included in his sample of black skulls more females than he included in the white sample. In his American Indian sample, he included more small-brained Inca skulls than large-brain Iroquois skulls. In addition, he omitted small-sized Hindu skulls from his white sample. In fact when Gould remeasured the skulls, he found that the average size of black male skulls was larger than that of white males. He know now that skull size or cranial capacity tells us nothing about intelligence, but in the nineteenth century such "scientific" efforts were used by people to legitimate their racial biases.

Most anthropologists no longer take the idea of race seriously. Human populations do differ in some respects in their genetic makeup (e.g. blood types), but there is little use in trying to lump groups into racial groupings

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based on often, physically meaningless characteristics (e.g. skin pigmentation). As such, the efforts of people such as Samuel George Morton to "prove" the superiority of one "race" over another, might be dismissed as historical oddities, except for one thing. There are still people who believe that they can scientifically prove that some groups of people are genetically superior to other groups. Much of this effort derives from eugenics, the attempt to distinguish between people with "good" and "bad" genes, and to foster social legislation that would encourage those with good genes to have more children than those with bad genes. You can find a wealth of information about the eugenics at the Image Archive on the American Eugenics Movement. Check out particularly the link on the social origins of eugenics. As the authors of the site note, eugenics, which was behind much of the Nazi effort to "eliminate" undesirable peoples such as Jews, Gypsies, Catholics, and Gays, has been thoroughly discredited. Yet it continually resurfaces in other guises, and may yet be fully revived with the discovery of techniques to modify the human genome.

Two of the most recent attempts to revive eugenic ideas about racial superiority are The *Bell Curve: Intelligence and Class Structure in American Life* by Richard J. Herrnstein and Charles Murray and the various writings of psychologist J. Philippe Rushton particularly his *Race,* . One can get both positive and negative views of The Bell Curve at Two Views of the Bell Curve and at Upstream. One of the more distressing facts about works such as The Bell Curve is the attention and legitimacy they are given by the media. Yet rarely do media commentators fully understand the fallacies that lie behind the basic building blocks of scientific racism (e.g. "intelligence," IQ, "race," etc), and manage to simply reinforce social biases that privilege some people and penalize others.

One of the best sites on the Web to get information about scientific racism is The History of Race in Science. Particularly valuable is the link page that will take you to some good examples of so-called "scientific racism." One of the more controversial theorists is psychologist J. Phillipe Rushton, whose work manages to reinforce virutally every racial stereotype common in American folk culture from the nineteenth century to the present. You can get a sample of his writings at Upstream.

The big question, of course, is why do people like Rushton continue to attract attention, and why does the racism that they reinforce persist. In our next Editor's Choice column we'll offer some possible explanations.

Ethnicity and Race in Anthropology - Franz Boas, Ethnicity, And Contemporary Physical Anthropology: Continuing Tensions, Cultural Fundamentalism And Instrumental Ethnicities

Ethnicity, as defined in the public domain, is "the cultural characteristics that connect a particular group or groups of people to each other". Twenty-first-century anthropologists, however, are likely to complicate simple notions of ethnicity, or they might refuse to accept a general definition of the concept without first demanding accounts of the particular formation of an ethnic identity in a unique place and time. Citizens of the United States, for example, are enculturated to associate ethnicity with a mapping of cultural or national origin to language, religious practices, styles of adornment, types of non (or un-) American foods, and oftentimes, physical looks. Anthropologists, however, are much more interested in ethnicity as a historically and politically situated set of identity practices, rather than as a state of natural or expected correspondence between physicality, behaviors, and attitudes. Such a nuanced view of ethnicity has not always been the norm in anthropology or in social science. A central story of ethnicity in anthropology is its labored disentanglement from now discredited biological and evolutionary notions of "race," ideas that continue to contribute to the general public's conceptualization of the "ethnic" as a physically distinct type of person.

In the mid-nineteenth century the terms *ethnic* and *racial* first came into common use, employed by pre-and post-Darwinian scientists, and later anthropologists, to construct human racial and cultural taxonomies. As the social corollary of race, "ethnic" (*ethnicity* as a unique term does not emerge in the United States until the 1950s) initially served to reinscribe physical notions of racial, and in some cases national, identity onto groups of people often naively assumed to have a shared cultural, historical, or even evolutionary past.

Throughout the nineteenth and into the twentieth century, race, was the dominant concept for the scientific, social, and political classification of human groups in the Western world. Though scientific and popular ideas of what makes a race, who embodies race, and what being of a certain race means have changed drastically through time, the eighteenth-century Enlightenment ushered in a very specific scientific classification of human beings, particularly the ordering of non-European and colonized peoples. Before the scientific community accepted the theory of evolution by natural selection in

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the nineteenth century, human races were thought to be the product of divine creation. Following the Adam and Eve story closely, monogenists believed in a single creation from which all people and consequently all races arose. Polygenists, though, believed that there was a separate creation for each major race. Furthermore, proponents of both of these pre-evolutionary models used continental labels for creating physical races as bounded groups. Anthropologists assumed that all Europeans, Asians, Africans, and Native Americans were naturally distinct from each other—even on the level of the species—through differentiation over time (monogeny) or through natural, created difference (polygeny). Cultural-historical notions of identity as essentially unchanging and the delineation of fixed anatomical categories also served to determine a certain people's "racial type."

The adoption of an evolutionary framework for understanding human difference, however, did little to transform early anthropologists' fixed ideas of race and racial connections to ethnic identity. Instead, scientists quantified human variation through morphology, mostly using erroneous comparisons of skull size and shape. These practices became the basis for the development of methodology in physical (or biological) anthropology in the United States and Europe. Based on misinterpreted and even falsified anthropometric data, physical anthropologists wrongly assigned labels of either "less evolved" or "degenerated" to nonwhite peoples. Evolutionary ideas initially applied to global racial taxonomies were also used to classify people on other physical, behavioral, or cultural scales. Anthropologists also considered white women, European and European-American people of lower socioeconomic classes, convicted criminals, disabled people, and people who practiced homosexual or other seemingly scandalous sexual practices (such as adulterers and prostitutes) lower on natural scales of evolution and development. Similarly, nonwhite people were wrongly thought to be more naturally susceptible to various behavioral and cultural vices, damaging stereotypes that persist among racists and cultural isolationists in the United States in the early twenty-first century.

Likewise, until ethnicity's emergence in the 1960s and 1970s as a term describing more fluid social processes of identity formation, social scientists used *ethnic* to describe a natural or fixed category of person. In the United States, especially during the intense debate over Eastern European and Mediterranean immigration in the early portion of the twentieth century, an "ethnic" was a person of a marked, lower, category, opposed to a bourgeois white identity. Ethnics, in this context, became lower-class whites, Jews, and nonwhite, colonized, or indigenous peoples.

Ideas of ethnicity have, therefore, consistently been relational; that is, one's ethnic identity, either physical or cultural, is defined through both assumed similarities within a group and assumed differences between groups. Franz Boas's 1912 study of cranial plasticity in American immigrants, however, was the first and arguably most influential anthropometric contestation of the fixed physical nature of ethnicity and racial identity. Among some anthropologists, though, fear of the negative or racially degrading influence of a large ethnic presence in America was used to argue against Boas'sconclusions, and for increasingly strict immigration policies and the establishment of eugenics programs across the nation.

Moreover, after the devastating results of such "Aryan" hysteria in Europe during World War II, greater anthropological use of the term *ethnic group* coincided with a general repudiation of biological determinism and racism within the social sciences as a whole, as well as with a particularly anthropological recognition of the emergence of anticolonial nationalism outside the First World. Specifically, anthropologists replaced *tribe* or *tribalism* with *ethnic group,* especially when describing African migrants to colonial urban centers. Still, all these early uses of the term *ethnic* in anthropology imply a bounded set of cultural traits, historical commonalities, or mental similarities between people of the same ethnic group, even if they later became delinked from physical or racial characteristics.

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GENETIC MARKERS

A trait can be used as a genetic marker in studies of cell lines, individuals, families and populations if :

(a) it is genetically determined

(b) can be accurately classified

(c) has a simple unequivocal pattern of inheritance

(d) has heritable variations common enough to allow to be classified as genetic polymorphism.

1. Hp – Haptoglobin

It is a protein in blood serum which combines with haemoglobin(Hb) in plasma and helps in transferring free haemoglobin to the liver where haemoglobin is broken down for the production of bile-salt(biliverdin). Hb-Hp complex can be detected by electroporesis.

Hp occurs in 3 types – Hp 1-1, Hp 1-2 and Hp 2-2 in which 2 genes – Hp1 and Hp2 are involved. Hp 1-1 and Hp 2-2 are homozygous while Hp 1-2 is heterozygous for Hp1 and Hp2. The two genes are codominant autosomal alleles.

In some tropical populations many individuals have no detectable Hp. This is usually a transient state due to haemolysis from malaria and other causes. The cases of haptoglobinaemia (due to Hp-null gene HpO) is relatively common in American blacks.

Good number of studies are conducted on Hp1 while the other gene Hp2 is being explored. High frequencies of Hp1 are found in many tropical African peoples, in central and south America, New Guinea and Polynesia.

Low frequencies of Hp1 are found in the regions from Middle East to India and SEAsia including Australia.

Different variants of Hp are studied by Anthropologists. HpáLs occurs in much higher frequency than Hpá1F in Europe. Hpá1F is same among Chinese, Japanese, Eskimos, Australian Aborigins while it is found in highest frequency in Nigerians. In Africa Hp1 occurs in highest frequency. It is found to be related to high occurance of malaria. But it is also found that among many Asian countries, even if Hp1 is low in frequency there is high occurance of malaria. These controversies are still being explored by Anthropologists.

2. Transferrin (Tf)

It is protein in blood serum, a â-globulin that transports iron from sites of red-cell destruction and from the intestine to the bone-marrow where Hb(Haemoglobin) is synthesized. Each such molecule can bind two atoms of iron.

About 20 inherited variants have been distinguished by differences of electrophonetic mobility (These variations probably depend on a series of alleles) Of these the commonest is Tf c that occurs in a single band while heterozygotes for one of the slow (D) variants or one of the fast (B) variants have an additional band. Transferrin bands can be identified specifically by adding radio active iron(59 Fe) to the serum and conducting electrophoresis later.

A slow variant Tf D is common in Africal populations, Australia, New Guinea and adjacent islands (1-5% gene frequency). In some Australian aboriginal groups the gene frequency is quite high (20%). Another variant Tf Dchi with an electrophoretic mobility similar to Tf D1 was 1st described in the Chinese and is found in S-E and East Asia in Veddas of Ceylon and a no. of American Indians.

Mostly the fast variants (B) are infrequent but Navajo Indians of S-W USA were found to have 8% of Tf B0-1. Experiments have so far failed to reveal any difference in iron-transporting proportions of Tf variants though different bacteriostatic properties have been claimed.

3. HLA (Histo compatibility system)

HLA is a region on chromosome 6. It is responsible for graft-rejection or acceptance in case of skin and other tissues grated from an individual to another. Inherited differences in HLA system of the tissues are being intensively studied with a view to improving graft-tolerance by matching the antigens of donor and the recipient.

The antigens are detected by a cytotoxicity test. The HL-A antigens are determined by 5 closely linked loci A,B,C,D and Dr located on the chromosome 6. HLAA,B,C are found on the surfaces of most of cells (except RBC, sperm and placental trophoblast). Dr has a much restricted distribution. For eg : it occurs on â-lymphocytes and not on T-lymphocytes.

A number of diseases are found to be strongly associated with HLA antigens. Eg : Multiple scleroris ankylosing spondylitis, haemochromatosis etc.,

There is a high variation with respect to occurance of genes related to HLA. In each category of HLA A,B & C there exist a large no. of Alleles.

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4. Pulse Rate / B.P level

Pulse rate refers to the rate at which heart pumps out and receives blood. It can be indicated in B.P levels. The pressure at which blood leaves heart is called SBP (Systolic) and estimates the contractility of heart. The pressure at which heart receives blood is called DBP (Diastolic) and measures expansion of heart. They are indicated in fraction SBP/DBP.

Increase in SBP –> Increase contraction
Increase in DBP –> Increase expansion } dangerous

Though 120/80 is considered normal, 115/70 is also seen as a normal range. Kaplan (1978) categorized populations into 3 groups based on variation of B.P.

1.	Hypertensive	SBP	> 160
		DBP	> 95
2.	Borderline	SBP	140-160
		DBP	90-95
3.	Normotensives	SBP	< 140
		DBP	< 190

V. Rami Reddy (1995) conducted the survey in rural-urban areas to identify composition of these categories in Andhra Pradesh for SBP & DBP respectively. He identified the proportion to be 96:2:2 & 97:0.3:2.7. He identified that in industrial societies the proportion of Normotensive to hypertensive is not so high, due to nutritional and life style reasons, absence of green vegetables and fruits, more of fat makes it happen.

Reddy, Celine, Mathew etc opined that BP depends on age, sex, heredity, body composition and social status.The variations with respect to age are very glaring.

Age	B.P.
Neonates (1st day)	70/50
Few months	90/60
Adult on wards	115/70

It is identified that 16% B.P. variation is due to environmental factors 48% due to additive genetic factors and 36% due to dominance. The environment at home, peaceful and stress free conditions also help in maintaining BP at normal levels.

These factors have been tested even outside India. Boyce Attenborough, Harrison, Thornbrook and Sinneff proved it in their studies in New Guinea.

5. Hb_s level

Hemoglobin present in blood is made up of Haem group (Fe) and four protein chains. Its level depends on various factors – nutrition, activity level, sex, age and ethnic group. The Hb level in RBC never rises above its metabolic limit of 34 gm/dl.

Hb level / Gender	Avg	Indians (Allan & Gridwood Study)
Adult males	16 gm/dl	12 gm/dl
Adult females	14 gm/dl	10 gm/dl

This clearly indicates that Indian population runs a heavy risk of being anemic Verma's study (1976) among Andamanese, indicates that 30% of Indian males and females are anaemic. Chattopadhya's (1992) study indicates 50% females being anaemic with Hb level less than 8.4 gm/dl. No cases of anaemic were reported among adult males. Among childern 57 % males and 62.5% females were anaemic Pandey's study among Oraon indicated anemia of 60.83% and 74.25% among males and females.

The higher nutritional status among developed countries clearly demonstrates in Hb levels and the peak Hb level attained at an early age. Pearson & Owen found among Quebec that, males attain the highest level of 15.3 gm/dl at 20 and maintain it up to 60. Among females 13.5gm/dl is attained at 10-14.American studies also indicate the same trends. Differences in male-female Hb-levels, particularly in post - puberty phase can be attributed to menstrual losses in females.

In developed countries practices like neglect or discrimination of girl child, less nutrition levels, practice of eating the left over food by mother/wives are the important factors. To address this problem at least partially GOI has introduced programmes like "JANANI SURAKSHA YOJANA"

6. Fat levels

1. In the body fat subcutaneous fat has a significant role to play in heat or cold tolerance. In warm climates it is thin and thick in cold climates. The subcutaneous fat depends on age, sex, heredity & environment.
2. The subcutaneous fat begins to be laid down in the foetus at 34 weeks and increases until the birth and from birth to 9 months when a peak is reached. From 9 months to 6-8 years of age fat

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level shorts a negative velocity as focus will be on growth.

3. Sex wise difference exists in volume of fat & velocity of its formation. The decrease in fat laying is less in females than in males. By 8 years they attain, most of the subcutaneous fat between 8 years and puberty fat level increases at which there is a brief halt.

4. The deposition of subcutaneous fat shows a definite-pattern on the body-subs copular, abdominal region, arms, thighs, buttocks, calf in the limbs are important places.

5. The fat level depends on genetics and environment. The genetic association of fat distribution was proved by Bushman & Hottentots, who have higher fat deposition in thighs and buttocks. Most often different ethnic groups despite inhabiting similar environmental regions possess difference in fat levels and manner of laying.

6. Fat levels also show correlation with B.P levels. Blair reported that upper body fat is more closely associated with high B.P levels than lower body fat. This influences coronary risk factors irrespective of age & sex.

7. Abraham Tylor proved that over weight and high fat in adulthood are directly linked to hypertension cardio vascular, renal diseases. Innumerate studies to prove these aspects are present. There is clear association of life style and body fat which is evident in blooming obesity especially in children in countries like USA.

ECOLOGICAL ANTHROPOLOGY

Some societies have proven to be sustainable for centuries or even millennia. Others have degraded their habitat and depleted the natural resources in it to the extent of undermining their economic and social viability thereby leading to sociocultural disintegration and even collapse. Examples of sustainable societies are the highlanders of New Guinea, Inca of the Andes, Menominee in Wisconsin, Norse of Iceland, Pueblo of the south western region of the United States, San of southern Africa, Tikopia in the South Pacific, Tokugawa era Japan, Tonga in the South Pacific, and Yanomami in the Amazon between Brazil and Venezuela. Examples of unsustainable societies that eventually collapsed are the prehistoric Anasazi of the south western region of the United States, central Classic Lowland Maya of Central America, Norse of Greenland, and Rapanui (Easter Island) in the South Pacific. However, there is considerable controversy surrounding some cases, especially the Maya and Rapanui.

Although many scientific and academic disciplines and professions provide information and insights into the important matters of sustainability, human ecology, and adaptation, anthropology is unique because of the breadth and depth of its perspective. Anthropologists can describe the changing niche of humans at the species level over more than four million years of prehistory andevolution. They can also describe the niche of humans at the population level for many of the more than 7,000 distinct cultures known historically.

Anthropology is the holistic study of human diversity and unity throughout time and space encompassing all aspects of being human, but with particular attention to culture. Culture is the system of socially patterned, shared, and learned ideas and behavior as well as their material manifestations that together distinguish a particular society or ethnic group. There is tremendous diversity in the ways that different cultures formulate and answer these four pivotal questions: What is nature? What is human nature? What is the place of humans in nature? What should be the place of humans in nature? Cultural diversity has been the key to the adaptation and adaptability of the human species. This has facilitated the global dispersal of Homo sapiens into a multitude of very different kinds of environments (biomes) from the tropics to the Arctic. Indeed, culture is more important than biology (morphology and physiology) in the adaptation of circumpolar peoples such as the Aleut, Chuckchi, Inuit, and Saami to their harsh environments. Each biome presents a different combination of limitations and possibilities for relating society to nature. Likewise, each culture also has a different combination of limitations and possibilities for relating their society to nature. Thus, culture can be a decisive factor influencing what individuals and groups identify as environmental resources, hazards, or risks, and how they deal with these and related phenomena. Given such considerations, ecological anthropology concentrates on how culture mediates the dynamic interactions between human populations and the ecosystems in their habitats. Accordingly ecological anthropologists have variously

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addressed each of the main categories of natural resources, including water, soils, plants, animals, minerals, andenergy, but each with special attention to how a particular culture influences daily decisions, choices, and activities in exploiting them. The primary approaches within contemporary ecological anthropology are cultural ecology, historical ecology, political ecology, and spiritual ecology. Building on these approaches is the applied dimension of ecological anthropology called environmental anthropology.

Julian Steward, far more than any other single anthropologist, is responsible for the development of cultural ecology. During the 1920s-30s, Steward conducted pioneering field research on the interaction of a particular human society and its natural environment in the Western United States working with Shoshone, Paiute, and other Native Americans. During the 1950s-60s, he developed formally his generic theoretical and methodological framework for cultural ecology. Steward recognized that there were multiple pathways for adapting to the same biome over time, a process he called multilinear evolution. The investigation of multilinear evolution was based on a comparison of the cultural ecology of societies at the same level of sociocultural integration (bands, tribes, chiefdoms, states) in the same biome (e.g., deserts, savannas, or tropical rainforests). In turn, this was based on detailed ethnographic investigations of the cultural ecology of several societies concentrating on identifying the natural resources that they depend on for their subsistence, the technology and organization of labor used to extract and process the resources, and how these factors in turn influence other aspects of culture. Steward's basic cultural ecology persists as an underlying general theme in subsequent anthropological research on human-environment interactions many decades later, even though new components and emphases have been added.

During the 1950s-1960s, Brent Berlin, Harold Conklin, Charles Frake, and others pioneered in the development of ethnoecologywhich may be considered as a research specialization within cultural ecology. Ethnoecological research concentrates on how different cultures view aspects of their environment, especially through identifying their classification of particular domains of nature such as soils, wild plants, birds, or insects. These classifications are pursued through vocabulary lists from the language of the culture in question. Accordingly, ethnoecology operates at the intersection of linguistics and cultural anthropology. Ethnoecology and cultural ecology have helped to demonstrate the profoundly detailed and reliable traditional environmental knowledge that indigenous and other societies have accumulated over centuries or even millennia of adaptation to the ecosystems in their habitat. For instance, William Balee has shown that the Ka'apor of the Amazon in Brazil recognize 768 species of plants from the seed to the reproductive adult stages which they utilize for food, medicines, crafts, building, and other purposes. Ethnoecology provides a most useful entry into understanding human ecology from the perspective of the people in a local community as they interact with the ecosystems in their habitat. However, researchers need to pay more attention to the relationships between what people say and what they actually do in their environment in order to transcend the limitations of purely linguistic inquiries and become more relevant to ecology.

In the 1960s-80s, cultural ecology was transformed into ecological anthropology by John Bennett, Roy A. Rappaport, Andrew P. Vayda, and others. This was accomplished by applying a systems approach to studying the role of a human population in the processes of energy flow and nutrient cycling within their ecosystem. Methodologically there was also far greater emphasis on the collection of quantitative data than in previous work. For example, Rappaport measured caloric input and output in the gardens of the Tsembaga people in the highlands of New Guinea who employ swidden or shifting horticulture. He related this in turn to their ritual and warfare beliefs and customs as mechanisms for maintaining the balance between the human and pig populations which could easily become competitors for many of the same food resources. The development of this systems ecology framework was further stimulated by the International Biological Program from 1964-1974. The IPB contributed to the subsequent movement to study the human dimensions of global environmental change.

Two additional theoretical and methodological frameworks were developed mainly in the 1980s and 1990s to try to render ecological anthropology more scientific. First, Marvin Harris vigorously pursued explicitly and systematically the development of cultural materialism as a research strategy to reveal and explain the ecological rationale underlying various aspects of culture. He divided the cultural system into three components: infrastructure as the product of the interactions among environment, population, andtechnology; structure as the local domestic and wider political economy; and superstructure as the ideational realm encompassing religion, myth, and the arts. Harris argued that the infrastructure is most basic and most influential because it functions as the ultimate adaptive mechanism for the very survival and maintenance of individuals and society as a whole. He asserted that infrastructure is probably the

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primary cause of much of the rest of the cultural system and accordingly assigned it research priority. Harris demonstrated the explanatory power of cultural materialism through his ingenious analyses of many cultural puzzles such as Aztec ritual sacrifice, the custom of the sacred cow in India, and Islamic and Jewish prohibitions on eating pork. For example, Harris forcefully argued that the cow was sacred in India because it was far more valuable alive for its milk, dung for cooking fuel and fertilizer for farm fields, and plowing than dead for its meat and hide. The Hindu religion reinforced this pragmatic infrastructural causality.

The second innovative framework is human behavioral or evolutionary ecology. Pioneered by Eric Alden Smith and Bruce Winterhalder, it shifts attention to individuals as the locus of adaptation with an emphasis on decision making in the use of natural resources ranked according to their relative costs and benefits (optimal foraging theory). This connects human ecology more directly with natural selection and other evolutionary theory.

Both of these special frameworks, cultural materialism and human behavioral ecology, have been criticized as simplistic andreductionistic. Nevertheless, both have proven to have some validity and utility in advancing the anthropological understanding of human-environment interactions.

In the 1990s, ecological anthropology diversified further by adding research variously focused on historical, political, or spiritual aspects of human ecology and adaptation. William Balee, John Bennett, and Carole Crumley, among others, developed a diachronic approach to examining the interactions between the sociocultural and environmental systems over extended periods of time as they transformed one another within a regional landscape. Previous research had been largely synchronic, examining a particular society as if it were isolated, traditional, static, and timeless, and also as if the society had no lasting cumulative impact on its environment and the latter was static as well. For instance, in the Republic of Guinea in Africa, foresters, botanists, andconservationists interpreted relics of forests near villages as having escaped the deforestation by villagers that created the surrounding savannas. However, James Fairhead and Melissa Leach demonstrated that these forest patches are actually generated and managed by the villagers as a focus for their rituals and other cultural practices. In other words, the forests are anthropogenic rather than purely natural. Other studies in historical ecology pay far more attention than cultural ecology used to do to the impact of external factors on local societies and ecosystems, such as cultural contact and change in the context of colonialism and, more recently, globalization.

The above are among the considerations contributing to the interest of some researchers in concentrating on political ecology; that is, how power differentials and struggles are involved in human-environment interactions. This in turn links with issues of economic, social, and environmental justice. For instance, Susan Stonichanalyzed how the economic development policies of the national government and international agencies impoverished the economy and habitat of farmers in Honduras. The interconnections between the abuses of human rights and the environment have been explored by Barbara Rose Johnston. Most recently she documented the radioactive contamination of the environment of various societies during the Cold War, such as the impact of U.S atomic bomb testing in the Marshall Islands of the South Pacific. Robert Hitchcock and other anthropologists are among those who have collaborated with the San hunter-gatherers of southern Africa to help them pursue community-based resource management to retain their subsistence economy and other aspects of their cultural identity in the face of external forces impinging on their traditional lands and resources including farmers, herders, miners, tourists, and even wildlife conservationists.

In some situations anthropologists concerned with ecology and the environment are faced with a real dilemma. They are committed to both nature and culture, yet sometimes the two interests can be in competition or even conflict. The most striking case in this regard is the Makah Indian Tribe of the Olympic Peninsula in the state of Washington who clashed with major environmental and animal rights groups when in the mid-1990s they sought to resume their annual hunting of a small number of grey whales as part of their cultural traditions. Yet in such cases, at least ideally, environmental anthropologists might well serve as mediators between the divergent cultural and environmental groups to try to secure a win-win solution since they are knowledgeable and concerned about both interests. Accordingly, more anthropologists need to acquire formal training and skills in the procedures for nonviolent conflict resolution.

Beyond historical and political factors, religion and spirituality can also be powerful influences in human-environmental interactions and in dealing with environmental problems and issues. This is revealed in the most recent approach to ecological anthropology that is sometimes called spiritual ecology. Religion is an ancient cross-cultural

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universal; no society is known that does not have one or more religions, although individuals within a society vary in the degree of their religiosity and spirituality if any. Spirituality is an integral component of religion, but also extends beyond organizations to personal relationships with other people, nature, and the supernatural. Those who pursue spiritual ecology as a scientific and academic endeavor, a personal path of spirituality, and/orenvironmental activism, share the conviction that the worsening environmental crisis can only be resolved by a fundamental change in the way humans relate to nature involving sustainable and green environmental worldviews, attitudes, values, behaviors, and institutions. No single religion is considered to be the cause or solution for the ecocrisis; instead; adherents to any religion or spiritual practice need to apply introspection to develop more environmentally friendly orientations and conduct.

Beliefs and emotions are often as important as reason in natural resource use, management, and conservation. For example, Bruce A. Byers and colleagues have shown that the Shona in the Zambezi valley of northern Zimbabwe consider some mountains, forests, and rivers to be sacred. Their sacred places and the biodiversity therein are more likely to be conserved in contrast to secular counterparts.

Part of this substantial diversification of approaches within ecological anthropology since the 1990s involves a growing emphasis on applied rather than basic research, although certainly the two are often interdependent. However, with the worsening ecocrisis and other factors, increasingly research has concentrated on identifying and solving practical environmental questions, problems, and issues. This is the arena of environmental anthropology per se. Researchers in this arena still pursue various approaches within ecological anthropology to investigate matters of survival, adaptation, and change with an emphasis on culture, communities, and fieldwork. However, their work concentrates on practical aspects like using the traditional environmental knowledge of indigenous peoples to promote alternative methods of economic development that are sustainable as they increasingly articulate with the broader regional, national, and global economies. As another example, other anthropologists deal with the relationship between local communities and protected areas established by the government and international environmental organizations to conservebiodiversity. Still others document and advocate the human rights of indigenous communities and other ethnic groups in the face of outside encroachment on their traditional territory and resources. Anthropologists also consider how variables such as gender, socioeconomic class, nationality, and identities relate to environmental matters.

Environmental anthropology is expanding further the realm of ecological anthropology to embrace research at scales ranging from the local to the regional, national, and global levels including the consequences of climate change; to collaborate in multidisciplinary research teams incorporating members from the host country; to use high tech resources such as geographical information systems, remote sensing, and satellite data imaging; to consider rights, politics, and policies as they are linked with social, economic, and environmental justice; to pursue nonviolent conflict resolution in areas where different groups compete for scarce or diminishing natural resources; to community challenges and responses with hazards and risk perception, whether natural or anthropogenic; and to environmental organizations, discourses, and identities.

In the process of the development of environmental anthropology as an extension of ecological anthropology that is more practical, applied research has not supplanted basic research nor have pragmatic concerns replaced those of the academic intellectual, instead these are all complementary. Furthermore, historical continuities persist, like the legacy of cultural ecology, this in spite of discontinuities such as the historicizing of cultural ecology through historical ecology as an additional component in the anthropological investigation of human-environment interactions.

By now anthropological interests and activities concerning ecology and the environment have reached maturity with distinctive organizations like the Anthropology and Environment unit of the American Anthropological Association that has a membership approaching a thousand and its own listserv; special periodicals (Human Ecology, Journal of Ecological Anthropology, andEcological and Environmental Anthropology); distinctive textbooks and anthologies; and special training programs at various universities.

Ecological anthropology and environmental anthropology are increasingly contributing research of broader relevance to the local, national, international, and global communities in coping with natural resources, hazards, and other environmental problems and issues. In various ways anthropologists have addressed pivotal environmental issues including the population explosion, natural resource depletion such as soil erosion, unsustainable economic development and consumption levels, habitat destruction likedeforestation, biodiversity loss, environmental mismanagement, pollution, and hazards.

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Furthermore, environmental matters are important politically, as well as ecologically and economically, because many of the conflict zones in the world are also regions of serious environmental problems including Afghanistan, Bangladesh, Burundi,Colombia, Congo, Ethiopia, Haiti, Indonesia, Iraq, Madagascar, Nepal, Pakistan, Philippines, Rwanda, Solomon Islands, Somalia,Sri Lanka, and Sudan. Moreover, climate changes, especially those connected with global warming, are destined to further aggravate many environmental problems as societies are increasingly impacted in all levels and aspects throughout the world.

No doubt ecological and environmental anthropology have developed the foundation, maturity, momentum, and achievements to continue to contribute to our understanding and advancement of human ecology and adaptation from the local to the global levels as long as humanity has a future. However, that future depends on replacing ecocide with ecosanity, and that in turn requires far more attention to the information and insights of the disciplines and professions that contribute to environmental studies including ecological and environmental anthropology. Ecocide is by far the greatest threat to the security of every being on planet Earth. Ultimately, any human population, economy, and society can only be as healthy as its habitat.

EPIDEMIOLOGICAL ANTHROPOLOGY

Both biological and cultural anthropologists work in medical anthropology, which focuses on disease health problems, health-care systems, and theories about illness in ditferent cultures and ethnic groups. Disease problems vary among cultures. Such epidemic diseases as cholera, typhoid, and bubonic plague are associated with dense populations and thus with agriculture and urbanization. Malaria spread as a result of forest clearing and population increase associated with plantcultivation. Certain diseases tend to accompany economic development. Schistosomiasis, or bilhorzias (liver flukes), is spread by snails that inhabit ponds. lakes, and waterways, often ones created by irrigation projects.

The incidence of particular diseases varies between cultures, and different cultures interpret and treat illness differently. All societies have "disease-theory-systems" to identity, classify, and explain illness. According to George Foster and Barbara Anderson (1978), there are three basic theories of causation: Personalistic, naturalistic, and emotionalistic. Personalistic disease theories blame illness on sorcerers, witches, ghosts, or ancestral spirits. Naturalistic disease theories (including scientific medicine) explain illness in impersonal systemic terms. This includes micro organisms and (in some traditional medical systems) unbalanced body fluids. Many Latin cultures classify food, drink, and environmental conditions as "hot" or "cold". People believe their health suffers when they eat or drink hot or cold substances under inappropriate conditions. Thus, one should not drink something cold after a hot bath or eat a pineapple (a 'cold fruit) wheri.one is menstruating (a 'hot" condition).

Emotionalistic disease theories assume that intense emotional experiences cause illness. For example, Latin American women are believed to be susceptible to 'susto', an illness caused by fright. Its symptoms lethargy. vagueness, distraction) are similar to those of 'soul loss,"a diagnosis of similar symptoms made by people in Madagascar. Modern psychoanalysis also focuses on the role of the emotions in physical and psychological wellbeing.

All societies have health – care systems beliefs, customs, and specialists concerned with ensuring health and preventing and curing illness. A society's disease-causation theory is important in regard to treatment. When illness has a oersonalistic cause. shamans'and other magicoreligious specialists may be good curers. They draw on varied techniques which constitute their specialized knowledge and expertise. A shaman may cure soul loss by enticing the lost spirit back to the body. In societies without the possibility of Caesarian section, shamans may aid difficult childbirths by calling on spirits to travel up the vagina and guide the baby out (Levi-Strauss 1967). A shaman may cure a persistent cough by counteracting a curse or removing an evil substance introduced by a sorcerer.

All cultures have health-care professionals. If there is a World's oldest profession" besides hunter and gatherer, it is curer, or shaman. The curer's role has some universal features (Foster and Anderson 1978). Thus curers become specialists through a culturally appropriate process of selection (parental prodding, inheritance, visions, dream instructions) and training (apprentice shamanship, medical school). Eventually, the curer is certified by established practitioners and acquired a professional image. Patients, who believe in the curer's special powers, consult and compensate him or her.

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Non - Western systems (traditional medicine) have certain lessons for Western medicine. Traditional practitioners may be more successful in treating mental illness than psychotherapists are. This may be due to non -Western systems which explain mental illness in terms of causes that are easier to identity and cure. It may be simpler to rid a body of a spirit possessor than to undo all the psychological damage that a Freudian might attribute to an unresolved Oedipus complex.

Another reason why non - Western therapy succeeds is thatthe mentally ill are diagnosed and treated in small, cohesive groups with the full support of their kin. Curing may be an intense community ritual in which the shaman heals by temporarily taking on and then rejecting the patient's illness (Levi - Strauss 1967).

In the impersonal mental institutions of contemporary North America, no preexisting social ties link patients to each other or to doctors and nurses. Mental illness is seen as the patient's lone individual burden. The social con text of treatment is on of isolation and alienation rather than intense participation by a social group in a ritual.

We should not lose sight, ethnocentrically, of the difference between scientific medicine and Western medicine perse (Lieban 1977) Despite adanvances pathology, microbiology, biochemistry, surgery. diagnostic technology, and applications many Western medical procedures have no clear

Justification in logic or tact. Over prescription of tranquilizers and drugs, unnecessary surgery and the impersonality and inequality of the physician-patient relationship are dubious features of our medical system. However there is no doubt that scientific medicine surpasses tribal treatment in many ways Although medicines such as quinine, coca, opium. ephedrine, and rauwolfia were discovered in non- industrial societies, traditional medicines are not as effective against bacteria as antibiotics are. Preventive health care has advanced significance:, during the twentieth century. Today's surgical procedures as much better than tnose of traditional societies.

Still, industrialization has spawned its own special health problems. Modern stressors include noise, air and water pollution, poor ntrition habits, dangerous industrial machinery, impersonal work organization, social isolation, homelessness, and substance abuse. With the conquestof most infectious disease, health problems in industrial nations, other than AIDS. are due more to economic, social, political, and cultural factors than to pathogens (Liaban 1977). In the United States, for example, poverty is linked to many illnesses. including arthritis, heart conditions, back problems, and hearing and vision impairment.

Medical anthropology, which is based on biological. social. and cross-cultural research, has theoretical and applied dimensions. Health programs must pay attention to native theories about the cause and treatment of illness. Once scientific medicine arrives, people tend to continue using traditional procedures while also accepting the new. Shamans may treat certain conditions (such as spirit possession), whereas M.D.s may deal with others. If both scientific and traditional specialists are consulted and thee patient is cured, the native curer may get as much credit as or more credit than the physician.

A more personal. social treatment of illness that emulates the non-Western curer-patient-community relationship could benefit Western systems. Western medicine tends to draw an overly rigid line between biological and psychological causaton Non Western theories usually make no such distinction, accurately recognizing that impaired health often has intertwined physical, emotional, and social causes. The mind body opposition is part of Western folk taxonomy, not of science.

Rather than seeking causes, non-Western practitioners often treat symptoms, and their aim is an immediate cure. Traditional curers have a high rate of success with health problems that our medical establishment classifies as psychosomatic (Foster and Anderson 1978) and dismisses as not needing treatment - despite the feelings of the patient. Non - Western systems tell us that patients should be treated as whole beings, to be treated with whatever combination of procedures may prove beneficial.

Illness and the desire to treat those who are ill are common features of all human societies. The types of diseases found, how illnesses viewed, and how people treat illness, however, vary considerably from society to society. Treatment of illness in Western industrial societies focus on curing specific diseased organs or controlling a specific virus. In the non-western societics greater emphasis is paced on the social and psychological dimensions of ilness,

Medical system means "the culturally based behaviour and belief forms that are in response to the threats posed by disease." Anthropologists study the medical systems of small-scale non western societies as well as those of large-scale industrial societies.

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The terms disease and illness do not refer to the same thing. Disease is a concept referring to a pathological condition of the body. or of some part of it, in which its functioning is disturbed or deranged. Illness is a cultural concept: a condition marked by pronounced deviation from what is considered a normal healthy state. Even though beliefs about disease and patterns of incidence may be viewed as sociological phenomena, diseases themselves are strictly physiological processes. Illness, on the other hand, is a broad concept referring to how people conceive of deviant mental and physical states.

EPIDEMIOLOGY

Epidemiology is the study of the relationships of the various factors that determine the frequency and distribution of diseases in societies. Thus on a world-wide basis the leading categories of disease that people dies from are

1. Cardiovascular
2. Respiratory
3. Infectious and parasitic
4. Forms of cancer.

In industrialized societies, cardiovascular diseases remain the number one killers, but the numbertwo spot is held by cancers, with communicable diseases relatively insignificant, communicable diseases are of much greater significance in non-industrialised countries. Diarrhoea is the leading cause of death among children world-wide (some 5 million a year), followed by a group of six other "Children' diseases (including measles, poliomyelitis, and diphtheria). These diseases re not only far more common among children that among adults, but their incidence of mortality is restricted largely to non industrialised countries.

Medical sociologists interested in epidemiology in complex societies study the effects of age, sex marital status, occupation, social class, and natural environment on disease patterns. By analysing statistical correlations, they seek to discover the causes of the occurrence of diseases. In general, anthropologists have tended to focus on epidemiological characteristics of disease among non-western peoples living in small scale societies. Unlike medical sociologists, anthropologists relied on direct observation and attention to particular cultural patterns in drawing their conclusions rather than on statistical correlation.

One of the best known examples of the contribution of anthropology to epidemiological research is thecase of kuru. a disease of the central nervous system that was discovered in the 1950s among the South Fore, New Guinea. No known treatment will cure or arrest kuru; which usually leads to death within six to twelve months. The disease was found to be-unique to the South Fore, and among them was limited to women, children ard occasionally young men. The disease first appeared during the early part of the twentieth century with its incidence increasing up to the late 1950s and declining thereafter.

Robert and Shirley Glasse found that the disease was linked to the cannibalism which was introduced among South Fore women around 1910. In particular. eating the brains of deceased kinswomen became part of their mourning ceremonies:

The brains were prepared and eaten primarily by women, with small portions occasionally being given to children of either sex. It was found a slow-acting virus was transmitted in the preparation. The decline of kuru was linked to the cessation of this ritual cannibalism in the 1 960s.

ENDEMIC DISEASES

A society's characteristic diseases are closely related to its physical environment, as well as to its cultural practices. Part of a population's adaptation to a particular environment entails biocultural adjustment to the local disease causing features. Especially when the populations and its adaptation are rather stable, relatively constant epidemiological patterns will emerge. One result is the development of endemic diseases: diseases that have a relatively low incidence but are constantly present in a given community.

EPIDEMIC DISEASES

Migration and contact between previously isolated human populations.are the prime causes of epidemics. An epidemic diseases is one that is not commonly present in a community and that is charaterised by high incidence and rapid and extensive diflusion.

Perhaps the most devastating series of epidemics in world history were those brought by the opening of regular contact between the New World and the old after 1492. The long isolation of the New World population made it especially vulnerable to diseases brought from Europe and Africa. Smallpox was one of the leading killers. Endemic to the Old World, it was a steady, predictable killer, responsible for less than 10% of the yearly deaths. But among people with no previous contact, smallpox will infect nearly every individual. Within two years after smallpox

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first appeared in Iceland in 1707, 18000 of the island's 50.000 inhabitants had died of it.

The modern epidemic diseases are skin cancer and AIDS.

Anthropologists are not interested simply in the effects of socio-cultural factors on disease, they are also concerned with its social consequences. Severe epidemics generally have highly noticeable effects on societies. Many authors have argued that the 16th century collapse of the various Indian empires in the New World, such as the Inca and Aztec empires was caused primarily by the ravage of newly introduced epidemic diseases. The more subtle effects of diseases are also significant. Endemic malaria in a region may kill only a small percentage of the population, but it may debilitate a much larger proportion, contributing to a general malaise that adversely affects economic and cultural life.

DISEASES OF DEVELOPMENT

Charles Hughes and John Hunter coined the terms "diseases of development" to refer to "those pathological conditions which are based on the usually unancipated consequences of the implementation of development schemes." Such diseases an be divided into three general categories.

There are those that result from integrating people into large-scale industrial societies: diabetes, obesity, hypertension etc. The people of Nauru found themselves with a dramatic increase in disposable wealth in the 1 960s and early 1 970s as a result of renegotiation of the royalties paid to them for mining phosphate on their isolated South Pacific island. This sudden increase in wealth led to high rise in food consumption, mostly imported which is of high in sugar and salt content and low in nutritious value. In 1983, out of a total population of 5000 Nauruan, about 400 were known to be suffering from diabetes.

The second category includes where development programs have upset existing environmental conditions and introduced or increased bacteria! and parasitic diseases. The slowly debilitating disease bilharziasis caused by a parasitic blood fluke found in certain water snails, has spread as a result of developmental projects. The number of persons suffering from bilhorziasis has grown to perhaps 200 million, and the disease continues to sfread perhaps at a rapid rate. This spread is almost entirely the result of the expansion of irrigated lands made possible by the construction of high dams.

Within three years of completion of the Naser High Dam in Egypt. the incidence olbilharziasis among the children in the areas affected rose from 5 to 25 percent to 55-85 percent. With the completion of dams elsewhere, the disease has spread throughout the world. Today it is found in Africa, the Near East, the Orient, and South America. Greater use of irrigation has also been responsible for increasing the incidence of malaria, since the mosquito that transmits malaria finds the waters of rice paddies and irrigation ditches ideal places to breed.

The third category are diseases resulting from the failure of developmental programs that seek to overcome the conditions that result from unsatisfactory integration of people into large-scale societies. These include the innumerable diseases associated with poverty. undernourishment, and poor sanitary living conditions.

MALNUTRITION

Hundreds of millions of people are undernourished, and trying to understand the causes and find solutions to malnutrition are among the most important problems in the world today. Minorcases of malnutrition may impair peoples ability to perform daily tasks. Malnutrition also resistance to disease. In extreme form malnutrition may lead to permanent brain damage or to starvation and death.

An initial difficulty in studying nutritional problems is determining human nutritional standards. Traditionally, criteria have been based almost exclusively on the perceived needs of "white" Europeans and North Americans. To apply this standard to all peoples may be a mistake, however.

For example, in Papua New Guinea while diet of coastal peoples includes protein derived from fish, many highland people subsist largely on sweet potatoes, a diet of almost carbohydrates. Yet these people are well muscled and able to carry heavy loads over long distances. They do not suffer from protein-related malnutrition. Such studies demonstrate the need for flexible, community specific nutritional standards.

MALNUTRITION AND SUBSISTANCE PATTERNS

Malnutrition is to a large extent a reflection of increasing social scale. It tens to be more a feature of large-scale agricultural and industrial societies than of small scale societies. Hunger was are among the traditional foragers a and their diets were usually sufficient to meet their nutritional needs. The extent to which malnutrition exists today among the contemporary foraging societies is primarily a function of their integration into a world system

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that has destroyed their customary subsistence bases while falling to replace them With new ones that are adequate.

In a study of nutrition among Australian Aborigines living at Edward River settlement of the Cape York Peninsula, John C. Taylor notes that since the Aborigines gave up their traditional nomadic existence in the 1 950s. their diet has been characterized by short coming in most important nutrients. This provides especiafly poor nourishment during pregnancy resulting in the birth of babies of less than optimal weight and impaired resistance to disease. Problems continue after birth and Australian Aboriginal today are noted for high disease and death and death rates.

The most significant finding of Taylor's study is that improvement in economic conditions for the Edward River Aborigines has not resulted in dietary improvements. In fact, just the opposite has occurred. Increases in wages have been diverted toward satisfaction of demands for non-food consumer items rather that foods. But even the Aborigines were not aware of this.

By employing a complex mixed foraging and agricultural adaptation, small scale farming societies also generally able to meet the subsistence needs of their numbers, as long as population pressure or external threats to their subsistence base do not become too great.

For many small scale societies. nutritional problems begin o emerge as they are integrated into large- scale market economies as peasant agriculturists, they find a large percentage of what they produce usurped by nonagriculturalists and what tley received in return for their products is rarely sufficient to meet their requirements, nutritionally or in any othersense. (give Indian examples)

Large scale societies produce a tremendous amount of food, but they have a lot of people to feed. In some countries, such as India, production simply doesn't keep up with population growth. Also the problems is not so much the total amount of food produced or marketed as it is the distribution of what is available. In the united States, a nation with an abundant supply of food, and an overfed and overweight middle and upper class, there are also millions of undernourished people occupying the lower socio-economic rungs of society.

CULTURAL PERCEPTIONS OF FOOD

Malnutrition can also develop because of poor dietary habits. But improving the dietary habits is not a simple thing.

MENTAL ILLNESS

All people recognize illness that are manifested by mental states perceived as abnormal. Mental illness includes disorders caused by physiological conditions; such as certain "birth defects" as well as those caused by individual adaptations to sociocultural conditions. In the later, mental illness is seen as resulting from stress produced by individual & attempts to deal with perceived inadequacies in the environment.

After decades of careful research on mental illness indifferent cultures, it now appears that the major patterns of mental illness known to Western psychiatrists are universal. In addition of universal forms of mental illness that vary in frequency from one society to another, particular environments and cultural traditions occasionally lead to unique manifestations. These are commonly referred to as culture-bound syndromes. One of such disorders is "arcc hysteria, found among a variety of circumpolar peoples. It is a condition marked by periods of silence followed by seizures characterized by and wandering about in the snow. The others are running amok, a frenzied violent behaviour by males in Malaysian society, and 'koro', a fear among Chinese that the penis in the case of males and breast and vulva in the case of females will withdraw into the body.

SOCIAL CHANGE AND STRESS

Rapid social change is a primary source of stress for many people. Migration commonly results in considerable stress. Many Okinawans who migrated to Hawaii during the immediate post - World War II developed high rates of psychoses. Migration and other forms or rapid social change may result in a state of anomie: a disorganization personality and a feeling of alienation. In his famous study of suicide (1951) Durkheim described anomie as a result of the abandonment of community goals and of the social rules for reaching them. For many indigenous peoples around the worlds, rapid social change has meant the destruction of the existing social order and all too often, its replacement with a social order based on impoverishment and exploitabon.

CONCEPTS OF ILLNESS CAUSALITY

1. MIND BODY DUALISM

Western medical traditions have posited a distinction between illnesses originating in the mind and those originating in the body. Other traditions have taken a more holistic approach, viewing health and illness as

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manifestations of the totality of a person's life. For example, most non-western medically systems do not have a separate category for 'mental illness". Mental illness and physiological illness in such systems are placed in the same general category. To the Azande, Sudan, mental illnesses are viewed as illnesses not unlike such diseases as malaria, smallpox, and dysentery and all are seen as ultimately caused by sorcery.

The traditional mind-body split ih.Western medicine increasingly has been called into question by Western medical practitioners. Many of them now recognize that a close interrelationship exists between psychological and physiological illnesses. Thus, on the one hand, it is now seen that many physiological illnesses are related to psychological problems influenced by environmental factors. On the other hand, research has shown that many psychological patterns have biochemical causes. Examples of the latter include some forms of depression, and perhaps schizophrenia.

2. PERSONALISTIC VS. NATURALISTIC EXPLANATIONS

So called "modern" or 'scientibc" beliefs. Nevertheless, there appear to be basic principles common to rnany of these belief systems. Foster and Anderson have distinguished two major systems of disease causality beliefs - personalistic and naturalistic - although this division oversimplifies matters somewhat.

In personalistic system, illness is thought to be purposely caused by a supernatural being (such as ghostor evil spirit) or a human being (such as a witch or sorcerer). According to Foster and Anderson, The sick person literally is victim, the object of aggression or punishment directed specifically against him, for reasons that concern him alone'

Personalistic belief systems predominate among non-western peoples living in small scale societies. One important aspect of such belief system is that they tend to lack a concept of accident. If a man falls from a tree and dies, it is assumed that someone or some being caused his death. This is not to say that people subscribing to such beliefs do not recognise more immediate causes of illness or injury, but they believe that there are other forces at work beyond the more apparent ones. It is not enough to say that a person fell out of a tree or caught a disease. One must then ask who or was responsible for what happened, and the answer is to be found in some malicious or noncorporeal being.

By contrast, belief systems based on naturalistic causes explain illness in impersonal, systemic terms. Such systems commonly posit that good health depends on maintaining an equilibrium within the body. When this balance is upset by natural external forces such as heat orcold or internal forces such as strong emotions, illness results.

Most contemporary naturalistic systems have been derived largely from the medical traditions of ancient classical civilisations, particularly Greece, India, and China. These include those naturalistic systems which are based on the concept of bodily humors, a belief which originated in ancient Greece. There were though to be four humors, or fluids, in the body: blood, phlegm, black bile (or melancholy) and yellow bile (or choler). Each was associated with certain qualities: blood with heat and dryness. For the ancient Greeks, illness resulted when one of these humors was eitherdefjcientor excessive, or if hey should fail to mix properly. Curing was largely a matter of restoring balance.

Today this humoral pathology remains an important concept in many parts of the world.

In the contemporary Latin America variant of humoral pathology, illness is ascribed to excesses of "heat' or cold". Foods are particularly important in maintaining a balance between hot and cold, so people adhering to such a concept of humoral pathology generally classify foods according to their emperature". In one Mexican American community, white beans, garlic and chile peppers are considered toe be very hot". Cucumbers and tomatoes to be "very cold". Illness is commonly attributed to eating an excess off hot or cold foods, and curing often entails eating things that will restore a balance.

Systems of causation behef are rarely so rigid as to ascribe all illness or misfortune to a single cause. Even the Azande recognise that not alt illness is caused by witchcraft. For example, they dont ascribe certain diseases that afflict infants exclusively to witchcraft or magic. There is a very high infant morality rate, and small children are barely considered social beings, their "sojourn into the world" at this stage being very tenuous. Much the same is about scientific medical system. A cold is not simply caught from "germs": such factors as anxiety, exhaustion, exposure to cold air, or living in damp or crowded conditions may also be part of the cause.

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CURING

Cultural beliefs about causality influence how people decide to treat illness. The Aztecs believed that head aches were caused by excessive blood in the head, curing therefore involved the use of medicines that produced violent sneezing and nasal bleeding to remove the excess blood by making an incision with an obsidian point. F one views the cause of an illness as being bacteria or other micro-organisms, the cure will probably include use of antibiotics. If sorcery is to blame, something may have to be done about the sorcerer.

All healing systems seem towork at lest part of the time. Methods and rationales used vary considerably, however, admit is hard to prove just what creates the cure. When an Aboriginal inhabitant of Wellesley Island. Australia comes down with malgri, an illness characterized by drowsiness and abdominal paid, said to be brought on by concern over violation of taboos, the Aboriginal curer kneels and begins massaging sweat from his armpit into the victim's body. Belt of grass or hair is then unraveled and tied to the person's foot. The cord is run down to the water to point the way home for the intruding spirit. The curer begins to sing, continuing into the night until a shooting star (the incarnations of malgri's eye) is seen. indicating the dispossession and banishment of the spirit. The string is then snapped and the victim recovers.

The basis for this cure is hard to explain in Western terms, though it can be argued that because the illness was essentially of a psychological nature, all that was required was performance that would alter the victims state off mind. Such an explanation can be applied to certain cures" too, such as the ability of placebo, or sugar pill, to cure a headache. It is rarely a simple matter to prove" that something specific has cured an illness. Most of our medical beliefs about cures are based on statistical averages. If something works often enough, we assume that it is effective. Sometimes, however, more clearly demonstrable physiological processes are at work, as in the case of antibiotics.

HEALTH CARE DELIVERY SYSTEMS

The social components of curing include aspects of the overall social structure that influence the curing process and more specificalty. what may be reffered to as the health care delivery system. A health care system mobilised resources to care for those who are ill. Patient,curer, and auxiliary personnal all have role expectations. The system also includes available technology and the physical setting with in which the curing process takes place.

There are two dimensions to the question of setting: privacy and specialisation, Among the Azande, diagnosis takes place in public. Heraded and accompanied by drums, diagnosis of patients are considered 'local events of some importance", drawing spectators from throughout the neighbourhood. F-lausa barber - surgeons in Ibadan, Nigeria place their roperating tables under a tree in public view. In contrast, in Western medical systems a premium is placed on privacy for both diagnosis and treatment. In many non-western setting curing takes place in a non-specialised locale, such as the village centre or the patience's or curer's own home. Among the Maya of southern Belize patients often move into the home of the curer for the duration of their treatment, sometimes bringing their entire family with them. In several instances the male patient died and his wife subsequently remained with the curer. Such events become the subject of gossip, but rarely do they influence the professional reputation of the curer.

In a large scale societies, health care often takes place in a specialised setting, such as hospital, hospitals have evolved into very complex institutions, functioning as small societies with many of their cultural attributes.

Every society recognises only limited number of peoples who are able to treat illnesses. He role of curer however, varies a good deal from society. In societies with personalistic views of disease casualty, there is a need for someone who is able to determine who caused an affliction and why - usually someone who to the supernatural. In such societies, the curer is usually a "shaman""sorcerer" or "witch doctor". On the other hand, societies with naturalistic beliefs require "doctors" who have learned curing skills through pfactice and observation of illnesses and the properties of medicines.

In many small-scale societies, curing is not a full time vocation. There simply is not enough work, and the curer will generally engage in the same primary economic activities as other members of the society. In large scale societies, curing is more often a full time occupation.

Most societies also recognise specialization within the medical profession. Although by far the greatest degree of specialisation occurs in western medical systems, even very small non-western societies tend to have different sorts of curers. Peasants in Philippines recognise midwives, bonesetters and general practitioners.

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Analysis of curers requires attention not only to their roles but also to other carer - related factors, such as recruitment, training and reputation" In personalistic systems recuitment may he personal dream or physical affliction, such as epilepsy. Recruitment in other instances may be linked to the would-be curer's personal inclinations, ability, or social status. Ins some cases only the children of curers can become curers: sometimes recruitment is limited to members of certain socio-economic classes. Often all the these factors are necessary.

For shamans and sorcerers, in many systems a period of apprenticeship is involved, sometimes lasting for a number of years. At the end of the training period, there is usuafly some rite of passage, and often a form of certification.

Eri Thomson describes the initiation of a Mayan curer in southern Beliz: "The instructor and the iniate retire to a hut in the bush for month so that there may be no eavesdropping. During this period, the initiate is taught by his master all the different prayers and practices used in causing and curing sickness. At the end of that period the iniate is sent to meet Kisin (a deity). Kisin takes the form of a large snake called Ochcan, which is described as being very big, not poisonous and having a large shiny eye.

When the ochcan meet face to face, the later rears upon his tall, and approaching the initiate will their faces are almost touching, puts his tongue in the iniates's mouth. In this manner he communicates the final mysteries of sorcery. To the north in Beliz. the snake is said to swallowthe iniate whole as the finale of the initiation, passing him out of his body with excrement. This leads to legitimisation as a recognized curer is sanctified by those who dominate profession.

The ability of a curer to continue working is in large part a matter of reputation. Curers generally must have are relatively good record of success, although with any system a good public relations effort can make up for lack of success in curing patients. In addition to chief curers, health care system often include a number of auxiliary persons who assist in the treatment process. Among Azande, assistants may include musicians orchanters who provide an accompaniment to the curer's activities.

An important part of disease treatment in modem societies involves promotion of public awareness of the nature of diseases and their cures. This involves the use of screening clinics, educational programs. and advertising campaigns. To be effective, such programs need to be sensitive to the cultures of those to whom thy are addressed.

Thus in campaigning against the spread of AIDS, public health workers have had to take into account how people view sex and appropriate means of discussing sexual maters in different cultures. An anti- AIDS advertisemmt in Austrlia aimed at Aborgness contained the following dialogue: What is this AIDS? This a white mans disease'

The reply "Nah! AIDS is a killer. like the pox. VD. Once yah got it, you are dead."

THE COSTS OF HEALTH CARE

Twenty of thirty years ago attention in Western mecicine tended to focus on discoveries of miraculous cures. More recently questions of cost have been dominant. Even in societies that are relatively poor in material goods, treatments can be quite expensive. Among the Inuit of North Alaska payment traditionally might involve the offering of a wife or daughter as a sexual partner or a gift of umiak (women's boat) and a substantial fee. Curers are not always wealthy, however, Fabrege and Silver found that the incomes of zinacantan cures were roughly the same as other villagers'.

While in small scale societies and many non western health care systems the primary cost of being cured is the curer's fee, with Western health care systems the curer's fee may n fact only be small part of the overall cost. Greater specialization has meant that a wider range of auxiliary personnel must be paid fo. There is also cost of maintain more elaborate settings for curing like hospitals. Then there is the costiy technology, for western medicine is of considerable concern to poorer, underdeveloped countries.

Building hospitals, developing technology, developing medicine, training the doctors all add to up huge costs.

The price of medicine, most of which is produced and sold by a few large multinational firms, is another issue. These drug firms have been accused by a number of Third World countries and by researchers of pushing drugs that are overly expensive, useless, and in some cases even known to be harmful. Medical technology is an additional problem in terms of cost, appropriateness and dependency. The question of dependency has both an economic and political dimension. As noted by Harvey William, "previous dependency on the US for the majority of its medical technology has made Nicaragua vulnerable to the Reagan administration's efforts to restrict credit and export permits."

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CHAPTER - 10

FACTORS AFFECTING HUMAN GROWTH AND DEVELOPMENT

Factors affecting Human Growth and development

Intrinsic Factors	Extrinsic(envtl factors)
	Nutrition
1) Genetic material	State of Health
-Twin studies	Psychological disturbance
-Parent-offspring correlation	Rural-urban divide
- chromosomal abnormalities	Socio-cultural factors
2) Hormones	Ecology
	Secular trend

Growth of an individual is affected by several factors. They can broadly be divided into (i) intrinsic factors (ii) extrinsic factors

Intrinsic factors refer to genetic components. They are observed between chidren-parents; siblings and within the same racial stock. But these factors are not totally distinct from the extrinsic or environmental factors. They depend on one other. For eg: the genetic factors do not express themselves unless a favourable environment is established.

1) Interinsic Factors

(i) Genetic material :

Genetic code is laid down very early (in zygotic stage) that has maximum impact on growth plan. The influence of genetic material is seen through the study of similarities among –

(a) Twins

1) The difference in height and weight is more among dizygotic twins than in mono zygotic twins.
2) The average age difference in reaching menarche among mono zygotic twins in 2 months while in dizygotic twins it is 10 months.
3) Skeletal maturity, eruption of teeth (both deciduous and permanent), sequence of eruption of teeth are more similar among mono zygotic twins.

(b) Parents - Offspring

Parents have more correlations in attainment of height, onset of puberty etc with their biological children than adopted children.

(c) Chromosomal Abnormalities

Suggest genetic control on growth eg : males with klinefelter syndrome (XXY) are taller than normal males.

(ii) Hormones

They are organic substances synthesized in endocrine glands. They promote growth. Their interaction with other hormones influences pattern of growth.

Eg : hormones cause changes during adolescence

Hypothelamus

↓ **Gn RH (Gonado Trophin Releasing Hormone)**

Pitutary gland

↓ **FSH – Follicile stimulating Hormone**
LH - Leutinising Hormone

Gonads

↓ **Testosterone**
Oestrogen

Secondary sexual features

Imbalance/disorder in secretion of hormones may lead to abnormal growth.

2) Extrinsic/Environmental Factors

They are capable of delaying the genetic features from getting exhibited or reducing their intensity. Some of the extrinsic factors are : -

(a) Nutrition : Adequate supply of proteins, carbohydrates, vitamins, minerals, water etc are essential for multiplication of cells and their maturation. Nutritional deficiency impairs normal growth during childhood and delay the appearance of adolescent spurt. This is proved in the studies on children malnourished during world war II. They attained normal growth with improvement in nutrition. But prolonged malnutrition leads to permanent retardation of growth.

(b) State Of Health : Illness influences health through genetic, endocrinal, physiological or environmental factors. Eg : defects in central nervous sys-

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tem produce severe mental retardation accompanied by short stature.

- Chronic anaemia is growth retarding
- Frequency and severity of diarrhea that could be associated with nutritional status results in poor growth.

(c) *Psychological disturbance :* Proper secretion of growth hormones is affected by the emotional and psychological stage of the individual. Hence, environment both at home and in the society are crucial in determining growth. Under favourable pshychological conditions endocrine system performs well.

(d) *Rural-Urban Divide :* The way of life, food & work habits, medical care & sanitation etc are different in rural and urban areas influencing growth pattern.

(e) *Socio-cultural conditions :*

- Children's growth is related to the occupation of their parents. Studies proved that children of upper classes are taller and heavier.
- Size of family indirectly influence children in a large family (with limited incomes) do not have proper care and their growth is affected.
- Cultural aspects like – breast feeding, child care, food habits, cleanliness, sanitation etc also influence growth.

(f) *Ecology :* Ecological factors affect the physique and growth patterns of individuals. Eg : people living in high altitudes (Andes, Himalayas etc) are shorter, lighter than people in plains. They have body dimensions suitable for hypoxia (lack of sufficient oxygen to the body), low humidity etc

Climatic factors like temperature and season influences growth. Eg : the rate of growth is more in summer than in winter.

(g) *Emergence of a secular trend :*

Countries like England, Germany, Poland, Norway, Sweden etc exhibit a tendency where in children show progressively larger sizes (taller&heavier) at all ages. This trend is called "*secular trend*". Though no clear reasons are identified for this, it can be stated that better nutrition, improved facilities, desirable environmental conditions etc can be the reasons.

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CHAPTER - 11

DEMOGRAPHIC ANTHROPOLOGY

BIOLOGICAL THEORIES OF POPULATION

Malthus gave a very important theory about population and it provided a food for thought to many economists and demographers of his days. It continues to be discussed with all seriousness even today. An improvement in this theory was made by the exponents of theory of optimum population. it has already been discussed how this theory proved an improvement upon the theory of Malthus and how the theory has been criticised by the critics. In our modern times biological theories about population have received considerable attention. Exponents of these theories make us believe that human beings live and die in the same manner as animals and plants. Some important theories in this regard are :

1. Michael Thomas Sadler's Density and Fecundity Principle.
2. Diet Theory of Thomas Doubleday.
3. Raymond Pearl and lowell J. Reed's Logistic Curve Theory.
4. Carrado Gini's Theory of Biological Stages.
5. Herbert Spencer's Analysis of Fertility Function.
6. Jouse deCastro's Theory of Protein Consumption.

These different theories have their own place and position in population studies and, therefore, need careful consideration.

MICHAEL THOMAS SADLER'S THEORY

Michael Thomas Sadler was born in 1780 and lived upto 1835. He was a British social reformer, an economist and a contemporary of Malthus. He brought out this ideas in his volume entitled. 'The Law of Population'.

Sadler was of the view that the nature's laws about population were just the opposite of those enunciated by Malthus. According to him, "The facundity of human beings is in the inverse ratio of the condensation of their numbers and still in direct contradiction to the theory now maintained (Malthus's) the variation is that facundity is effectuated not by the wretchedness and misery but by the happiness and prosperity of the spices."

Sadler believed that for maintianing fertility it was essential to maintian both labour as well as privacy. With the growth and advancement of civilisation, he opined, desire to put labour and hard work was decreasing. The work which was being done till yesterday by human beings is today being done by machine. In other words, labour is becoming only a mental exercise. Due to shortage of accomodation, there is also no privacy and as such fertility rate is bound to come down, when there is neither privacy nor desire to put labour. He has, therefore, come to the conclusion that nature itself is helping mankind, rather than behaving in the negative way, which Malthus makes us believe. Thompson and lewis while expressing their views have said that, "He held that man's tendency to increase in number would cease to grow at the exact point where the number of the people would enjoy the greatest happiness, all other things being equal." He himself also said, "At every steps the principle of increase contracts, and as I contend, would pouse at that precise point where it had secured the most possible degree of happiness to the greatest possible of human beings.

Sadler was of the opinion that as population increases with that fertility decreases. On the other hand, when population begins to decrease with that fertility increases. He felt that population, by which the increase of mankind has been and still is in all cases, regulated is simply this. This facundity of human beings under similar circumstances varies inversely as their number increase on a given space."

Sadler did not agree with Malthus that there was any fear that growth of population will result in checks to be applied by nature. He was also not in favour of restrictive method of population control. he believed that as population would grow with that people's capacity to produce more will also grow. If population will grow, the people will surely produce what they need.

He does not agree with malthus that population grows in geometrical proportion whereas food grows only in arithmetical proportion. According to him this just cannot happen because when population will grow with that capital too will increase and with that fertility rate is bound to come down. It so happens then, because the capacity to produce children is reduced. He therefore, believed that if people made sincere efforts everything was possible.

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Sadler also believed that if death rate increased, with that birth rate was also bound to increase and vice versa. According to him "The prolificness of an equal number of individuals, other circumstances being similar, is greater when the mortality is greater, and on the contrary smaller when the mortality is less."

Criticism of the theory : Sadler theory has, however, been criticised on several accounts. It is said that he did not distinguish between fecundity and fertility. According to Thomson and Lewis, ".... he did not distinguish clearly between fecundity and fertility and he believed that there was no antagonism between man's natural capacity to produce, which he never defined, and his capacity to produce, adequate subsistence for all who would be born."

Then another point of criticism that has been advanced is that Sadler believed that when capital would grow with that the population would come down. But this has not come out to be true. In countries like India, capital has considerably increased but still rate of population growth has not come down.

But in spite of all this the theory has its own value. Its greatest contribution is that it removed that pessimism which Malthus had created in his theroy. He brougth considerable hope to the economists, sociologists and demographers.

DOUBLEDAY'S DIET THEORY

Doubleday (1790-1870) was an English economist and social philosopher. He gave his ideas about population growth almost two decades after Sadler. In his theory he has laid great stress on food supply. He felt that as the food supplies will improve with that rate of population growth will also come down. According to Lewis and Thompson, Doubleday believed that, "Whereas Sadler believed that the density of population caused a decrease in fecundity and hence, largely eliminated the influence of the positive checks to population increase. Doubleday held that man's increase in numbers was inversely related to his food supply. The better the food supply, the slower the increase in his numbers." He himself enunciated his theory in the following terms :

"There is in all societies a constant increase going on amongst that portions of it which is the worst supplied with food.

Amongst those in the state of affluence, and well supplied with food and luxuries, a constant decrease goes on. Amongst those who form the mean and medium between two opposite states, that is to say, amongst those who are, tolerably well supplied with good food, and not over-worked, nor yet idle, population is stationary. Hence, it follows that it is upon the numerical proportion which there three states bear to each other in any society that increase or decrease on the whole depends."

Doubleday has discussed both about Plethoric and de-Plethoric States. In the former case one can get good food whereas in latter state there is its shortage and one even does not get sufficient to eat. In de-plethoric state fertilityis always higher as compared with plethoric state. he believed that fertility of all living beings depends on their bulky or thin build up. Experience has sown that fat animals or birds give birth to less children whereas thin to more. Same can even be said about plants and trees. He said that, "An over-fed plant can be revived only if depletion of plant i.e., ringing the bark, extreme lopping and the trending the roots is done." He, therefore, believed that whole population can be divided into three categories in the first category come people of plethoric state, who are rich and can afford to have good food. They have less fertility. But if the people of this category begin to work hard and instead of lazy life begin to live a busy life, then their fertility can also increase. Then comes deplethoric state in which the poor live. They have more capacity to produce and as such population rapidly increases here.

Those belonging to average income group neither produce more nor less and population with them remains almost the same. Those who get very little food, population in that class very rapidly increases. In society whether the population will increase or decrease will very much depend on the number of people who get sufficient to eat, who get very good and excellent food and those who do not get sufficient to eat. he, therefore, said that, "There is in all societies a constant going on amongst that portion of it which is worst supplied with food, in short among the poorest."

According to Doubleday non-vegetarians, have less capacity to produce. That is the reason that in pastoral countries density of population is low. But where the people are both vegetarians as well as non-vegetarians, the fertility is bound to be somewhat higher. In countries where the population is purely vegetarian, fertility is bound to be the highest. Same can also be said about the people consuming rice.

Criticism of the Theory : Doubleday's theory has been criticised on several grounds. There is of course a

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relationship between food consumption and fertility but it cannot be accepted as a rule that in rich countries where people can consume more food, the fertility will in any way be less, as compared with other countries. Experience has shown that population in rich countries has never become stationary but has always been rising and growing.

Spencer has said that Doubleday has proceeded on the presumption that every rich person is fat and that fatness brings down fertility. We know that every rich person is not always fat. Fatness comes not with good food but with bad digestion. According to him, "The cases Doubleday cites of infertility accompanied by fatness are not cases of high nutrition properly so-called, but cases of such defective absorption or assimilation as constitute low nutrition." These days when the rich are more conscious about their health and they can purchase medical facilities, they avoid fatness and try to have normal health.

According to Doubleday in de-plethoric state population gets reduced because then people do not have sufficient eat, whereas population in plethoric state comes down because the people have no reproductive capacity. As such in both cases the population will come down. This has, however, not happened in actual life and the population has all along been going up.

Doubleday makes us believe that all the rich persons have only few children, whereas the poor are bound to have many. But we find that in quite a large number of cases the rich have many and the poor very few children.Doubleday was of the view that Nature automatically distributes wealth. According to him the wealthy are likely to produce less children and with the passage of time they will not have heir to own property, which will pass into the poor. In this way the wealth will get distributed. But this has neither come true, nor appears to be very scientific distribution about wealth distribution.

Then another criticism levied against this theory is that the author has failed to distinguished between fertility and fecundity and as such has created a confusion.

Thompson and lewis while critising the theory have said, "The only comment needed on these theories is to say that there is no scientific basis for the belief that the density of population, the porportion of protein in the diet, or the relative abundance of calorie in-take have any noticeable effect upon fecundity... we know that the fertility of any population, i.e., the actual bearing of living children, may be effected substantially by preventive checks of various kinds while facundity remains unchanged."

PEARL AND REED'S LOGISTIC CURVE THEORY

Raymond Pearl and J. Reed have given their logistic curve theory. The theory was first of all enunciated by Verhulst in 1838, but that did not receive the attention of the people. It was again revived in 1844 but it came to prominence only when Raymond developed it in his book entitled, 'The Biology of Population'.

According to him population grows like fruit flies. After his experiments he came to the conclusion that to begin with these flies very rapidly increase and thereafter with the passage of time the rate of production slows down. After some time it again begins to increase. This can also be seen in the growth of human population. In his own words, "Population wanes and waxes, rises and falls and increases rapidly and slowly but the whole of it is ever increasing."

Both Pearl and Raymond do not agree with the idea that the population grows in geometrical proportion but on the other hand maintains that population growth is a mathematical representation. They also believe that the field of population growth is limited and we cannot have unlimited land and in this limited land there cannot be unlimited population. Graphically curve can be represented as follows:

From this graphic presentation it becomes clear that population grew between 10–15 and then come down between 15–20. It went up between 20–25 and tried to stabilise thereafter. According to Raymond, "The point of maximum rate of growth is the point of inflection of the population growth curve. After that point is passed the rate of growth becomes progressively lower, till finally the curve stretches along nearly horizontal, in close approach to the upper asympatote which belongs to the particular cultural epoch and the area involved." They found population growth in cycles of stages and according to them. "To determine the cycle by change in the growth of population is obviously circular." Raymond in his Biology of Population Growth, has also said that, "The rate of reproduction permeated female per day declines as the density of population increases at first extremely rapidly and then more slowly at higher densities."

This theory also believes that the poor couples like to come frequently closer to each other with the result that they have more children as compared with the rich. They have also given biological reasons of death as well when they say, "As the different groups of people differ in structure, habits, mental outlook and thought, they fight as group for they get mad at each other."

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Critical Analysis of the Theory : Like all other theories, this theory too has been criticised on several grounds namely:

This theory is quite different from the one expounded by Malthus, who has said that population increases by geometrical means and proportion. This theory on the other hand, believes that population increases in a cycle. According to this theory, when population reaches a certain stage, after that it begins to come down. The value of this thery lies in the fact that it has proved considerably true. But this theory too is not absolutely correct.

Pearl believed that as the wealth grows with that fertility rate comes down. But in a actual practice this has not proved very true.

Pearl has tried to lay down his conclusions on the principles which were found true in respect of fruitfiles. But as we know that there is not much analogy between fruit flies and human beings.

According to this theory it is not possible to find out when optimum stage has reached, when population will begin to come down and also when that stage will reach when population will begin to go up.

According to this theory when population grows with that fertility comes down. But they have failed to explain that how in labour colonies where the population increases and with that fertility also remains very high.

Then another point of criticism agianst this theory is that whereas it has laid stress on biology only, it has almost completely ignored role of social, economic and political factors in population growth.

It is also said that though in population growth environmental factors play their part, yet this theory has completely ignored this factor.

CORRADO GINI'S THEORY

Corrado Gini was born in 1884. He was an Italian Sociologist and had particular interest in population problems. He believed that every country at the prime has almost the same sort of composition and simplicity of growth. During this period fertility rate is high. Accordingly when the population of a country grows with that economic and social problems of the country also become complex. Simultaneously industrial as well as trade problems also become very serious. In society new classes begin to emerge. But when social and economic conditions become serious then fertility rate of the population also begins to come down.

He also believes that evolution of a nation and for that matter even that of a society is very closely linked with the changes in the rates of population growth. In the words of Thompson and Lewis, "He thought that the evolution of a nation or any society was closely linked to the changes in their rates of population growth and to the varying propositions of this growth coming from the different social classes."

According to him basic factor in population growth was always biological and not social or economic change. He believed that different rates of increase which were withnessed in different classes or sections of society were because of biological traits of the entire population. According to him in every society comparatively small proportion of one generation produced the majority of the people in the succeeding generation. He felt that whole process of growth of nation was cyclical rise and fall of population.

Cycle of Growth : According to Gini there is no difference between the cycle of growth of the individual as well as that of the population. In this cycle, the first period is that of extremely rapid growth. Then comes the next stage in which period of growth is comparatively slow. This slow period is followed by a period of senescene. It is this period in which numbers decline and the quality of a civilisation considerably deteriorates. In the words of Lewis and Thompson, Gini believed that, "Every nation in its youth is simple and undifferentiated in its structure and has a high rate of fertility, because each generation springs from the people who are hereditarily most prolific, i.e., highly fecunded." As already said he believed that as the numbers grow with that organisation becomes complex. As more and more industrial and commercial activities grow, with that the structure becomes more complicated. Attempts are made to check pressure of population either through war or colonisation or both.

In the second stage population decreases because (i) energetic people are lost either in the war or by way of colonisation and (ii) due to increase in the population of upper class population which is always less prolific than the lower class people. But he said that basic cause for this lower rate of growth was always biological because in that really under laid the influence of economic and social factors. In other words, he believed that decline in fertility was due to decline in fecundity. Decline in fertility first of all becomes visible among the rich and within a short period it is seen among the people belonging to other classes as well. When people belonging to lower

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class get themselves absorbed in the upper classes, they also begin to acquire their habits and become sterile and at no stage give up the fertility of their old class. According to Prof. Gini, "Even the sterility of these climbers is not a consequence of social conditions surrounding their climbing; it is, rather a result of the weakneing of the reproductive instinct and is an inevitable phase of the cycle of the population growth."

While discussing his theory of population Prof. Gini says that decline of population always results in decline in the qualities of individuals. It is because of biological change in the heredity qualities of the individuals. According to him there was some inevitable natural force which determined the rise and fall of population in all societies. He thus invoked some mystical biological changes, which were not within human control, which determined both quantitative growth as well as quality development of human population.

HERBERT SPENCER'S THEORY

Herbert Spencer (1820–1903) was an English philosopher. He considerably wrote about social and biological changes and proposed a natural law of population growth in his "The Principles of Biology". The theory of population as expounded by him, in some respects is on the similar lines as those expounded by Sadlar and Doubleday.

According to him there was a natural law that absolved human begins from every responsibility about increase in population. This is done because nature in itself weakens man's interest in reproduction and the interests are gradually more devoted towards personal, scientific and economic development. According to him in nature there is antagonism between individuation and genesis. In the words of Lewis and Thompson, he believed that, "Hence, the more strenous the efforts the individual must make to ensure his personal advancement in any field of endeavour, the weaker will be his interest in reproduction." According to his philosophy reproductive capacity of human being gradually decreases, especially among women. In the words of Spencer, "That absolute or relative in fertility is generally produced in women by mental labour carried in excess, is more clearly shown.... Most of the flat – chested girls who survive their high pressure education are incompetent to do this. Were their fertility measured by the number of children they could rear without artificial aid, they would prove relatively very infertile."

He believes that as our social and economic problems will go on increasing with that rate of production will also start coming down.

According to him there is always an antagonism between individuation and genesis. From individuation he meant any attempt by any human beings to maintian the life of their members, whereas genesis means producing new members and that both could not go side-by-side. he said that if in a society life expectancy of human beings is higher, then the purpose of society will be served with less children. the genesis will be less and birth rate will also be low. Similarly if birth rate is high, then due to many reasons accordingly life expectancy will also be low and death rate will go up.

In his theory of evolution, Spencer has said that those human beings who are small, will have more and higher fertility rate. he said that, "The minutest organisms multiply asexually in their millions." Many small cells do not mature, otherwise they can grow in number twice or thrice and the population will very rapidly multiply itself. The small compound type cells are born in thousands, whereas more compound in hundreds and still bigger lose their productive capacity. On this basis Herbert Spencer has given his views about productive capacity of the human beings. According to him human beings can also be divided into three categories; namely, (i) The poor and those who live a very simple life; (ii) middle class people and (iii) those who live developed and complex life. he was of the view that as life and living becomes more and death rate comes down. In poverty physical and more mental labour, their reproductive capacity comes down. it is quite often seen that men of unusual mental ability leave no offspring.

Spencer believed that growth of population was not always socially harmful because that helped in more exploitation of natural resources, which in turn helped in raising economic, social and cultural standard of the people. Changes in social economic and cultural outlook always help in bringing about world revolutions. He believed that if birth rate is to be brought down, for this it is essential that expectancy in life should be taken up.

Spencer also believed that increase in complexity in life is likely to decrease fertility. For him fertility was inversely related to the size of living beings. The smaller the animal, the greater the fertility. He assumed that compound type small species multiply in thousands whereas other increase only in hundreds.

Critical Evaluation of Spencer's Theory : Spencer's theory appears to be considerably reasonable. We find that fertility among western women came down only when education became widespread there. But so far statistically it has not been proved that birth rate of

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those who are mentally more developed is in any way is less than those who are mentally less developed. his theory, however, cannot become true law of population because we find that in many rich and intellectually sound families birth rate is very high. Even many educated and employed women have many children.

CASTRO'S THEORY OF PROTEIN CONSUMPTION

Jouse-de-Castro has expounded his theory of Protein consumption in this famous book entitled, 'The Geography of Hunger' in which he has linked birth rate, with the consumption of protein. In order to prove his contention he started his experiment in 1920 on the rats. On the basis of these experiments he came to the conclusion that when a female rat consumed 10% protein, then the birth of the rats was 23.3% but when protein consumption was 18% then the number of rats produced was 17.4%. Again when the quantity of protein increased was 22% then the number of rats produced was 13.8%. On the basis of this he came to the conclusion that fatness helps in reducing birth rate and capacity to produce children goes down.

According to him, those who are poor do not have any balanced food and they are always short of protein in food and their lever does not properly work. Under the circumstances oestrogens in women body get neutralised and increase in their number and their fertility also increases and with that more number of children are produced. That is the reason as to why the poor produce more children.

He said that usually for higher fertility in poor countries, societies or people are blamed which is not desirable. in fact, such a blame should lie on the people of affluent societies and on rich nations. The imperial and colonial powers have never tried to improve living standard of their people and made any attempt to provide them good food. On the other hand, they have always tried to purchase food supplies and raw material at cheap rates and sold finished product at high costs in their colonies. They gave them very low wages with the result that they got imbalanced food and protein content in their diet was almost missing. Absence of protein in food results in capacity to produce more children and their poverty considerably increases and imbalance in food still more increases resulting in still more capacity to produce still more children.

He was pained to know that when the question of starvation in the world comes to the front, then the problem of imbalanced food is not given its due weightage. Why no importance has been given to the poor so far? According to him it is because of the cleverness of the capitalists, the rich, imperialists and even the scientists.

He also said that starvation does not mean non-supply or non availability of food but in fact that means absence of protein in the food. He came to the conclusion that today almost 2/3rd of the world is suffering from this type of starvation and thus becomes victim of diseases or early death. In the absence of proper quantity of protein in food, both the person who consumes food and the food itself gets ruined. In a country like India in fact mortality is very high and many die without producing any children. He felt that when the people are starved of food, they try to satisfy their desire by indulging in more sexual relationship. He said that, "Chronic hunger increases sexual activity. Thus primary need is to emphasize and to compensate for the deminution of the others. But sexual activity goes down with acute hunger." That is the reason why the poor have more children.

He suggested that in order to provide adequate food to the people we should bring more lands under plough. He also stressed that we should try to produce food from inorganic matter. He said that, "When deserts of ice and impenetrable jungles are being turned to gardens and orchards, when the lands we farm and the plants we grow are being made to multiply theirs field, and while we are barely learning to tap the great food reservoirs of waters, the wild flora, and of artificial synthesis, the Malthusian go on setting up their sinister scarce crows. It is nothing to us, since we have no reasons to fear them."

He believed that economic development and growth can result in bringing down population and birth rate, provided proper care is taken in this regard.

Critical Evaluation of the Theory : This theory too has been criticised. There is no doubt that there is relationship of diet with fertility but again it can be said that this relationship cannot be expressed in terms of any law. There are many countries of world which saw population growth when these were quite rich. the season for this is that in these countries employment opportunities increase and as such the parents do not hesitate to marry their children. similarly, the number of love marriages also increases and a married couple does not bother to strictly observe to follow family planning norms. This results in increased fertility.

Castro has also been criticised by Leibenstein when he says that, 'The reason why this approach is fallacious is that the economy might not have experienced sustained

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development if fertility rates had not declined at some crucial stage during the expansion. One need not impute the absence of a barrier simply because the race is of such a nature that the barrier cannot be seen when it is successfully overcome".

Thompson and Lewis while commenting on this theory have said that, "The only comment needed on it is to say that there is no scientific basis for the belief that the density of population and the proportion of protein in the diet or relative abundance of calorie intake have any noticeable effect upon fecundity."

In this way biological theories about population growth have not found much favour in society, primarily because no theory can take the shape of law. Every theory has its own value and significance of course, but not as a theory of population growth or even that of decline.

THEORY OF DEMOGRAPHIC TRANSITION

Theory of Demographic Transition is comparatively recent one and has been accepted by many nations and scholars of the world as reasonable one about population growth. It has been propounded on the basis of statistics collected in many European countries. It appears to be scientific as well as rational theory. Almost all sociologists agree that population process in every state passes through many stages. Each stage has its own peculiarity. If some countries are in the first stage, then there are many which have crossed that and reached second or third stage. The scholars on the subject are, however, divided about classification and groupings of these stages. According to Thompson and Prof. Bogue there are only three stages, while some scholars believe that there are as many as five stages.

Prof. O.P. Walker's Views About Classification : Prof. Walker has classified stages into five.

First Stage : It is high stationary stage. During this stage birth as well as death rate is very high, but the former exceeds the latter and as such population does not very much increase. Before 1920, china and India were at this stage. This stage is usually found in countries which depend on agriculture which is main source of income of their people. Countries like Nigeria, Ethopia, Tanzania, angola, etc., fall under this category.

Second Stage : It is early expanding stage. During this stage birt rate does not come down but death rate very much declines and population very rapidly increases. this happens because it becomes possible for the nation to provide better public health services, though at a slow rate.

Third Stage : It is called late expanding stage. During it both birth as well as death rates descrease. But birth rate does not exceed to the extent of death rate. This happens when the country has attained certain level of agricultural development and is stepping towards urbanisation. There is also some sort of industrialisation. Japan, Chile, Canada and erstwhile USSR, etc., fall under this category.

Fourth Stage : It is low stationary stage. During this stage death rate is at the lowest stage and at the same time birth rate also slowly decreases. There is no growth rate in population and both fertility and mortality rates are low. In such a society per capita income is high and there is higher industrial growth rate. In the economy there is no unemployment and the masses have considerably high real income. The people now being more educated very much care for the education and good quality food for their children. The countries like the U.S.A., Germany, Britain and Australia fall under this category.

Fifth Stage : It is declining stage, when death rate, as compared with birth rate, is high. France is at this stage. In some cases such a stage reaches when there are unforeseen calamities.

Views of Thompson and Notestein : Thompson and Notestein have given their own views about these stages. According to them first and fifth stages are unusual. Neither the population is at high stationary stage nor it is at declining stage. According to him rapid growth of population during the past three centuries was mainly due to decline in death rate, because of modernisation. Throughout modern West, birth rate reached very low levels by the middle of 1930's because of wide spread use of contraceptives and desire to have small family size. According to them there are as such only three intermediary stages namely when :

(a) Birth rate is high but on account of reduction in death rate, population rapidly increases.

(b) Both the death and birth rates are decreasing but death rate has decreased more as compared with birth rate and population increases but slowly and steadily.

(c) Both birth and death rates have equally come down and population remains almost unchanged.

Thus according to them stages are : (a) Pre-transition stage, (b) Transition stage and (c) Post-Transition stage. In the first stage, there is little control over mortality and

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fertility rates and fertility is stationary. In the second stage, both birth and death rates decline but birth rate falls slowly. In the third stage, both mortality rates are at a very low level. At this stage living standard of the people is also very high. The people like quality and small family sizes. There is optimum utilisation of resources.

Views of Donaldolen and Cowgill : They have given the five stages of population as growth cycle. According to them :

First Stage : At this stage birth rate is high and to begin with death rate is also high and thereafter it begins to reduce itself. After some time it again begins to increase and this way due to increase and decreases in death rate, there are fluctuations in population. when there are good food crops death rate comes down and when there are famines, pressure of death rate begins to increase. This type of cycle is found in agrarian societies and is dependent on food supplies. It is even today found in such societies which are industrially backward and have agricultural economy.

Second Stage : At this stage there is reduction both in death and birth rates but death rate, as compared with birth rate, comes down very quickly. But gradually both death and birth rates try to come at equal level. The people now start getting employment and there are visible changes both in food habits as well as in family size. Per capita income now begins to increase and diseases are brought under control. Traditional views start yielding to new values and there is more use of conceptives. There is stability in population.

Third Stage : It is a stage during which both the birth and death rates are on the increase but as compared with death rate, birth rate increases more rapidly. The result is that there is higher growth rate of population, but this, it is believed very rarely happens.

Fifth Stage : This is a stage during which there is high fertility and low mortality. During this stage growth rate of population is very slow because of low fall in mortality rate in the beginning. There are increases in social health services due to which birth rate declines and growth rate becomes slower.

Laudry's Views : This view is based on Cantillan's views about population. He establishes relationship between food and population. According to this view point there are three regimes in the development of a country. First is Primitive Regime during which availability of food very much depends on the population growth rate. In other words, food supply and population are directly linked with each other. Then comes Intermediate stage. During this stage role of food in population somewhat gets reduced and now economics begins to play its role. The people now wish to maintain their living standard. Economy begins to move upward and population control methods become popular. The people now start marrying at late stage. Then comes last stage, which is also known as modern epoch, in which population growth rate has no link with food. There is definite fall in birth rate. There is maximum development in nation's economy and the people are least interested in having big family size.

Ausley J. Coale and Edger M. Hoover's Views about Demographic Transition : According to them everywhere in agrarian economy there is high birth and death rate which varies in harvests and incidents of epidemics. The reasons for high death and birth rates are poor diet, poor sanitary conditions, lack of preventive and curative medical and public health programmes, etc. As the agricultural economy becomes inter-dependent on other ecnomies nation gradually becomes highly industrialised with, marked oriented urbanised economy. When this happens death rate considerably comes down because of supply of adequate food and medieval facilities.

After sometime, they believe even birth rate also begins to fall and in the urban areas because of socio-economic conditions, small family size becomes popular. This then passes on to rural areas. Whereas decline in death rate comes first, birth rate follows subsequently. After sometime both birth and death rates follow parallel down ward trend. the result is that population growth rate becomes very slow; but whereas death rate does not fluctuate, birth rate continues to fluctuate from year to year.

Karl Sax Views About Demographic Transition : Karl Sax of the view that population growth comes in stages. First stage is found in societies where development is yet to start and in these societies both mortality and birth rates are bound to be high. In the second stage, there is higher mortality rate, whereas there is no change in the fertility rate. Thus, there is increase in population. There is slow economic development in the society. In the third stage, death rate reaches minimum level, whereas birth rate starts declining. then comes last stage of equilibrium. Whereas second and third stages are known as those of population explosion, the remaining two stages are called those of stationary population.

U.N. Views About Demographic Transition : On the basis of population U.N. has classified societies as follows :

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a. Societies which have high birth and death rates.
b. Societies which have death rates but there is declining high death rate.
c. societies which have high birth rate but fairly low death rate.
d. Societies which have declining birth rate but low death rate.
e. Societies with fluctuating birth and death rates.

Analysis of Views : Of course, these thinkers have given their own view points about population growth, but one thing is clear that every country must pass through different stages. First stage is applicable to backward countries where both death and birth rates are high. In these countries agriculture is the main source of income. the poeple live in the rural areas and either there is no industrialisation and if there is any, that is very insignificant. Per capita income is very low and the children are considered more a source of income rather than anything else. The children of all ages find work in agriculture and thus even a small child becomes source of income. In fact, not much attention is paid to the health and education of the children and as such bringing up of children is considered no problem at all for the parents.

During the second stage, economy begins to proceed towards development. Along with agriculture, industry also begins to play its own role. In other words industrialisation takes place. Along with industrialisation, urbanisation starts and many transport facilities are provided to the society. Education, food, health and similar other facilities take place. In social life, hold of orthodoxy begins to relax itself. At this stage there is almost population explosion.

At the third stage, living standard improves and along with that there is improvement in the physical standard of the people. The women also begin to get educated and so also employment. These wish to have less number of children and find time to play their role in other walks of life. On the other hand, aspirations in the economic field go very high. The parents wish to give very high education to their children. There is definite trend towards urbanisation and industrialisation and birth rate definitely falls down.

According to this theory all the countries of the world are passing through these stages. Broadly speaking, some African countries are in the first stage. Asian in the second and European in the third stage. But there is no country in the world which does not pass through all the three stages. Thus, these three stages are unavoidable for the country as a whole.

Criticism of Theory of Demographic Transition : There is no doubt that this theory is quite useful in describing demographic history. But at the same time it has its own limitations. It has been said that it is not a theory in the strict sense of the term because it is only a broad generalisation and eos not encompass the experience of even all the western countries. it even does not fully explain the phenomenon of 'Baby Boom' which came in western countries after economic recovery and Second World War. It is also pointed out by the cities that the theory does not provide a theoretical explanation of fertility which is so necessary for any demographic study. it is not a theory in the sense that it does not extract fundamental processes from a phenomenon and identify crucial variables. Because of this it does not have any predictive value. A serious limitation of this theory is that is cannot be applied with confidence in the developing countries. In brief, it can be said that though this theory provides good framework for wider empirical generalisations, yet is cannot really be considered a theory.

SOCIAL AND ECONOMIC THEORIES OF POPULATION

Like biologists, many thinkers in the field of sociology and economics have also given their own views about population growth. All these scholars have tried to attribute growth and explosion of population to social and economic causes. According to them population of a country is much dependent on economic and political conditions of that country on the one hand and social on the other and that both cannot be separated from each other. Some of the important theories which have been expounded include:

1. Surplus Population Theory of Karl Marx.
2. Surplus Population and Social Mal-adjustment Theory of Henry George.
3. Social Capilarity Theory by Dumont.
4. Voluntarism by Fetter.
5. Principle of Individuality by Nitti.
6. Brento's Theory of Increasing Prosperity.

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MARX'S THEORY OF SURPLUS POPULATION

Karl Marx (1818-1883) was a German social philosopher, who became founder of modern communism. Marx did not believe that all hardship with which the people were suffering were due to man's tendency to grow in numbers faster than his production of subsistence would permit. On the other hand, he believed that man's tendency to press on the means of subsistence was due to evils of capitalism which would disappear with the emergence of communism. He related population growth with present economic system and for him both were inseparable. To quote him :

"It is the working population which, while effecting the accumulation of capital, also produces the means whereby it is itself rendered relatively superfluous, is turned into a relatively surplus population and it does so to an increasing extent. This is a law of population peculiar to capitalist method of production, and in fact, every method of production that arises in the course of history has its own peculiar, historically valid law of population. it is only for plants and animals that there is law of population in the abstract and that only in so far as man does not interfere with them."

He believed that poverty and unemployment were not due to increased population, but on account of capitalist system which failed to provide jobs. It was this system which was responsible for real production and uneven distribution of wealth and for providing jobs to only few persons.

Marx did not agree with Malthus's Theory of Population. According to him Malthus was plagiarist and poured ridicule over him. According to him work of Malthus was that of school children. According to Marx, 'His work in the first form was nothing than a school boyish, superficial and parsonic, declamatory plagiarism from Dofoe, Sir James Stuart, Townsend, Franklin Wallace and others and did not contain single sentence thought out by Malthus himself."

According to Marx in no country of the world population increases on account of fertility but it increases only on account of capitalist policies. The capitalists make labour part of their production and steal something out of that and also spreads unemployment out of that as well. In this way he creates an army of unemployed persons and thus labour wages go on decreasing. The poor population cannot nourish their children on account of their poverty and as such the question of surplus population arises. he thus came to the conclusion that main cause of surplus population was nothing else but wrong policies of the capitalists. There is less production of food supplies from the lands on account of zamindari system where there is uncertainty about ownership of land. According to him, It is working population which, while effecting the accumulation of capital, also provides means whereby it is itself rendered relatively superfluous, is turned into a relative surplus population, and it does so to an ever increasing extent. This is law of population peculiar to the capitalist method of population.

Marx also believed that in every period of production there is a separate law of population. That law suits that condition alone. Thus, a law of population prevailing in a capitalist society cannot be made applicable to the socialist society and so on.

The socialists, who closely follow Marx include Urlanis, Boyarski, Shusharin, Malishev, Riabouckin and Raichko. They all believe that in socialist system there will be fullest opportunities of employment and as such there will be no section of population which in any way will be surplus. In socialist countries there is no need to reduce birth rate and that in any way will be surplus. In socialist countries there is no need to reduce birth rate and that in these countries production is increased by controlling nature and exploiting natural resources. There is no need to check birth rate because whole system of distribution is rational and to the advantage of the society as a whole. They, therefore, believe that present system of unemployment which prevails in a capitalist society is responsible for increased or over-population.

He was of the view that one important cause of population growth was food deficiency, which was due to evils of capitalist. he supported his argument by saying that in communist countries where all were getting proper food, there was no population problem. In no communist country there is compulsory family planning. He was convinced that fertility differentials were due to economic inequalities.

The socialists believe that population in a socialist society will increase slowly than in a capitalist society because of the superior status of women under socialism. Marx was of the view that in a socialist society reproductive behaviour would develop a complete harmony between the individual and the society. The socialists today believe that birth control contributes to the emancipation of women. Modern socialists believe not only in birth-control but allow abortion as well.

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Critical Evaluation of Theory : It is to a great extent true that in capitalist societies there is surplus population on account of unemployment, but it is not very true to believe that under socialist system there will be no need to check population growth at any stage. Even in socialist countries population growth is checked on the plea that no mother should have more children so that her health does not deteriorate. Even in erstwhile U.S.S.R. population was decreasing. Whereas in 1917, birth rate in that country was 40 per thousand, in 1960, it came down to 23. Factory workers were provided contraceptives in their factories so that birth rate was kept low.

Not only this, but in all socialist countries economic inequalities have come to an end. A question which, therefore, needs consideration is as to why even socialist countries have different birth rates. If economic inequality is main cause of birth rate, then in these countries these rates should not differ. The fact, however, remains that need and necessity of family planning is felt in these countries as well and if there is full-employment in these countries, main cause for that is that sufficient manpower is engaged in defence establishments and in active military service.

In his theory of population Marx has criticised Malthus but his criticism does not appear to be very reasonable. In the words of Jewis and Thompson, "There can be no question of sincerity of Marx's belief in the evils of capitalism as the sole basis of poverty, but he should not have confined his efforts to discredit Malthus's views to personal ridicule and vilification. Apparently Marx must have had deep-seeded doubts that Malthus's views could be sufficient disposed of by an appeal to facts and reasons."

HENRY GEORGE AND THEORY OF SURPLUS POPULATION

Henry George (1839–1897) was an American economist. His ideas are available in his book entitled 'Progress and 'Poverty' which was first published in 1879. He did not believe that there was any basic antagonism between man's natural tendency to increase in numbers and his ability to provide subsistence for them. He believed that increase of people involves increase of his food. According to him, "In other words, the law of population accords with and is subordinate to the law of intellectual developments and any danger that human beings may be brought into a world where they cannot be provided for arises not from the ordinances of nature, but from the social mal-adjustments that in the midst of wealth condemn men to want."

George, also agreed with Spencer that fecundity and/or fertility will decrease as individual intellectual development becomes more widespread.

He was very much convinced that in case zamindari system was abolished with that uncertainty about possession of land would come to an end. Abolition of zamindari will also solve many population problems. He was in favour of voluntary use of contraceptives on the one hand and reduction will go up and population problem will be solved considerably.

He also agreed with Spencer that as the society will intellectually grow with that birth rate will come down. he is however, not clear whether that situation will arise because of reduced fertility or by adopting family planning means and methods.

THEORY OF SOCIAL CAPILLARITY

Theory of Social Capillarity was given by Arsene Dumont (1849–1902), who was Professor in the University of Strasbourg. he carried out population studies in France in, later part of 19th century. Thompson and Lewis have said that, "Briefly stated, his belief was that the individual tends to amount to higher levels in his social environment by a process similar to physical capillarity. In this process of climbing upward, he becomes less and less likely to reproduce himself; he is drawn out of his natural mnieu and away from the family." According to him in this process he loses interest both in the family as well as race. he then desires to move upward keeping his personal interests into consideration not caring whether those clashed with those of his community or race or not. According to his belief in a society where movement from one class to the other is easy, social capilarity is as inevitable as gravity. In his own words, "What gravity is to the physical world, social capillarity is to the social order."

Dumont was of the view that movement from one class to the other was direct cause of decline in the birth rate. he came to the conclusion that, "The development numbers in a nation is in inverse ratio to the development of the individual." He, however, did not agree with Spencer that change in number was due to biological changes. According to him social capillarity is more effective in a country where movement from class to class is weak and hindrances very few.

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As regard democratic society Dumont felt movement was rapid and birth rate should be low, particularly in the areas which were near the centre of attraction. In the words of Thompson and Lewis, Dumont believed that, "People at a distance from centre of attraction and in occupation where individual ambition has little opportunity to develop are not drawn into this capillary movement so rapidly and hence are not likely to reduce their birth rate to the same degree. They will continue to incease and thereby, to supply new candidates for capillary action; these in turn, will become ambitious social climbers and will soon die out."

He believes that in societies where caste system is rigid there can be no tendency for the birth rate to decline; because a rigid social structure is bound to prevent upward social movement. In a rigid society individual development cannot be very engrossing.

He did not believe that Malthus's theory was very correct or rational. The population does not increases with increase in food supplies, but increases in wealth only increases family prestige.

Dumont did not agree with Malthus in so far as his theory of population was concerned. According to him his theory was most suited to animals and those who lived like animals rather than human beings. Thus his theroy was for uncivilized people. it is only in that society that it can be thought that the population will increase by geometrical proportion and not in any other society.

He also did not agree with Quillar's Principle's of population. According to him when a civilised society goes little forward then it begins to believe that where there is food, there are people to eat as well. In such a society population increases keeping in view food production and as such need and necessity of positive checks does not arise. he asserted that in every civilised society in matters of population growth principle of social capillarity must act. Everybody wants to have some prestige in life, which is not confined to social or economic but every walk of human life. According to his view point, "From the moment when the imagination and the attraction of the ideal enter in scene we find ourselves in the presence of a new principle of population."

According to Dumont, where there is a developed society, i.e. where there are developed cities and there is an enlightened government, there is keen desire on the part of the people to go up. On account of social capillarity, the people who migrate to cities have less fertility. he said that, "Just as a column of liquid must be thin in order to rise under the force of capillarity so must a family be small to rise in the social scale". among civilised people there are less children, with the result that they are more civilised.

He said that at present the poor utilise their time and energy in producing more and more children. In case they use their time, energy and resources for production of wealth they can also obtain social mobility. That can be possible only when such people produce only one or two children. The poor will have to labour hard, according to the extent of their poverty, in order to rise in life.

Dumont came to the conclusion that in cities fertility is always low whereas in the villages it is very high. it is because in cities people produce less children because they have desire to go up and rise in life. In cities not only the people have capacity to think but also avenues and opportunities to go up. Birth rate in the cities is also low because cost of bringing up the children is very high and there is keen desire to go up as quickly and speedily as possible. On the other hand, in the villages birth rate is high because there is orthodoxy, rigid caste system, lack of vigour, poverty and illiteracy. he was convinced that poverty in itself, is not the cause of high birth rate but it is because of ignorance and lack of vigour.

He was also opposed to socialist viewpoint about population. According to him, "The socialist principle of equality of function if entirely applied, would necessarily imply the destruction of social capillarity and cause a high birth rate."

Critical Evaluation of the Theory : Like other theories this theory too is only partly just. It is undoubtedly true that in the cities birth rate is usually less as compared with the villages. it is also true that urbanisation and industrialisation is responsible for low birth rate, but this cannot be accepted as a rule. In India e.g., the labourers keep their families in villages and that is the reason why birth rate in the cities is low. If these families had been in the cities, it is doubtful the rate would have been so low.

One also finds that rising in life is not directly or even indirectly related to number of children. If this would have been true, then ladies without children would have risen hightest in life because they have no children. In fact, as we know rising in life depends very much on social, economic and political conditions of the society in which one is living or working or the circumstances in which one is placed. There are many persons in society who

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have many children before they rise in life and continue to produce children even after they have risen high in life. There are many more in society who have less number of children and still they do not rise in life and remain in the lower strata of society.

While discussing the importance and position of Dumont's theory, Thompson and Lewis say, "On the whole this statement of relation between individuation and genesis seems a decided improvement over Spencer's. It has the merit of directing more attention to the actual conditions under which people live as important factors in developing preventive checks to population growth, checks which could render the subsistence checks less operative, Malthus hoped for this result although he did not approve of contraception as Dumont clearly did.

FRANK FETTER AND THEORY OF VOLUNTARISM

Frank Fetter was another social and economic thinker who thought about population growth. He did not agree with malthus about his conclusion about population growth. He was convinced that malthus had given more weight to sexual relationship than anything else in his theory. He also did not agree with Malthus that death rate remained high because there was shortage of food. He said that there were many societies in the world where the children or the aged were killed not because there was shortage of food but because the people were victim of orthodoxy or they wanted to have only beautiful or healthy children. According to his view point fertility had inverse relationship with economic development, in so far as capitalist societies were concerned.

BIOLOGICAL FACTORS AFFECTING HUMAN FERTILITY AND FECUNDITY

Reproduction is a quintessentially biological process, and hence all fertility analyses must consider the effects of biology. Such factors, in rough chronological order, include:

- the age of onset of potential fertility (or fecundability in demographic terminology);
- *the degree of fecundability—i.e., the monthly probability of conceiving in the absence of contraception;*
- *the incidence of spontaneous abortion and stillbirth;*
- *the duration of temporary infecundability following the birth of a child; and*
- *the age of onset of permanent sterility.*

The age at which women become fecund apparently declined significantly during the 20th century; as measured by the age of menarche (onset of menstruation), British data suggest a decline from 16–18 years in the mid-19th century to less than 13 years in the late 20th century. This decline is thought to be related to improving standards of nutrition and health. Since the average age of marriage in western Europe has long been far higher than the age of menarche, and since most children are born to married couples, this biological lengthening of the reproductive period is unlikely to have had major effects upon realized fertility in Europe. In settings where early marriage prevails, however, declining age at menarche could increase lifetime fertility.

Fecundability also varies among women past menarche. The monthly probabilities of conception among newlyweds are commonly in the range of 0.15 to 0.25; that is, there is a 15–25-percent chance of conception each month. This fact is understandable when account is taken of the short interval (about two days) within each menstrual cycle during which fertilization can take place. Moreover, there appear to be cycles during which ovulation does not occur. Finally, perhaps one-third or more of fertilized ova fail to implant in the uterus or, even if they do implant, spontaneously abort during the ensuing two weeks, before pregnancy would be recognized. As a result of such factors, women of reproductive age who are not using contraceptive methods can expect to conceive within five to 10 months of becoming sexually active. As is true of all biological phenomena, there is surely a distribution of fecundability around average levels, with some women experiencing conception more readily than others.

Spontaneous abortion of recognized pregnancies and stillbirth also are fairly common, but their incidence is difficult to quantify. Perhaps 20 percent of recognized pregnancies fail spontaneously, most in the earlier months of gestation.

Following the birth of a child, most women experience a period of temporary infecundability, or biological inability to conceive. The length of this period seems to be affected substantially by breast-feeding. In the absence of breast-

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feeding, the interruption lasts less than two months. With lengthy, frequent breast-feeding it can last one or two years. This effect is thought to be caused by a complex of neural and hormonal factors stimulated by suckling.

A woman's fecundability typically peaks in her 20s and declines during her 30s; by their early 40s as many as 50 percent of women are affected by their own or their husbands' sterility. After menopause, essentially all women are sterile. The average age at menopause is in the late 40s, although some women experience it before reaching 40 and others not until nearly 60.

CONTRACEPTION

Contraceptive practices affect fertility by reducing the probability of conception. Contraceptive methods vary considerably in their theoretical effectiveness and in their actual effectiveness in use ("use-effectiveness"). Modern methods such as oral pills and intrauterine devices (IUDs) have use-effectiveness rates of more than 95 percent. Older methods such as the condom and diaphragm can be more than 90-percent effective when used regularly and correctly, but their average use-effectiveness is lower because of irregular or incorrect use.

The effect upon fertility of contraceptive measures can be dramatic: if fecundability is 0.20 (a 20-percent chance of pregnancy per month of exposure), then a 95-percent effective method will reduce this to 0.01 (a 1-percent chance).

ABORTION

Induced abortion reduces fertility not by affecting fecundability but by terminating pregnancy. Abortion has long been practiced in human societies and is quite common in some settings. The officially registered fraction of pregnancies terminated by abortion exceeds one-third in some countries, and significant numbers of unregistered abortions probably occur even in countries reporting very low rates.

STERILIZATION

Complete elimination of fecundability can be brought about by sterilization. The surgical procedures of tubal ligation and vasectomy have become common in diverse nations and cultures. In the United States, for example, voluntary sterilization has become the most prevalent single means of terminating fertility, typically adopted by couples who have achieved their desired family size. InIndia, sterilization has been encouraged on occasion by various government incentive programs and, for a short period during the 1970s, by quasi-coercive measures.

FACTORS AFFECTING MORTALITY

Overall human mortality levels can best be compared by using the life-table measure life expectancy at birth (often abbreviated simply as life expectancy), the number of years of life expected of a newborn baby on the basis of current mortality levels for persons of all ages. Life expectancies of premodern populations, with their poor knowledge of sanitation and health care, may have been as low as 25–30 years. The largest toll of death was that exacted in infancy and childhood: perhaps 20 percent of newborn children died in their first 12 months of life and another 30 percent before they reached five years of age.

In the developing countries by the 1980s, average life expectancy lay in the range of 55 to 60 years, with the highest levels in Latin America and the lowest in Africa. In the same period, life expectancy in the developed countries of western Europe and North America approached 75 years, and fewer than 1 percent of newborn children died in their first 12 months.

For reasons that are not well understood, life expectancy of females usually exceeds that of males, and this female advantage has grown as overall life expectancy has increased. In the late 20th century this female advantage was seven years (78 years versus 71 years) in the industrial market economies(comprising western Europe, North America, Japan, Australia, and New Zealand). It was eight years (74 years versus 66 years) in the nonmarket economies of eastern Europe.

THE EPIDEMIOLOGIC TRANSITION

The epidemiologic transition is that process by which the pattern of mortality and disease is transformed from one of high mortality among infants and children and episodic famine and epidemic affecting all age groups to one of degenerative and man-made diseases (such as those attributed to smoking) affecting principally the elderly. It is generally believed that the epidemiologic transitions prior to the 20th century (*i.e.*, those in today's industrialized countries) were closely associated with rising standards of living, nutrition, and sanitation. In contrast, those occurring in developing countries have been more or less independent of such internal socioeconomic development and more closely tied to organized health care and disease control programs developed and financed internationally. There is no doubt that 20th-century declines in mortality in developing countries have been far more rapid than those that occurred in the 19th century in what are now the industrialized countries.

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INFANT MORTALITY

Infant mortality is conventionally measured as the number of deaths in the first year of life per 1,000 live births during the same year. Roughly speaking, by this measure worldwide infant mortality approximates 80 per 1,000; that is, about 8 percent of newborn babies die within the first year of life.

This global average disguises great differences. In certain countries of Asia and Africa, infant mortality rates exceed 150 and sometimes approach 200 per 1,000 (that is, 15 or 20 percent of children die before reaching the age of one year). Meanwhile, in other countries, such as Japan and Sweden, the rates are well below 10 per 1,000, or 1 percent. Generally, infant mortality is somewhat higher among males than among females.

In developing countries substantial declines in infant mortality have been credited to improved sanitation and nutrition, increased access to modern health care, and improved birth spacing through the use of contraception. In industrialized countries in which infant mortality rates were already low the increased availability of advanced medical technology for newborn—in particular, prematurely born—infants provides a partial explanation.

INFANTICIDE

The deliberate killing of newborn infants has long been practiced in human societies. It seems to have been common in the ancient cultures of Greece, Rome, and China, and it was practiced in Europe until the 19th century. In Europe, infanticide included the practice of "overlaying" (smothering) an infant sharing a bed with its parents and the abandonment of unwanted infants to the custody of foundling hospitals, in which one-third to four-fifths of incumbents failed to survive.

In many societies practicing infanticide, infants were not deemed to be fully human until they underwent a rite of initiation that took place from a few days to several years after birth, and therefore killing before such initiation was socially acceptable. The purposes of infanticide were various: child spacing or fertility control in the absence of effective contraception; elimination of illegitimate, deformed, orphaned, or twin children; or sex preferences.

With the development and spread of the means of effective fertility regulation, infanticide has come to be strongly disapproved in most societies, though it continues to be practiced in some isolated traditional cultures.

MORTALITY AMONG THE ELDERLY

During the 1970s and 1980s in industrialized countries there were unexpectedly large declines in mortality among the elderly, resulting in larger-than-projected numbers of the very old. In the United States, for example, the so-called frail elderly group aged 85 years and older increased nearly fourfold between 1950 and 1980, from 590,000 to 2,461,000. Given the high incidence of health problems among the very old, such increases have important implications for the organization and financing of health care.

MENARCHE AND MENOPAUSE

Menarche is a biological term which was first introduced by Kisch in 1910. As a biological phenomenon, it gets expression in women's body which points out an important stage of life cycle. Menarche indicates the specific stage of first periodical regular flow of blood from woumb in all the healthy normal females. At a particular level of biological developments, the girls begin to menstruate for the firsttime which is continued for a period of four to five days. The menstrual cycle is usually one of twenty eight days and the estimated blood loss is between 50ml. to 200 ml. (Barua, 1996). Menarche – the first menstruation, is a well defined milestone that denotes the functional development of the female reproductive system, although a few initial cycles are anovulatory. It is a clear indication of puberty and much of the adolescent growth spurt has already passed by the time of its onset. It is not a final phenomenon of sexual maturation, but probably because of endocrine regulation is connected with other phenomena of puberty and growth processes including influence on adult body size.

With increasing age, various morphological and physiological changes are accompanied in women's life. The reproductive life of women ends with varying degree of suddenness with the menopause – a largely human phenomenon. In other words, menopause is considered as the most critical biological event in the body and from this stage, the reproductive function of women is completely ceased by consequently arresting the menstruation f low of blood and ovulation. Sometimes, the term 'menopause' is acquainted as the symbolic end

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of womenhood because the women who attained menopause are not capable for conception.

However menopause is a very complex process of changes that occur in the biological structure of women, accompanied by a variety of events, sometimes resulting in extremely unpleasant symptoms which include hot flushes, headaches, insomnia and itching, etc.

Menarche and natural menopause are two important physiological events in a woman's life. An early onset of menarche and later menopause are well-established risk factors for the development of breast cancer and endometrial cancer, the researchers explain. On the other hand, early menopause increases risk of osteoporosis and cardiovascular disease.

Previous studies have suggested both menarche and menopause may be partially under genetic control. To identify common genetic variants influencing these states, the researchers analyzed more than 317,000 gene variants in a total of 17,438 women from the Nurses' Health Study (NHS) and the Women's Genome Health Study (WGHS) based at BWH.

"At these newly identified loci, fine mapping or sequencing might lead to identification of the causal variants, and thus expand our knowledge of the underlying physiology and biological regulation of these traits," said lead author Chunyan He, an assistant professor at Indiana University School of Medicine who was a doctoral student in the HSPH Department of Epidemiology while conducting the research. "Insights into the genetic factors influencing the timing of menarche and natural menopause might shed light on normal reproductive function and the prevention of the diseases associated with these two traits."

Daniel Chasman, Director of Computational Biology in the Center for Cardiovascular Disease Prevention at BWH, said: "The collaboration of the WGHS and the NHS represents a great example of how large cohorts with genome-wide data can complement each other. While only one locus reached near genome-wide significance in the NHS alone, the meta-analysis of combined data had much more statistical power and revealed a total of two loci for timing of menarche and four for timing of menopause." Chasman, also an assistant professor at Harvard Medical School, added, "Of the loci for timing of menopause, three were already strong but not proven candidates in the NHS; these loci reached genome-wide significance in the WGHS alone, supplementing the meta-analysis by a second mode of validation through replication. The remaining locus, for timing of menopause, would not have been identified, even as a candidate, without the joint power of these two cohorts working together. Future collaborations will hopefully continue to leverage the combined power of the two cohorts for association studies directed at other clinical characteristics."

The NHS was begun in 1976 to investigate the potential long term consequences of the use of oral contraceptives. The studies were soon expanded to include diet and nutrition, in recognition of their roles in the development of chronic diseases. The research continues today with more than 116,000 women enrolled in the study.

The WGHS was announced in 2006 as a survey of genetic differences among 28,000 initially healthy American women who had already been tracked for over a decade for the development of heart disease, stroke, diabetes, and other common health disorders. Stored genetic samples from each participant undergo a fully confidential "genome-wide scan" evaluating more than 317,000 potential genetic differences.

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CHAPTER - 1

ETHNO - ARCHAEOLOGY

Since archaeologists essentially deal with the *products* of human activity they need an instrument of analysis which can decode informations regarding the *nature* of activities. For long the antiquities have been looked at as 'fossilized' culture traits and it is only during the last four decades that we are trying to look at them more closely. Since all activity can not be reduced to ethnicity we are trapped in a logical fallacy if we consider all antiquities as representing culture. Further, we need not be looking for only cultural traits when we are uncovering a past civilization. Archaeologists during its *formative stage* (1800-1930) in their enthusiasm of having unearthed spectacular remains classified all antiquities as products of by-gone cultures. The *second phase* that followed roughly after the thirties of this century has been a descriptive stage. This stage seems to be still going on in many countries and an *analytical stage* would appear a far cry for them. The *descriptive stage* attempts to describe newer sites and compare them with the already available information and scale of reference available from the *formative stage*. Thus, we have such terms as *Oldowan , Acheulian* and *Clactonian* used in the manner of cultural archetypes in our descriptions.

This does not enable us to reconstruct a total picture of the past human societies, neither does it sustain strict scientific scrutiny of creating law-like generalizations. Ethno-archaeology was initiated at the mid-point of the *descriptive stage*. With an aim to steer away from this conservatism where classification of antiquities formed an end in itself. Around the thirties the totemistic rock paintings done by Australian aborigines were used to explain the Franco-Cantabrian Palaeolithic art. During, fifties Chang, a famous Prehistorian from the United States, addressed himself to the specific issue of Ethno-archaeology as an extremely meaningful approach to explanation of antiquities. But he also warned against the dangers of using this as an established methodology. In the late sixties Binfords came out with a comprehensive discussion on the entire methodology of archaeological reconstruction. This also attempted to remove many of the built in problems of Ethno-archaeology.

Ethno-archaeology, in very simple terms, means explaining the past society on the basis of a body of generalised knowledge constructed from ethnography of contemporary living but simple societies. May be here an example will illustrate the point. The ground plan of primitive huts from Upper Palaeolithic and Mesolithic sites in Ukrania have consistently yielded large stones or elephant skulls transported from elsewhere and arranged along the boundary of the hut. These objects in themselves can yield no cultural information. Hence traditional methods in archaeology do not even pretend to seek an explanation of this feature. Available methodology can go as far as identifying if this practice is common over a larger area and hence different from another area where such stones and bones are not arranged. If, however, ethnography can be conducted on simple societies we might start observing the entire process of hut building in actuality. Such observations may not only enable us to fix a range *of functions* for archaeological *structures*, but also yield numerous other informations regarding relationship between persons, division of labour, ideology and cultural cognition of a dwelling unit etc.

The theoretical assumption underlying such a methodology is based on the fact that *socio-cultural changes in mankind occur under the same operative rule irrespective of time and space.* There may be serious problems in proposing -such blanket assumptions in social sciences. It is probably because of this that numerous scholars had opposed Ethno-archaeology in the *analytical stage*. 'New Archaeology' as proposed by L.R. and Sally Binford, on the other hand started getting more popularity. Binford proposed hypothetico-deductive logic on the basis of Leslie White's cultural models. This required the enquiries from archaeological excavations to be geared to the reconstruction of the activities that took place at a site. Naturally the manner in which these activities took place becomes easier to approach as more

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and more antiquities start comming from the younger levels. Once a proper comprehension of the activities is attained an archaeologist comes very close to understanding culture. Thus, instead of taking a ceramic type as a cultural attribute a new archaeologist might discover that the house type is closer to cultural cognition. It is quite likely that ceramic types cross-cut various community boundaries and hence indicate a larger and erroneous cultural boundary.

Ethno-archaeology may not be surviving today as an established method of archaeological reconstruction. But no body can deny that ethnography forms an important and all too powerful knowledge base for all archaeologists. This is more so because what we are essentially seeking is not generalization but fuller understanding of culture. For instance one can ask the members of a simple community to verbalize the meaning of activities and also to explain the semiotic value of the products of these activities. One may also directly observe members of a community in a series of activities involving manufacture and value ascription of a variety of objects which play a role in the activities observed. In either case the archaeologist gathers a larger insight into human nature and the role and function of objects discovered through excavations. If not anything ethno-archaeology certainly prevents an archaeologist from attempting over-simplistic interpretation of antiquities.

Finally it must be emphasized that ethno-archaeology is never useful in drawing cross-cultural comparisons. It is a tool of analysis which should be restricted to explain the contents of a dig within the limits of its own context. *Large area comparison on mono-causal ethnographic logic can totally destroy the advantages of this tool.*

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CHAPTER - 5

ANALYSIS OF THE IMPACT OF GLOBALISATION ON THE TRIBALS

Adivasis are the most backward section of the population in India, who are facing a lot of problems all over the country, particularly in Kerala. In this state their live issues have come to the limelight along with their longstanding struggle for land. Since their employment opportunities are decreasing their means of livelihood are affected and their education, health and food security are being threatened. This study tried to examine how globalisation has affected the tribals in Kerala. Both primary and secondary data were used for this purpose. Primary data was collected using survey method, case studies of three tribal villages in three districts, and personal unstructured interviews.

The data collected from these primary sources were presented in the previous chapters. This chapter will analyse these data, which will be complimented by secondary data. The analysis is to study the overall impact of globalisation on the livelihood and sociocultural life of the tribals in Kerala. Before this is done, responses of the tribals to two other questions and their analysis are given. The first question tried to find out whether the tribals had any benefit from globalisation and the second one was to give suggestions to solve the problems the tribals are facing due to the policies of globalisation. Therefore, the analysis of data will include the opinions of tribals regarding the benefit they have received through globalisation policies of the government and their suggestions to solve the problems supposed to have been caused by globalisation like its impact on their livelihood, on their cultural and religious practices, on health, education, and on tribal women. Each section will analyse the findings from the study with the support of secondary data.

The objective of the study was to find out how globalisation has affected the life of the tribals in Kerala. A survey method was used to enquire from the tribals the effect of globalisation on their life. As part of the survey, the tribals were asked about the benefits they received from the policies and programmes of the government during the last ten years viz, the period of globalisation. This question was aimed at an objective evaluation of globalisation with its positive and negative impact on the tribals.

Table 5.1

Benefits Received from the Policies and Programmes of the Government During the Last Ten Years

Description	Number of Respondents	Percentage
Nil	355	44.4
Free ration	3	0.4
Housing	254	31.8
Housing and free ration	23	2.9
Housing and electricity	20	2.5
Housing and latrine	4	0.5
Housing and road	7	0.9
Electricity and drinking water	2	0.3
Housing, road and electricity	30	3.8
Solar lamp, free ration and fertilizer	91	11.4
Housing and solar lamps	11	1.4
Total	**800**	**100**

As is shown in the table, majority of respondents (55.6%) reported that they were recipients of some kind of government benefits. Among those who got some government benefits, a majority (32%) received housing benefits. This is followed by a sizeable section (11%) of tribals who got solar light, free ration and fertilizer together.

The study reveals that majority of the tribals received some benefit or other from the government in the last ten years. The benefits they received included housing, solar light, free ration, roads, electricity and drinking water. That the government programmes have benefited the tribals, the most marginalised group in Kerala, is positive and encouraging. But it is to be remembered that these programmes existed much before the ten year period of globalisation. Significantly none of them have mentioned

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any benefits connected with globalisation like information technology and cell phones. This means that the real benefits of globalisation have not touched the lives of tribals. Reports from case studies and interviews also confirm this factor.

The tribal respondents were also asked to give their own suggestions to solve the problems faced by the tribals in connection with globalisation e.g., the new policies of the government on privatisation and liberal import.

Table 5.2

Suggestions to Solve the Problems the Tribals Face due to Policies of the Government like Privatisation and Liberalisation

Description	Number of Respondents	Percentage
No suggestions	583	72.9
Distribution of promised land	59	7.4
Increase in the price of agricultural products	62	7.8
Use of Swadeshi products	8	1.0
Decrease in the price of ration items	7	0.9
Protest against the high price of food items	67	8.4
Change in the import policy	6	0.8
Change in the tribal policy of the government	1	1
Government should change the new economic policies	1	0.1
Total	**800**	**100**

We found that a notably large proportion of the tribal respondents could not give any suggestion regarding this question (73%). Among those who offered some suggestions, 8 percent wanted to oppose the high price of food items, another 8 per cent demanded increase in the price of agricultural products and 7 per cent wanted distribution of land promised to them by the government. Other suggestions given by a few included use of local products ("swadeshi" goods), reduction in the price of civil supply items, a change in the export and import policy, and a change in the present economic reforms.

It is understandable that most of the tribal respondents were not in a position to answer the question as to how to solve the problems arising from globalisatio, since it is a technical question. At the same time the responses of the minority who answered this question are revealing, since they seem to have understood the implications of globalisation. The highest percentage of tribals wanted the price of food items to be reduced. If the answers of other respondents are taken into account it could mean they were referring to prices of essential commodities distributed through the Public Distribution system (PDS). As will be shown later, the government had increased the price of these food items, and the tribals were not able to pay the new prices. The respondents may also have meant prices of goods sold in the open market as they have to buy them for their survival. This can also include goods manufactured and sold by private companies as well as government companies. As the government decreases its subsidy to essential goods are services, price of these goes on increasing. The message from the tribals seems to be that subsidies given to essential commodities should be continued and that government should not withdraw from areas connected with the basic needs of the poor. The next highest group of respondents suggested increase in the price of agricultural products. This can mean that they understood the connection between the decrease in their price and the liberalisation policy of the government. They implied that imports should be curbed and support price should be given to agriculture so that the farmers get a remunerative price. This is corroborated by the suggestion of others to change the import-export policy of the government, which is directly linked to the policy of liberalisation. If there is adequate price for farm products, Adivasis who have land and cultivation can benefit. Those who do not have land also can benefit as the farmers will restart agricultural operations which will give employment to the tribals who are mostly dependent on the agricultural sector. A small percentage of respondents were for "Swadeshi goods". This could mean that they were against flooding the market with foreign goods often imported from other countries, which are beneficial mainly to the rich and middle-class and not to the marginalised groups like the tribals. This is also connected with the above-mentioned suggesion to change import-export policy of the government. A significant suggestion by a small proportion of respondents was that the present economic

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system should be changed, meaning that the globalisation policy of the government should be reverted. This can mean that they are dissatisfied with the globalisation process which has made their life very miserable. A quarter of the respondents suggested that the government should distribute land which it had promised to the tribals. This means that they are conscious of their right to land.

The villages of Mulli and Marakadavu, two out of the three villages selected for case study, do not find any hope for Adivasis in the future. They assert that the government has destroyed their life, livelihood and culture totally, especially with its policy of globalisation. The politicians and the government officials steal all the funds meant for tribal development. Tribals will gradually die of illness and starvation, and their very survival is in peril. The tribal respondents at Marakadavu feel that they can survive if they get land for cultivation. They will have to organise themselves and work jointly for this, since real liberation is possible only through the tribals themselves.

In contrast, the tribals of Paliyakudy in Idukky district, the third village included in the case study, believe that the Adivasis have a future if they get education at all levels. Information technology, if made available to the tribals, will help in their development. The other two villages were skeptical about the relevance of information technology for tribals. They said, "What is the use of information technology for those of us who are illiterate? We cannot even dream of it since it involves a huge expense."

ANALYSIS OF THE IMPACT OF GLOBALISATION ON THE LIVELIHOOD OF THE TRIBALS IN KERALA

It is interesting to note that none of the tribals interviewed mentioned any benefit of globalisation for tribals. Tribal respondents in the survey study and case study also did not mention any merits of globalisation for tribals. The data regarding the negative effects of globalisation as revealed in the study are analysed below. This section on livelihood will include land, employment, wage and availability of essential commodities.

LAND AND CULTIVATION

The study shows that 59 per cent of the tribals do not have land at all. A great percentage (45%) has less than one acre of land. Case studies of the three villages, Paliyakudy in Idukky district, Mully in Palakkad district and marakadavu in Wayanad district confirm these findings of the survey. Though more than half of the poeple of these villages have land, it is less than 50 cents mostly owned by joint families. The study confirms the reality of the tribals in the state where a good percentage does not have land. Even when they have a bit of land it is not a real source of income and so they have to go for wage labour. The tribals lost their land over the last many years, though only 1 per cent lost land during the last ten years of the globalisation period. land is not only a source of livelihood but also a symbol of sociocultural life for the tribals everywhere. Naturally providing land for the future existence of tribals is a must, and it is mainly for land that the tribals have been agitating but have not yet got it.

Among those who have land, 28 per cent were practicing subsistence farming have been reduced to 13 per cent. In its place those having commercial farming increased to 22 per cent from 3 per cent ten years back. This means commercialisation of susistence economy is increasingly taking place in kerala within the globalisation period. This is a reflection of the increasing tendency of cultivation of cash crops in the state as a whole. But it has its disadvantages. First of all, tribals who have a bit of land are affected by the decrease in price of cash crops due to globalisation. Secondly, when the price of cash crops like pepper and coffee crashed, the tribals lost not only the only means of income but also the food crops on which they could survive. That is why the question whether the tribals with a little land should go in for commercial crops or should be encouraged to cultivate food crops, becomes very relevant. if they cultivate food crops they can survive on them irrespective of fluctuations in price of cash crops. In short, distribution of land to the tribals is very basic to their welfare, which can be ensured only with the assistance and encouragement of the government.

TRIBAL FOREST BILL

The Scheduled Tribes and other Traditional Forest Dwellers' Recognition of Forest Right Act, 2006 was passed by the Parliament in 2006 and has become a law throughout the country. This law gives certain rights and privileges connected with forests to the tribals. Since tribal land and life are very much related to forest it is pertinent to see some of the advantageous provisions of this law for the tribals:

1. The law gives rights to the forest dwelling scheduled tribals to hold and live on forest land and to cultivate it for livelihood.

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2. The tribals are given rights to collect minor forest produce as a means of livelihood.
3. Rights of settlement and ownership of traditionally owned forest villages by tribals can be converted into revenue villages.
4. If forest land occupied by tribals is taken away for common purposes, they will have right to full rehabilitation with alternative land.
5. Grama Sabha of the area can decide on the rights of tribals over forest land where they are living, and determine the nature of resettlement and rehabilitation of tribals evicted for common purposes.

Since the Tribal Forest Act is about rights and occupation of forest land by tribals, it is relevant for the tribals in Kerala also. The tribals who are living within the area of reserved forest or close to it as in Idukky or Plakkad district, could be given forest rights and title deed so that they are not evicted on the basis of forest laws, and are also able to get the benefit of electricity and other public utilities. Secondly, tribals are allowed to collect minor forest produce as permitted by the Act so that they have supplementary income for survival. Thirdly, tribals are to be properly resettled and rehabilitated in case of eviction for common purposes. The central and state governments should make necessary rules and legal structures to implement the provisions of this act so that the tribals in Kerala get all the benefits of this Act at the earliest. Tribal organisations and NGOs working among tribals can facilitate in getting this long awaited law operationalised in the state.

IMPACT OF GLOBALISATION ON THE EMPLOYMENT AND INCOME OF THE TRIBALS

It was the availability of employment of tribals, which was most directly affected by globalisation. According to the survey 29 per cent of men have only 50-100 days of work, while the figure was 5 per cent ten years back. While 34 per cent had up to 300 days of work a year ten years back, only 3 per cent of tribals have that much at present. among women, 31 per cent had 100-200 days of work a year ten years back while only 11 per cent have that many days at present. While 16 per cent of women had 200-300 days of work ten years ago their percentage had dwindled to 2 per cent day. If we take into account the fact that 21 per cent of men and 29 per cent of women do not have any work, we can gauge the pathetic situation of tribals in kerala. The drastic decrease in the number of working days has made the tribals more miserable during the period of globalisation. This is confirmed by the rate of working days for the trigbals in the villages chosen for case studies. At Marakkadavu in Wayanad district, the average working days for men was 100-120 days while earlier they were getting 250-300 days of work. In Mully village of Palakkad district, 100 tribals worked in the nearby tea estate ten years ago, while at present only ten get work there. This is because there are no agricultural operations in the tea estate due to fall in price of tea. Some tribal youth were getting jobs in textile mills in Coimbatore, Tamil Nadu but they had lost them since many of these factories had closed down rendering thousands jobless. This is apparently due to globalisation since the traditional small scale factories were not getting the support and subsidies during the pre-reform period that they used to get. We can also notice the much lower number of working days a year for tribal women than for men. This reveals the poor economic situation of tribal women.

The daily wage for 39 per cent of men was Rs 51-100 ten years back, while only 1 per cent receives that much wage at present. Only 0.4 per cent of women get the same range of wage now compared to 4 per cent ten years back, at present only 25 per cent receive that much. In the matter of wage also tribal women were discriminated against and had greater misery. The average daily wage for men and women at present is Rs 68 and Rs 40 respectively. Though there was no decrease in the wages of tribals there was no increase either. The villages selected for case study also did not report any decrease in wage during the last ten years. Since the price of essential commodities had increased considerably during the last few years, the tribals got less quality of life with the same money than in the previous years. This is all the more so since the number of working days has come down drastically. Thus the wage of the tribals was adversely affected during the period of globalisation.

The data from the survey and case studies reveal that the average working days per year for the tribals had decreased drastically both for men and women. The reason cited by the respondents themselves is the crash in prices of practically all the cash crops, especially coffee, tea and pepper. Fall in prices of cash crops was attributed by

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the respondents to the import of the same products from other countries during the post-reform period. Import of these agricultural products was mainly due to the reduced rate of excise and import duty on these products. In the post-reform period this is related to the liberalisation policy of the government based on the regulations of the World Trade organisation. According to the policy of the government, it had to lower the tariff for imports and allow import of many goods liberally. Thus coffee and tea from other countries especially Sri Lanka were imported to India at a much cheaper rate than that of Indian products, resulting in drastic fall in the price of these commodities. This happened during the last ten years when the government vigorously implemented the policies of globalisation.

When the price of cash crops fell below the remunerative price, the farmers either stopped or drastically reduced agricultural operations for these crops, especially coffee and tea. Since the tribals were dependent on these farms and estates for wage labour, they lost their regular employment there. Thus the average working days of the tribals per year decreased as was shown in our study, resulting in less income, leading to adverse effect on their livelihood. Thus the policies of globalisation followed by the Indian government directly impacted on the livelihood of the tribals, the most marginalised group in India and Kerala, and made their life more vulnerable than earlier.

Almost all those who were interviwed personally asserted that there was decrease in job opportunities for tribals as a result of globalisation. "Adivasis face the greatest threat from shortage of employment which is related to the policy of globalisation", said one tribal woman leader. In the words of another one who was interviewed, "as a result of globalisation the Adivasis face unemployment, poverty, disease, starvation and death. They are on the verge of extinction." another remarked, "Adivasis are the first victims of globalisation." Another remarked, "Adivasis are the first victims of globalisation." thus according to the respondents of personal interviews globalisation has caused poverty and misery for tribals in Kerala.

Adivasis who have some land are also adversely affected since they do not get remunerative price for their cash crops. Thus income from their land, which is often less than one acre, is also considerably reduced. Since tribals totally depend on agriculture or agriculture - related occupation, their livelihood is seriously affected. As one tribal said, "When there is no cultivation due to crash in price of agricultural products we are badly hit". This is all the more so since their traditional sources of livelihood like forest produce and cottage industry are not existing now. "The crisis in agriculture which has come as part of globalisation has affected the tribals most", said one social activist.

The survey study revealed that the number of working days for tribals during the last ten years of globalisation has considerably decreased. The village selected for case study also confirmed the decrease of working days for tribals. Thus decrease in employment of tribals during the post-reform period is a fact identified and confirmed by the survey, case study and interviews. So the most glaring effect of globalisation on tribals in Kerala is the reduction in their working days leading to deprivation of their source of livelihood.

GLOBALISATION AND AGRICULTURE IN INDIA

From the survey study, personal interviews and case studies it was seen how most of the tribals in Kerala lost their land, their most important source of livelihood. When they lost their land, the only means of livelihood available to them was wage labour. Thus most of them started working in plantations of cash crops and paddy fields as coolie workers. When the price of agricultural products drastically came down, farming operations got reduced, resulting in decrease in employment. The Adivasis who were depending on agriculture were thus badly affected. The study showed how the employment available per year for the Adivasis decreased, and thus their economic well-being was affected. Globalisation appears to be the main reason for the crisis in agriculture in India, particularly in Kerala. If there is a drastic fall in employment in the agricultural sector, how is it related to the liberalisation policy of the government regarding agriculture? This section examines whether globalisation has affected agriculture in India and in Kerala in order to see its impact on the tribals in Kerala.

Studies show that during the period of globalisation from 1991 onwards agriculture in India was facing a crisis in production and profits for the farmers and there was an overall decline in agricultural growth. The status report of the agricultural ministry says that the annual average farm

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growth rate has declined from 4.9 per cent in the eighth plan to 1.3 per cent in the tenth plan. The share of agriculture investment in Gross Domestic Product (GDP) delcined from 1.99 per cent to 1.66 per cent during the same period (Parsai, 2006). The period referred to is the period during which the policy of globalisation through structural adjustment was implemented. This is explained by the pattern of the production of agricultural crops during the last five years.

Table 5.3

Agricultural Growth in India

	1998-1999	1999-2000	2000-2001
Food Production (in cr. tonnes)	20.35	20.89	19.90
Change from previous year (%)	5.8	2.7	– 4.7
Rate of agriculture growth (%)	178.0	176.8	170.6
Change from previous year (%)	7.7	–0.7	–3.5

***Source** : Economic Survey 2000-2001 (Kunjikannan, 2001:30)*

The table shows that the average growth of crop index fell from 5.8 per cent during 1998-99 to – 4.7 per cent during 2000-01. This was due to the fall in investment in irrigation projects, which subsequently affected the productivity of small and marginal farmers. To a certain extent, the drastic fall in growth rate can be explained by the drastic reduction in the investment in agriculture. Other indicators in agriculture explain it further. "The annual growth rate of food grain yield declined from 16 per cent in 1991-96 to 2.9 per cent in 2001-05". This is the period in which the policy of liberalisation was implemented. The area under cultivation declined from 128 million hectares in 1981-86 to 120-125 million hectares in 2001-2005. May be due to the fact of decline in growth rate of food grain and its area of cultivation, 27 per cent of farmers surveyed found farming unviable. As a result of these developments the number of cultivators declined from 110.70 million in 1991 to 103.63 million in 2001. it was also pointed out that more than 20 per cent of farmers were below poverty line in India in 1999-2000. No wonder M.S. Swaminathan, Chairman of the National Commission on Agriculture said, "Indian agriculture is in a state of serious crisis. The rate of growth in food production has fallen below the population growth rate".

The impact of crisis in agriculture was seen by the increasing suicide rate among farmers, of which the highest rate was in Rajasthan – about 2000 suicides in three years of 2000-2002, and 500 in 2001 alone. Thus among the several reasons for farmers' suicide, one is the economic reforms that were a part of globalisation. "Their tale of woe began in 1990s when India took its first step to liberalise, and worsened progressively from 1995". The farmer's suicide in Andhra Pradesh, Karnataka, Maharashtra and Kerala indicates the desperate situation of farmers due to the crisis in agriculture. According to Chalam, "The first victim of the new economic policy appears to be the agricultural sector. The subsidies of the sector have been withdrawn and prices of most of the inputs have been raised".

Thus we find from authoritative sources that agricultural growth declined drastically in the 1990s with the decrease in area of cultivation and in the number of cultivators in the farm section. This created a real crisis in agriculture, and a concrete expression of this crisis was the increasing number of suicide cases in a number of states in India. The reason for this crisis was a declining trend in investment in agriculture in the form of irrigation, subsidies etc. This phenomenon was part of the government's policy of reducing subsidies and withdrawal of government support to agriculture, which was a part of its policy of liberalisation and globalisation based on regulations of WTO and World Bank. Reduction in subsidy to agriculture was an important regulation of WTO enforced in most of the countries. Thus the crisis in agriculture due to decline in production and investment was found to be caused by the policy of globalisation followed by the government.

GLOBALISATION AND AGRICULTURE IN KERALA

If there was a crisis in agriculture throughout India as a result of globalisation, it was particularly so in Kerala with its over-dominance of cash crops. Special reasons related to cash crops have aggravated the problems in Kerala, which will be analysed in this section.

Cultivable land in Kerala got reduced from 5.03 lakh acres in 1994-95 to 3.36 lakhs in 2001. The price of practically all the major cash crops in Kerala crashed to an unprofitable level during this period. "The major tea and other plantations have been shut down throwing thousands out of work in the past two years". Because it was not profitable to cultivate tea, many tea estates were closed down in Wayanad and Idukky, the major tea

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producing districts. many coffee estates in Wayanad also stopped agricultural operations. This resulted in massive loss of job for thousands of labourers, mostly tribals and Dalits. As one farmer from Wayanad put it, "the wages are down, pepper, coffee and everything is down. only debt burden is up".

The impact of globalisation on cash crops like tea and coffee in kerala was enormous. In Peerumedu taluk of idukki district, an important taluk for tea, 50 per cent of the estates are in crisis with the owners leaving the estates or declaring lockout. Price of one kilogram of tea that fetched Rs 14 in 1996-97 got reduced to Rs 3 in 2000. Due to slash in price the estates got paralysed. Out of 36 estates in the taluk 22 got closed down and About 40,000 workers went out of job. Water, electricity, health care, education, accomodation and other benefits which were due to them as estate workers were no more available to them. So they were facing starvation and poverty. "In Peerumedu, poverty is writ large on faces here. Malnourished women, near naked children, disillusioned men, all stand for that ugly reality".

Tea industry is very important for kerala and for the Indian economy. Kerala produced 10 per cent of the national output, in 2000-01 and 34 per cent of tea produced in South India. It is spread in 37,000 hectares of land, which is 8 per cent of the total area of cultivation in the country. Around 84,000 workers are employed in this industry. One of the main reason for the crisis in tea industry was the fall in price due to import of cheap tea from Sri lanka, which was on account of the liberalisation policy of government. Cash crops like tea and coffee were liberally imported to India since tariffs and import duties on these products were drastically reduced as part of the liberalisation policy of the government. Free flow of goods among countries with low import duties was a requirement of the globalisation process. Therefore, crisis in agriculture in India, particularly in Kerala, is due to globalisation specifically enunciated by WTO and GATT.

As many as 64 per cent of the farmer households in kerala were in debts in 2005 compared to the national average of 49 per cent, according to national Sample Survey Organisation (NSSO, 2003). The debt amount of farmers in the state stood second in the entire country. The total outstanding loan per-household amounted to Rs. 33,907 while the national average was Rs. 12,585. The consequences of the agricultural crisis was the debt trap of farmers about which the above data speak. Farmers committed suicide in large numbers due to unrepayable loan. Over 600 farmers committed suicide between 2001 and 2005 in Kerala.

According to a statement by Chief Minister, A.K. Antony in 2003 in the Legislative Assembly, the number of suicide cases in kerala increased from 17,048 in 1996-97 to 19,774 in 2000-02, which corresponds to the period of globalisation in India. The suicide rate went up from 15 per 1,000 population in 1997 to 27 per 1,000 in 2003. The reason given by the State Commission for Women is globalisation. Kerala is "at the receiving end of adverse effects of globalisation, commercialisation and privatisation". In other words the economic crisis triggered by globalisation is one of the major causes of suicide in Kerala.

CRISIS IN AGRICULTURE AND WORLD TRADE ORGANISATION

"WTO and GATT agreement have very complex conditions and laws. Agricultural agreement and intellectual property rights implemented as part of these, has created very adverse impact on Indian agriculture". The price of coconut, pepper, ginger, tea, nutmeg etc. fell due to the agreement of the Central Government with countries like Sri Lanka, Indonesia and Malaysia. These countries were allowed to import these products duty free. This was admitted by the Central Agricultural Minister, Ajit Singh, when he said that Kerala was the most badly affected state in Idnia, especially since Kerala depended on on crash crops the price of which crashed due to WTO and globalisation. Due to the import policy of the Central Government the price of major cash crops – coconut, rubber, pepper, coffee, tea, arecanut, ginger, cardamom, turmeric – fell steeply in the years 2000-2012. The agricultural sector in the state thus incurred a loss of Rs 6645 crores. "Drought and disease hurt us. But it is free trade policies that have driven prices down" said a farmer from Wayanad.

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The price decline of major cash crops in Kerala due to the import policy of the government during 1995-2003, the post-reform period, is given below :

Table 5.4

Average Price Rate of Major Cash Crops in Kerala (Rs/ quintal)

Crops	1995	1999	2003 (till Nov.)
Dressed Coconut (1000)	3,104	4,933	4,601
Rubber (RSS–4)	5,204	3,148	4,770
Pepper	7,411	19,055	7,375
Cardamom	24,800	57,700	27,000
Ginger	6,169	6,776	5,066
Arecanut	5,410	10,411	4,017
Coffee	9,000	6,800	2,300
Tea	4,100	6,200	4,950

Source : State Agricultural Crop Price Fixing Board, Eco-state Department.

Except coconut all other crops faced a fall in price. thus we find that globalisation has adversely affected agriculture in Kerala more than in any other state. That most of these crops are grown in the three districts we have chosen for study and where Adivasis depend on these crops for their livelihood, is again telling on their plight due to liberalisation. Due to lack of education and other reasons. Adivasis in Kerala did not have skills to take advantage of globalisation. Rather they suffered from deprivation since the employment opportunities decreased for them due to the crisis in agriculture, that was on offshoot of globalisation.

GLOBALISATION AND ADIVASIS IN KERALA

The National Sample Survey (2003) states that one in two Adivasis lives below poverty line in rural India, which is almost double the figure for the general population in rural India. "What is worse, the only social category whose below-the-poverty line population increased in the 1990s, is the Adivasis". "In terms of changes in poverty in the 1990s, it is found that while scheduled caste, agricultural labour (rural) and casual labour (urban) households experienced decline in poverty on par with the total population, scheduled tribe households fared badly in both segments" (Sundaram and Tendulkar, 2003 : 5263). In other words, during the post-reform period the economic situation of tribal population in India got worsened. This is confirmed by the present study about tribals in Kerala.

Because of the crisis in agriculture; the worst sufferers were the SC/ST population who mostly depend on agriculture. "This would definitely hit the reservation groups particularly scheduled castes and scheduled tribes and backward classes as these are the sections which largely rely on agriculture" (Chalam, 1994 : 37). This is what happened particularly for Adivasis in Kerala.

No wonder the life of Adivasis became miserable due to the problems in agriculture, which was a direct consequence of globalisation. "For tribals, globalisation is associated with rising prices loss of job security, lack of health care and tribal development programmes" (Raj, 2005 : 9). This is true of Adivasis in other parts of India too. They were adversely affected by globalisation. "In the economic sphere globalisation had a negative impact on the Adivasis causing migration of youth to big cities, effecting not only a brain drain but also a generation drain" (Tirkey, 2004 : 6). The reference is to Adivasis in Jharkhand and madhya Pradesh, who migrated to the cities in the north when they lost work in the agriculture fields in their own places. In Kerala also a similar phenomenon has happened to Adivasis in Wayanad district, who have migrated, not to the cities for jobs but to places like Gudalur in Tamil Nadu for agricultural operations. This was destroyed even their sociocultural life.

Wayanad, with the highest concentration of tribals and with major cash crops of coffee, pepper and tea, is a typical example of how agricultural crisis has affected the livelihood of tribals. "The huge fall in prices has savaged the district of Wayanad. Pepper price has fallen from Rs 27,000 a quintal in 1997 to about Rs 5,400 in 2004. Grown in 70,000 hectors, farmers and their labourers suffer" (Sainath, 2004). In real terms the collapse in price of pepper and coffee affected the labourers who depended on them and Adivasis were the worst sufferers. One of the main reasons for this situation was the dumping of imported goods, which was directly due to liberalisation. "In 2002-03 India imported 6,204 ton pepper without any import duty. It was a big increase from 1241 tons in 2001-02. Besides, 15,100 ton pepper was imported from different countries"; remember pepper is a major cash crop in Wayanad (Deputy Minister"; Lok Sabha, 2003).

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"The agrarian crisis and farmers' distress in Kerala over the past one decade are loosely linked to the neo-liberal policy regime implemented in the country in the recent past. The association between the two is more in the regions of the state which are heavily dependent on export oriented crops, such as coffee and pepper. The worst affected are the small farmers, as they are more vulnerable to crop and price fall. They find it extremely difficult to pay back the loans they have incurred, in order to grow crops and survive" (Mohankumar, 2006 : 1553).

The present duty has shown that the annual working days of tribals drastically decreased since there was no employment in farm states of crash crops where most of them worked. the farm estates were not operating since there was no remunerative price for these products. Price fell due to liberal imports of the same crash crops as part of liberalisation policy of the government. Thus this policy has directly reduced the employment opportunities of the tribals. Thus globalisation has adversely affected the livelihood of the tribals in Kerala.

AVAILABILITY OF ESSENTIAL COMMODITIES FOR THE TRIBALS

Food security by means of availability of essential commodities is very important for the people, especially the poor. So, essential commodities like food grains and kerosene are provided to the people through the Public Distribution System (PDS) at subsidised rates. This has been a great source of food security for the marginalised people like Dalits and tribals. As part of livelihood issue for the tribals, the study has examined the availability of essential commodities for them during the last ten years.

It was found from the study that 83 per cent of the tribal house-holds had ration cards and most of them regularly purchased essential commodities from PDS. this was confirmed from the villages selected for case studies. In other words, the tribals fully depended on PDS for their food requirements especially since their traditional sources of food from forests and their own farmland became scarce. But during the past few years the situation has changed.

The study has shown that 57 per cent of the tribals get cereals (rice/ wheat) from private shops and only 49 per cent get it from PDS. The rest get from their own farms. The reasons given are high price and poor quality of food grain. So tribals buy from privte shops at high price but of good quality and sufficient quantity. This means that the tribals either spend more on food or they do not buy it enough. Either way their nutritional needs and living standard are adversely affected. Thus they do not get the benefit from PDS as they used to get earlier, and by depending on the private shops for essential commodities, their food security is badly affected. The study has also revealed that the tribals spend more on essential commodities like kerosene and rice than on fish, meat, milk and grains, since about half of them (49%) purchase them from private shops.

GLOBALISATION AND AVAILABILITY OF ESSENTIAL COMMODITIES

The experience of the tribals with regard to essential commodities through PDS is confirmed by the national level situation of PDS as followed by the government in the post-reform period. The study analysed from secondary sources the macro-situation of PDS and how it has affected the people, especially the marginalised groups all over the country, and how it is related to the globalisation policy of the Central Government. Although there was increased in food production and food stock, there was decrease in distribution through ration shops.

Table 5.5

Agricultural Production and Distribution in Kerala (in crores of tonnes)

Year	LAP per cent	Food Production	Food Stock	Ration Distribution
1991-92	– 2.0	17.84	1.96	2.08
1992-93	4.1	17.95	1.79	1.88
1993-94	3.6	18.43	2.81	1.64
1994-95	4.5	19.15	2.60	1.40
1995-96	– 2.1	18.04	2.26	1.53
1996-97	9.5	19.94	1.98	1.83
1997-98	6.2	19.23	2.36	1.78
1998-99	8.2	20.30	2.63	1.84
1999-2000	– 1.5	20.89	3.08	1.70
2000-01	– 4.6	19.61	3.55	1.21

Source : CMIE, May 2001, Economic Survey 2000-01 (Kunjikannan, 2001 : 64).

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The table shows that the distribution through ration shop has been steadily decreasing during the period 1991-2001, though there was steady increase in food procurement and even production. Food-stock steadily increased from 1.96 crore tonnes in 1991-92 to 3.55 crore tonnes in 2000-01, while the distribution through ration shop has steadily and steeply decreased from 2.08 crore tonnes in 1991-92 to 1.21 crore tonnes in 2000-01. This means there was a decline in the purchase of food grains from the ration shop by the people. "After the reform, off-take of wheat and rice as a percentage of the allocation has shown considerable delcine. In the years from 1990-91 to 1992-93 this percentage was generally over 80 per cent but in the next three years from 1993-94 to 1995-96 it came down to below 50 per cent in the case of wheat and to a little over 60 per cent in the case of rice". "..... since, the difference between the Public Distribution System (PDS) retail prices and market prices became marginal, the not so poor left the PDS, and owing to unaffordable prices the poor cut their comsumption. The resultant effect on nutrition and hunger had study revealed that the families of landless labourers and marginal and small farmers reduced their off-take by over 50 per cent" (Teltumbde, 2005 : 5). "..... During two years of reforms there was 8.43 per cent fall in per capita availability of cereals and 12.02 per cent fall in that of pulses" (Teltumbde, 2000 : 5).

One main reason was that as part of the globalisation policy the Central Government reorganised the ration distribution among people on the basis of Below Poverty Line (BPL) and Above Poverty Line (APL) and steadily increased the price of ration goods. As a result, people did not have purchasing power to buy even the ration food grains. It is a clear case of how the poorest people in India are hit by the policy of reducing the subsidy as part of liberalisation.

The table 6.6 shows the steady increase of price of ration rice and wheat. In 2001 the government increased the price by 30 per cent after creating the BPL – APL categories. This price increase has seriously affected the food security of SC/ST people.

Table 5.6

Wholesale Price Increase of Ration Food Grain in India (Rs/ quintal)

Year	Rice			Wheat
	Normal	Fine Super	Fine	
1.4.91	289	349	370	234
28.12.91	377	437	458	280
11.1.93	437	497	518	330
1.2.94	537	617	648	402
1.6.97 APL		650	750	450
BPL	350	350	250	
1.12.97 APL		700		250
BPL	350	350		250
29.1.99 APL		905		650
BPL	350	350		250
1.4.99 APL				682
1.4.2000 APL		1180		900
BPL	590	590		450
1.7.2001 APL	830			610
BPL	565			610
	300			610

Source : FCI documents (ibid.).

Retail price of food grains given through Public Distribution system (PDS) also got increased during the reform period.

Table 5.7

Retail Price Increase of Ration Food Grain in India – 1991 – 2001 (Rs/ kg)

Items	1990	1991	1992	1993	1994	1997	1999	2000	2001*	2001**
Rice	3.28	3.78	3.94	5.02	6.84	7.20	9.65	12.50	12.50	9.00
Wheat	2.26	2.56	3.02	3.80	4.35	5.00	7.20	9.65	12.50	9.00
Sugar	5.25	5.56	6.90	8.40	9.15	–	–	12.60	13.25	
									BPL ONLY	

Source : Ration shops (Kunjikannan, 2001 : 69)

According to the table the retail price of rice and wheat increased four-fold and of sugar three-fold during the said period. Between 1991-2001 the price of ration food grain increased ten times. This had impact on the intake of ration food grain since the card holders did not have the purchasing power to buy ration food

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at the increased price. The government increased the price of ration food continuously so that it was able to reduce its subsidy on food as part of the requirements of globalisation process guided by WTO.

In the case of Kerala, the price of ration grain was raised nine times in the 1990s. The price of Rs. 3.20 per kilogram of rice in 1990 was increased to 13.20 in 2000. This is clearly the globalisation period. As a result, sale of food grain decreased considerably. The people who are not able to purchase food grain are the marginalised people like Dalits and tribals. The tribals particularly were affected by this as was shown from the case studies. It was said that the tribals did not get the full benefit of ration shops due to high price, less quantity and poor quantity. In other words, the globalisation policy directly affected the in-take of essential commodities of the tribals, and their poverty increased.

Table 5.8

Sale of Ration in Kerala (in metric tonnes)

Year	Rice	Wheat
1995	11.32	4.23
1996	14.04	4.92
1997	16.17	3.52
1998	16.39	4.58
1999	13.58	2.71
2000	6.14	0.62

Source : *FCI documents (Kunjikannan, 2001 : 101).*

The decrease in the purchase of ration goods is shown in the table. The sale of rice through Public Distribution system (PDS) got reduced from 16.39 metric tonnes in 1998 to 6.14 metric tonnes in 2000. Similarly sale of wheat through PDS has came down from 4.58 metric tonnes in 1998 to 0.62 metric tonnes in 2000. The reason is that the purchase of these commodities by consumers was low.

"Wherever economic reforms were implemented, they have worsened the situation of the masses of poor people in absolute as well as relative terms" (Teltumbde, 2000 : 17). The Dalits and the tribals were hit the hardest in India. "While those at the centre have benefited from globalisation, the already marginalised have often been left further behind" (Fernando, 2006 : 19).

The government was consistently increasing the price of ration goods so that it could reduce its subsidy on PDS. For this, they also classified people as BPL and APL and reduced a substantial quantity of essential goods for APL category. reducing the subsidy for and withdrawal from PDS by the government is necessitated by the conditions of WTO and IMF. As government's withdrawal from the socioeconomic sphere is part of the proclaimed policy of globalisation, it is enforced on PDS meant for the poorest of the poor in our country. Thus they lose the food security they have been enjoying so far without substituting alternative source of income or food. The tribals in kerala are mostly illiterate and unskilled, and their traditional coolie work has been substantially reduced as an aftermath of the liberalisation policy of the government. When their purchasing power is lowered even to buy the existing commodities in PDS, they will be all the more hard-pressed to buy at a higher price. Naturlaly they are compelled to buy essential commodities from private shops at a highe rprice. This badly affects their spending on other essential needs of the family like health and education. Thus globalisation has adversely affected the tribals, the most vulnerable section of the poeple of Kerala.

It was shown how livelihood of the tribals in Kerala were adversely affected by the policy of globalisation. Earlier land, employment, and essential commodities through PDS were means of livelihood and life support systems for tribals. Land is no more a source of livelihood as they are deprived of it. Their wage from employment in farmland and estates of cash crops got reduced substantially as agriculture was not remunerative with low price for agricultural products. With limited income, the PDS was a great source of support for the tribals especially when their employment and other source of livelihood got reduced. But as the government was steadily increasing the price of food grains distributed through PDS, they were forced to depend on private shops for the same, probably at a higher price and with reduced quantity. With their low employment and subsequent low purchasing power, the in-take of the tribals naturally was less than in earlier times. This had an impact on their life and health. Thus the policy of globalisation has adversely affected the food security of the tribals, and in turn their life itself.

Poverty and misery of tribals have increased as an outcome of globalisation. When there is lack of sufficient

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work as mentioned above, their main source of income is lost. since they have hardly any other source of income, they face deprivation and starvation. Besides, their suffering further increased due to the rise in price of essential commodities including those provided through the Public Distribution System. Since the tribals do not have purchasing ower they are not able to procure food even from the ration shop. Naturally they suffer from poverty and malnutrition. I the words of one who was interviewed, "as a result of globalisation the Adivasis face unemployment, poverty disease, starvation and death. They are on the verge of extinction." Another remarked, "Adivasis are the first victims of globalisation." thus from the data from survey case study, interview and secondary sources it can be concluded that globalisation has caused more poverty and misery for tribals in kerala than in the pre-reform period.

ANALYSIS OF THE IMPACT OF GLOBALISATION ON THE SOCIOCULTURAL LIFE OF THE TRIBALS IN KERALA

This section analyses the result of the study on the impact of globalisation on the cultural and religious practices of the tribals in Kerala together with data from secondary sources. There are five areas covered in this section – cultural and religious practices, health, education, tribal women and organisation of the tribals.

CULTURAL AND RELIGIOUS PRACTICES OF THE TRIBALS

To understand the effect of globalisation on culture and religion, questions were asked in survey method about the effect of television and cinema on the culture, values and religion of the tribals. More than half of the respondents (51%) stated that television and cinema had negative impact on the family structure and customs of the tribals. The areas influenced by them are their lifestyle, fashion, dressing, language and food pattern. Forty-three per cent of the respondents also said that the traditional tribal values like sharing, common ownership, and cooperation had declined due to the modern media. The study also showed that more than half of the respondents (55%) are of the opinion that they had lost the religious and non-religious tribal customs and practices due to the influence of media. This was also confirmed by the tribals from the villages chosen for case studies.

In the village of Mully in Palakkad district, people used to watch video films throughout the night for days together so that "Koothu", their main cultural form had fotally gone out of use. The same village reported that tribal values like cooperation and sharing had got diminished due to the influence of modern media and life. In Paliyakudy in Idukky district the tribals admitted that modern media and life had destroyed their customs so much that the new generation did not even own them up. They added that only a traditional dance connected with death has survived. The village of Mully and Paliyakudy specially reported that advertisements in television and films had developed consumerism among the tribals by motivating them to buy non-essential commodities with their meagre income, thus curtailing the consumption of essential food items. In Paliyakudy many tribals had got into the debt trap by buying such non-essential consumer goods. Since advertisements in electronic media are very much part and expression of globalisation it can be said that globalisation has adversely affected the simple lifestyle, customs and religious practices of the tribals.

The respondents of personal interviews also confirmed that tribal culture has been adversely affected by globalisation and its vehicle of media and communication. "Given the sociocultural beckground, the tribals are influenced by consumerist culture," said a social activist. As a result of their contact with the mainstream culture they have started losing interest in their traditional lifestyle and religious practices. Tribal values of cooperation, sharing and family bond have been the first casualties. The process has been accelerated with modern media like television and cinema. The respondents also feel that modern media as part of globalisation has had a bad impact on the art and artefacts of the tribals. Thus the personal interviews largely confirm the result of the survey schedules, namely, that globalisation has had an adverse impact on the tribal culture. This is also endorsed by authors on tribal culture. This is also endorsed by authors on tribal culture: Globalisation is oppressive and would suppress local languages and cultures with its market forces" (Seabrook, 2001).

Some of those personally interviewed said that it was not possible to protect tribal culture from globalisation due to the following reasons : (1) Tribal identity and culture are already lost or are in danger of being lost to the mainstream culture. (2) Religious practices, dress style

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and even body features are changing. (3) The power of media can destroy the culture of the tribals. (4) People in power impose their culture on tribals. (5) Forest and forest produce, which were closely related to tribal culture, are disappearing. (6) Destruction of agricultural sector also has an indirect impact on their culture.

Most others who were personally interviewed of the opinion that tribal culture could be protected from the impact of globalisation. They have given various suggesions to protect tribal culture and identity.

1. Tribal language, lifestyle and customs, arts and artefacts, and festivals should be encouraged and protected. The tribals should be helped and encouraged to value their tribal identity and culture.
2. The living environment of tribals should be restored. Land, soil, trees, forest, water, air and other natural resources should be made available to them. Land and food security should be provided based on sustainable development through farming based on natural manure, and bio-diversity agreement and benefit sharing should be implemented.
3. Developmental activities, which endanger their traditional way of life, should be given up.
4. Programmes envisaged in Panchayat Raj for tribals including self-rule should be implemented.
5. Tribals language should be used in lower primary school and tribal culture should be included in the school syllabus.
6. Cultural and artistic organisations should be formed to protect tribal culture. they should organise cultural 'melas', and NGOs and government could support them in such activities.
7. Encroachment and occupation of tribal land and areas by outsiders who destroy tribal culture should be prohibited.
8. Tribals should be motivated to give up emulating the bad habits of upper class and caste and mainstream culture, particularly the use of drugs and drinks.

Taking into account the two views of the respondents about tribal culture given above, one would say that integration of tribal life into modern life should be fostered while maintaining the identity of tribal life and culture so that the tribals live and develop as tribals. If the good customs and practices of tribals are not kept up the tribals will lose their identity as tribals, which can create problems for the indigenous people as has happened in other parts of India and the world. Here the remark of an author is relevant. "If the unequal exchange that marginalises these tribals is to be reversed, then they must not be left in isolation, not even in the mistaken notion of preserving their tribal identity. This only marginalises them still further. For ethnic identity is dynamic, not static..... We believe that like every human identity, tribal ethnic identity too must be dynamic, and it actualises the human potential that is present in every human group. The danger, however, in romanticising them is to condemn them to primitivism.....". In other words, preserving the tribal identity and culture does not mean that they are kept as museum pieces. Like all dynamic societies, tribal society also should interact with mainstream culture so that the dynamic aspect of the tribal culture is kept up.

What is the study has revealed about the evil effects of globalisation on tribal culture and practices is confirmed by the experience of tribals in other places. According to jeremy Seabrook, "Globalisation is oppressive and would suppress local languages and cultures with its market forces" (Seabrook, Deccan herald, 2001). "Folk songs which are hundreds of years old and related to particular seasons and cycles of nature are disappearing under the impact of the commercial culture promoted by electronic media and cinema" (Bidwai, 2006 : 52). The same author, while speaking of the evil effects of globalisation, also enumerates countless crafts, languages, music, notions of aesthetics and beauty of traditional communities that are getting lost under the impact of globalisation all over the world. Thus the effect of globalisation on the cultural life of the tribals in Kerala is a reflection of what has happened and is happening all over the world.

HEALTH SITUATION OF THE TRIBALS

The study has brought out the health situation of tribals in Kerala in the post-reform period vis-a-vis the pre-reform period. This section analyses the data with the support of secondary data.

Health is an important parameter for social development. From the survey study we understand that 93 per cent of the household members of the respondents are in good health reflecting the good health index of Kerala. A negligible number of tribals suffer from rheumatic fever

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and other ailments. Sixty-eight per cent of the respondents' families witnessed death due to old age, while 32 per cent did not have death at all in their families during the last ten years. The case study of three villages also confirmed this good health index of the tribals in Kerala. But one aspect which the government and NGOs should give attention to is that a significant 44 per cent of the respondents did not know about immunisation of their children. Greater awareness needs to be created among the tribals about this.

In spite of the good health index, a pertinent question related to globalisation is how the tribals maintain this good health index and where they get medical treatment from. while 56 per cent go to private hospitals. If we add the 12 per cent going both to government and private hospitals, more than 25 per cent go to private hospitals in total. In contrast ten years back hardly 3 per cent of the respondents went to private hospitals for treatment. While more than 65 per cent went to the government hospitals ten years back, the fiture has reduced to 59 per cent today. The reasons they have given for going to private hospitals are non-availability of medicines, and lack of medical facilities and medical care. The case study of three villages also confirms the fact that tribals are increasingly depending on private hospitals. This increasing dependence of the tribals

Lack of medicines and of proper medical treatment are the main reasons cited by tribals for not going to government hospitals. This is a phenomenon experienced by people in most government hospitals in Kerala. This means that the government may not be spending as much money for medicines and other facilities as it used to ten years back. Though the government has not decreased its expenditure on health it has not substantially increased the allocation either. With rise in prices of medicines and medical facilities the government will not be able to provide the same facilities and medicine without substantially increasing the allocation for health. The main reason is that as in the case of education, the government seems to be withdrawing slowly from health sector as part of the policy of globalisation. Thus, when the policy of globalisation is implemented it is the most marginalised sections like tribals and Dalits who are the worst sufferers.

GOVERNMENT POLICIES AND EXPENDITURE ON HEALTH SECTOR IN THE 1990s

The study shows that there is less medicine and medical care in government hospitals for tribals at present than there was ten years ago. The pattern of government expenditure on health may explain the reasons for this phenomenon. For this the annual expenditure of the central and state governments for the health sector during the period 1990-91-96, the period when globalisation process started, is examined.

Table 5.9

Expenditure for Health Sector in the Annual Plans of Central Government in the 1990s

Year	Total Developmental Cost (in crores)			Health Expense (in crores)		
	Central and States	Central	States	Central and States	Central	States
1990-91	40062	23511	16551	659 (1.65)	189 (0.8)	471 (2.84)
1991-92	39075	22839	16236	629 (1.61)	183. (0.8)	447 (2.75)
1992-93	39946	23958	15988	665 (1.67)	210 (0.88)	455 (2.85)
1993-94	44573	27942	16631	658 (1.48)	203 (0.73)	454 (2.73)
1994-95	48482	31186	17296	780 (1.61)	273 (0.88)	507 (2.93)
1995-96	54662	33518	21144	924 (1.69)	285 (0.85)	639 (3.02)

Note : *Figures given in brackets indicates percentage.*

Source : *National Coordination Committee, "Arogyavum Agolavalkaranovum" (Malayalam), Health and Globalisation, Kerala Sastra Sahitya Parishad, Kozhikode, 2001 : 51*

Though the percentage and the amount of expenditure for health has remained steady and has even increased slightly, it is not a substantial increase when we consider the rise in prices of commodites. With a marginal increase in expenditure for health, it would be difficult to maintain the existing health facilities in government hospitals.

Table 5.10

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Share of Health and Education in Total Budget Expenditure in India

Five-Year Plan	Years	% of Total Plan Expenditure		
		Education	Health	Total
1. First Five - Year Plan	1951-56	7.86	3.32	11.18
2. Second Five-Year Plan	1956-61	5.83	3.01	8.84
3. Third Five-Year Plan	1961-66	6.87	2.63	9.40
4. Plan Holiday	1966-69	4.60	2.11	6.71
5. Fourth Five-Year Plan	1969-74	4.90	2.12	7.02
6. Fifth Five-Year Plan	1974-79	3.27	1.92	5.19
7. Sixth Five-Year Plan	1980-85	2.70	1.86	4.56
8. Seventh Five-Year Plan	1985-90	3.70	1.88	5.58
9. Eighth Five-Year Plan	1992-97	4.50	1.70	6.20

Source: *PIRG, 1992 : 16, Government of India, 1998; Raghava, 2002:57.*

The budget for education and health in Five-Year Plans upto the Eighth Five-Year Plan in 1997 reflects the decrease in their allocation. In fact, the percentage of plan expenditure for health decreased especially in the 1990s, the period of globalisation.

With increase in prices of medicine and medical equipments, decrease in expenditure on health will mean less medicine and medical facilities in government hospitals and primary health centres. The table shows a similar trend in education too. This indicates that the government is reducing its expenditure on these sectors. The health expenditure of the Central and state governments together was more than 1 per cent of GDP in mid-1980s, but since then it has been declining and is now only 0.9 per cent. In fact, the expenditure of Central Government alone at present is only 0.1 per cent of GDP (Ghosh, 2007). In the post-reform period this might be connected with the policy of globalisation under which the government is withdrawing more and more from social sectors. This was especially the case of states with poor economic growth. Kerala was one such state which was struggling to keep up even the earlier standard of good spending on health. Naturally poor people, especially the tribals, did not find even the minimum medicines available in government hospitals. The experience of the people has been that instead of getting medicines from the government hospitals and health centres, they were getting prescriptions to get medicines from medical shops. This shows that the government's policy of globalisation has adversely affected the health status of tribals in Kerala.

PRIVATISATION OF HEALTH SECTOR

The study shows that even the tribals are increasingly depending on private hospitals. Many reasons like lack of medicine and proper treatment facilities were pointed out for this phenomenon. Their dependence on private hospitals is confirmed by the manifold increase of private health institutions especially in the 1990s. The following table shows the increasing trend of private hospitals in India:

Table 5.11

Private Involvement in Health Sector in India

	1951	1971	1981	1996
Hospitals	2,694	3,862	6,805	15,097
% Private	8	14	43	68
Beds (000s)	117	349	505	870
% Private	–	20	29	56
Doctors (000s)	61	153	266	504
% Private	70	68	71	76

Source : *Dr. K.P. Aravinda, 'Kerala Arokya Mathruka Puthiya Nootandeleku', (Malayam), Kerala Sastra Sahitya Parishad, Kochi, 2001 : 123 :*

Though there was steady increase in hospitals, beds and practising doctors from 1951-1981 in India, the increase and its rate in the private sector was significant from 1981-1996. The same trend can be seen in Kerala also. While beds in government hospitals increased from 36,000 in 1986 to 38,000 in 1996 with 5.5 per cent growth, the increase of beds in private sector was from 49,000 in 1986 to 67,500 in 1996 with a growth rate of 40 per cent. During the same year there was 64 per cent increase in the number of doctors and 81 per cent increase in para-medical personnel.

with quantitative increase in health sector, modern scientific facilities like CT scan and magnetic Resonance Imaging (MRI) scans, and the most modern surgical methods are provided much more in private-hospitals than in government hospitals. No wonder an increasing number of people in Kerala approach the private hospitals for treatment. Sixty-two per cent of people in rural Kerala depended on private hospitals for medical treatment in 1996.

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The study has already shown that even the tribals are increasingly approaching private hospitals not only for the above reasons but also because of lack of even basic medicines and treatment facilities in government hospitals. The consequence is that they spend a substantial portion of their income on private hospitals, which should have been spent on food and basic needs. This means privatisation as part of globalisation has adversely affected the health needs of the tribals. in short, globalisation has made the life of the tribals more miserable than earlier. If the expense for health care of households in Kerala is analysed it can be found that expenses for treatment has gone up considerably, especially in the 1990s.

Table 5.12

Expense of a Patient for One-Time Treatment in Kerala (in Rs)

Items	1987	1996	Increase %
Medicine	8.24	83.48	913
Fees	2.90	27.87	930
Others	5.41	51.87	859
Total	16.56	165.22	898

The above table shows that expense of one patient for one-time treatment increased from Rs. 16.56 in 1982 to Rs. 165.22 in 1996, which is 898 per cent. The increase in expense is due to a big increase in price of medicine, fees for the doctors and medical check-up.

Table 5.13

Per Capita Annual Expense of People of Kerala for Medical Treatment (in Rs)

Items	1987	1996	Increase %
Medicine	44.20	182.36	539
Fees	15.60	99.06	535
Others	29.12	167.44	475
Total	88.92	548.86	517

If the annual per capita expenditure for medical treatment for people of Kerala is examined, it is found that per capita expenditure increased from Rs. 88.92 in 1987 to Rs. 548.86 in 1996, an increase of 517 per cent.

Table 5.14

One-Time Expense for Medical Treatement of a Patient in Kerala in Different Health Centres (in Rs)

Treatment Place	Expense
House	46.55
Government Hospital	183.96
Private Hospital	195.71
Others	171.10

An interesting finding is that there is not much difference between expenses in a private hospital and in a government hospital, as is shown in the table 6.14. While the one-time treatment expense in 1996 for a patient in a private hospital in Kerala was Rs. 195.71, it was Rs. 183.96 in a government hospital. The main reason is that the patients have to buy most of the medicines from outside with their own money.

If this is the situation of an average patient in Kerala, the situation of the tribals is worse. They spend a substantial amount of money for treatment even when they go to the government hospital, since they have to spend money on purchase of medicine. it means even the government hospitals are not affordable for the tribals. Naturally they will go to private hospitals for better treatment.

The withdrawal of the government from health sector and subsequent dependence of the poor on private hospitals is also confirmed at the national level. The slow but steady withdrawal of the state from comprehensive primary health care supported by secondary and tertiary levels, has had an impact on the confidence of the people to access medical facilities that are being provided to them by the state. many studies reveal that 'increasing number of individuals from poor and lower income households are accessing private care facilities, irrespective of the cost they have to pay for the same, thanks to the dilapidating quality of public health care" (Raghav, 2002 : 42).

Privatisation of even social sectors is a characteristic of globalisation. Since the Government of India is following the policy of globalisation with privatisation, it is slowly withdrawing from social sectors and is allowing private

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sector to take over those responsibilities. The people too will be forced to approach private institutions for basic services. In this case, the tribals will have to depend more on private hospitals than on government hospitals. Thus globalisation has seriously affected the health needs of the tribals in Kerala.

As the tribals started increasingly going to private hospitals, they were forced to spend more money in the 1990s than in earlier times when they dependend on the government hospitals with much less expense. thus the globalisation period has adversely affected the tribals in their health needs. "Kerala which was a model for achievement in public health is facing a crisis. We had this achievement with moderate expenditure. we were model for providing health for weaker sections. But now we are facing a crisis". (Aravindan, 2001).

In spite of good health index of the tribals the study has shown that they are increasingly depending on private hospitals due to lack of medicines and medical facilities, compared to the pre-reform period. This has not only affected the health of the tribals but also their living standard as they have to spend more on health and less on other essential needs. One main reason for poor health facilities in government hospitals is failure of the government to increase the spending on its hospitals since it is gradually withdrawing from health sector as part of its policies of privatisation and globalisation. The secondary data substantiate and support this conclusion. The expenses of individuals on medical treatment have increased manifold, while the spending on the health sector by the government has not increased proportionately. People worst hit by this situation are the tribals. So it can be reasonably said that globalisation has adversely affected the health system of the tribals in Kerala.

EDUCATION OF TRIBAL CHILDREN

Since quality education is a must to cope with globalisation, the study examines the effect of globalisation on the education of tribal children. While the state prides itself on a literacy rate of above 90 per cent, its tribal population has only 57 per cent literacy. If the 43 per cent of illiterate tribals have to be made literate at least for the future generation, quality elementary education should be given to tribal children. All the more needed is education and training to equip them to face globalisation creatively. But the situation of tribal education is pathetic. The study shows that the average drop-out rate of tribal children from 1st to 10th Standard was 39 per cent. The main reasons the respondents gave for the dropping out were low standard of government schools, financial difficulties and high fee structure in private schools. Most of the tribal children were studying in government school since the tribals cannot afford private schools. These are of poor quality without even basic facilities. The school system in general is not adapted and made attractive to tribal culture and life. And if they are not even of reasonably good standard, tribal children will not have any incentive to attend these schools. These reasons explain the high drop-out rate among tribal children. The government and NGOs can remedy this problem by providing good schools and remedial education.

In the three tribal villages chosen for case study, the drop-out rate in lower primary schools is considerably low since they function in the village itself and since lumpsum grant and noon-meals are provided regularly. But in higher classes it is goes up to 50 per cent. One reason cited was the long distance from home to school since high schools are usually located far off from the villages. in one village, children have to travel 30 milometres to reach a high school, that too without regular and proper transport facilities. Thus schools have to be set up nearer to tribal villages to attract and retain children in schools. Secondly, the government should see to it that tribal children get lumpsum grant at the beginning of the academic year as the tribal families will not be able to buy the study materials for the children otherwise. Besides, noon-meals also should be provided from the beginning of the academic year since it will be a great incentive for the tribal children.

Privatisation of professional colleges is a special feature of the post-reform period in Kerala. When the government permitted engineering and medical colleges to be started by private agencies, many such institutions came up in Kerala. They charged very high fees, which made these professional institutions inaccessible groups like Dalit and tribals. This has become a very controversial issue which has polarished the leftist government on the one side and the private professional instituions on the other. What is obvious in this privatisation process, which is part of the policy of globalisation of the Central government and which is followed by the state government, is that tribals will not be able to access professional education which equips

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people with information technology and bio-technology. Thus globalisation, which has led to privatisation, has deprived the tribals of any chance of professional education. They will be all the more marginalised when they are not able to get even elementary education in government schools, which are getting neglected due to increasing privatisation of school education.

Those who were personally interviewed expressed their opinion about the possible benefits of professional courses and information technology, which are an integral part of globalisation. Some of them said that the tribals were not able to get these benefits since they were not equipped for them and since their basic education itself was of low standard. Today those who are educated and qualiified forge ahead, while those who are not, fall back in the labour market. Since tribals are not educated they are pushed back further. Besides, according to the respondents, tribals will not be able to compete with others since competition is alien to tribal culture. "Tribal do not have the capacity to face the competitive aspect of globalisation", said one social activist. So culturally, socially and educationally they are handicapped.

As part of the policy of globalisation there is a tendency in the government to withdraw from the social and the educational fields. This is seen in the reduction of the expense on education or not increasing it (Table 5.10). This can affect the standard of education, especially in government schools. since most of the tribal children go to government schools, their education is affected and they become victims of globalisation. Secondly, as part of globalisation, privatisation is taking place in every field including education. It is a common sight in Kerala that private English medium schools are functioning very close to the government schools, with the result that the government schools do not have enough children and slowly get closed. It is obvious that tribal children are not able to go to private schools due to high fees. Thus privatisation as part of globalisation is adversely affected the education of tribal children.

In a globalised India and Kerala, educationally and even technically qualified people are needed for any employment and economic advancement. In the present situation, the tribal children are not able to get even basic school education, leave alone technical or higher education. Naturally, they will be pushed back and will become even more economically poor than at present. Thus the poor people like tribals and Dalits are becoming poorer as a result of globalisation. To remedy this situation and to help the tribals, especially their future generations, to face the challenges of globalisation they have to be given quality education including technical education. That is how education can be made a source of socioeconomic progress even for the tribals.

SITUATION OF TRIBAL WOMEN

The high status of women in any society is an indicator of the healthy social situation of the society. In tribal societies in general, women had a high status. But with the modernisation of these societies in relation to the mainstream society, the situation has changed. Women are losing the socioeconomic and cultural position that they enjoyed earlier. This study examined the impact of globalisation on tribal women.

By many socioeconomic characteristics, tribal women are much behind tribal men. In our study while 32 per cent of men were illiterate, 41 per cent of women were illiterate. The education at school level for boys was higher than that of girls. There were more widows (10%) than widowers (2%) in the tribal society in our study. Occupational distribution of tribals showed that 32 per cent of men were unemployed while the corresponding figure for women was 45 per cent. As a result, 57 per cent of tribal women did not have non-agricultural income, in comparison to 44 per cent of men who did not have such income. These data show that in comparison to men, many tribal women do not have family and financial support systems.

With regard to livelihood, our study found that employment for tribals in general has decreased ocnsiderably during the last ten years, more so for tribal women. While women had on an average 107 working days per year ten years back, they have only 48 days at present. The reason for this drastic decrease was the same for women as for men. They were totally dependent on agricultural farms and estates for work. When the price of agricultural products crashed due to the liberalisation policy of the government, agricultural operations and subsequently employment declined. The tribals in general, especially women, were adversely affected. when women had work, they had some income to manage teir own

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needs and those of the family. This gave them certain independence, freedom and self-respect. But when they did not have income from wage they had to totally depend on men whose income also got reduced due to the same reason. This means that globalisation has adversely affected the life and livelihood of tribal women more than of men.

Although there was no decrease in wages of women during the last ten years but there was no equal pay for the same type of work. Our study reveals that average wage per day for men was Rs 68 while for women it was only Rs 40. this put the women at a more disadvantageous position than men. Besides, though there was no decrease in wages, there was no increase in wages either during the last ten years. Since the price of essential commodities was regularly rising, the tribal women and men would not be getting the same amount of commodites with the same income. as a result, the quality of life of women was more affected than for men, as women often took greater responsibility over the household needs than men. Besides, the substantial reduction of working days during the last ten years has added misery to the tribal women.

The study of three villages also confirms the above data of considerable decrease in employment of women during the last ten years. Women are not as economically independent as they used to be since they do not get enough work or equal pay. Forest produce used to be a source of income particularly for women in olden times. Now they hardly get any such income. Even for firewood and fodder they have to travel long distances since the forests are destroyed, and in hilly areas they have a hard time collecting water. As a result, all the three villages reported that the workeload of women had increased during the last ten years of globalisation with the resultant effect on their health too.

The data given above showed how tribals are increasingly depending on private sops for essential commodites, since the quantity and quality of these commodities given through PDS had considerably reduced and their prices were steadily increasing. This was particularly affecting women, since women have greater responsibility than men to provide food for the family members. Since their meagre income from reduced working days is spent on buying essential commodities from private shops they are forced to forgo many other household things, and this in turn adversely affects their quality of life.

Because of the increasing hardships caused by the economic reasons described above, the social life of tribal women is also badly affected. The status they enjoyed in tribal society does not exist anymore. They are also exploited in many ways. One of the villages of the case study reported sexual exploitations in many ways. One of the villages of the case study reported sexual exploitation and a few cases of unwed motherhood. The health of women is also getting affected more than that of men as they are increasingly depending on private hospitals. When women spend their meagre income on medical treatment in private hospitals, their basic needs of essential commodities for the family are getting compromised and thus it is often the women who suffer most in the family.

The analysis of the study shows how tribal women are adversely affected by the policy of globalisation. Their livelihood is seriously hampered since they do not have sufficient employment opportunities as a result of the liberalisation policy of the government. They suffer more than men in the family since essential commodities at moderate prices are not available through PDS as they used to be. So they are forced to go to the private shops to get these essential goods at higher prices. This is due to the reduction in subsidy for PDS as the government is gradually withdrawing from the field of food security for the poor. Under same policy of privatisation and globalisation the government is withdrawing from the field of health also. Here again women are worse sufferers than men for their medical and health needs. In short, the liberalisation, privatisation and the policy of government's withdrawal from health and educational fields, have adversely affected the tribal women.

ORGANISATION OF TRIBALS TO FACE THE CHALLENGES OF GLOBALISATION

The earlier chapters tried to find out what could be done to help the tribals to face the evils of globalisation. The role of the government NGOs and the non-tribal people was mainly discussed. now the question as to what could be done by the tribals themselves to face the challenges of globalisation becomes relevant. To answer this question, it is important to know the nature and extent of collective action that the tribals have in their life. In other words, the study tried to find out the present level of sociopolitical participation of tribals, and to suggest the future collective

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action the tribals could take up based on that. This section will include data about participation of tribals in sociopolitical organisations and possible kinds of collective action that the tribals could take up to face the challenges of globalisation.

With regard to social participation, the study has revealed that the vast majority of tribals (61%) are not involved in any social participation. This shows the low level of sociopolitical consciousness among tribals in Kerala where the general population has one of the highest political consciousness and participation in India. therefore, it is not easy for tribals to organise themselves for any political action. They have to be made aware of it and to be initiated into it gradually.

Among those who have some sociopolitical participation a large proportion (39%) are involved in political parties, not as active workers of political parties but as affiliates to political parties or their ideologies. Other areas of social participation on a priority basis are "kundumbashree", self-help groups (SHGs) organised by the Panchayat, cooperative societies and SHGs organised by NGOs. Since a sizeable number of tribals participate in SHGs organised either by the Panchayat or NGOs, they could be made available for largest regions and for a greater number of people, and tribals could be encouraged to join them.

With regard to grass-roots level political participation the study has shown that a substantial portion of tribals (80%) attended Gram Sabha meetings and majority (54%) attended "Oorukoottam", the tribal Panchayats. The case studies in three villages in three districts also confirmed the result of the survey study that vast majority of the tribals attended the Grama Sabha and "Oorukoottam". This shows that Panchayat Raj was beneficial for tribals especially under the People's Plan Campaign. The reasons could be that tribals benefited from schemes like housing and drinking water, and the Panchayat was localised to the tribal situation. For more active and meaningful participation in Panchayat bodies, training programmes in Panchayati Raj can be conducted for tribals. Also tribal developmental programmes can be channeled through the Panchayat.

The study has found that only 44 per cent of the respondents participated in any tribals organisations other than Panchayat and SHGs. This means that the organisations other than Panchayat and SHGs. This means that the organisational strength or experience is low among the tribals of Kerala. It further means that it is difficult to build up tribal organisations or movements among them. But the heartening fact is that vast majority (73%) are willing to join such tribal organisations. Therefore, efforts could be made to organise the tribals. As part of the organisational participation, the respondents were asked whether they had participated in any tribal agitations. It was reported that only 34 per cent of tribals participated in agitations at either local or state level. This further confirms the low level of political consciousness and participation of tribals to demand their rights. Again it is an indication that it is difficult for the tribals to get organised on the tribal issues. But the hopeful sign is that a great proportion of the respondents (77%) were ready to participate in agitations.

About the question whether they wanted to work in a political party or an a non-party political organisation to fight for the tribal cause, 63 per cent wanted to work in a non-party political organisation. The tribals in all the three villages chosen for case study also expressed their willingness to join non-party political organisations. This shows that tribals are more inclined to work in a non-party tribal organisation to achieve their development and welfare. Since there is no state-level non-party tribal organisation other than Adivasi Gothra Maha Sabha (AGMS) known as established in Kerala, it is easy for AGMs to get acceptability and accessibility among the tribals in Kerala. All the more so since most tribals were willing to work collectively and organisationally for their rights and justice in a tribal organisation and not in a political party. In other words, a non-party political organisation can help the tribals in their needs especially to counter the evil effects of globalisation.

The respondents of the personal interviews also had a similar opinion regarding the type of organisation tribals should have in order to face globalisation. Many said that tribals should be organised to form a powerful sociopolitical movement to resist globalisation and defend themselves against its onslaught. For this, awareness should be created about the adverse effects of globalisation, and the tribals should be motivated to work collectively. They should be organised as far as possible on a non-party political basis. Their organisational strength should be used to pressurise the government to correct

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the policies of globalisation harmful to them and make policies that are helpful to them. The government should be pressurised to give the benefits of globalisation to the weaker sections like tribals and Dalits.

There were opinions among the respondents that tribals themselves should work in their own way to counter globalisation. They should try for diversification of jobs instead of depending on the traditional occupations. For this, they should develop their human resources through alternative forms of life and labour. They should also avoid consumeristic tendencies of the West, which are a by-product of globalisation. Those who have land should cultivate local products for food self-sufficiency and sustainable development, with green manure for food crops. Almost unanimously all the respondents agreed that tribals should organise themselves and collectively work to face the effect of globalisation by protecting their rights over their livelihood and culture. There is need of a powerful movement of tribals to face the challenges of globalisation. However there was disagreement regarding the type of organisation.

In the survey and interview among the tribals and non-tribals the respondents were otherwhelmingly of the opinion that a non-party political organisation (NPPO) would help tribals to face globalisation in an effective way. The Adivasi Gotra maha Sabha (AGMS) led by Ms. C.K. Janu, a NPPO, was the most known Adivasi organisation which brought up the tribal issues and problems in the forefront of Kerala's political scenario. Therefore, the respondents were asked to assess the role and function of AGMS.

Most respondents said that AGMS as a NPPO organisation of tribals was good and it should be welcomed. C.K. janu took the tribal issue to the national and even international scene, and she thus represented the emerging power of people. it has relevance and great scope to work effectively for tribals. "Janu really raised the issues of Adivasis in Kerala before the public in the state and even India", said a social activist in the interview. "Janu emerged as the voice of the suffering and exploited Adivasis in Kerala", said another. AGMS can give hope to the Adivasis in the state for coming together on a common tribal identity. However the respondents were divided in their opinions about AGMS.

While appreciating the great contribution AGMS has made, the respondents also mentioned its limitations and its failures in certain areas. Many felt that it should have a stronger district/ state-level organisational structure and network. Some felt that it has not succeeded in bringing up a second level leadership of tribals while many others said that its activities do not reach the grass-roots level and therefore its popular base and support have to be widened. As regards Ms. C.K. Janu, many respondents opined that she had not a got wide influence in all the districts. These felt that the AGMS should have a broad outlook and it should work for the unity of all tribals and tribal organisations. many suggested that they would like to cooperate with good organisations in civil society for the development of tribals.

Most of those who were interviewed were of the view that AGMS had not got a popular mass base. The organisation was concentrated mainly in Wayanad district, that too among one or two tribal communities only. "Even in Wayanad AGMS has not got much root", said one activist from Wayanad. "AGMS keeps a distance from other sections of people", said another in the interview. What a group of tribal students from Attappady in Palakkad district said in the interview is striking: The tribals at Attapady do not know who Janu is and what she does. The will be effective only if she goes to the people in other districts and mobilises them for AGMS". This was confirmed by tribals included in our survey and case study. Thus many thought that AGMS did not have enough organisational base to lead and sustain a powerful movement of tribals, especially to face the dangers of globalisation. However it could succeed if it mobilised at least the major tribal communities in various districts and strengthened its grass-roots level base and organisational structure in a democratic manner. But given the fact that there were over thirty- five tribal communities it was not easy to bring all of them together in one organisation. The identity of each community was too strong to come under the banner of one organisation. In fact, there were about a dozen organisations of tribal communities in Kerala. So, as a practical solution, many suggested a federation at the state level, of various organisations of tribal communities, on the lines of the Adivasi Ayika Samithi, which is a federation of several organisations in Wayanad district.

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However, given the limitations of the most marginalised group in the state, AGMS will find it extremely difficult to do this work of building up a strong state level organisation. Support and assitance of existing organisations of tribal communities, social activists, academicians and people's organisations of other marginalised groups are vital for such an organisation. AGMS should be open to this vision if it has to make a break through in building up a powerful organisation.

This section analysed the data from survey, case study and personal interviews about participation of tribals in sociopolitical organisations and found that the vast majority of tribals in kerala do not have any experience in participating in sociopolitical organisations including tribal organisations. Those who had some experience were involved in SHGs and Grama Sabhas of Panchayat. But most of them expressed willingness to work together and even participate in agitations organised by tribal organisations to face the challenges of globalisation. They also preferred a non-party political tribal organisation, and not political parties, and so they were open to work with Adivasi Gothra maha Sabha (AGMS), which hand led the state-level agitation for land, provided it built up a wider and deeper popular base and was open to collborate with all other tribal organisations with the support of various organisations of civil society.

This chapter analysed the data collected by the study about the impact of globalisation on tribals in Kerala with the support of secondary data. The analysis covered the data on the effect of globalisation on the livelihood of tribals including employment and availability of essential commodities, and sociocultural life of the tribals including religio-cultural practices, health education, condition of tribal women, and organisation of the tribals. The secondary data confirmed the major findings of the study and showed how the results of the survey study were related to the policies of globalisation adopted by the government across the country at both micro and macro levels.

STRATEGIES AND SUGGESTIONS TO COUNTER THE EFFECTS OF GLOBALISATION ON THE TRIBALS

The present study has shown indisputably that globalisation has adversely affected the life and livelihood of the tribals in Kerala. Most of them had lost their land long ago and even those who have land do not find it a source of livelihood since cash crops which many of them have cultivated do not fetch sufficient income for them. Besides, since land is already used for cash crops, it is not available for food crops for their sustenance.

Since there is no profitable price for cash crops like coffee and tea as a result of the liberalisation policy of the government, the tribals who depended on agricultural farms or estates for their employment found their working days drastically reduced and their means of livelihood adversely affected. Compared to the pre-reform period their wage for agricultural labour also did not increase during the post-reform period. Added to this, prices of essential commodities from public distribution system increased so much as a result of the policy of globalisation that the tribals increasingly bought them from private shops with adverse impact on their food availability and security. As a result poverty and misery leading even to starvation deaths increased among the tribals. As was shown in the study their sociocultural life have also got adversely affected, by the modern media like television and videos, which have become powerful vehicles of globalisation.

Health and education are badly affected as the government is showing tendencies to withdraw from these sectors by reducing its spending on them. Since privatisation is the policy of the government, health system in India and Kerala is getting more and more privatised with the result that even the tribals, the most marginalised group in kerala, are forced to depend on private hospitals for their medical treatment. With their already meagre income, tribals are being further squeezed by the mounting medical expenses. With low employment, low income and shrinking availability of forest products for fodder, food and water, the tribal women are the worst sufferers due to lack of proper food and free medical treatment. In such a situation they are also being exploited by non-tribals. Thus the tribals were found to be badly affected by the policy of globalisation initiated by the Indian government. Now the natural question is how the tribals can face globalisation

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and save their life, livelihood and culture. In other words, what could be the alternative strategies for tribals to counter the effects of globalisation in this situation? The study sought the opinion of tribals themselves about this question. Data was collected through personal interviews of tribals in the villages cosen for case studies and from their responses in the interview schedules. Based on the analysis and data the study in this section explores the alternative strategies and suggestions for the tribals to counter the effects of globalisation. This will be done under the broad headings which were followed all through this study namely, alternative strategies for livelihood, sociocultural life including health, education, status of women, and organisation of tribals.

Means of livelihood is vital and basic for any people, especially the marginalised, who often need to struggle to live and survive. The study has examined land, employment and availability of essential commodities as the source of livelihood for the tribals in Kerala. The suggestions will cover all these areas.

DISTRIBUTION OF LAND TO THE TRIBALS

Since land is a necessary means of livelihood for tribals and since they were unjustly deprived of it over the years, land should be an important factor for their development. This is all the more so in a globalised world when they do not have any of the other source of livelihood which other sections of the pupulation have.

Almost all the respondents of our study were of the opinino that land should be distributed to the tribals to help them to face globalisation at least in a limited way. The reasons stated by those personally interviewed are the following: Firstly, much of the land of the tribals in Kerala was alienated because of non-tribals and the faculty policy of the government. The people and the government of the state have an obligation to give back what was taken away. Secondly, till the tribals get equipped for alternative avenues of job, land can be a viable means to cultivate their food crops. This is all the more so, since the globalisation. Therefore, land can be a means of minimum subsistence to tide over the bad effects of globalisation. Thirdly, land is the basis of tribal culture, and so it is needed to preserve and sustain their culture and customs, which are threatened by globalisation. Therefore, the study recommends strongly to the government to distribute land to the tribals at the earliest as it had promised long ago. Land should be given in close-by places to form a village, not in scattered places. The land should also be sufficient and viable to meet part of their food requirement.

Merely distributing land is not enough to fulfil the obligations of the government towards the tribals. They need to be supported to cultivate it and to use it creatively, since they do not have enough capital to invest in land. Some of the tribals communities may not be familiar with developed methods of cultivation and their traditional methods may be too primitive to be economical today. this means the government should make suitable arrangement for them to get easy financial and technical help for a proper method of cultivation. This need not be a big problem for the government since there are many schemes and projects of the Central and state governments which are meant for the tribals and which are often not made use of. If the government can make a package using various tribal schemes, sub-plans and tribal component plans and coordinate them creatively, it can be of real assistance to the tribals. The Department of Tribal Welfare and the "Oorukootams" of the tribals can implement such schemes for the development of agriculture.

Out study also discussed the nature and type of cultivation. Since our study showed that 53 per cent of tribal households resorted to commercial farming, many respondents of personal interview suggested that the tribals should be encouraged to cultivate food crops. Subsistence farming has been reduced to 13 per cent at pesent from 28 per cent ten years back. This has is implementations in a global economy. Tribals who depended on their land for their food crops started to cultivate cash crops and to purchase all food products from the market. Even the villages chosen for case study had increasing tendency to cultivate cas crops. Since the price of cash crops fluctuates and often falls below the remunerative level as it has happened now, cultivating cash crops cannot be a reliable source of income especially in a globalised market. Secondly, the tribals can get essential food items from their land if they cultivate food crops and the land can then become a source of food security. Though the choice of crops should be left to the tribals, they could be motivated to go for food crops rather than cash crops.

There were differing opinions in the study about the

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method of cultivation suited to the tribals. Some advocated purely traditional methods of cultivation, while others were for modern methods of cultivation. The study showed that only a small section of tribal households use modern methods of cultivation; the rest use traditional methods of cultivation. However, since most tribal communities are still using very primitive methods of cultivation, a combination of traditional and modern methods of cultivation is recommended. Progressive and self-sustaining methods of cultivation traditionally used by the tribals should be preserved and organic farming with green manure could be encouraged. Better varieties of seeds and suitable crops could be slowly introduced. Mechanisation of agriculture should be preferably avoided since land has to provide jobs for the family members and should also be financially viable. Constant guidance from the Tribal Welfare Department and "Krishi Bhavan" (agricultural centre) will be very useful for the tribals.

In short, distribution of land will be only the first step for the sustainability of tribal life. Durable programmes for helping and guiding them for viable cultivation are equally important otherwise, given the background of the tribals in Kerala, the purpose for which land is distributed will not be served. Though this is primarily the duty of the government, it can also get the support of NGOs and tribal organisations. The latter can cooperate with and complement the efforts of the government for ensuring greater effectiveness and people's participation. In fact, the study revealed that NGOs have assisted the tribals to modernise their agriculture much more than the government agricultural department. Therefore, NGOs have a creative role to play in helping tribals adapt modern methods of cultivation to their life and culture. However it is the government which should have a greater role in developing an integrated plan to improve tribal land and its mode of cultivation.

IMPLEMENTATION OF TRIBAL FOREST RIGHTS ACT, 2006

It is recommended that the government should make necessary rules and legal structures to implement the provisions of "The Scheduled Tribes and other Traditional Forest Dwellers' Recognition of Forest Rights Act, 2006", so that the tribals in Kerala get the benefits of the Act, including ownership of forest land and the right to collect minor forest products. As part of forest rights, tribals could be employed as forest guards. Tribal communities could be allowed to participate in protecting forests and its resources as it being done successfully in the state of Orissa.

CREATING EMPLOYMENT OPPORTUNITIES

The study has found that a glaring effect of globalisation on the tribals in Kerala is the drastic decrease in their employment opportunities during the last ten years of the globalisation period. The survey, case studies and personal interviews revealed and reinforced this fact. Since the tribals lost their employment they lost their source of livelihood, especially since they do not have other sources of income. Thus poverty and starvation increased among the tribals. It was also shown from the analysis that decrease in the number of working days of the tribals was due to the policy of liberalisation of the government. If so, the government has a duty to provide alternative job opportunities for the tribals, the most vulnerable section of society. This is because the right to livelihood is closely related to the right to employment. Providing employment will be one way of helping the tribals to face the adverse impact of globalisation.

In a globalised economy the government seems to be shirking its responsibility of creating jobs for people. The industries increasingly controlled by the private sector will not be able to provide jobs for the tribals since they are not skilled to work in industries. This means that the tribals can find employment mainly in the primary sector, viz., agriculture. One example of employment creation is the Attapady ills Areas Development Society in Palakkad district, which is a huge scheme initiated by the government to generate employment for the tribals. To generate employment for the tribals, agriculture which was shown to be facing a crisis, should be revived. In Kerala, for the tribals, this refers to cash crops like coffee and tea. A basic requirement for the revival of agriculture is ensuring remunerative price for agricultural crops. As the price of agricultural crops crashed due to the imports of foreign agricultural products as part of liberalisation policy of the government, such policies should be changed to support agriculture, which can in turn provide employment to marginalised groups like tribals and Dalits.

It was shown that agriculture is facing faces a cris throughout India. Large-scale suicide of farmers in one clear indication of this crisis. To revive agriculture, which

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has been the backbone of Indian economy, the Central government needs to make drastic changes in its agricultural policy. Economic reforms that are more in favour of industries and are at the expense of agriculture will lead to pauperisation of large sections of people, especially Dalits and tribals. There should be a multi-pronged strategy to revive agriculture. Firstly, liberalisation policy of the government allowing duty-free or dutyless agricultural products should be changed to support agriculture in our country. For this sufficient control in import and export duties and customs should be introduced so that national agriculture is protected. The government should not hesitate to support its farmers in spite of the globalisation policy. Western countries are massively subsidising their agriculture and if they can do that in spite of the regulations of WTO and IMF, India too can support its farmers for whom it is a vital means of livelihood. No globalisation policy can be pursued leaving the large sections of farmers to their fate. Secondly, to make the farmers competitive in a globalised world, they should be given the necessary support and subsidies from the government side. Thirdly, increased investment in agriculture, at least at the rate of the pre-reform period, for providing irrigation, soil conservation measures, and remunerativ eprice for crops, should be ensured. The government which gives many facilities to industry through export promotion measures and through creating investment climate, cannot neglect agriculture, the life-line of millions of people in the rural areas of India.

ENSURING ESSENTIAL COMMODITIES AT LOW PRICE THROUGH RATION SHOPS

Since land is not a source of livelihood for the majority of the tribals, and since their employment opportunities have been considerably reduced because of globalisation, they have no means of livelihood. as a result, the tribals in Kerala are facing poverty and in some cases even starvation. At this juncture, the government has to step in to provide food security to the tribals by making essential commodities available through PDS at affordable prices. since the tribals are rated below poverty line, they deserve to get essential commodities distributed through ration shops.

Our study revealed that the vast majority of tribal households in Kerala depended on Public Distribution System (PDS) for their essential commodities before the reform period. This means that PDS was a real support for their life, especially when traditional sources of food from forests and land became scarce. The situation changed after economic reform was introduced in India. According to the study majority of the households at present buy cereals from private shops instead of ration shops. This is revealing especially since the eocnomic situation of tribals deteriorated due to decrease in their working days. In fact, more tribals should have depended on PDS but just the opposite has happened. The reasons given for moving away from PDS are high price of food grains distributed through fair price shops and their poor quality. The analysis has shown that after the reforms period started the government has been consistently raising the price of essential commodities due to the policy of globalisation, so much so that the tribals did not have the purchasing power to buy food grain even from ration shops. This means that tribals spend either more on food or do not buy enough of it. Thus they do not get the benefit of PDS and their food security is affected. This explains the high poverty and even starvation deaths among the tribals. Such a grim situation demands that the government provides essential commodities to the tribals at affordable price, especially since they belong to the BPL category. Since it is the policy of globalisation pursued by the government has brought about this deteriorating food situation, it is duty-bound to help the tribals to face the adverse impact of globalisation by ensuring sufficient food materials at reasonable price.

PROTECTING THE TRIBAL CULTURE FROM THE IMPACT OF GLOBALISATION

Our study has shown that cultural life of the tribals in Kerala has been adversely affected by globalisation. When they were alienated and in many cases unprooted from land, the basis of their culture was destroyed. Through their contact with people of mainstream culture they not only lost many tribal practices and values like community living and sharing, but they also tended to imbibe the bad habits of mainstream culture like consumerism and drinking. The modern media, particularly television and cinema, replaced tribal cultural forms, religious practices and lifestyle so much so that the modern generation of tribals had no interest in and even knowledge of their traditional practices, customs and traditions.

Three main suggesions which camp up from the

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respondents of the study to preserve tribal culture are, first to keep away tribals from mainstream culture, second, to help them to fully integrate which mainstream culture, and, third, to help them to adapt themselves to mainstream culture. The proposal to keep the tribals away from the mainstream culture is not possible as they are already living and working with and among people of mainstream culture. Majority of them find their livelihood in employment provided by the agricultural farms and estates of non-tribals. Once they are kept away from them they will not have any means of livelihood, especially since the majority does not have land for cultivation.. That means they cannot have any life and existence without the help of the non-tribals. Besides, in the globalised world where countries and culture and interdependent, tribals in one state cannot be kept separate from the rest of the people. Their economic, political and social life being closely inter-mingled with that of the non-tribals, they cannot have a separate territorial and cultural identify.

The second view is that they should be integrated fully into the mainstream culture so that they are also able, as was said by a respondent, "to enjoy the good things of modern life". This was only the opinion of a small minority and does not take into account the values of tribal culture and the need of preserving it. The third view is that they can be helped to adapt themselves to modern life evn by preserving the culture and practices of tribals. In other words, they will be adapting themselves to mainstream culture rooted in their own culture and identity. For this, ways and means should be devised for preserving the traditional practices and customs of tribal culture and equipping the tribals to face modern life in a globalised world. The study subscribes to this third view, i.e., helping tribals to preserve their culture and at the same time equipping them to face modern life in a globalised world. It recommends the following concrete suggestions based on the survey and case study methods and on the suggestions from the respondents during personal interviews.

1. Tribal customs and life styles, art, artifacts and festivals should be protected. For this the tribals should be taught to value their tribal identity and culture through studies, seminars and training programmes. In addition to these, cultural and artistic organisations should be formed to protect tribal culture which should organise cultural 'melas' consisting of traditional tribal cultural forms, art, music. NGOs can play a vital role in this effort and the government should support and supplement the efforts of the NGOs and tribal organisations in this regard.
2. Tribal culture and cultural forms should be included in the school syllabus so that the educational system itself nurtures and supports tribal culture.
3. The living environment of the tribals should be restored. For this land, soil, trees, forest, water and other natural resources should be made available to them and they should be allowed to live close to nature and the natural habitat. The cultivation of land by the tribals could be based on sustainable development by using organic faming and natural manure.
4. Developmental activities which endanger the traditional way of life and culture of tribals should be avoided.
5. "Oorukootam", the traditional tribal Panchayat should be established wherever it is possible, so also tribal self-rule. These sociopolitical structures will supervise the activities and life of tribals according to tribal culture and customs and will also be able to resist the onslaught of globalisation on tribal culture.
6. Encroachment and occupation of tribal land by non-tribals who destroye tribal culture should be prevented by legal measures enforced strictly. Under no circumstances should outsiders be allowed to control the life and culture of the tribals. This is important to check the adverse influence of globalisation on the culture of the tribals
7. Tribals should be encouraged and motivated to give up and resist bad and harmful habits and practices, values, and lifestyle typical of the globalised world, which have become part of the mainstream culture. Consumerism is one example. The study showed that even tribals get influenced by advertisements on television and they buy non-essential consumer goods with their meagre income, thus often falling into a debt trap. they should also be made aware of the evil effects of drinking and taking drugs which are very much prevalent in the mainstream culture. Modern media like video and television which are the concrete mirrors of globalisation should be discreetly used without replacing the religious and cultural practices essential to the tribals.

While the tribals should be encouraged to keep away

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from the evil effects of global culture and even the mainstream culture, they should be helped, as far as possible, to adapt themselves to a globalised world. As the world is changing very fast tribals cannot be expected to remain isolated. In fact, the study has shown that an overwhelming majority of tribals do not want to be isolated, but to be integrated with the mainstream society. Secondly, any society including tribal society needs to keep its dynamic character. However this dynamism should primarily come from within by using and developing its own sociocultural life. At the same time such sociocultural changes cannot take place without integrating changes and realities outside. Thus the tribals should be helped to preserve their tribal identity and culture, and also to integrate the good aspects of mainstream culture. The government and NGOs can do a lot in this matter.

PROVIDING HEALTH FACILITIES FOR THE TRIBALS

The study indicated good health index of the tribals reflecting the good health of the population of Kerala. But there is sufficient indication to show that globalisation as shown in the privatisation policy of the government has affected the health situation of the tribals. This is seen by the increasing number of tribals going to private hospitals at present whereas ten years back very few of them went to private hospitals. The main reasons for the shift are non-availability of medicines and lack of medical facilities and medical care. This is because the government is not spending enough to maintain government hospitals; rather it seems to be withdrawing from health field as part of the policy of globalisation. It is instead allowing private sector to take care of the health need of the people. When the tribals with their frugal means go to the private hospitals, they spend the money which they would have otherwise spent on food and other essential commodities. Thus the tribals have been adversely affected by globalisation and the government has a duty to help them to tide over this adverse impact. The suggestions emerging from the study are the following :

1. As an effect of globalisation medical facilities in government hospitals are found to be less available to the tribals now than pre-reform period. So the government should allot sufficient funds especially for tribals so that medicines and facilities are available to them, who are the most marginalised group in our society. If this is not done, the tribals will depend even move on private hospitals as they do now and will be spending more on health than what they can afford, which will lead to poverty and starvation. Therefore, the government should actively help the tribals to overcome the adverse impact of globalisation on their health.
2. Where there are no public health centres the government should start one for the tribals, so that their dependence on private hospitals could be reduced.
3. The study has revealed that almost half of the tribals did not know about immunisation. This is disturbing in a state where the health index is very good. The government should take special care to spread awareness about immunisation among the tribals. NGOs can help the government in this effort.

PROVIDING BASIC AND TECHNICAL EDUCATION FOR TRIBAL CHILDREN

All the respondents of the study of the survey and case study emphasised the need of both basic and technical education for the tribals to face the globalised world. But they differed on the type of education suited for the tribals. It is a fact that technical skills and quality education are necessary to cope with a globalised world. Those who have them thrive both in India and in the world. But the tribals have neither, and they cannot acquire technical skills unless they have quality basic education. The study has found that quality education is not available for most of the tribal children in Kerala. As a result, illiteracy rate among them is very high compared to that of the state. Worse still is the drop-out rate among tribal children from 1st to 10th standard. The reasons given by the respondents are low standard of government schools, financial difficulties and high fee structure in private schools. Siince most tribal children depend on government schools, their standard should be of high quality. The government should give utmost attention to providing good infrastructural facilities, ensuring regular teachers and building up a conducive cultural atmosphere in schools where predominantly tribal children study.

BASIC EDUCATION

The mainstream education is not at all adapted to the culture and life of the tribals. Besides, the quality of education available to the tribals is of very poor standard.

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No wonder there is high drop-out rate among the tribal children. It is in this background that the people have suggested 'Kanavu' type of education for tribal children. This is based on tribal culture and lifestyle, it has, to a certain extent, succeeded and proved to be a model education. But the study showed that there is also one section of the people who are critical of the 'Kanavu' model of education. Their main concern is whether such a type of education, though very effective, can be replicated for the tribals all over Kerala since there is a practical difficulty in spreading such a type of education. Secondly, when modern education in the era of globalisation is becoming more and more sophisticated and complicated, the question is raised as to how far students with 'Kanavu' education can cope with globalisation. The answer, based on the study, is that a combination of both tyese types of education could be adopted for tribal children. For this, the good elements of 'Kanavu' type of education can be integrated with the modern education. Besides, some important elements of tribal culture, customs and life style can be brought into the mainstream education.

Specific recommendations to improve the education of tribal children based on data from survey and case studies and personal interviews are the following.

1. Lumpsum grant and noon-meal should be given right at the beginning of the academic year. If they are delayed, as it often happens, the two most important incentives for the tribals will be missing, and naturally children will be dropping out of schools.
2. Adequate infra-structural facilities like good classrooms, furniture, audio-visual aids and playground should be provided generously to all the schools where tribal children are studying.
3. There should be high schools reasonably close to tribal hamlets so that accessibility becomes by itself an attraction, especially since tribal habitats are in hilly and forest areas.
4. Tribal language be used in primary schools since Malayalam is a foreign language for most tribal communities. This possibility should be explored and studied by anthropologists. NGOs and the government.
5. Tribal culture and customs should be included in the school curriculum and syllabus so that an atmosphere in which tribal children feel at home prevails in he schools which they attend. Tribal arts, cultural forms and games could also be included in the youth festivals in schools and colleges. These measures will help tribal children are predominant.
7. Once quality education is ensured, modern techniques like computer and communication skills could be introduced simultaneously.
8. The above mentioned measures should be adopted for the tribal children in aided schools also.
9. In unaided private schools in tribal areas preference should be given to tribal children through a policy of reservation, fee concession, extra coaching and other facilities. Private schools canot shirk their responsibility to the socially and economically disadvantaged sections of our population.

Many of these suggestions need not be a big problem for the government since there are already provisions for the education of tribals under tribal sub-plan and special component plans of both Central and state governments. What is needed is determination of the government and its machinery to implement them in letter and in spirit with a consideration for a people who are the most deserving as they are the most exploited in Kerala society. The government is morally obliged to do this in order to help the tribals to counter the adverse effects of globalisation and to fact it creatively and productively.

TECHNICAL EDUCATION

Once basic education is ensured, tribals will be prepared and equipped to take on technical education which will give them the skills to face a globalised world. There was general agreement among all the respondents that technical education should be given to the tribals as this will enable them to diversify their employment potential. But what is the type of technical education most suited to the tribals? Highly sophisticated technical education will not be suitable since the tribal children at present are not getting quality education to equip them for professional education. Therefore, technical education related to their traditional forest-based occupations should be fostered like basket-making and bee-keeping, which are the most suited to the tribals. Technical training in agriculture-related occupations and even agro-industries could be the other areas that can be explored for tribal children.

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Based on the data from personal interviews the study also recommends regular training and capacity building for the tribals. It can include learning about tribal identity and culture which can help them to respect and value their own culture. Capacity building consists of providing various skills related to the traditional occupations of the tribals like basket-making and bee-keeping, agriculture-related occupations like vermin compost and mushroom farming, and industry-related training like welding and wiring.

When a specific question was asked about the use of information technology for the tribals to face globalisation, especially since it is closely linked to the process of globalisation there were divergent views among the respondents, both tribals and others. Some said that it was totally alien to them while others recommended it to avoid their being further marginalised. Even if the latter view is taken, the tribal children with their poor quality of education may be ill-equipped for information technology.Therefore, the first step should be to impart quality basic education. Together with it information technology through computer education could be gradually introduced for the tribal children.

ADOPTING SPECIAL PROGRAMMES FOR TRIBAL WOMEN

Our study has shown that the socioeconomic situation of tribal women has deteriorated during the last ten years due to negative effects of globalisation. Therefore, the following proposals are made based on the study :

1. *Providing employment opportunities for tribal women:* The study has revealed that women have only a small number of working days per year at present compared to the pre-reform period. So their income has decreased and alongside their independence and sociocultural status has got reduced. That means globalisation has adversely affected the status and economic freedom of tribal women. Therefore they should be provided with employment which is basic to all their other needs. Any effort to increase the means of livleihood for tribal women would benefit all the tribals since women are responsible for the entire family.

2. *Distribution of land :* If the tribals are given land, as was promised by the government, women can cultivate essential food materials in them. working in their own field is better suited to women than to men who can go to far off places for work. So land is both a source of employment and livelihood.

3. *Properwages for women :* Payment of equal wage both for men and women for the same type of work will be another measure which will benefit women. The average wage of women at present is much lower than that of men in many areas of Kerala. This gender disparity should be rectified with equal wage, and the government should take the initiative to fix a just wage for tribal men and women, especially since there was agricultural depression in Kerala due to globalisation.

4. *Giving greater access to forest products :* Out study showed that the workload of women has increased manifold since forest produce and fuel and fodder are not available. Tribal women should be permitted to access forests for firewood and fodder, where ever there are forest. Speedy implementation of provisions of Tribal Forest Rights Act 2006 will particularly benefit tribal women.

5. *Ensuring essential commodities at moderate prices through Public Distribution System* (PDS) : Providing essential commodities through ration shops is relevant for tribals in general, but it helps women more than men since women have the responsibility of preparing food for all the family members. If they have to depend on private shops for these commodities, their share of food will be reduced. The government should ensure sufficient food materials made available through PDS.

6. *Providing sufficient medical facilities for women :* The study has shown has tribals are increasingly going to private hispitals instead of government hospitals, since there are not enough medicines and health care facilities in government hospitals. Women and children need more medical facilities than men, and so they should be adequately provided with these facilities. The government should provide medicines and good medical facilities in government hospitals and primary health centres rather than withdrawing from health field as part of the policy of globalisation.

7. *Giving education for tribal women :* The study has shown that tribal women are less educated than men.

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Also the educational level of tribal boys is higher than that of girls. This is different from the situation in kerala where girls have equal level of education as boys. Adult literacy programme could be taken up for tribal women by the government and NGOs. Facilities for education of girls should be increased by setting up schools near their villages, infrastructural and academic facilities should be ensured in schools, and supplementary education for tribal students, particularly girls should be provided. In fact, if proper facilities and conducive atmosphere are provided girls will do equally well or even better than boys education, as is the experience in Kerala.

8. *Providing social protection for tribal women :* The study has also pointed out social problems of tribal women like unwed motherhood and widowhood. Though they are not widespread there is need to take measures to protect tribal women who are increasingly vulnerable to insecurity from ousiders in a globalised society. The basic solution lies in providing livelihood, education and health care which have already been discussed above. Once they are taken care of, the government should give legal protection to the tribal women from non-tribal intruders. NGOs and tribal organisations also can take measures to protect tribal women and to make them aware of outside exploitation

STRENGTHENING THE ORGANISATION CAPACITY OF THE TRIBALS

Almot all the respondents of the study agreed that tribals should organise themselves and work collectively for their rights over livelihood and culture. Most of the remedies suggested above can be effected only if the tribals are organised. So there is need of a powerful organisation or movement of tribals at the state level to work for the house of the tribals.

Now the question arises as to what type of organisation would be effective for the tribals – an organisation affiliated to a political party or an organisation on a non-party basis. Most of the respondents of the study are of the opinion that the tribals should organise themselves on a non-party political basis. The situation of India and Kerala today seems to support this view of the tribals due to the following reasons: Firstly, if the tribals are organised under the banner of political parties they will not be able to keep their unity and identity. since there are many political parties each will try to bring the tribals under its fold. Thus tribals will be divided on party line as is happening all over India. This naturally weakens the cause of tribals as tribals. Given the low economic and social status of tribals, non-tribals will be controlling the party and the tribals too. Tribals will not be able to get into leadership positions and they will be used as mere vote banks. Besides, the tribals cause will be only one among many issues for a political party. Secondly, the trend today in India and the world is that the marginalised people organise themselves on their cultural identity. Thus Dalits organise themselves as Dalits and tribals as tribals. In other words, liberation and development of tribals should be in and through tribals themselves.

Though the tribal respondents preferred to work in a non-party political organisation of the tribals, it does not mean that tribal organisations based on political parties should be irgnored. Political parties have a definite role to play in a democracy, and in Kerala most major political parties have their own tribal organisations. In fact, the survey has revealed that the largest number of Adivasis (12%) belong to the Adivasi Kshema Samithi of CPI (M) and they are active among the tribals. Besides, non-party political organisations will have many practical problems like organisational network, structure, function and leadership, and therefore, on their own they may find it difficult to country the effects of globalisation. So it is advisable that tribal organisations on a non-party basis collaborate with tribal organisations of political parties on issues connected with globalisation. issue-based collaboration and networking among all organisations working with the Adivasis including those of political parties is the ideal. This need not be an impossible task since fishworkers in kerala collaborate with unions of political parties and non-party organisations on issues relevant to the fisheries sector, especially those connected with globalisation. In other words, effort should be made to muster all organisational strength available among the tribals to defend themselves to face the adverse impact of globalisation.

If the tribals are to be organised under a non-party political organisation, the question as to which organisation would be able to lead them becomes pertinent. Most of the respondents acknowledged that Adivasi Gothra Maha Sabha (AGMS), which spearheaded the state-level agitation in 2001 in Trivandrum and the agitation at Muthanga in Wayanad District under the leadership of

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Ms. C.K. Janu, could be one such non-party political organisation to lead the tribals in Kerala to face the onslaught of globalisation. Based on the study it looks that AGMS could be a rallying point for the tribals to organise themselves. There are many reasons for this argument:

First of all, AGMS took up the basic issue of land before the government, organised agitations at the state level and made the government and the general public aware of the plight of the Adivasis. For the first time it forced the government to give an assurance to distribute land to the tribals. This was not a mean achievement considering the acute unorganised and economically backward situation of the tribals in kerala. Thus it has opened up the possibility of the tribals in kerala to organise at the state-level and demand their right. No other tribal organisation has so far emerged as an organisational force in Kerala, much less the tribal unions of the political parties. Secondly, AGMS brought name and identity for the tribals in Kerala. Earlier the public in Kerala were not even aware of the existence of tribals in the hill ranges of Kerala. Through AGMS and its agitations the people came to know how they were alienated from their land. Thirdly, it brought about a certain amount of networking of tribal organisations in a tribal style, though in a loose form, throughout the state. In short, AGMS for the first time gave hope and scope for a state-level tribal organisation in Kerala and thus crossed the first hurdle of forming a state-level organisation.

While acknowledging the importance of AGMS and apreciating its historical role in leading the tribals in Kerala the respondents are not unaware of its limitations. First of all, AGMS is concentrated in Wayanad district, that too among a few communities only. It is a great limitation that, being a state-level organisation, it has not got sufficient organisational base in other districts. It should have had its unis and activities at least in the major districts where there is tribal population. Being an association of the most marginalised group which does not have any resource and personnel, it is not easy for AGMS to establish its presence in all the districts but being a state-level organisation it should represent tribals from major districts so that it is able to take up the issues of tribals in those districts.

Secondly, AGMS does not seem to have organisational network to lead and sustain a powerful movement of tribals against the dangers of globalisation. This is related to the first limitation. If there is a good mass base, there is a possibility of at least a loose organisational structure or network, which can further help to mobilise the tribals in wider areas and at a deeper level. For this, it could at least mobilise major tribal communities in various districts and strengthen its grass-roots level organisation in a democratic manner. Even this is not easy or an organisation of a marginalised group. It is here that the support and assistance from existing organisations, academicians, people's organisations of other marginalised groups and social activists become imperative. In other words, various organisations or forums of civil society should be made use of to build up a strong state-level organisation. AGMS should be open and willing to solicit cooperation from all like-mined people to make the organisation a powerful state-level movement.

Will this non-party political organisation be able to unite all the tribal communities in Kerala? Accoding to the official figure, there are at least 35 tribals communities in Kerala and there are more than a dozen organisations based on community identities. Some of these communities are Hinduised so much that they consider themselves superior in status or purity than others based on caste. So there may be difficulty in bringing all these communities under one umbrella. in fact, Adivasi Gothra Mahasabha tried to do that but with limited success. Though such a unity of tribals could still be aimed at, a more viable option would be a federation of various community-based tribal organisations to be visualised at the state-level. For this to heppen the leadership of AGM and of other tribal organisations should be open and flexible to accept and assimilate various tribal organisations on an equal footing based on the criterion of tribal identity and unity.

If the tribals are organised as a non-party political organisation what should be the issues it should take up in the era of globalisation? To answer this question the respondents of our study were asked how to solve the problems of globalisation. On a priority basis they are to reduce the price of food items, to increase the price of agricultural products, to change the import-export policy of the government, to change the present economic system meaning the globalisation policy of the government and to distribute land to the tribals. These are the issues to be taken up by the tribal organisation with the

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government, and public authorities, so that the government changes the policy of globalisation in favour of the tribals. This will be the way the tribals themselves could be working to face the adverse impact of globalisation.

The first suggestions for solving the problems of globalisation is to reduce the price of essential commodities distributed through PDS. The government has been steadily increasing the price of essential commodities during the post-reform period. It also restructured the PDS with the two categories of BPL and APL. The study has shown that these measures forced the tribals increasingly to depend on private traders thus endangering their food security and availability. So it is natural that the tribals want the government to reduce the price of essential commodities distributed through ration shops. The government had increased the price of these commodities since it wanted to reduce the subsidy for food materials, which is part of the policy of privatisation and globalisation. Since the food security of the tribals is affected due to the policy of globalisation, the government should take steps to protect a vulnerable section like the tribals from this adverse impact of globalisation. The tribals want the government to continue giving subsidies for essential commodities and not to withdraw from the areas directly connected with the basic necessities of the poor.

The next suggestion of the respondents is to increase the price of agricultural commodities. The tribals seem to have understood the relation between the decrease in price of agricultural commodities and liberalisation policy of the government. For the farmers to get a remunerative price, import should be curbed and support price should be given to agricultural products. As was indicated earlier, the liberalisation policy of the government has created a crisis in agriculture which employs about 65 per cent of the population in India. The government should rectify these lopsided policies and thus protect agriculture and its dependents. If there is remunerative price for farm products, farmers will restart agricultural operations, and Adivasis who depend on farming will get employment, and thus they will benefit. This is especially so in the case of cash crops like coffee and tea in Kerala, in which the tribals find work. If there is good price for the farm products, Adivasis who have land also will benefit from the sale of their own products. Adivasis who have land also will benefit from the sale of their own products. Therefore, the tribals want the government to change the export– import policy which is directly linked to globalisation so that the farmers and the marginalised groups like tribals and Dalits are able to face the negative effect of globalisation.

Another significant suggestion of the tribals is to have the present economic system, meaning the system of globalisation with privatisation and liberalisation, changed. This means the tribals are dissatisfied with the globalisation process which has made their life more miserable than earlier times. As the tribals from the villages selected for case study asserted, the policy of globalisation has destroyed their life, livelihood and culture. It should be drastically changed in favour of the marginalised and the majority of the people. A sizeable section of the respondents wanted land to be distributed to the tribals as a means of countering the effects of globalisation. When the tribals have lost employment and other means of livelihood, they feel that they could at least fall back on cultivation of food crops in their own land. This can also be a means of preserving their culture against the onslaught of globalisation.

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CHAPTER - 7

RISE OF TRIBAL MOVEMENTS

MEANING OF SOCIAL MOVEMENT

According to Oxford dictionary 'movement' means 'a series of combined actions and endeavours of a body of persons for special object'. Specialists on the study of social movements differ as to what constitutes a social movement and what are the forms generated by a social movement. We often hear about various movements like religious movements, nationalist or race movements, reformist movements, revolutionary and resistance movements, intellectual movements, women movements and so on.

Each of these movements arises under special circumstances peculiar to time and temporal features of its own. Since our principal objective is to understand the freedom movements among the ethnic groups (i.e. the tribals) discussions on other forms of social movements would be ignored.

Dealing with the forms of ethnic movements in India a noted ethnographer of India observes:

"Broadly, there are two types of ethnic movements among the tribals; one marks a phase of expansion of tribal entities by creating united symbols and organs of command among erst-while isolated tribal communities. In other type, the tribals are in close interaction with the larger nation society and are aware of national goals and symbols."

These movements may be grouped into three phases:

Phase I : Movements upto 1857, that is during the regime of the East India Company. These were basically in the form of crude agitations of primitive character for securing some concessions from the rulers.

Phase II : Movements from 1857 upto the beginning of the 20th century against the British economic and political policies in the background of new land revenue system, civil and criminal regulations and widespread famines. Both tribals and their surrounding non-tribals rose against the colonial power.

Phase III : Tribal struggles during the first three decades of the present century. These were economic struggles against the non-tribal intruders - moneylenders, traders, zamindars, administrators who tried to exploit the tribal way of life. During this period, in some parts of the country, tribals joined the groups led by Indian National Congress taking up specific issues and struggles launched by Mahatma Gandhi and his associates.

As is clear from above all tribal struggles were not politically motivated. They were close to what social scientists term 'agrarian reform and religious revitalization' type of social movements.

It would not be out of place here to outline briefly the views held by some specialists including anthropologists and sociologists on the agitations in tribal regions during the British period.

It is a well-known fact that a series of revolts spearheaded by India's war of Independence of 1857 rocked the British Empire to its foundations. Along with these there were several agitations and uprisings in tribal areas also. A famous social worker who did a lot for tribal welfare writes, "The tribals too initiated struggles to safeguard their honour, to protect their cherished freedom, and to get redress against money-lenders, the zamindars, and other parasitic landholders, who tried to deprive them of all they had".

During the period 1778-1820 there occured at least ten major disturbances in tribal regions of Bihar, viz. revolt of the Pahariya Sirdars against the British Government in 1778, revolt in Tamar regions of Chotanagpur in 1789 and 1794, revolt of the Chari movement in Bihar in 1795, 1801, 1807; and in 1820 Chotanagpur area of Bihar saw tribal revolts against the administrators and local zamindars. The causes underlying these initial disturbances in Bihar were the resentment among the tribal people against exploitation by their more advanced neighbours.

A noted philanthropist Dr. Verrier Elwin who wrote several monographs on the various tribes of Central India remarked in his book *A Philosophy for NEFA*:

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'In other parts of India (than NEFA) where the tribal communities have declined in many ways, the first cause of their depression was the loss of their land and forests. This had the effect of so enervating the tribal organism that it had no interior resistance against infection by a score of other evils.... To the tribal mind, Government's attitude about land and forests is as important as any scheme of development or education. If we look back over the long series of tribal rebellions against authority in other parts of the tribal India, we see that the majority of them arose over this one point. Thus the Kol insurrection of 1833 was caused by encroachment on tribal land. The Tamar rebellions, repeated seven times between 1789 and 1832, were primarily due to the illegal deprivation of their rights in land, which the Hos, Mundas and Oraons had suffered'.

Even some British administrators accepted the fact that the revolts in tribal areas occurred due to land ownership which passed into the hands off outsiders.

One must not overlook the fact that the revolts of the tribal people were not completely isolated; they had many characteristics in common with the revolts of non-tribals peasants. In fact the common elements between the tribals and non-tribals were so close that one may not assess correctly whether a particular struggle was initiated by the tribals or non-tribals.

TRIBAL MOVEMENTS IN BIHAR

BACKGROUND

The earliest known tribal movements in India originated in the regions of Chotanagpur and Santhal Paraganas which lie in the Southern most part of Bihar. This geographical area consists of series of plains, hills and forests. Both terrain and rich resources made it possible for the original settlers such as the Munda, Oraon, Santhal, Ho to live comparatively in isolation. Presently, the region comprises three administrative divisions, viz. North Chotanagpur, South Chotanagpur and Santhal Paraganas. Of the total tribal population of 66.17 lakhs in Bihar, nearly ninety one percent tribals are concentrated in various districts of Chotanagpur and Santhal paraganas. Among thirty three scheduled tribes in the State, the dominant groups are the Santal (also spelled Santhal) (20 Lakhs), the Oraon (11 Lakhs), the Munda (8.50 Lakh) and the Ho (5 Lakh). These major and other smaller tribal communities belong to the Proto-Australoid racial stock displaying physical characteristics of average height, dark brown complexion and healthy body. Though the Munda and Oraon, mainly of Ranchi district, had been in peaceful contact with each other for hundreds of years and they speak mutually unintelligible languages. Though their languages differ, both the Munda and Oraon have basically similar socio-cultural features and similar mode of subsistence, i.e. rice cultivation.

The Santals, mainly of Santhal Paraganas area, too have many things in common with the Munda and Oraon, though they are much closer to the Munda so far the speech is concerned. The Ho, numbering over 5.35 Lakh, reside in the area popularly known as Kolhan (presently parts of Singhbhum district of Chotanagpur Division). Both in racial and cultural characteristics they are closer to the Munda.

TRIBALS OF CHOTANAGPUR : BEFORE AND DURING THE BRITISH RULE

On the basis of tribal legends the specialists, including earlier ethnographers like S.C. Roy, have maintained that the Munda were the first settlers of Chotanagpur plateau. The Oraon came to this region about the beginning of the Christian era. As it would be imagined, the entire Chotanagapur region must have been covered by the dense green forest. Both the Munda and the Oraon must have had cleared the forest for shelter and cultivation. In Munda's native term Khunkatti means those who cut the first post. Original settlers and their descendants were locally designated as Khunkattidar and were given prestigious positions in the society. They alone were eligible to the posts of village headman and village priest.

While the Munda were living in the south-eastern parts of the plateau, the Oraon soon after their migration occupied north-western and central parts of Chotanagpur. The descendants of the founding Oraon families were known as Bhuinhars. The individual Bhuinhar Oraon families were the real owners of the village lands, their headman (Mahto) and village priest (Baigaf) also hailed from the Bhuinhari Khunt i.e., lineage of the founders.

As was noted earlier, in spite of the fact that the Munda and Oraon differed remarkably in speech pattern

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they had developed similar economic base for their livelihood. Both these tribes evolved a democratic way of administering their societies. In course of time they developed a system, known as Parha, a territorially organised tribal panchayat for settling inter-village disputes and for regulating tribal laws. The Parha head among the Munda is called Munda Parha and his counterpart for the Oraon is known as Parha Raja.

Around the sixth century, both the Munda and Oraon jointly selected a leader or 'Raja' and Chotanagpur was made a tributary to the Mughals in the year 1585. The Mughals were attracted to Chotanagpur by its reputation for diamonds. A new system of revenue administration was introduced. Also, a system of granting Jagirdari to the members of royal families was started by the Mughals.

'The above system was operative till the arrival of the British. In Ranchi district all the tribal communities virtually lived peacefully. In the year 1765 A.D., on the grant of Diwani of Bengal, Bihar and Orissa to the East India Company movement of the British personnel started in Chotanagpur. A British headquarter was established in Rajgarh in 1772. This further accelerated the movement of non-tribals from the neighbouring plains. By 1830 many tribals of Chotanagpur had already come to the forefront. Tribals hailed the "outsiders" as Dikus. Between 1770 and 1810, the British had already introduced their form of administration throughout Ranchi district. Because of new form of administration and judicial system, totally strange to the tribals, the so-called Dikus gained a sounder base in tribal regions.

Around 1820 both Hindu and Mohammaden traders from North Bihar and Northern India were introduced as lessees (thikadars) over the heads of tribal villages. To depict the situation in Ranchi one may quote the following passage of Major Sunderland who wrote in the year 1832:

"The hatred of the Kols (tribals on Ranchi and surrounding parts) seems to have been excited by the conduct of the Hindu Mohammaden inhabitants of . their (tribals) country, whom they called Sud or foreigners, in a degree hardly inferior to that which they feel towards our police and tax gatherers. The Mohammadens were mostly the farmers or thikadars of the villages which had been resumed by the Raja or his chiefs, or which were mortgaged to others; .... the Hindus were mostly traders and money-lenders. Long stories are told of the enormous profits made by the tax farmers and of the usurious interest levied by the money-lenders with the impossibility of the simple Kol ever getting out of the clutches of either, backed as they were by our police and Adalat (court)...."

Thus goaded to desperation, the entire tribal population of Chotanagpur rose up in arms against the Dikus in 1831-32, known as the Kol insurrection.

We are told, on December 20, 1831 the Mundas resolved to carry into effect the resolutions made at a series of tribal meetings. By January, 1832, the tribals (the Oraons, the Mundas and the Hos) joined the insurgents who decided not to leave a single Diku alive in Chotanagpur. Bloodshed started soon. Houses of the Dikus were burnt and plundered. The British administration though not prepared for the outbreak of such magnitude, put down the revolt eventually.

It goes without saying that the outcome of the above rebellion was even more far-reaching and effective control by the British than before. The South-West Frontier Agency (SWFA) was created in 1834. High ranking British officials were posted in crucial areas. Increased police force helped landlords to continue their oppressive tasks. Because of language barrier even the court took the side of Hindi speakers (Hindus) and Muslim moneylenders, landlords and government employees.

In order to improve the situation SWFA was abolished in 1854, and Chotanagpur came under a Commissioner. Survey of *Bhuinhar* lands started in 1858, and to an extent it pacified the tribals. The Chotanagpur Tenure Act of 1869 came into existence.

ARRIVAL OF CHRISTIAN MISSIONARIES IN RANCHI

In 1845 the Christian Missionaries appeared on the scene and began to implement their conversion programme slowly but successfully. Many elderly members of Munda and Oraon communities especially who were in distress situation saw immediate benefit in accepting the new religion. Also, the mission, was seen by some tribals as a powerful institution for social welfare. So the number of tribal converts increased several folds by mid 1860s. The Christian mission offered assistance to tribals in many forms in the fields of education, health and economy. A few educated tribal Christians formed the agency of Christian Tribal Organisation which latter flourished into the

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'Chotanagpur Unnati Samaj'. The new Christian leaders approached the administrators for the return of alienated Khunkatti and Bhuinhari lands from the hands of Diku landlors. The administrators generally took the side of the latter since they (landlords) by now were paying revenues to the Government. This movement started by the tribals is known as 'Sirdar' or 'Sardar' movement. The Lieut. Governor Sir Steward Bailey severly dealt with this unrest and brought the situation under control. The Sardari agitation was followed by the Birsa insurrection in 1895 to be described separately.

BRITISH RULE IN THE DISTRICT OF SANTHAL PARAGNAS

The district of Santhal Paraganas in Bihar had no Santhal inhabitants till the late 18th century. The entire region was, by and large, being ruled by Hindu chiefs. It was only around 1700 that the area was peopled by the new inhabitants from the South and South-east. They were Santhals. This happened due to the great famine of Bengal, and in subsequent years the land went out of cultivation. During the next fifty years the vacated land was occupied by the Santhals. In course of time due to population pressure the Santhals extended their habitat by clearing the barren forests lying at the base of the Rajmahal hills. Concentration of agriculturalist Santhals in this region is popularly known as Damin-i-Koh, i.e. the land occupied by the Santhals.

From 1778 upto 1827, the Rajmahal hill tracts which included the Damin-i-Koh region were under the control of the British administration. Separate rules under the Regulation I of 1796 were made for administration of the area. Damin-i-Koh was made a non-regulation area and was put under the control of the Collector. In 1827 the Damin-i-Koh area was placed at par with other areas by the Regulation I of 1827. Soon after the Kol disturbance in 1831, the British reviewed its policy towards the tribals. Because of enactment of the Bengal Regulation XIII of 1833 the entire Chotanagpur area was declared a non-regulation area and was put under the charge of an agent of the Governor General of India – Capt. T. Wilkinson.

A CLOSURE LOOK AT THE SANTHAL REBELLIAN

In the beginning the Santhals did not pay rent to the Hindu chiefs, but due to penetration of Diku they began losing their lands. Initially, the tribals did not pay much attention to Diku but when their traditional economy began falling apart they stood against the alien landlords and administrators. Since more powers were given to the administrators over the issues of tribal land alienation and indebtedness the landlords and moneylenders started taking advantages of the new system. They employed all sorts of tricks to grab Santhal's land and agricultural produce. Once a Santhal took a loan from the moneylenders he had little chance of escape. He was charged exorbitant interest ranging from 50 to 500 percent. Unable to repay the loan many Santhal families became the bonded servants of their creditors. This practice was known as Kamioti (or Agrestic Serfdoms) system.

The above noted circumstances first led to bitterness against the prosperous money lenders and later on the alien Government. Further, the system of Justice through the Law Courts was beyond the reach of simple and ignorant tribals. The Indian employees of the Executive and Revenue departments were thoroughly corrupt and oppressive. They not only supported the wealthy money-lenders and landlords but also exploited the tribals for personal gains. Such socio-economic situation prevailed in Domin-i-Koh for several years. The result was obvious. The Santhals became desperate and stood against the alien Government and the Dikus.

It is about this time (around 1855) that two Santhal brothers - Sido and Kanhu, assisted by two other Santhals -Chand and Bhairab came to the forefront providing leadership for mass uprisings. Their slogan was "we should slay all the Dikus (aliens) and become rulers of ourselves..... We should only pay eight annas (fifty paise) for a buffalo plough and four annas for a bullock plough and if the rulers (both British and Indian aliens) did not agree we should start fighting..."

The subsequent results can well be imagined. Both old and young Santhals equipped themselves with bows and poisoned arrows, spears, axes, etc. rose enmasse in a bloody rebellion in the second week of July, 1855. It began in Bhaganadihi, a place close by Barhat, the hearquarter of Damin-i-Koh, and quickly spread to the neighbouring regions, viz. Bhagalpur, Hazaribagh and Manbhum.

The slogan of the Santhal rebels was "death to the police, to the Mahajans, to the civil court officers, to the railway staff and to the zamindars." The insurgents propitiated the supernatural powers by killing five money lenders in the first week of July, 1855 and set out on their

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long march plundering public and government places — spreading terror all round. Both postal and railway services came to stand still. The Commissioner of Bhagalpur got the news of "general rising and indiscriminate plunder of villages." A body of troops under the command of Major F.W. Burroughs sent to subdue the rebels, were defeated in armed encounters on 16th July, 1855. Soon a large force was deployed. In a fierce encounter between the 200 strong British regiment equipped with warfare weapons and 4000 Santals with traditional weapons the latter lost the battle. The Santhal brothers were wounded with bullets.

On 29th July, 1855 Captain Sherwill leading 150 Indian sepoys of the 40 Regiment toured the disturbed areas and destroyed more than a dozen Santhali settlements. It was reported that the armed insurgents numbering about 30,000 were still active in parts of Rajmahal hills. To meet the situation the Government issued proclamation on 17th August, but that went unnoticed by the insurgents. Soon martial law was declared and the same was suspended on 3rd January, 1856. One of the principal leaders, Kanhu was captured and was executed sometime in the third week of February. Thereafter the movement began to slow down.

Though the Santals lost the battle the British Government realising the pressing need of the tribals reviewed its policy and introduced several beneficial measures for the inhabitants of Damin-i-Koh. Some of them are noted below:

(1) By the Act of XXXVII of 1855 the Damin-i-Koh area was separated from the districts of Bhagalpur and Birbhum and was formed as a separate district of Santhal Paraganas. This district was excluded from the operation of the general regulation and a special system of administration was introduced. The district of Santhal Paraganas was subdivided in four subdivisions, viz. Dumka, Godda, Deoghar and Rajmahal; each kept incharge of a separate Deputy Commissioner assisted by four Assistant Commissioners. (Presently, these four sub-divisions are districts; Rajmahal is renamed as Sahibganj).

(2) The Act of 1855 was followed by Police Rules of 1856 which reorganised the village headman system; accordingly, the police duties were entrusted to the Paraganas (territorial) heads. Thus, an atmosphere of direct communication between the tribals and their rulers was made possible.

(3) The appointment and removal of the Paraganas heads was reserved by the Deputy Commissioner.

(4) No intermediary was allowed to intervene between the Santals and the government.

(5) Verbal complaints made by Santals were allowed to be treated as valid.

(6) All criminal work was to be transacted with the help of Santals themselves who were to bring in the accused with the witness to the courts.

(7) The bonded labour system (Kamioti) was abolished.

CAUSES OF SANTALS UNREST

Opinions differ about the causes of Santhal rebellion. On the basis of the official documents and ethnigraphic notes written by Culshaw, Archer, Martin Orans four main causes attributed to mass uprising were:

(a) Oppression of the money-lenders and big landlords;

(b) Personal and hereditary debt bondage;

(c) Corruption and extortion by police helping non-Santals; and

(d) Impossibility of Santals getting court redress.

"A deeper cause", as Me Pherson observed, "was the Santal yearning for independence, a dream of the ancient days when they had no overlords, perhaps a memory of prehistoric times when according to some speculators they were themselves masters of the Gangetic valley and had not yet been driven back by the Aryan invaders".

Between the period 1872 and 1878 land ownership rights were settled throughout the district of Santhal Paraganas. Thus, Santals to a large extent got security of land and certainity about rents. In spite of these measures, the peoples dependence on money-lenders remained as before. Thinking that probably their socio-economic position cannot be regained by mere agitation the Santals turned towards religion.

RELIGIOUS MOVEMENTS AMONG THE SANTALS

Between the period 1871 and 1879-in the Santhal region many frustrated tribals who had lost their lands associated themselves with a new kind of religious order coming from their headman, named Bhagirath. Calling himself as the representative of the people he ordered his followers to slaughter their domestic animals and birds. It

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is interesting to note that many Santals paid rent to Bhagirath instead of Government via Hindu landlords. Bhagirath was arrested soon. The followers of Bhagirath called themselves Kherwars, an old name for Santal. The movement is also called Kherwari. Missionaries started asking people to embrace Christianity. Those who did were acquitted. In Maranghada, Soeko and Sarwada many Mundas were converted in this way.

Government announced a reward of Rs. 500/- for the arrest of Birsa. The Christian Missions took active part in assisting the police to arrest Birsa and his co-leaders. Some of the tribals themselves came forward to know about the whereabouts of Birsa, who still evaded the police and was changing his shelter every time. The search party was moving from one area to another of Singhbhum, Torpa and Khunti. Gaya Munda, a close associate of Birsa, was caught in Etkee village. It is held that two tribals — Kande and Bir Singh Munda attracted by the cash reward informed the police about Birsa's hiding. On 3rd February, 1900 Birsa was found in dense forest of Singhbhum alongwith his two wives. After arrest he was brought to the Bangaon government rest house. Here thousands of people were waiting to have a glance at their beloved leader. Covered his head and body with a white sheet smiling Birsa told his followers not to gel dissipated and promised "to return one day and win his kingdom."

On 20th May Birsa was brought to the court room. He was ill and was sent back to the Ranchi jail, where he died on 9th June due to cholera. Thus, ended the life of a great freedom fighter.

According to the official documents, 482 persons including a few tribal women were arrested for trials. Of these, 98 were convicted, 68 were bound down to keep the peace and 296 were acquitted. Capital sentences were awarded to Gaya Munda and his son for killing two constables and to Sukhrun for murdering a chowkidar. Others numbering 34 were sentenced to imprisonment. Gaya Munda's wife was sentenced to two years imprisonment.

Today Ranchi Munda tell their children the story of the brave Birsa. They hail him as Bhagwan. Birsa has become a theme of tribal songs and folk tales.

Thus we see that within a very short period (1895-1900) Birsa became the embodiment of reform among the Mundas. He gave a call for open rebellion against the outsiders in support of "no rent and better religion". Sachchidananda, a noted ethnographer and historian of Bihar writes:

"Birsa movement was an embodiment of the socio-economic and religious unrest among the Mundas. Buffeted by the waves of repression and oppression and torn by despair, the Mundas saw in Birsa a prophet and a saviour. Birsa's religion is a combination of Munda belief, Hinduism and Chiristianity. The movement had such a tremendous impact that the foundation of British rule in Chotanagpur was shaken for some time".

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TWO TRIBAL HEROES OF RANCHI! BIRSA AND JATRA

Both Ranchi and Santhal Paraganas were in the grip of exploitation in the closing years of past century. The so-called 'Diku' dominated the tribals in many ways. Most of the Munda families had already lost their agricultural lands and were living in a state of starvation. About the same time Christian missionaries had established a firm footing in tribal regions. The result was a feeling of discontent and unrest among the tribals. At this juncture Birsa, a Munda youth, raised his voice against the oppression and exploitation of landlords and British rulers who, in general, stood by the side of landlords.

Today, the Munda and other tribals of the Ranchi district hail Birsa as their God. They call him Birsa Bhagwan. It is interesting to note how Birsa was inspired and how he kindled the flame of revolt among his fellow members. A glimpse of Birsa's early years is essential for it reveals how he became a true revolutionary.

Born on 15th November 1875 in a small village (Garevia Ulihatu) in Khunti, 40 Kilometers from Ranchi city, Birsa was the fourth son of Sugna Munda. Since Sugna Munda had very meagre means of livelihood he sent his five-year old Birsa to his mother's native village Ayubhatu. After some time he was taken by his mother's sister to village Khatanga. After some years he was married in the same village.

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Birsa came into contact with missionaries who used to visit the tribal villages. The missionaries taught the inquisitive villagers reading and writing. Smart Birsa soon got into the system and learnt reading and writing. He was so much involved in reading that once while tending cattle he forgot completely about his goats. It is said that all goats were lifted by wolves. As a result Birsa was severely beaten by his uncle. Ultimately he left that village and came to reside with his eldest brother.

During his early years Birsa was very much influenced by a Hindu Brahman, named Anand Pande. Birsa joined him along with other villagers and regarded him as his Guru. He used to spend most of his time in his Guru's house and fields, and besides earning, he took lessons in the evenings. He learnt a lot about Hindu epics and was much impressed by the story of Rama, Lakshmana, Bhima and Arjuna, the legendary Hindu heroes. At that time he came into contact with a reputed Vaishnava saint who used to give discourse on Bhakti cult. He was so much inspired by the saint that he gave up hunting and eating non-vegetarian food.

In May, 1886, Birsa came into contact with a preacher of the German Lutheran Mission who took him to a Mission School from where he got through the primary education. He then came to the Chaibasa Mission School to continue his education. Being a vegetarian, Birsa sometimes got annoyed with non-vegetarian food which was the principal diet in the Mission kitchen. It is told that several days he went without food. In spite of this, his zeal kept him involved in educational training for sometime.

Later on, he joined his parents in Chalkad, the village to which they had migrated. Because of the training in Guru's Ashram and the Mission School, Birsa got a feel of the darkspots of his society. He found Munda people steeped in ignorance, worshipping numerous spirits of unknown world. Sometimes Birsa's family had nothing to eat in the house and the entire family had to starve. Once he saw an old woman being buried in the village graveyard. Along with the corpse were some silverware and ornaments. At dark of night he exhumed the grave and took out all the ornaments. The intention was clear. He sold them in the market and bought food for his family. Soon this action of Birsa was discovered and all the elderly Mundas began hating him. Birsa left his home to wander for several weeks in forests seeking divine light. He returned to his village but still was treated as an outcast.

Birsa began spending most of his time in meditation and *Kirtans*. Soon several young Mundas started coming to his place to listen to his devotional songs. After Kirtans he gave discourse on religion. He told his people to worship one tribal diety, *Singbonga*, the principal Sun God of the Munda. He asked them not to indulge in violence and animal sacrifice. "All the creatures including human beings are the children of Singbonga", was the message of Birsa. Some of the converted Mundas got so much inspired by Birsa that they left Christianity and re-entered the tribal religious fold.

Once small-pox broke out in the village. Since the disease is attributed to a diety in many parts of rural India involving animal sacrifice to please the goddess Shitla, Birsa was held responsible for deaths due to small-pox in the village. People forced him to leave the village, but even after his departure the epidemic spread unabated. Birsa returned to his village because his parents also got smallpox. He rendered selfless service to his people. This act earned him respect and restored his position in the village.

The real revolutionary figure of Birsa emerged when he saw that the tribals were being looted by the outsiders (Diku). The solidarity of the tribals was shaking. He was much moved by this and raised his voice against the growing influence of Christian missionaries and oppression of landlords and British administrators. As long as foreign dominance was not uprooted, the tribals could not be saved, Birsa thought. He exhorted his followers to get united and agitate against the Diku-cum-rulers. The influential Christian preachers requested the government to take action against him. A force was sent to Chalkad and after a scuffle an arrest warrant was issued. Ultimately Birsa was arrested and brought to Ranchi. After trial he was sentenced to 30 months imprisonment. On appeal the period of imprisonment was reduced to two years.

Finally, Birsa was released in 1897 on the eve of Diamond Jubilee of Queen Victoria's rule.

By now Birsa had become a popular figure and people from far and near came to his house for his darshan. With the help of his fellowmen Birsa once again began reorganizing his team - this time with greater zeal. Since police kept watch on the Birsa groups constantly, all the party meetings were held at night outside the village. Beginning in 1897 Birsa sent revolutionary messages to

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many of his co-workers in distant villages. With the advent of Holi festival he called for an open revolt.

The period between 1857 and 1899 was very unfortunate in the Munda region as famine, scarcity and disease showed their ugly faces. This 'gave Birsa and his fellow workers an excellent opportunity for social service. While serving the people they gained their confidence. Meanwhile, the Christian missionaries seized the opportunity of the then prevailing situation and stepped up their activities of helping and converting the tribals. The local landlords continued to oppress the natives.

Under these circumstances Birsa could not remain silent and began building up a force comprising of tribal youths of nearby villages. All the recruits received training in use of primitive weapons–bows and arrows, swords, etc. Gaya Munda was made the Minister and Commander-in-Chief of Birsa Dal. It is said that Birsa travelled as many as two dozen villages of-Khunti and Chakardhpur, crossing the barriers of hills and forests. Khunti remained the headquarters of the rebel forces.

On the eve of the Christmas day of 1899 Birsa Dal decided to launch their movement on a grand scale. The mission houses at Sarwada, Murhu and Birju were attacked by the rebels on an appointed day. Gaya Munda led his group and attacked the Sarwada mission and set it on fire. The/missionaries of Birju called the police. The encounter with the police lasted some hours and five policemen were killed. Birsa himself with 300 armed men attacked the Khunti Police station and killed a sub-inspector of police.

Within no time the news of the rising reached the authorities in Ranchi. Initially, the administrators wanted to make peace but the Birsa Dal was not ready to listen to them. Ultimately, a well armed police party surrounded Dumbari Hill where the rebels along with women and children had taken refuge. On the 9th January, 1900 the police started firing. Birsa Dal replied with a shower of arrows. But they could not stand long against the firearms. Ultimately, the rebels dispersed. Several people were arrested on charge of suspicion for aiding the rebels.

BHAGAT MOVEMENT

'Bhagatism' was essentially a religious movement which sprang up around 1914 in the western part of Chotanagpur when the Oraon tribals were undergoing deprivation and oppression by the local zamindars and policemen. An Oraon Bhagat, Jatra by name, of Chingari village close to Bishnupur Police station (Gumla District) proclaimed that in a vision he had received a revelation of Dharmas or God for the fellow Oraons. His message soon swept through Oraon country and people from far and near began to come for his darshan. By 1915 this movement spread to parts of Ranchi and neighbouring areas.

Jatra's message was that it was Dharmes (principal Oraon's God, the Sun) order that the Oraons should totally give up worshipping spirits (ghosts) andi should stop animal sacrifice. People should lead an ascetic life and give up non-vegetarian food, wine, tobacco and group songs and dances. They should not pay rent to the Zamindars and should not work for the aliens. Jatra warned his people in strong words that if they did not obey His (God's) order they would soon perish.

Acting upon Jatra's advice people began refusing to work for the landlords and disobeyed rules and regulations imposed by the British rulers who in turn issued orders for the arrest of Jatra and his close disciples. Jatra was brought before the sub-Divisional Court and sentenced to imprisonment. After completing his term in jail Jatra lived for a very short period.

The followers of-Jatra are called 'Tana Bhagats', because 'Tana' means 'pulling together' and 'Bhagat' implies a devotee. Jatra was trying to pull together all Oraons into his fold.

'Although Tana Bhagat movement was a movement to reorganize people on the basis of new socio-religious ethics it acquired political overtones in due course.

Around 1921 the followers of Tana Bhagat took part in the Non-cooperation movement. They had heard of Mahatma Gandhi and Congress, and some of them actively participated in Congress Sessions outside Bihar. It was on their persuasion that the Congress Government of Bihar passed a special bill, The Bhagat Agricultural Lands Restoration Act' with a view to return the lands that were lost by them on their refusal to pay rent to the British Government. At present all the Tana Bhagat political sufferers receive a monthly stipend from the government. Today the Bhagat movement is leaning more towards the adoption of papular Hindu cults. Although the followers have built shrines for Hindu deities they have not given up their tribal faith.

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CHAPTER - 9

SECULARISM, COMMUNALISM AND FUNDAMENTALISM

SECULARISM AND SECULARIZATION

In purely sociological term, Secularism stands for an ideology and has strong ideological connotations but the usage of this term in India does not conform to this ideological position. First, let us take up the ideological implications of this term in pure sociological terms. On a theoretical plane, as Emile Durkheim puts it, all that is there in the world - beliefs, myths, dogmas, actions, persons and beings - are either sacred or secular. All that is related with religion is covered under 'Sacred' and according to this scheme of things all traditional societies are basically 'Sacred Societies' because they are governed, in crucial segments of social life and in varying degrees, by laws of religion. It is the religion that controls the activities of the members of such socieities. Among the characteristics of secularism the following are the most important ones:

(i) Separation of authority between 'church' and the 'state',

(ii) Emphasis on 'this world' as opposed to the 'other world', and

(iii) Rationalism and scientific outlook.

Secularism implies that the state will not be governed by the church i.e., by religious authorities such *as Bishop, Pundit, Mullah* or *Khalifa*. The doctrines promoting social inertia such as fatalism and divine retribution have no place in secularism. A secular mind is characterized by rationality. A secular (modern) society is not against religion as such but it has to fight superstition and intolerance as well as bigotry and obscurantism. Societies are not born secular. Instead, the societies that are practising varying degrees of secularism had to undergo transformation from traditional (religious/sacred) to secular. The process of secularization has played its' part in this transformation. Secularization is, thus, a process of social change through which the public influence of religion and religious thinking declines as it is replaced by other ways of explaining reality and regulating social life. A number of western societies are examples of this phenomenon where religion still survives but it is largely confined to the private domain of individuals.

Secularism is inscribed into the preamble of our constitution together with democracy. But two concepts of secularism have been operating side by side. The 'western' concept of the separation of religion and politics, especially separation of religion and the state, which virtually relegates religion to the realm of the private and on the other hand *Sarva Dharma Sambhava* (equal respect to all religions) go together. The 'true Hinduism' promises and pays equal respect to all religions but the 'political *Hindutva*' thinks and acts otherwise. Thus, secularism in the Indian context is opposed to communalism ideologically. It signifies religious non-discrimination and equal liberty for all citizens, believers and non-believers. A secular polity is one in which the state does not discriminate between citizens on the basis of their religious conviciton. It grants equal civil rights to all citizens, respects the religious liberty of each and protects even the liberty of those who do not believe in religion.

Secularism today is facing crisis world wide. In India, as Gurpreet Mahajan (2003) points out, the critics of secularism make two arguments: (a) the concept of secularism, a separation of religion and politics, is alien to Indian society and way of life, hence it has no place in the Indian context; (b) separation of religion from politics is an idea that comes with the Protestant ethic - as a 'gift of Christianity', it has limited applicability. The advocates of secularism regret the fact that the Indian State is not sufficiently secular as it allows the use of religion in the public domain. The critics of secularism, on the other hand, feel that this concept has no relevance for India as religion cannot be separated from politics in this country. But to

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what extent the intermixing of religion and politics can play havoc with the peace and prosperity of the country is amply clear from the politics of *Hindutva* using Ayodhya as a means of achieving political power. The prevailing conditions in our neighbour Pakistan should be a lesson to us.

Secularism is understood as antithesis of communalism. This view also promotes the protection of minorities from a dominating majority. Indian society, since its' beginning, was able to set traditions under which an integrated social, cultural and political life was possible in a multi-religious, multi-ethnic and multi-lingual framework. Imposing it's own cultural and religious ideas and symbols on *others* has been alien to Hindu tradition and true Hinduism. This is amply demonstrated by one of it's greatest exponents Swami Vivekanand.

Broadly speaking, the process of secularization operates through three factors:

(i) Education;

(ii) Legislation; and

(iii) Social reform

Education, when imparted on scientific and rational lines, gives rise to rationality in the thinking process. If we take the case of India, education was imparted through religious institutions - *Pathshala, Madrasa* and seminaries - attached to *temples, maths, mosques* and *churches*. When the scope of acquisition of formal education was widened and secular educational institutions were opened, it brought about a sea change. The change in the content of education and medium of instruction was largely responsible for it. That is why the role of formal education in social transformation cannot be minimized. The legislative action is also crucial factor in the process of secularization. But mere legislative action not supported by proper motivation and social environment may not achieve the desired results. Many developing countries may be cited as examples of this situation. Take the case of India. Our country has gone secular by legislative fiats without the backing of sufficient infrastructure for the emergence of true secularism. Behind our country's facade of secularism and national unity lurk a bewildering variety of parochial, narrow minded, obscurantist and unscientific feeling that threaten to tear the fabric of our 'secular' country apart. Casteism, tribalism, linguism and religious communalism are" painful facts of our day to day existence. The so called 'do gooders', 'Godmen' and religious bigots are flourishing at will. In the face of these factors secularism is reduced to just a paper tiger. In reality, secularism should be a way of life. It is not a slogan but an article of faith. Social movement for social reform is a potent weapon for the promotion of the process of secularization. It does mental conditioning. It prepares us to accept social change. Whatever degree of secularism our society has achieved and absorbed owes a lot to various social reform movements launched from time to time.

Two subsidiary factors in the secularization process are urbanization and industrialization. Both these factors own a lot to technological innovations. It has been observed that as the area of scientific knowledge and technology widens, the area of religion shrinks. This hardly means that religion and science are incompatible. As S.C. Dube (1974) rightly points out - "on a higher spiritual and moral plane, religion is not necessarily inconsistent with the idom of science and the ethos of progress. Moreover, urbanization and industrialization help loosen the grip of 'in group' feeling and present newer alternatives to man. Further, it promotes rational cause-effect explanations and the hold of religious beliefs over different aspects of life declines, although interest in religion as a phenomenon persists. It sheds much of its' collective and communal overtones' and becomes largely a personal concern".

The State of India seeks to promote the secularization process by:

(i) abolishing such symbols of ritual inequality as untouchability, the practice of which has been declared a penal offence;

(ii) throwing open all religious places and institutions to all classes and sections of the people;

(iii) reform of the Hindu Personal Law eventually aiming at evolving a common personal law for all communities in the country;

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(iv) forbidding all such agencies as shops, public restaurants, hotels and places of public entertainment from discriminating against any citizen of the state;

(v) throwing open all educational institutions to members of all communities;

(vi) secularizing the content of text books by including such material as could be helpful in the promotion of a scientific-rational outlook by deleting such material as was likely to poison young minds, which smacked of traditionalism, obscurantism and religious orthodoxy; and

(vii) developing a sound economic base so as to eradicate poverty and ensure an equitable distribution of wealth among the people.

Secularism is one of the basic elements in the process of modernization. Thus, if our society is to become modernised to cope with the worldwide changes we are left with no other option but to adopt 'secularism in practice' by speeding up the process of secularization.

COMMUNALISM, FUNDAMENTALISM AND COMMUNAL VIOLENCE

The most striking feature of India's current socio-political scenario is the 'explosion' of communalism in the face of New Economic Policy and Globalization. It is threatening the very social fabric of Indian Society.

Communalism is a highly complex phenomenon in pluralistic India. The academics from different disciplines in social sciences have produced a variety of definitions - some vague, some based on certain static preconceived notions while others on the basis of the ever changing socio-political dynamics. In some of these definitions communalism has been explained as 'above all an ideology', to some others, 'a false consciousness', 'a struggle for scarce resources', 'competition for jobs', 'an instrument of ruling class polities' and so on. On the basis of it's highly intolerant and anti-damocratic nature, it is also seen as a precursor to fascism. Among the plethora 61 definitions, W.U. Smith's (1979) definition presents a fairly generalized and popularly accepted scenario'.

"Communalism is that ideology which has emphasized as the social, political and economic unit the group of adherents of each religion, and has emphasised the distinction, even antagonism, between such groups."

Even at the risk of being seen as over-simplification of a complex phenomenon the same definition may be rephrased as: "Communalism is an ideology based on the belief that society is divided into religious communities whose interests differ and are opposed to each other and hence antagonistic in nature". The above rephrasing makes Smith's definition clearer and more understandable in the context of India's experiences of religious communalism carrying social, economic and political dimensions. Here, since religion and religious identity (or may be communal identity) is central to the understanding of communalism, and India being a traditional society having religion as the highest value in life, no discussion on communalism shall be worth while without the role of religion and religious identity. Here, one can say that the sense of religious affiliation, which has commonly been central to the sense of one's identity, is ordinarily acquired during childhood in the course of socialization.

Let us go back to the basis of communalist ideology- that the society is divided into religious communities whose interests differ and opposed to each other - to see the empirical and logical fallacies the communalist organizations are proud of and their misinterpretation of the past. Are the problems of poverty, unemployment and lack of education and healthcare not common to all the religious communities? Do their interests in these matters differ or opposed to each other?

Communalism can be said to have had more than a hundred ears old history, affecting aspects of the socio-cultural and political life of India in myriad ways. During certain times, it lays dormant, while being potentially explosive. Bipan Chandra (1987) in his acclaimed work on communalism argued that, "communalism was the false consciousness of the historical processes of the last one hundred fifty years, because, objectively no real conflict between the interests of Hindus and Muslims existed". It was a false view of history. He further argues that "communalism was an expression of and deeply rooted in the interests, aspirations, outlook, attitudes, psychology and point of view of the middle classes in a social situation characterised by economic stagnation and

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the absence of vigorous struggle to transform society - the communal question was petty bourgeois question par excellence". K.N. Panikkar (1991) said that "communalism is a state of consciousness..... which primarily draws upon certain assumptions of distinct cultural identity for member professing the same religion". Gyanendra Pandey (1990) suggested that "communalism is a form of Colonialist knowledge, which in it's Indian usage, means a condition of suspicion, fear and hostility between members of different religious communities". All these definitions broadly touch upon the antagnostic relationship whether 'imagined' or real between the two major religious communities such as Hindu-Muslim, Hindu-Christian, Hindu-Sikh, Muslim-Sikh relations.

The demand for converting India into a *Hindu Rashtra* is the most obvious and popular manifestation of communalism and V.D. Savarkar, Shyama Prasad Mukherji and M.S. Golwalkar are seen as the major ideologues of this school of thought. Savarkar, in his attempt to arrive at his concept of 'Hindu Rashtra' used the medium of *Hindutva* and proceeded to argue that *Hindutva* is different from Hinduism and the *term Hindutva* defying all attempts at analysis. "Forty centuries if not more had been at work to mould it as it is .... Hindutva embraces all departments of thoughts and activities of the whole being of Hindu race" (Savarkar, 1989 edition). While Savarkar tries to perfect the rhetoric of a "Hindu Rashtra" where Hindus would rule and others would be subordinated, M.S. Golwalkar does not permit any space to the non-Hindu. He looks at all the non-Hindus as 'foreigners'. He declares in clear terms that "there are only two courses open to the foreign elements, either to merge themselves in the national race and adopts its culture or to live at it's mercy so long as the national race may allow them to do so and to quit the country at the sweet will of the national race. That is the only sound view on the minorities problem". Golwalkar is very clear in his view that if the minorities (non-Hindus) do not go by the religion and culture of the National race (Hindus) they may have to lead a life of ignominy. "The foreign races in Hindustan... may stay in the country, wholly subordinated to the Hindu Nation, claiming nothing, deserving no privileges, far less any preferential treatment.... not even citizen right" (1939).

Jamaat-e-Islami, the dominant face of Muslim communalism and fundamentalism is interested to see India and other countries as Islamic states, obviously theocratic state. In the Islamic state' of the chief ideologue of Jamaat-e-Islami Maulana Maududi, the non-Muslims could live but only as subjects *(zinimi)* with certain defined and specific rights and privileges but would not be allowed to influence the policy of the state. In the wake of anti-Ahmdiya riots in Pakistan in 1953, he was arrested. When confronted by Justice Munir Commission probing the riots as to what he would feel if a Hindu India decides to go for a theocratic Hindu state threatening to persecute the Muslims living there the way Maududi wanted to persecute the Ahmedia Muslims because of religious differences. Maududi's candid reply unveiled his fanatic mind. He deposed that he would not object if the Muslims living in India are tre'ated by the Hindu India as *Shudra* and *malecha* and Manu's laws are applied on them depriving them of all share in government and rights as citizens. Perhaps, that is why, Jamaat - e - Islami was bitterly opposed to Jinnah and his vision of a modern Pakistan. Jamaat-e-Islami was also opposed to the Jamat-e-Ulema and the Deoband line of thought based on Indian nationalism and Muslim being part of a composite Indian nationalism. The striking resemblance of ideology of Golwalkar's Rashtriya Swayam Sevak Sangh (R.S.S.) and Jamaat - e -Islami is worth noting. Both of these had no vision of a liberal, secular, democratic state bestowing equal rights on all the citizens irrespective of their caste, colour and creed. Interestingly, Maulana Maududi, after partition, fled to Jinnah's Pakistan. It's Indian counterpart Jamaat-e-Islami Hind remained in India. Though it remains wedded to the concepts of an Islamic state but has no courage to advocate it openly and takes a defensive stand. Each fascist system of thought and politics necessarily looks for or 'creates' an enemy or scapegoat which Golwalkar conveniently found in Muslims the way Kilter's Nazism created an enemy in the form of Jews.

When we look at the class character of communalism we find that during Colonial period the landlords and bureaucratic elements (like *jagirdari* elements), moneylenders and merchants apart from the petty bourgeoisie provided the internal base for communalism. The Hindu Mahasabha, R.S.S., Jamaat-e-Islami and Muslim League flourished on this social class. However in the post-

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colonial and post-independence period, apart from the petty bourgeoisie, it is not easy to assign class roots to communalism today. The social class character and base of communalism have undergone a major change since 1947". Is it easy to assign a social class base to Bharitya Janta Party, the political front of R.S.S.?

Communalism is neither inspired by religion nor is religion an object of communal politics, even though the communalist bases his politics on religious differences, uses religious identity as an organizing principle and in the mass phase of communalism uses religion to mobilize the masses. Communalism, especially in it's violent form, makes a ready appeal to urban poor and the urban lumpen elements whose number has grown rapidly as a result of lop sided economic development and large scale migration into cities. Rootless and improverished, millions today live on the margins of society in urban clusters. Their social anger and frustration often find expression in spontaneous violence and loot and plunder whenever opportunity arises. A communal riot provides an opportunity for both.

COMMUNAL VIOLENCE

Violence between religious communities in the form of riots, pogroms or terrorism draws our attention in a dramatic manner but the underlying and long term cause of this violence is the spread of communalism. As Bipan Chandra (ibid) remarks, while the communal riots give credibility to the basic communal ideological precepts among the ordinary people and enlist further support for communal politicians, it is communal ideology and politics, which the communal politicians and ideologues preach in normal times, which form the real basis on which communal tensions and violence occur. In other words, communal ideology and politics are the disease, communal violence only it's external symptom. "Communal riots and other forms of communal violence are only a concrete conjunctural manifestation of the communalization of society and politics. Communal ideology leads to communal politics and psychology differentiation, distance and competition along religious lines".

Communalism and communal violence are the products of the overall social, economic and political situation of the society. Communal violence has been, by and large, an urban phenomenon where the deprived and poor people have been involved. But more than religion, it is communal propaganda; rumour mongering and communal mobilization have been responsible. Any type of violence and lawlessness provides an opportunity to the interested people to indulge in loot and plunder largely for economic reasons. But the recent happenings (of 2002) in Gujarat have shaken all the sensible, justice loving, democratic and secular people of the country. The involvement of the state in a communal violence has never been so blatant and naked. It was virtually a state-sponsored violence against the minority community as brought out by several citizen's organizations and forums. Some of these reports even described it as 'ethnic cleansing' done under state supervision. Significantly, the dirty work of violence is left to a certain type of communalised militant youth especially belonging to the *dalit* and backward castes, and sometimes to the tribal groups, while the ideological work evolves with great subtlety. This may be an intelligent tactic on the part of communal leadership to divert the attention of the traditionally deprived sections from their problems. It may also be a ploy of blocking the deprived sections of the society from finding a rightful place in the power structure which till now has been tightly held by the upper social strata. But in the Gujarat carnage the involvement of the upward mobile, prosperous middle class in loot and plunder is a phenomenon that required an urgent analysis by the sociologists.

COMMUNALISM: THE NEW FACE OF INDIAN POLITICS

Indian State was concieved in a secular, non-communal manner. In the secular-democratic state an important role was assigned to politics and the political parties. Unfortunately elections in independent India became ends in themselves and narrow self and political interests rather than larger national interests became more important. In this scenario, the communal implications of the numbers game in a pluralistic society became apparent. The political parties conceive of their interests in terms of caste and religion. "Such an ethnic orientation of the numbers game has two prominent features. One is the rise of new political

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organizations that are sometimes blatantly and sometimes not so blatantly communal. This happens on the local levels or starts at that level and then moves up" (Rajni Kothari, 1998). He looks at Shiv Sena as a blatant example of this but there are many others too. The capture of political organizations by cultural and sectarian organizations is also a case in point. What starts as non-political organizations take on political roles. Rajni Kothari gives the example of R.S.S. and Jainat-e-Islami of this phenomenon or process. Such organizations declare that they have nothing to do with politics but they go on spreading their tentacles. So that, in course of time, political organizations like parties, trade unions and professional associations like those of students and teachers, even women's organizations, become increasingly dependent on sectarian and communal organizations which are not supposed to be political.

Lumpenisation of politics also ultimately gets linked to communalism. But the role of middle class in communal politics is an alarming phenomenon. Substantial sections of middle class become the torch bearers of communalism and get involved in communal politics as activists or supporters. Rajni Kothari says that the Hindu middle class has, for long, thought of itself as being besieged by the minorities (religious as well as social, for example, the *dalits* and the tribals). The typical attitudes have been, why do we 'pamper' them all the time? What has the State done for us? They have the reservations, they are getting more prosperous, the state helps them in all this; only we are left behind? So that the majority community has acquired a sense of hurt and a sense of being discriminated against by the policies of the Congress party all these years. Meanwhile, a whole lot of 'cultural' Hindu organizations of a militant messianic kind have emerged that aim at bringing to the Hindus a sense of belonging to one united homogenous, centralised entity. Among these are the Vishwa Hindu Parishad, the Hindu Ekta Manch, the Hindu Raksha Samiti and of course of the older organizations like the R.S.S. which were lying low politically for sometime after the murder of Mahatma Gandhi but have been reactivated of late". Even the premier secular party, the Congress party, has also made a dramatic shift to 'soft *Hindutva*' in order to take on the avowedly communal parties. In this scenario, the left failed to apply any restraint on this trend because of their restricted influence. New types of 'opposition' parties or old ones that find a new power base, like the Ittihad-ul-Muslimeen in Hyderabad and the BJP are getting further communalised taking advantage of new levers of power.

FIGHT AGAINST COMMUNALISM

The long term remedy for fighting communalism lies in initiating the process of de-communalising the people at all levels. This means waging a continuous and intense ideological-political struggle against communalism. If communalism is basically an ideology then it cannot be suppressed by force. An ideology is to be fought at the level of ideas. An honest educational and cultural policy alongwith the use of media are required to educate the members of the society towards the danger of communalism. That is why controlling the departments of Education, Home and Information has been central to the scheme of things of the communal political parties whenever and wherever they are able to have their share in running the government.

Sometimes, even the positive factors of development such as literacy may be exploited by an interested ideology and it's practitioners. "Illiteracy, combined with traditional/folk culture protected the mass of Indian people from the ideologies of hate. Literacy has also exposed them to the influence of communalism, obscurantist and irrational ideals purveyed through newspapers, pamphlets and posters; even in children's books communal interpretation of Indian history, particularly of the ancient and medieval periods, forms the bedrock of communal ideology in India". Communalization of law enforcement agencies is becoming a major problem in India today. In a landmark study, V.N. Rai, a senior IPS Officer of Uttar Pradesh cadre - *Perception of Police Neutrality in Communal Riots,* has come out with the partisan role of police personnel (especially at the lower levels in various communal riots. The role of the media is immensely heightened during the course of communal violence. They can greatly contribute in checking the inflamed tempers as well as pouring oil to the raging fire. Communalism has to be fought right from the socialization of children in the families and here parents would be playing a crucial role.

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FUNDAMENTALISM, COMMUNALISM AND GLOBALIZATION

Fundamentalism is frequently used in the same breath as communalism. But fundamentalism is more closely associated with the notion of a religious fanaticism. It is an attempt to go back to the 'fundamentals' of a religious tradition in a glorifying, non-critical and absolutising way. "It is unquestioning faith in the infallability of a doctrine, usually religious, accompanied by a readiness to act in it's defence and/or promotion. The word is most widely used in modern times in relation to Islamic faith. The rise of Hindu militancy with a similar attitude of intolerance has brought the phrase Hindu fundamentalism into current usage" (T.P.G. George, 1998).

Fundamentalism may be seen as a religious movement that emphasizes the absolute truth of essential or 'fundamental' aspects of faith. Fundamentalism is sociologically important not only because of its' unique place among religions, but because it easily extends itself into the political realm. Whatever the religious colouration, fundamentalists are opposed to modern ideas and denounce all forms of intellectual arguments. "Though strictly speaking, fundamentalism is more a characteristic of the religions of the (divine) book like Judaism, Christianity and Islam. However, what can today be observed is an attempt to 'fundamentalise' Hinduism as well, by reducing it to certain basic scriptures like the Vedas, Upanishad, Gita and certain "unifying' myths like the martial and militant version of Ram" (Gabriele Dietrich, 1995).

If we take a close look we find that fundamentalism associates itself with political conservatism and with heavily patriarchal values. At the same time it associates "itself with aggressive forms of capitalism which can be seen among the Christian fundamentalists in the 'electronic church' in the USA" (Dietrich, ibid). Fundamentalism has closer links with religion than Communalism. While communalism today claims to speak the language of secularism (secular jargon) and nationalism, fundamentalism tries to argue about the fundamentals of religion. It may, therefore, at times, open up inter-religious arguments but tends, however, to be frightened of inter - religious dialogue. Under the present pressure of communal politics which creates a feeling of 'Hinduism', 'Islam', 'Christianity' etc. being in danger, fundamentalist forces are gaining ground in all the religious communities. "Occasionally, fundamentalism can also be a response to the uprooting effects, accelerated by capitalist development, as the socially reformist beginnings of Bhindranwale, the present face of Sikh fundamentalism. Jamaat-e-Islami in India represents Muslim fundamentalism because it continues to overemphasise, as core ideology, certain fundamentals of Islam which do not seem to be compatible with modern times. The fundamentalist elements in Pakistan and the *Talibans* of Afghanistan are the worst forms of Muslim fundamentalism threatening to take their societies into medieval or ancient times.

Rise of ethnicity as a political response to globalization is also an important dimension of communalism and fundamentalism. Samir Amin, a powerful voice on the ill effects of globalization on the third world countries, in his major work (1997) comments that the "present epoch is surely characterized by an awakening or reawakening marked by collective identifications... Regionalism, linguistic and cultural assertion, tribal or ethnic loyalities, devolution to a religious group, attachment to a local community, are some of the multiple forms this reawakening has taken". The communalised version of 'Cultural Nationalism' put forward by RSS in India may be seen as the reluctance to adopt anti-imperialist nationalism or 'people-centric' nationalism. "What threatens India, (in their view) is not the imperialist invasion of today but the so called 'Muslim invasion' of medieval times, .not foreign corporate capital but 'foreign religions', arid the demolition of Churches and Mosques is presented as acts of redemption of national honour precisely at a time when the nations' publicly owned assets (and resources) are being sold to foreign investors and domestic capitalists alike" (Aijaz Ahmed, 2002). The attempt to 'cdmmunalise' and 'fundamentalise' the civil society and the state is against the very grain of Indian civilization as well as the process of modernization and should be fought with full force.

COMMERCIALIZATION OF RELIGION

The rapid expansion of markets and market forces have also contributed to the growing communalisation of Indian society. The market forces are pushing out

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spirituality from the religion and transforming religion into a saleable commodity. Significantly, religious festivals in different areas are undergoing a change. They used to take place in a localised manner. There were local rituals, local deities and local, austere, peaceful and simple celebrations. "What gradually one finds emerging is that not only are the festivals becoming longer and longer, their very nature seems to be undergoing a profound change too. *Ganesh Chaturthi* is not observed on a particular day; it becomes *Ganesh Mahotsaua* which goes on for ten, twelve, fifteen days. *Durga Puja* was always celebrated for a few days, but is now being extended, with all commercial interests catching on, with blaring loud speakers all over the place at every street corner, with lumpen-type orators haranguing the crowds. One festival or another is always on... The local deities and local rituals are gradually being replaced by what may be called 'universalised Hinduism'. So that idols of Durga and Ganesh are first *nationalised* and then propagated to replace what used to be local festivals around local deities" (Kothari, ibid). The *Navratri* festival in Gujarat where the *garbha* pandals were transformed into high-tech, pop, commercial centres in which entry fee escalated to hundreds and thousands of rupees may be seen as the most obvious example of this phenomenon. *Moharrum,* a solemn and simple festival of mourning is also being transformed into a commercialised occasion with the entry of high-tech devices such as audio-video cassettes, VCDs and VDDs.

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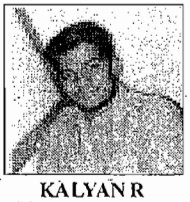
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